

GENDER EQUALITY AND ECONOMIC DEVELOPMENT IN SUB-SAHARAN AFRICA



Editors

LISA KOLOVICH

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Foreword

In recent decades, sub-Saharan Africa has made great progress in addressing gender inequality. Maternal death rates have fallen, education gaps have narrowed, and access to health and finance has generally increased. Many countries have reformed their legal systems, removing or altering a large share of the laws that treat women differently than men. Countries such as Rwanda, South Africa, and Uganda are among the pioneers that adopted gender-responsive budgeting; several more have recently started following their path.

A lot more needs to be done, however. In many countries in the region, gender inequality in its variety of forms is still excessively high. This is, of course, a moral issue. But from an IMF perspective, it also has significant bearing on a country's economic development. For instance: fewer women than men in the formal labor market, more women than men in vulnerable jobs, substantial earnings gaps between men and women—all of these are visibly linked to the economy. Women also face higher barriers to starting a business, in part because of the significant financial and digital divide. Issues outside the labor market equally undermine economic development. For instance, when gender inequalities lead to a high incidence of child marriage or to gender-based violence, they put a heavy toll on women, families, and the well-being of children. They diminish the returns from education and lower productivity by destroying human capital that the region's rapidly growing and young population is counting on to build a prosperous future.

At a time when the traditional engines of growth have weakened globally, and as sub-Saharan Africa grapples with a shock-prone environment amid a significant funding squeeze, promoting gender equality can help boost and sustain inclusive growth over the medium term. It can also help to better address other key challenges of our time, such as climate change, demographic trends, and digitalization.

Despite the oftentimes macrocritical relevance of gender issues, macroeconomic, structural, and financial policies often fail to take gender into account, and gender equality is mostly treated separately from core macroeconomic issues. This is a gap that needs to be addressed.

In 2022, the Fund adopted a gender strategy to precisely ensure that, when macrocritical, gender issues are integrated in our core macroeconomic policy analysis.

We are committed to supporting our 190 member countries in designing policies with a gender lens. This includes, for instance, implementing expenditure and tax policies to achieve better macroeconomic outcomes and narrow gender gaps, incorporating gender into public financial management, and removing legal obstacles to women's economic empowerment. By integrating gender into core macroeconomics, we can better complement the efforts of many

partners that are working towards gender equality. This book showcases some of the emerging analytical and policy work, with a focus on sub-Saharan Africa, that can help in policymaking.

Sub-Saharan Africa's development and sustainable growth would benefit from women's further empowerment. The stakes are high, but history shows that, with sustained commitment, substantial change is possible.

Antoinette M. Sayeh
Deputy Managing Director
International Monetary Fund

Acknowledgments

As this book is released, we are looking back on more than 10 years of formal and informal collaboration on gender with each other. It's not always the case that colleagues become good friends, but when they do, a world of opportunity presents itself. For us, this opened doors to being part of the institutional change that has now brought gender to the IMF's core work. So, first of all, high five to us!

More seriously, we are grateful to all the brilliant colleagues who were trailblazers at the IMF before the approval of the IMF's first Gender Strategy—Kalpana Kochhar, Daria Zakharova, Stefania Fabrizio, Sonali Jain Chandra, Rasmané Ouedraogo, Carolina Rentería Rodríguez, Vincent Tang, Priyanka Chaturvedi, to name only a few—and those who have elevated the case for the IMF's role on gender and macro, ensuring that it is enshrined and implemented institutionally—Rishi Goyal, Ratna Sahay, Valentina Flamini, and Anna Fruttero, among many others across the institution under the great leadership of our Managing Director Kristalina Georgieva and Deputy Managing Director Antoinette M. Sayeh. We thank our dedicated colleagues in the Inclusion and Gender Unit in the Strategy, Policy and Review Department—your passion in moving the needle motivates us every day.

We would like to thank the leadership at the African Department, in particular David Robinson and Cathy Pattillo, for spearheading the work in the region. The chapters in this book are evidence of the enthusiasm with which teams have approached tackling gender gaps in sub-Saharan Africa. The chapters also show how much more work is needed. We count on you, our readers and colleagues, to take these lessons forward.

This book reflects the hard work of many colleagues inside and outside the IMF, and we are grateful to each of them. We are fortunate to work with a growing cadre of mission chiefs, desk economists, directors of IMF training and capacity development centers, and technical experts—all of whom are dedicated to examining gender issues. Our external partners and colleagues offer a wealth of expertise and have warmly welcomed collaboration and innovation opportunities.

Our outstanding colleagues in the IMF Communications Department have kept us on track and done a fantastic job supporting us throughout the process. Many thanks in particular to Lorraine Coffey and Patricia Loo for their patience and guidance.

Finally, we would like to thank our families and cover several generations at once: Monique would like to thank her (great)-grandmothers for being an inspiration (you know what you did). Lisa would like to tell her daughter that her unique perspective, resilience, and, most of all, kindness are gifts to be celebrated (and that half of those come from her dad).

Lisa Kolovich and Monique Newiak

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Abbreviations

AE	advanced economies
AFR	African Department
AIDS	acquired immunodeficiency syndrome
APD	Asia and Pacific Department
BAZ	BMI-for-age z-score
BMI	body mass index
CEDAW	Convention on the Elimination of Discrimination Against Women
CLRA	Communal Law Reform Act
CRED	Centre for Research on the Epidemiology of Disasters
DAC	Development Assistance Committee
DHS	Demographic and Health Survey
EAP	East Asia and Pacific
ECA	Europe and Central Asia
ECOWAS	Economic Community of West Africa States
EM-DAT	Emergency Events Database
EMDC	emerging markets and developing countries
EMDE	emerging markets and developing economies
EME	emerging market economies
EU	European Union
EUR	European Department
FAO	Food and Agriculture Organization of the United Nations
FAS	Financial Access Survey
FE	fixed effects
FEMNET	African Women's Development and Communications Network
G7	Group of Seven
G20	Group of Twenty
GB	gender budgeting
GBI	Gender Budgeting Index
GBS	Gender Budget Statement
GDI	Gender Development Index
GDP	gross domestic product
GGI	Gender Gap Index
GIA	gender impact assessments
GII	Gender Inequality Index
GLS	generalized least squares
GMM	generalized method of moments
GMO	Gender Monitoring Office
GSMA	Global System for Mobile Communications
HCI	Human Capital Index

HIC	high-income countries
HIV	human immunodeficiency virus
IBFD	International Bureau for Fiscal Documentation
ICT	information and communication technology
ICU	International Communication Union
IFC	International Finance Corporation
ILO	International Labour Organization
ILOSTAT	International Labour Organization Department of Statistics
IMF	International Monetary Fund
IPU	Inter-Parliamentary Union
IPUMS	Integrated Public Use Microdata Series
ISO	International Organization for Standardization
IV	instrumental variable
LAC	Latin America and the Caribbean
LFS	Labor Force Survey
LIDC	low-income and developing countries
LSL	Lesotho Loti
MCD	Middle East and Central Asia Department
MENA	Middle East and North Africa
MSME	micro, small, and medium enterprise
N1	Nigeria Naira
NBER	National Bureau of Economic Research
NEET	not in education, employment, or training
NIDS	National Income Dynamics Survey
NST	National Strategy for Transformation
ODI	Overseas Development Institute
OECD	Organisation for Economic Co-operation and Development
OLS	ordinary least squares
PEPUDA	Promotion of Equality and Prevention of Unfair Discrimination Act
PPP	purchasing power parity
SACU	South Africa Countries Union
SADC	Southern African Development Community
SALDRU	Southern Africa Labour and Development Research
SAS	South Asia
SDG	Sustainable Development Goal
SIGI	Social Institutions and Gender Index
SME	small and medium enterprise
SSA	sub-Saharan Africa
STEM	science, technology, engineering, and mathematics
SWAPO	South West Africa People's Organisation
UEA CRU	University of East Anglia Climate Research Unit
UK	United Kingdom
UN	United Nations
UNAIDS	Joint United Nations Programme on HIV/AIDS

UNDP	United Nations Development Programme
UNDESA	United Nations Department of Economic and Social Affairs
UNESCO	United Nations Educational, Scientific, and Cultural Organization
UNFPA	United Nations Fund for Population Activities
UNFCCC	United Nations Framework Convention on Climate Change
UNICEF	United Nations Children's Fund
USAID	United States Agency for International Development
VAT	value-added tax
WAEMU	West African Economic and Monetary Union
WAZ	weight-for-age z-score
WBES	World Bank Enterprise Survey
WBL	Women, Business and the Law
WDI	World Development Indicators
WEF	World Economic Forum
WEO	<i>World Economic Outlook</i>
WHD	Western Hemisphere Department
WHO	World Health Organization
XOF	West African CFA Franc

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Introduction: How Gender Equality Can Shape Sub-Saharan Africa's Development

Lisa Kolovich, Monique Newiak, and Catherine Pattillo

The vast expanse of sub-Saharan Africa is filled with remarkable diversity. Ancient cultural traditions from a multitude of ethnic groups blend with modern technologies. Hundreds of languages can be heard from the coasts of Madagascar to the lush rainforests of the Congo Basin to the sweeping vistas of the Atlantic from Senegal, Sierra Leone, and São Tomé and Príncipe. A burgeoning youth population brimming with an entrepreneurial spirit builds on these foundations, experiences, and wisdom from parents, grandparents, and ancestors, and is working towards a prosperous future.

As this book goes to press, however, the continent is still waiting for a “light on the horizon” (IMF 2023). Sub-Saharan African economies are emerging from a multitude of shocks, including the COVID-19 pandemic, which have been a setback to the region's economic development. They are still grappling with economic scarring from the pandemic and the ongoing repercussions of several external conflicts that have put pressure on domestic prices and exchange rates, while fiscal and external buffers to deal with new shocks are generally low.

While economic growth for the region is set to rebound, additional policy efforts are crucial to secure a sustainable increase in per capita incomes and make progress towards the United Nations' Sustainable Development Goals (SDG). Sub-Saharan African economies are not alone in their quest for growth—worldwide, policy makers are looking for new sources of sustainable, inclusive, and increasingly green growth (Sayeh, Badel, and Goyal 2023), especially since traditional sources of growth, such as trade, while still critical, may not be sufficient to produce reliable growth and prosperity dividends going forward. In developing economies, there is a concern whether attainment of the SDGs by 2030 is feasible.

In this book, 20 chapters—written by staff at the IMF, the World Bank, UN Women, and others—make the case that empowering women and girls and putting a gender lens on policies to tackle the significant challenges of our times will be critical to the region's economic development.

GENDER EQUALITY—AN ENORMOUS UNTAPPED SOURCE OF PROSPERITY

The region's prospects must be seen in the context of how equal—or unequal—opportunities and outcomes have been for the main share of the population.

The good news is that sub-Saharan African economies have made tremendous progress on the SDGs, including fostering gender equality (SDG 5), that cuts across all other development goals. Progress has been significant in eliminating gender inequality in legal rights, increasing education for both boys and girls, improving health outcomes for children and mothers, and boosting financial access and digitalization. Several sub-Saharan African countries, including Rwanda and Uganda, have been best practice examples on how to mainstream gender into public financial management via gender budgeting—a practice that incorporates a gender lens into public financial management aimed at improving gender equality—and others are increasingly claiming this field (Tanzania, Togo, Sierra Leone).

However, long-standing gender inequality persists, and recent shocks have exposed vulnerabilities. By many composite measures, gender equality in sub-Saharan Africa remains high compared to other regions (Chapter 1). Maternal mortality rates significantly exceed global averages. Adolescent pregnancies are frequent, with health risks for both mothers and children, and can result in higher school dropout rates for girls. The region has the lowest expected years of schooling compared to other regions, with a significant gender gap in literacy, and still substantial gaps in educational attainment in many countries. There are significant gender gaps in financial inclusion (Chapter 2), and a substantial digital divide, with women less likely to use mobile internet or own mobile phones (Chapters 18 and 19). Many laws that are “on the books” discriminate on the basis of gender (Chapters 12 and 13). The incidence of gender-based violence remains high (Chapter 6), with one-third of ever-partnered women in the region having experienced violence in their lifetime. Harmful practices, such as child marriage and female genital mutilation, have yet to be eradicated (Chapter 7). Women and girls are overrepresented in informal and vulnerable employment, and are shouldering disproportionately the responsibilities in providing paid and unpaid care services (Chapter 17). The COVID-19 pandemic has exposed some of these vulnerabilities and disproportionately impacted women in many countries (Chapters 4 and 5).

The root causes of this inequality are complex, interwoven with socioeconomic challenges, cultural and social norms, long-standing laws on the books, political factors, and at times distortionary policy. More often than not, macroeconomic and other policies remain blind to their gendered impacts.

Gender equality has enormous macroeconomic consequences—addressing them will benefit growth, economic and financial stability, a more equal income distribution, and other development outcomes (Kochhar, Jain-Chandra, and Newiak 2017). A long-standing vast literature highlights that gender equality can boost GDP per capita (Ostry and others 2018; Cuberes and Teignier 2016, 2018), benefit economic growth and productivity (IMF 2015, Chapter 11;

Hakura and others 2016), lower income inequality (Gonzales and others 2015), support economic diversification in low-income contexts (Kazandjian and others 2019), and foster financial stability (Sahay and Čihák 2018), firm profitability (Christiansen and others 2016), and development outcomes more broadly (Duflo 2003, 2012).

This book builds on this literature to highlight specific gains from gender equality in outcomes that are important for sub-Saharan Africa. With the incidence of gender-based violence among the highest in the world, Chapter 6 argues that a decrease in violence against women by 1 percentage point is associated with a significant 9 percent increase in nightlights per capita, a proxy for economic activity. Ending child marriage today would result in a more than 1 percentage point increase in the long-term annual per capita real GDP growth in emerging and developing countries (Chapter 7). Chapter 8 shows that closing gender gaps in education for different income groups could yield GDP gains of 5 to 11 percent in West African economies, while boosting formality and revenue mobilization. Economically empowering women is also one of the most impactful means for building the human capital of the next generation, with a strong association between women's empowerment and control of resources with children's health, especially in households with less resources (Chapter 9). Chapter 10 guides the reader through the reasons why gender equality can also provide for peace in the region, including by reducing electoral violence.

INCLUDING A GENDER LENS IN POLICYMAKING TO BOOST REFORM IMPACT

There is a significant body of evidence on which macroeconomic and structural policies can promote gender equality (IMF 2022). While the specifics of policies may differ across countries, this book sheds light on the key areas in which areas policymakers can reap quick wins to set the foundation for equality and where they can leverage existing policies and frameworks to integrate a gender lens into the policy design. To focus the discussion on the areas in which the IMF has built expertise, we collect evidence and policy examples around the areas of legal reform, public financial management, and fiscal policy, while touching more briefly on other reform areas.

Fair Play: Equal Laws and Women's Economic Empowerment

Creating a fair system of rules is critical. Chapters 12 and 13 call attention to how legal frameworks can serve as potent instruments for enshrining equal opportunities, shifting societal perceptions of gender roles, and ensuring accountability for achieving gender equality outcomes. The importance of adopting laws that promote gender equality in labor markets; access to finance and assets, including through equal inheritance rights; and decision-making roles, and that disincentivize and eliminate gender-based violence should not be underestimated. Legal frameworks provide a binding foundation for implementation and enforcement. Five case studies from the region—Rwanda, Namibia, South Africa, Mauritius,

and Cabo Verde—countries that have achieved some of the most gender-equal outcomes in the region on a range of dimensions—provide examples of how legal reforms have been implemented across the region.

Effective legal frameworks for public financial management can also play a crucial role in facilitating gender-balanced fiscal policies (Chapter 14). Legal reforms can introduce gender-budgeting practices, including gender-balance principles, gender-budget statements, and gender-impact assessments, into the budgetary system. These legal provisions can strengthen the mandate of audit institutions in conducting gender-related audits and establish enforcement mechanisms. Promoting gender equality by applying a gender perspective to fiscal policies and the budget process, a practice known as *gender budgeting*, can help address gender inequality concerns in various aspects of the economy and society. This approach ensures that laws, policies, and budgets promote the well-being and economic participation of women.

Large Role for Fiscal Policy

There is a significant role for tax and expenditure policy to tackle gender equality in the region (Chapter 15). Gender-responsive tax policies can support women's economic empowerment, and policymakers should more generally aim to promote formal employment and ensure that revenue mobilization contributes to funding social programs that foster gender equality. There are still countries in sub-Saharan Africa where there is explicit gender discrimination in the tax code, mostly in labor income taxation. Such biases need to be eliminated. Personal income tax reforms should focus on progressivity and individualization (that is, taxing individuals' incomes rather than families' incomes), while capital income taxes, which predominantly benefit men, could be raised by abolishing preferential tax treatments. A single-rate value-added tax and sales-tax system with minimal exemptions can enhance revenue mobilization and fund social programs for gender equality. Excise taxes, when set to correct for externalities, for instance, to disincentivize the consumption of harmful goods and services, can reduce gender inequality. They can also mitigate climate-induced income shocks on women.

The role for a gender lens in expenditure policy is also high. For instance, policymakers aiming to enhance both access to and the quality of education should also consider designing gender-responsive complementary policies addressing social assistance, infrastructure development, and school dropout rates, particularly for girls facing pregnancy-related challenges. Providing access to quality care services, increased public investment in social care and rural infrastructure, and a redistribution of care responsibilities can support greater equality and women's economic empowerment.

Gender budgeting is critical in the context of limited fiscal resources, rising debt pressures, and exacerbated gender disparities (Chapter 16). It not only showcases a government's dedication to promoting gender equality but also enhances the effectiveness of resource allocation through improved design of responsive fiscal policies. Sub-Saharan African countries, including regional leaders like Rwanda,

South Africa, and Uganda, were early adopters of gender budgeting and have introduced many innovative practices. While lessons from advanced economies, such as Canada, highlight that fully adopting gender budgeting takes time, incremental steps towards the adoption, for instance by starting the work with pilot ministries, is useful and can drive the reform process.

In all these efforts, working with reliable data is essential. For example, empirical work shows that epidemics can have a disproportionate, significant, and lasting negative impact on girls' education (Chapter 3). This finding underscores the importance of collecting gender-disaggregated data to monitor and tailor policies effectively. New data collection efforts, and currently still-infrequent surveys, should be set up with a gender lens. Moreover, collecting higher-frequency data, such as monthly or quarterly data, can provide more nuanced insights into where gender gaps are occurring and what are the impacts of a gender-responsive policy response. High-frequency data, for instance collected from mobile money transactions, can improve the understanding of gendered economic patterns.

EMPOWERING WOMEN, TACKLING MEGATRENDS

The recent experience with a global pandemic and megatrends, such as digitalization and climate change, which are also closely interlinked with gender gaps and gendered outcomes, demand a concerted and tailored effort around transformative fiscal, financial, and structural policy measures.

The COVID 19 pandemic has exposed many issues with unpaid work, which is often undervalued and disproportionately falls on women and girls, with consequences for their economic empowerment (Chapter 17). Transforming social norms and reducing unpaid caregiving burdens on women are essential steps toward achieving gender equality and sustainable development in the face of global challenges like climate change and economic shocks. In this context, tackling long-standing cultural norms and traditions is crucial for achieving a gender-equal society, but this requires patience and persistence. Programs may focus on role modeling and changing stereotypes, but they must engage men and boys and partners on many fronts, such as traditional leaders, for these efforts to succeed.

Financial technology, or "fintech," has had a positive impact on women's welfare, increasing both the number of female employees in the workforce and the female-to-male employee ratio (Chapter 18). However, the positive effects of fintech are more pronounced in countries with strong governance and institutional quality so that addressing the digital divide and promoting good governance, laws, and regulations are essential steps to maximize fintech's impact on gender equality. One promising avenue for increasing female political representation is through boosting digital access, such as through internet access and social media use, which is correlated with more favorable perceptions of women's electoral chances in sub-Saharan Africa (Chapter 19).

Chapter 20 shows that, under strong and timely policies, women can benefit from the green transition in sub-Saharan Africa. As policymakers are designing

appropriate policies to encourage a transition to the green economy, they should apply a gender lens to so that women can benefit from emerging green jobs. The large female entrepreneurial capacity, in turn, could be a strong force of innovation, an incentive for making climate and gender concurrent, not competing, goals. At the same time, policymakers should be attentive to unintended consequences of well-intentioned policies. For instance, they should consider the distributional impact along gender and income lines of increased public transportation, carbon pricing, and taxes. When well designed, reforms can concurrently advance gender equality and protect the vulnerable.

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PART I

Where Things Stand: Gender Equality in Sub-Saharan Africa

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Revisiting Trends in Gender Equality

Chiara Broccolini, Anna Fruttero, and Saanya Jain

Sub-Saharan Africa has significantly reduced gender inequality over the past decades. However, the region still grapples with various dimensions of gender inequality, posing ongoing challenges. Progress has been significant in narrowing gender gaps in education, especially at the primary and secondary levels, and improving health outcomes for women and girls. Nonetheless, the COVID-19 pandemic has exposed the fragility of prior gains, emphasizing the need for continued efforts. Significant challenges remain in many areas—an inclusion issue and a key detriment to achieving the Sustainable Development Goals and better macroeconomic outcomes. This chapter sets the stage for the remainder of the book by providing an overview of the landscape of gender inequality in terms of opportunity, outcomes, and representation.

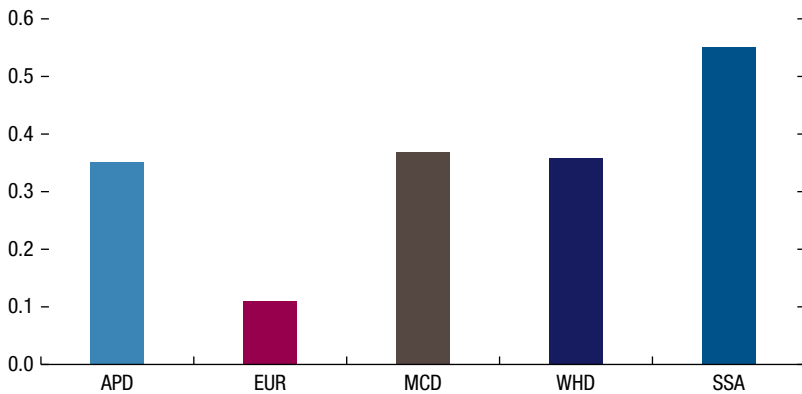
INTRODUCTION

This book makes the case that gender equality is a crucial development goal and that closing gender gaps will support favorable macroeconomic outcomes and long-term economic and social goals. Understanding the potential of closing gender gaps in sub-Saharan Africa requires understanding the types of gaps the region still faces.

The statistics presented in this chapter draw on rich data collected or made available by many institutions, such as the International Labour Organization (ILO), the World Bank (including through its World Development Indicators; Findex; and Women, Business and the Law), and the World Health Organization, as well as indices developed by the Organisation for Economic Co-operation and Development (OECD) and the United Nations. Increasing availability of this information at the country level helps provide a stocktaking of gender gaps in sub-Saharan Africa and lays the foundation for macro-gender analysis. The findings motivate the regional and country analysis in the remainder of the book and the call for action for policies to close the gaps.

Where does sub-Saharan Africa stand overall? Looking at gender indices offers a sense of gender inequality in sub-Saharan Africa (see Annex 1.1). The Gender

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Figure 1.1. Gender Inequality Index, 2021*(Index, scale 0–1)*

Source: United Nations Development Programme 2021b.

Note: The Gender Inequality Index measures gender disparities in health, empowerment, and the labor market. It ranges from 0, where women and men fare equally, and 1, where one gender fares as poorly as possible in all the measured dimensions. APD = Asia and Pacific; EUR = Europe; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Inequality Index for 2021—which measures inequality of opportunity through education and health gaps, labor market outcomes, and inequality of representation—suggests that gender inequality in sub-Saharan Africa remains substantial in absolute terms and compared to other regions (Figure 1.1).¹ In particular, women’s labor force and political participation in sub-Saharan Africa lag behind those in other regions. Moreover, women face challenges in achieving their full human development potential due to high levels of maternal mortality, adolescent fertility rates, and limited access to secondary education.

The Global Gender Gap Index from the World Economic Forum—capturing economic participation, wage equality, educational attainment, health and survival, and political empowerment—provides a similar assessment (Annex Table 1.1.2).² Sub-Saharan Africa bridged just 67.9 percent of its gender gap and surpassed the Middle East, North Africa, and South Asia. According to the World Economic Forum, it would take the region 98 years to close the gender gap at the current rate of progress.

¹ A low Gender Inequality Index value indicates low inequality between women and men.

² The Global Gender Gap Index uses a 0-to-100 scale. The scores can be interpreted as the distance covered toward parity (this is the percentage of the gender gap that has been closed). It has four main components: Economic Participation and Opportunity (labor force participation rate; wage equality for similar jobs; estimated earned income; legislators, senior officers, and managers; professional and technical workers); Educational Attainment (literacy rate, enrollment in primary education, enrollment in secondary education, enrollment in tertiary education); Health and Survival (sex ratio at birth, healthy life expectancy), and Political Empowerment (women in parliament, women in ministerial positions, years with female/male head of state) (World Economic Forum 2022).

The OECD's Social Institutions and Gender Index offers a more comprehensive perspective by assessing both legal and practical forms of discrimination that women encounter across different facets of life. It illustrates how ingrained social structures continue to undermine women's empowerment.³ The overall index score in sub-Saharan African countries is higher than the global average, indicating a higher level of gender inequality.

The remainder of this chapter explores gender inequality in the region in terms of three dimensions: inequality of opportunity, inequality of outcomes, and inequality of representation.

INEQUALITY OF OPPORTUNITY

Gender gaps persist in various domains, including access to opportunities, education, health care, and finance. Violence against women and traditional practices remain significant challenges. These gaps and barriers, influenced by legal systems, societal norms, and sometimes discriminatory policies, are still pronounced in sub-Saharan Africa, often surpassing those observed in other regions. Despite progress in some areas, significant disparities remain. This section provides an overview of the gender landscape of critical opportunities.

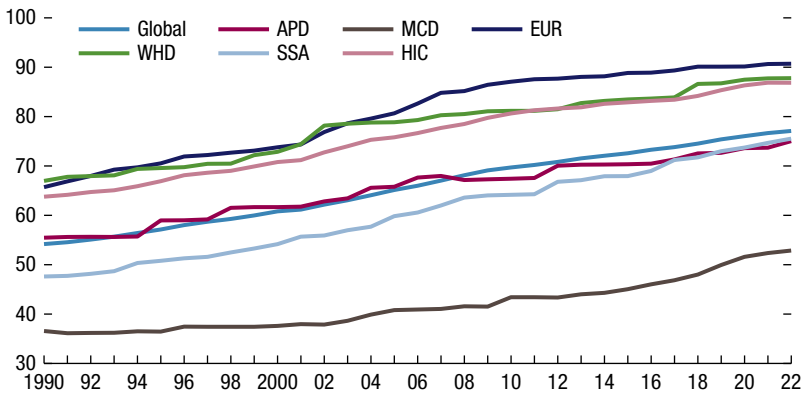
Legal Rights

Robust legal frameworks and institutions protecting women's economic rights are essential for promoting economic empowerment and economic development, as discussed in Chapters 12 and 13.⁴ However, the most recent data from Women, Business and the Law (2023) reveal that globally, women possess only three-quarters of the economic rights enjoyed by men. This gap means that a staggering 2.4 billion

³ The OECD Development Centre's Social Institutions and Gender Index (SIGI) measures discrimination against women in social institutions in 179 countries. By considering laws, social norms, and practices, the SIGI captures the underlying drivers of gender inequality to provide the data necessary for transformative policy change. A value of 0 indicates no discrimination, and 100 indicates absolute discrimination. The SIGI is also one of the official data sources for monitoring the Sustainable Development Goals (SDGs) 5.1.1: "Whether or not legal frameworks are in place to promote, enforce and monitor gender equality and women's empowerment." The SIGI covers four dimensions of discriminatory social institutions, spanning major socio-economic areas that affect women's lives: discrimination in the family, restricted physical integrity, restricted access to productive and financial resources, and restricted civil liberties (OECD 2022).

⁴ Evidence has shown that improvements in legal equality between women and men are associated with many positive outcomes across many domains of gender equality. For example, research has documented a link between legal equality and women's educational attainment (see, for example, Deininger and others 2019; Harari 2019), women's health outcomes (Anderson 2018; Harari 2019), women's participation in the labor force (Hyland, Djankov, and Goldberg 2020), women's employment in the formal sector (Hyland, Djankov, and Goldberg 2021), better jobs for women (Islam, Muzi, and Amin 2019), strengthened women's intra-household bargaining power and decision-making (Voena 2015; Harari 2019; Haldar and Stiglitz 2013; Heath and Tan 2020), female entrepreneurship (Paoloni and Lombardi 2020; Strawser, Hechavarría, and Passerini 2021), and women's political representation (Hyland, Djankov, and Goldberg 2021).

Figure 1.2. Legal Barriers (1990–2022)
(Score, scale: 1–100)



Source: World Bank 2023.

Note: The figure reports the Women, Business and the Law Index score, which measures how laws and regulations affect women's economic opportunities. Overall scores are calculated by taking the average score of each index (mobility, workplace, pay, marriage, parenthood, entrepreneurship, assets, and pension), with 100 representing the highest possible score. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

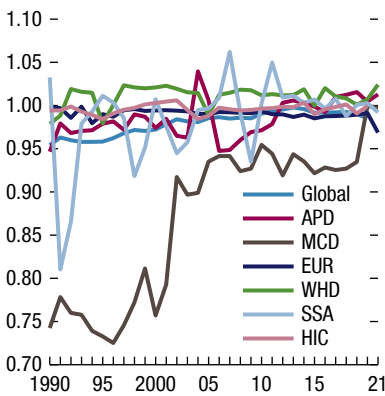
working-age women lack the same legal rights as men. When analyzing the global landscape, economies in Europe, Central Asia, Latin America, and the Caribbean exhibit scores above the global average of 77.1. In contrast, the Middle East, North Africa, and South Asia exhibit the lowest average scores, indicating the need for further improvement in legal frameworks and institutions to enhance women's economic rights in these regions.

Where does sub-Saharan Africa stand? Encouragingly, sub-Saharan Africa accounts for more than half of all reforms in legal systems introduced in 2022. Seven economies have enacted 18 positive legal changes. Remarkably, for the first time in 53 years, two economies in sub-Saharan Africa have achieved scores above 90.⁵ Consequently, the region's average score now surpasses that of the Asia and Pacific region (Figure 1.2). Despite progress in de jure-based discrimination over time, women are granted, on average, less than three-quarters of the legal rights afforded to men, as indicated by the average score of 72.6 in 2022 (Figure 1.2). In the African region, progress across the topic areas measured has been uneven, with most reforms issued in domains that historically lagged behind. The catch-up effect has been strongest in the laws affecting equal treatment in the workplace, parenthood, and pay indicators.⁶

⁵ See Elefante and others 2023.

⁶ Workplace score for sub-Saharan Africa increased from 17.2 percent in 1990 to 80.5 in 2022. Similarly, parenthood and pay indicators moved respectively from 20 percent and 38 percent in 1990 to 45 and 64 percent in 2022.

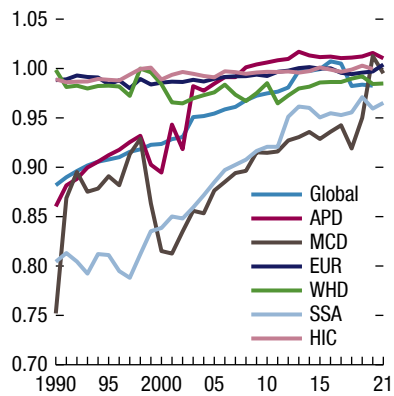
Figure 1.3. Gender Parity in Preprimary Education, 1990–2021
(Index)



Source: World Bank 2021b.

Note: Gender Parity Index for gross enrollment is the ratio of girls and boys enrolled in preprimary levels in public and private schools. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Figure 1.4. Gender Parity in Primary Education, 1990–2021
(Index)



Source: World Bank 2021b.

Note: Gender Parity Index for gross enrollment is the ratio of girls and boys enrolled in preprimary levels in public and private schools. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Despite improvements in the past 20 years, the parenthood⁷ and pay⁸ indicators still have the lowest scores in the region, revealing stubborn legal barriers to more and better jobs for women.

Education

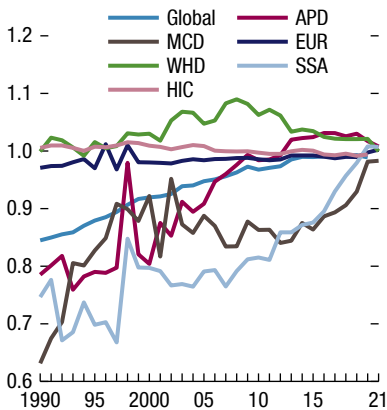
In the past decade, significant progress has been made in achieving gender equality in education, although global improvements conceal notable regional disparities. Access to preprimary education has reached parity (Figure 1.3). There has been a substantial increase in the proportion of African girls receiving primary education and improvements in the gender parity index since 1990 (Figure 1.4) and in secondary education since 2000 (Figure 1.5). However, the primary and secondary education gender parity indices for sub-Saharan Africa still range from 0.8 to 1.6 for primary and 0.6 to 1.3 for secondary.

Despite these gains, girls face more barriers than boys when accessing tertiary education. While sub-Saharan Africa has witnessed rapid progress in tertiary

⁷ The parenthood indicator examines laws affecting a woman's work during and after pregnancy, including paid leave, and laws prohibiting firms from dismissing workers because they are pregnant.

⁸ The pay indicator examines whether laws are in place to ensure equal remuneration between men and women for work of equal value and whether they allow a woman to work at night, in industrial jobs, and in jobs deemed dangerous in the same way as a man.

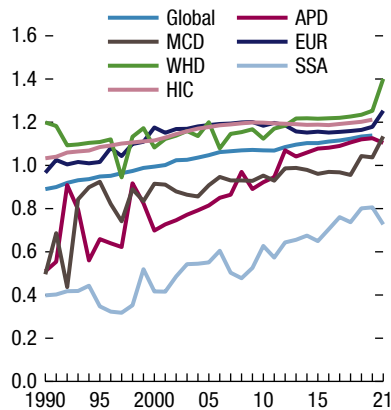
Figure 1.5. Gender Parity in Secondary Education, 1990–2021
(Index)



Source: World Bank 2021b.

Note: Gender Parity Index for gross enrollment is the ratio of girls and boys enrolled in preprimary levels in public and private schools. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Figure 1.6. Gender Parity in Tertiary Education, 1990–2021
(Index)



Source: World Bank 2021b.

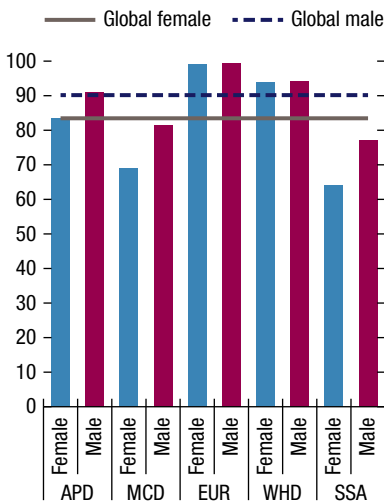
Note: Gender Parity Index for gross enrollment is the ratio of girls and boys enrolled in preprimary levels in public and private schools. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

education enrollments, with the gender parity index increasing from 0.40 in 1990 to 0.73 in 2021, it is the only region where women do not have an advantage over boys (Figure 1.6). The gender parity index in tertiary education varies across countries, ranging from 0.3 to 1.7.

Sub-Saharan Africa also has the lowest expected years of schooling compared with other regions, with 9.2 years for women and 8.6 years for men as of 2020–21.⁹ Moreover, while Europe and the Western Hemisphere have almost achieved universal and equal adult literacy, and Asia and the Pacific are making progress in closing the gender gap, sub-Saharan Africa, along with the Middle East and Central Asia, significantly lag at 64 percent for women and 77 percent for men as of 2020, reporting the widest gender gaps. Again, within sub-Saharan Africa, there is substantial variation along these two dimensions: expected years of schooling for girls ranges from 5.7 to 12.5, and only 40 percent of countries report an adult female literacy rate above 75 percent (Figures 1.7 and 1.8). Additionally, sub-Saharan Africa exhibits a significantly larger proportion of population between the ages of 15 and 24 who cannot read and write with

⁹ Expected years of schooling is the number of years a child entering school age is expected to spend at school or university, including years spent on repetition (World Bank 2021b).

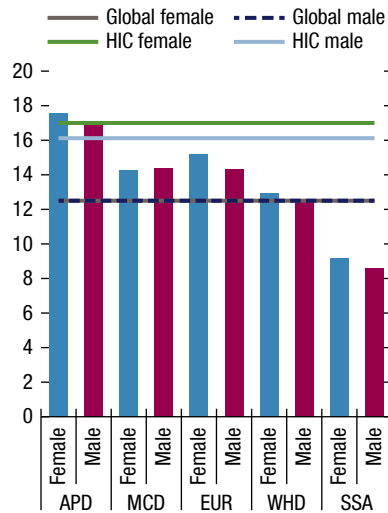
Figure 1.7. Adult Literacy Rate, 2020–21
(Percent)



Source: World Bank 2021b.

Note: Adult literacy rate is the percentage of people aged 15 or above who can both read and write with understanding a short simple statement on their everyday life. APD = Asia and Pacific; EUR = Europe; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Figure 1.8. Expected Years of Schooling, 2020–21



Source: World Bank 2021b.

Note: Expected years of schooling is the number of years a child of schooling entrance age is expected to spend at school or university, including years spent in repetition. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

comprehension compared with the global average, and it demonstrates a higher prevalence of illiteracy among women within this youth demographic (21 percent for young men and 26 percent for young women, compared with 7 and 9 percent, respectively, worldwide). The lower years of schooling and literacy rates contribute to the persistence of inequality between men and women, as laid out in more detail in Chapter 8.

School closures due to COVID-19 threatened progress made in education, with girls expected to be at risk of paying the highest toll (Chapter 3). UNESCO 2020 projected that worldwide 11.2 million girls and young women were expected to be at risk due to the COVID-19 pandemic and UNICEF 2021 estimated that up to 10 million more girls will be at risk of becoming child brides because of the pandemic. Several studies also expressed concern that, in the context of developing countries, adolescent girls' attendance would be more negatively affected than boys,¹⁰ and UNESCO 2021 argued that girls were expected to be more strongly

¹⁰ See, for example, Azevedo and others 2020 and Giannini and Albrechtsen 2020.

affected by the increased burden of unpaid work within households while boys were expected to participate more in income-generating activities to substitute for pandemic-related reductions in household income.

While evidence continues to emerge, according to the World Bank (2023b), there has not been an increase in gender gaps in education during the pandemic in sub-Saharan Africa. In fact, these gaps have narrowed. However, the findings indicate that COVID-19 may have had extensive effects on education, affecting factors that typically play a role in school attendance, such as parental education and socioeconomic status. This suggests that the economic aftermath of the pandemic could have increased vulnerability among households, making it challenging for them to provide the necessary resources to support their children's education.

Health

Regarding health indicators, sub-Saharan Africa faces more significant challenges than other regions in areas such as maternal, infant, and under-5 child mortality rates, fertility rates, and adolescent fertility rates.

Maternal mortality rates improved, declining from 792 per 100,000 live births in 2000 to 511 in 2020.¹¹ Nevertheless, the region still exhibits alarmingly high levels of maternal mortality compared with the global average of 223 and it remains five times higher than in high-income countries, where the rate is only 12 per 100,000 live births.¹² The primary causes of maternal deaths are obstetric hemorrhage, hypertensive disorders during pregnancy, non-obstetric complications, and pregnancy-related infections. However, deaths resulting from incidental causes may not be accurately reported in many countries in the region (Musarandega and others 2021).

Between 1990 and 2021, infant and under-5 mortality rates have significantly decreased for both boys and girls. However, rates remain 10 times higher than those observed in high-income countries. In 2021, the female infant mortality rate was 43 per 1,000 live births, while for males it was 53. These rates starkly contrast with those in high-income countries, which stood at 3.8 and 4.5, respectively (Figure 1.9). Furthermore, in 2021, the probability of a newborn not surviving the age of in sub-Saharan Africa was 15 times higher than in high-income countries, and gender disparities persisted. Boys were at a higher risk than girls, with mortality rates of 75 and 64, respectively, compared to 5 for boys and 4 for girls in high-income countries (Figure 1.10).

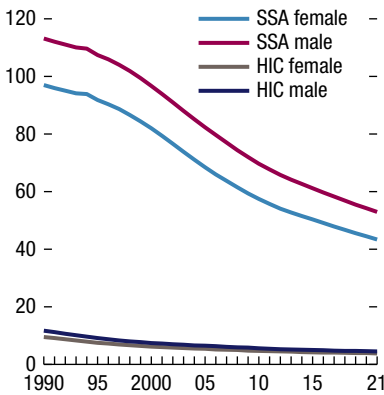
Adolescent girls who become pregnant face elevated health risks, including a higher probability of maternal death, childbirth complications, and low birth

¹¹ Sources: WHO, UNICEF, UNFPA, World Bank Group and UNDESA/Population Division. Retrieved from the World Bank Gender Data Portal.

¹² In 2020, the maternal mortality rate was 74 deaths per 100,000 live births in East Asia and Pacific, and 13 in Europe and Central Asia.

Figure 1.9. Infant Mortality Rate, 1990–2021

(Number of deaths per 1,000 live births)

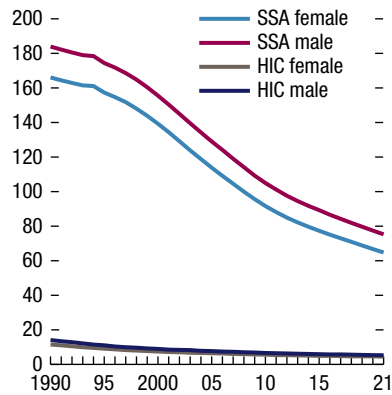


Source: World Bank 2021b.

Note: Infant mortality rate is the number of infants dying before reaching one year of age per 1,000 live births. HIC = high-income countries; SSA = sub-Saharan Africa.

Figure 1.10. Mortality Rate of Children under Age 5, 1990–2020

(Number of deaths per 1,000 live births)



Source: World Bank 2021b.

Note: Under-5 mortality rate is the probability per 1,000 that a newborn baby will die before reaching age 5. HIC = high-income countries; SSA = sub-Saharan Africa.

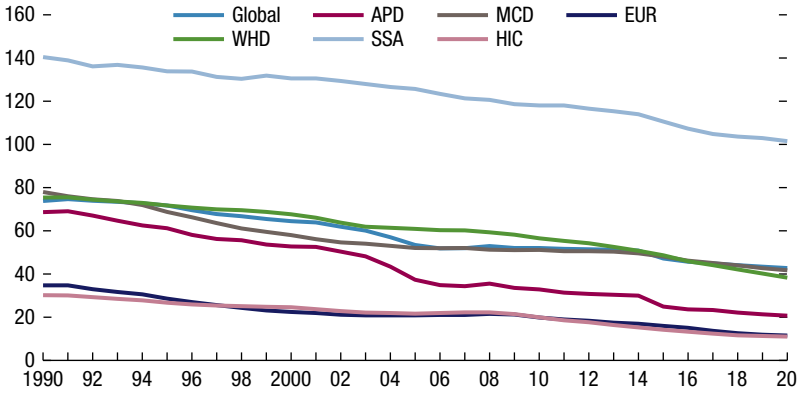
weight for their babies. They also endure social stigma and often have limited opportunities to access higher education and employment. Furthermore, adolescent pregnancies can have intergenerational effects, increasing the risks of teenage pregnancies in future generations. In 2020, teenage fertility rates in sub-Saharan Africa, while displaying some decline, remained the highest among all regions, with 101.6 births per 1,000 women aged 15 to 19. This is more than twice the global average of 42.7 births per 1,000 adolescents and nearly 10 times higher than in high-income countries (Figure 1.11). There are some success stories: Ghana has made significant progress in reducing these births over time, and other countries have managed to halve their rates within 40 years. It is also worth highlighting the correlation between teenage pregnancy and school dropout rates. In countries with the highest rates of adolescent fertility, girls' expected years of schooling are typically lower than seven.

Sub-Saharan African countries have made remarkable progress in raising life expectancy, mainly through significant reductions in infant mortality. Such improvements typically result in a substantial drop in fertility rates. However, as of 2020, the region's fertility rate was 4.7 children per woman of childbearing age, twice the global average and three times higher than in high-income countries (Figure 1.12).

In the 1950s, fertility rates in Asia and Latin America mirrored those in Africa. However, with advancements in health care, living conditions, and

Figure 1.11. Adolescent Fertility Rate, 1990–2020

(Number of births per 1,000 women ages 15–19)



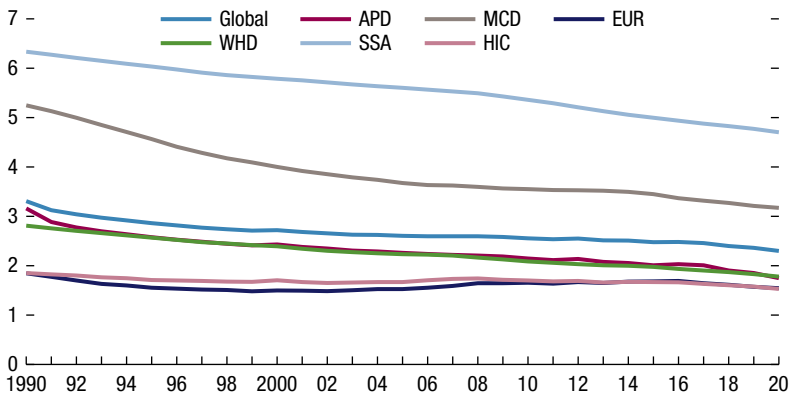
Source: World Bank 2021b.

Note: APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

overall economic prosperity, fertility rates in these regions steadily declined, eventually falling below replacement levels in 2020. In stark contrast, sub-Saharan Africa has followed a different trajectory. High fertility rates persist, despite a 40 percent increase in GDP per capita from 1990 to 2015. This suggests that progress in educational attainment and income growth may not have kept pace with those in other regions (Shapiro and Hinde 2017; Bongaarts 2017).

Figure 1.12. Fertility Rates, 1990–2020

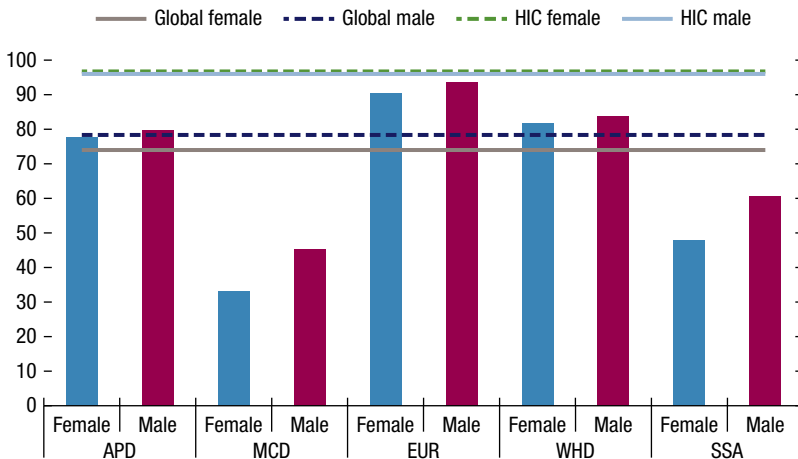
(Births per woman)



Source: World Bank 2021b.

Note: APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Figure 1.13. Accounts at Financial Institutions, 2021
(Percent)



Source: Global Findex 2021.

Note: The figure shows the percentage of respondents (ages 15+) who report having an account (by themselves or together with someone else) at a bank or other type of financial institutions in the previous year. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Adding to this challenge, as of 2019 modern contraceptive prevalence rates in sub-Saharan Africa were significantly lower than the global average: 28.6 percent and 55.7 percent, respectively.¹³

Financial Access

In 2021, globally, 78 percent of men and 74 percent of women held financial accounts (Figure 1.13). However, while in high-income countries there were no significant differences between men and women, a more pronounced gender gap was evident in developing economies, where 74 percent of men and 68 percent of women had accounts. Sub-Saharan Africa faced the most significant gender gaps with 47 percent of women holding an account compared to 60 percent of men.

Notably, in 2021, there was hardly any gender gap among young adults (15 to 24) who exclusively used mobile money accounts. However, the take-up of mobile

¹³ Contraceptive prevalence, any modern method, is the percentage of married women ages 15 to 49 who are practicing, or whose sexual partners are practicing, at least one modern method of contraception. Modern methods of contraception include female and male sterilization, oral hormonal pills, the intrauterine device, the male condom, injectables, the implant (including Norplant), vaginal barrier methods, the female condom, and emergency contraception. Source: Household surveys, including Demographic and Health Surveys and Multiple Indicators Cluster Surveys. Largely compiled by UN Population Division. Retrieved from the World Bank Gender Data Portal.

money accounts is lower for older women than men in the same age group.¹⁴ A lack of access to a mobile phone was a common reason cited by 35 percent of unbanked adults for not having a mobile money account.

In 2021, approximately 50 percent of female and male adults in sub-Saharan Africa borrowed money. However, only 13 percent of women and 16 percent of men used formal channels such as obtaining loans from a financial institution. The percentage was significantly lower compared with high-income countries, where more than half of adults (56 percent of women and 58 percent of men) had borrowed money formally in the previous 12 months. Conversely, in sub-Saharan Africa, informal borrowing from family and friends continued to far outstrip formal channels, with 39 percent of women and 43 percent of men receiving financial assistance informally, in contrast to just 14 percent of men and women in high-income countries.

The spread of mobile money accounts in certain regions has created new opportunities to cater to traditionally marginalized groups, including women, poor people, and other groups traditionally excluded from the formal financial system. Access to finance can empower women by providing greater economic and entrepreneurial opportunities and increasing independence within the household. Chapter 9 explores this issue in more detail.

Digital Finance

The direct receipt of payments into an account plays a crucial role in providing individuals with access to various financial services. In developing economies, the proportion of adults involved in digital payments as senders or recipients has significantly increased from 35 percent in 2014 to 57 percent in 2021.¹⁵ However, this figure is low compared to high-income countries where digital payment adoption is almost universal, with a rate of 95 percent for both men and women. Encouragingly, in developing economies, 83 percent of adults who received digital payments also made digital payments, a substantial increase from 66 percent in 2014 and 70 percent in 2017.¹⁶

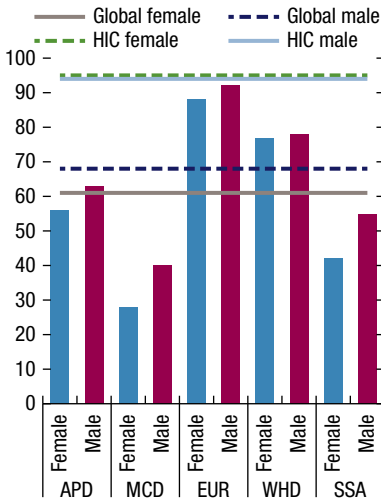
Worldwide inclusion in digital finance is progressing rapidly and with smaller gender gaps than in traditional finance systems. In sub-Saharan Africa, in 2021, 42 percent of women and 55 percent of men aged 15 or older made or received digital payments in the previous year. While the gender gap in digital payments remains higher than in other regions (Figure 1.14), the share of women involved in digital payments has grown significantly since 2014, surpassing the growth rate in account ownership during the same period (Figure 1.15).

¹⁴ Source: Global Findex 2021.

¹⁵ Ibid.

¹⁶ Ibid.

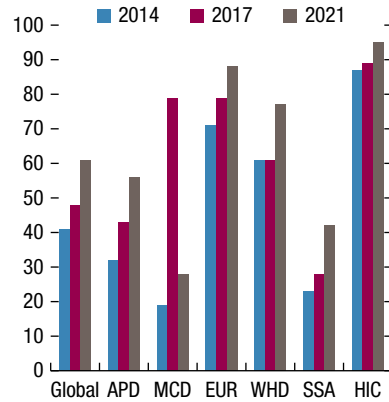
Figure 1.14. Respondents, Ages 15+, Making or Receiving Digital Payment in the Previous Year, 2021 (Percent)



Source: Global Index 2021.

Note: The figure shows the percentage of respondents (ages 15+) who report using mobile money, a debit or credit card, or a mobile phone to make a payment from an account in the previous year. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Figure 1.15. Respondents, Ages 15+, Making or Receiving Digital Payment in the Previous Year, 2014–2021 (Percent)



Source: Global Index 2021.

Note: The figure shows the percentage of respondents (ages 15+) who report using mobile money, a debit or credit card, or a mobile phone to make a payment from an account in the previous year. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; WHD = Western Hemisphere; SSA = sub-Saharan Africa.

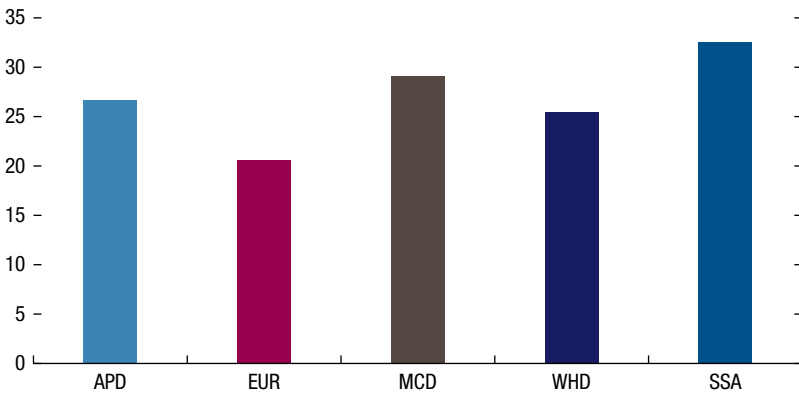
Violence against Women

Sub-Saharan Africa faces significant challenges in addressing violence against women. On average, approximately 33 percent of ever-partnered women in the region have experienced violence in their lifetime (Figure 1.16), and 20 percent have experienced violence over the previous 12 months (WHO 2021).¹⁷ None of the countries in the region fall into the group with the lowest prevalence estimates for lifetime physical and sexual intimate partner violence among ever-married/partnered women aged 15 to 49.

It is essential to acknowledge that these estimates may not capture the full extent of non-partner sexual violence due to underreporting influenced by societal stigma and measurement challenges. The true prevalence of non-partner sexual violence is likely to be higher than the reported and estimated prevalence figures.

¹⁷ Middle East (23.7), Europe (19), Asia Pacific (31), and Western Hemisphere (25).

Figure 1.16. Women, Ages 15–49, Subjected to Physical and Sexual Violence by an Intimate Partner as of 2018
(Percent)



Source: The Global Health Observatory, WHO.

Note: The figure shows the share of ever-partnered women (ages 15–49) subjected to physical and sexual violence by a current or former partner in their lifetime. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Harmful Traditional Practices

Eliminating harmful traditional practices like female genital mutilation¹⁸ and child marriage is a target of SDG 5 (United Nations Department of Economic and Social Affairs 2022). These practices hinder women's economic opportunities and economic growth. Female genital mutilation is internationally recognized as a violation of human rights and an extreme form of gender discrimination with immediate health consequences. While rates have decreased in sub-Saharan Africa, with fewer girls under 14 experiencing female genital mutilation compared to those aged 15 to 49, there are still countries with high prevalence.

Ending child marriage is estimated to increase long-term annual per capita real GDP growth in emerging and developing countries by 1.05 percentage points (Mitra and others 2020). Despite a decline over the past decade, child marriage remains widespread, affecting approximately one in five girls globally. Ongoing crises, such as conflicts, climate-related disasters, and the lingering effects of the COVID-19 pandemic, threaten progress.¹⁹ Child marriage primarily stems from gender inequality, disproportionately affecting girls. Globally, child marriage rates for boys are only one-sixth of those among girls.

¹⁸ Female genital mutilation comprises all procedures that involve partial or total removal of the external female genitalia or other injury to the female genital organs for nonmedical reasons.

¹⁹ Child marriage is any formal marriage or informal union between a child under age 18 and an adult or child (UNICEF 2023).

In 2022, an estimated 640 million girls and women were married in childhood. Child brides face higher risk of domestic violence, reduced education, increased pregnancy complications, and poorer economic and health outcomes. These consequences often extend to their children. Child marriage also isolates girls from their families and friends, resulting in significant mental health challenges (UNICEF 2023). Generational trends estimated by UNICEF suggest that despite recent declines, pandemic-related risks have offset previous progress.²⁰

Sub-Saharan Africa is a region of primary concern, with the highest global risk of marriage before 18. The progress observed has been mainly among wealthier women (UNICEF 2023). Bride price payments historically prevalent in Africa²¹ incentivize child marriage and early marriages, resulting in stifled human capital accumulation as girls are taken out of school to be married for payment in cash or other goods (see, for example, Corno and Voena 2015; Corno, Hildebrandt, and Voena 2020).²²

INEQUALITY OF OUTCOMES

Promoting women's economic empowerment is crucial for achieving sustainable development and inclusive economic growth. Yet, women continue to confront barriers when it comes to accessing financial resources, credit facilities, and land ownership. These barriers, in turn, stifle entrepreneurship and hinder income-generating activities. Therefore, it is essential to give priority to bridging the gender gap in financial inclusion and providing tailored support for female-led businesses. These efforts are critical for fostering inclusive economic growth and harnessing the untapped potential of women in the region.

Labor Force Participation and Employment

Over the past three decades, sub-Saharan Africa consistently maintained the highest average female labor force participation rate among all regions. As of 2021, this rate stood at 62 percent, marking a slight decline from 65 percent in 1990. Meanwhile, the global average also witnessed a slight decline, dropping from 51 percent in 1990 to 47 percent in 2021 (Figure 1.17). Notably, during

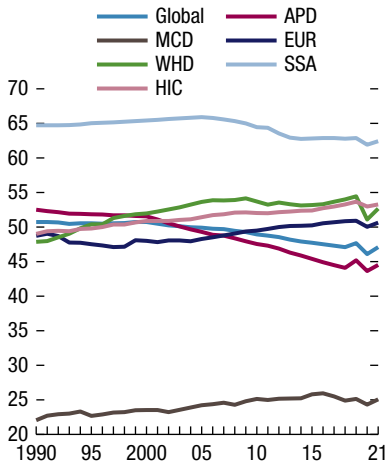
²⁰ Prevalence estimates reflect child marriages that occurred at least two years and as many as six or more years before the reported year. For this reason, the impact of recent changes, such as those arising from the COVID-19 pandemic, is not expected to be evident yet in reporting that relies on the standard indicator.

²¹ Bride prices and dowries are the most well-known types of marriage payments. Bride price payment is a cash or in-kind transfer given by or on behalf of the groom to the bride's family upon the marriage. Dowry payments involve a transfer from the bride to the groom's family upon the marriage. The Women Stat Project estimates for 2016 track bride price and dowry practices worldwide; in all the African countries, net assets move from the groom's family to the bride's family.

²² This topic is explored further in Chapter 4.

Figure 1.17. Female Labor Force Participation, Ages 15+, 1990–2021

(Percent of population)

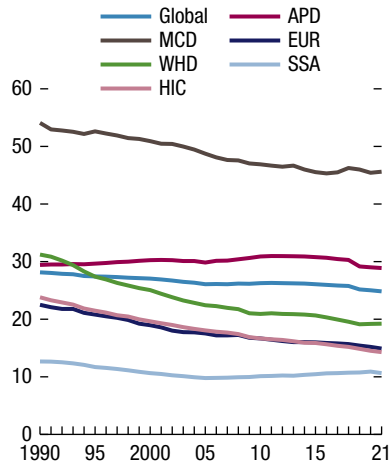


Source: World Bank 2021b.

Note: APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Figure 1.18. Gender Gap in Labor Force Participation, Ages 15+, 1990–2021

(Percentage points)



Source: World Bank 2021b.

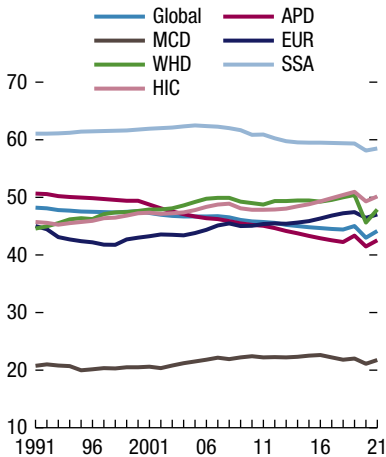
Note: The figure shows the difference between male and female labor force participation. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

this period, only one other region, Asia and Pacific, experienced a decrease. At the same time, male labor force participation declined in all regions, with the global average decreasing by 7 percentage points. Sub-Saharan Africa saw the smallest reduction in male labor force participation, contributing to the relatively stable gap in labor force participation between men and women, which remained at 11 percentage points in 2021 (see Figure 1.18).

Trends in employment outcomes mirror those in labor force participation. Between 1990 and 2021, the global average for the female employment-to-population ratio declined from 48 to 44 percent (Figure 1.19). In sub-Saharan Africa, this ratio went from 61 percent to 59 percent over the same period. The gender gap in employment has remained relatively stable, standing at 10.4 in 2021 (Figure 1.20). Sub-Saharan Africa maintains its position as the region with the lowest gender gaps in employment. In 2021, the female employment-to-population ratio was 58.5 percent, while the male employment-to-population ratio was 68.8 percent.

A smaller proportion of women is employed, and among those who are, a significant percentage are in precarious positions. Globally, there is a prevailing trend

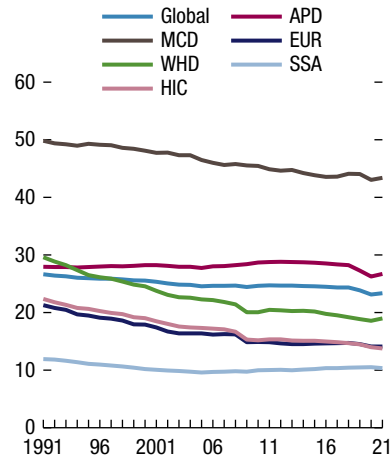
Figure 1.19. Ratio of Female Employment to Population, Ages 15+, 1991–2021
(Percent)



Source: World Bank 2021b.

Note: The figure shows the proportion of a country's population that is employed. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Figure 1.20. Gender Gap in the Ratio of Employment to Population, Ages 15+, 1990–2021
(Percentage points)



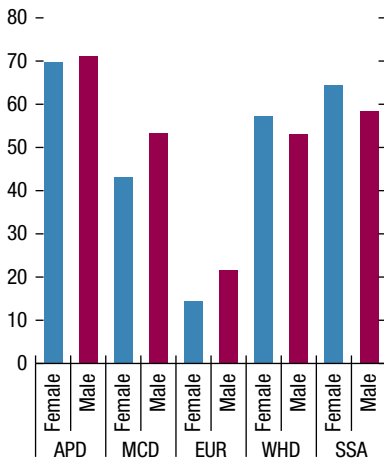
Source: World Bank 2021b.

Note: The figure shows the difference between male and female employment-to-population ratio. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

of higher male representation in informal employment.²³ However, in sub-Saharan Africa, women are more likely than men to find themselves in informal and vulnerable employment. Specifically, 64 percent of women work in the informal sector, while the figure for men is 58 percent, 79 percent of women are in vulnerable employment compared to 67 percent of men (Figure 1.21). The higher incidence of informality can be due to various factors, including the need for flexible work hours or reduced working hours to accommodate caregiving and household responsibilities. Engaging in the informal sector work leads to lower earnings for women and restricts their access to non-monetary benefits and labor law protections. Consequently, this contributes to a widening gender gap in lifetime earnings (Sahay 2023).

²³ Informal employment includes all jobs in unregistered and/or small-scale private unincorporated enterprises that produce goods or services meant for sale or barter. Regardless of size, self-employed street vendors, taxi drivers, and home-based workers are all considered enterprises. However, agricultural and related activities, households producing goods exclusively for their use (for example, subsistence farming, domestic housework, care work, and employment of paid domestic workers), and volunteer services rendered to the community are excluded.

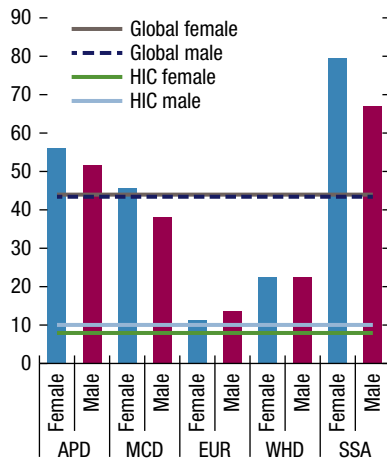
Figure 1.21. Informal Employment, 2015–19
(Percent)



Source: World Bank 2021b.

Note: Informal employment is defined as a percentage of total nonagricultural employment. APD = Asia and Pacific; EUR = Europe; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Figure 1.22. Vulnerable Employment, 2019
(Percent)



Source: World Bank 2021b.

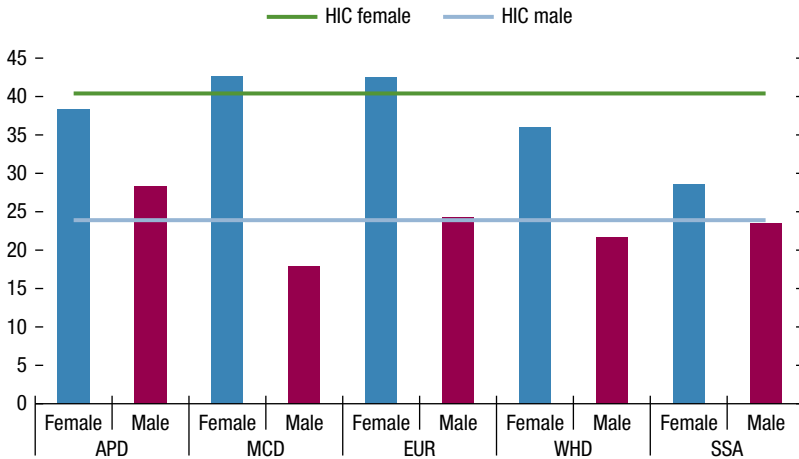
Note: Vulnerable employment is defined as contributing family workers and own-account workers as a percentage of total employment. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Across most regions in developing countries, women are more likely to be in vulnerable employment (Figure 1.22). Although a significant portion of workers is self-employed women, these women are more likely to be unpaid workers in family enterprises and less likely to become own-account workers (Bue and others 2022; Gindling and Newhouse 2014). This situation leaves them vulnerable in the labor market, as they have a lower chance of having formal work arrangements, access to benefits, or participation in social protection programs (ILOSTAT 2013).²⁴ This disparity is particularly pronounced in sub-Saharan Africa, where 79 percent of women and 67 percent of men are engaged in vulnerable employment. In some countries, over 95 percent of women (and 85 percent of men) work as unpaid or own-account workers. These gender disparities in vulnerable employment underscore the need for targeted interventions to improve women's working conditions and livelihoods in the region. Furthermore, in nearly all countries, women are more likely than men to engage in part-time employment.²⁵ While only 50 percent of

²⁴ ILOSTAT (2010) defines vulnerable employment as the sum of contributing family workers and own-account workers as a percentage of total employment.

²⁵ Part-time employment refers to regular employment in which working time is substantially less than normal. Definitions of part-time employment differ by country.

Figure 1.23. Part-Time Employment, 2021
(Percent)



Source: ILOSTAT 2021.

Note: Part-time employment refers to regular employment in which working time is substantially less than normal. Definitions of part-time employment differs by country. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

women are employed in high-income countries, approximately 28 percent of them work part-time, compared to 23 percent of employed men (Figure 1.23).

Cultural norms play a significant role in shaping these patterns, as women are often expected to shoulder a larger share of unpaid household work than men. Across Africa, young women bear a heavier burden of unpaid domestic work, including caregiving for family members, than young men. Consequently, they often must opt for less formal or part-time employment opportunities closer to home.

Comprehensive time-use surveys²⁶ conducted in several African countries indicate that women's unpaid care work is, on average, 2.86 times that of men. In contrast, in Europe and North America, women spend twice as much time as men on such activities, while in South Asia, it is almost nine times. When including paid work, the total workload for women in sub-Saharan Africa is 1.2 times that of men, meaning that African women bear the heaviest work burden across all regions (African Development Bank 2020; Charmes 2019). The lack of affordable childcare is a significant barrier that can impede women's access to formal employment opportunities. This obstacle often forces women into a difficult dilemma where they must choose between pursuing a formal career or

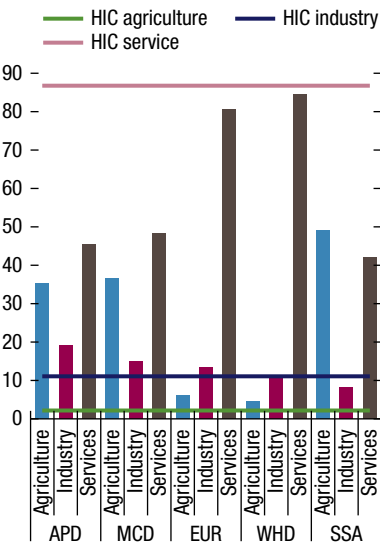
²⁶ Benin (2015), Cameroon (2014), Cabo Verde (2012), Ethiopia (2013), Ghana (2009), Madagascar (2001), Mali (2008), Mauritius (2018), South Africa (2010), Tanzania (2018), and Uganda (2018). See also UN Women 2021.

prioritizing their responsibilities as mothers. Without accessible and affordable childcare options, women face challenges in striking a balance between work and caregiving, limiting their ability to fully participate in the workforce while fulfilling their roles as mothers.

Sectoral Segregation

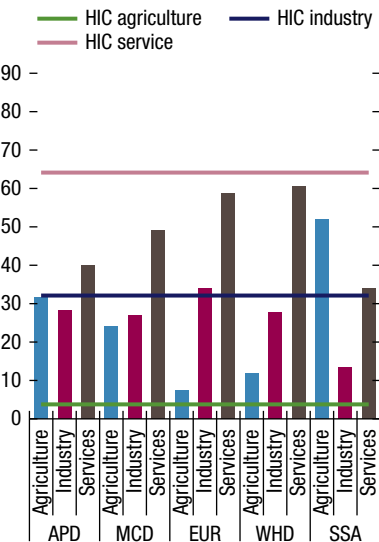
Sectoral segregation refers to the pattern in which men and women tend to be concentrated in different sectors or industries of the economy. This segregation typically results in men and women having unequal access to employment opportunities, in wage disparities, and in differing levels of representation in various sectors. Sectoral segregation is widespread across regions, with women more likely to be engaged in the service sector than men and less likely to be involved in agricultural and industrial positions (Figures 1.24 and 1.25). *Occupational segregation* refers to the unequal distribution of men and women across different types of jobs or professions. It arises from a combination of self-selection by girls and women into gender-conventional career paths and the societal pressure that explicitly or implicitly dissuades them from pursuing non-gender-conforming paths. The interplay between individual choices and external influences intersects with

Figure 1.24. Share of Women Employed by Sectors, 2021
(Percent)



Source: ILOSTAT 2021.
Note: APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Figure 1.25. Share of Men Employed by Sectors, 2021
(Percent)



Source: ILOSTAT 2021.
Note: APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

disparities in skills or resources, such as lower levels of expertise or capital, which further restrict the range of occupational options available to women. In Africa, agriculture serves as a primary sector of employment. However, women are 3 percentage points less likely to be employed in agriculture, working on their plots or someone else's than are men.

Share of Youth Not in Education, Employment, and Training

Despite sub-Saharan Africa having a higher ratio of female-to-male labor force participation rates than other regions, these seemingly positive figures conceal underlying challenges. Specifically, girls face greater obstacles to accessing education, employment, or training than do their male counterparts. The obstacles faced by young women can have far-reaching consequences since the employment of young women contributes to economic growth and enhances their voice and agency, resulting in positive effects on the individuals, families, and society at large (Klugman and others 2014). However, the school-to-work transition is slower in sub-Saharan Africa than in high-income countries, and young people in the region require assistance in entering the labor market after completing their education (Elder and Kring 2016). In 2021, among individuals aged 15 to 24, 30 percent of girls and 19 percent of boys were not in education, employment, or training (NEET). The gender gap in this regard has shown no improvement between 2004 and 2021 and is significantly wider than in high-income countries (Figure 1.26). In 2021, sub-Saharan Africa ranked second in the world for having the highest gender gap in this aspect, following the Middle East and Central Asia (Figure 1.27).

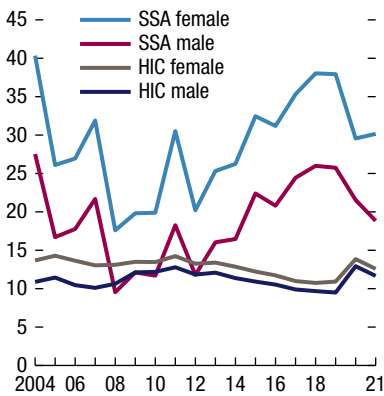
Between 2017 and 2021, NEET rates ranged from 77 percent to below 20 percent, with gender gaps as large as 28 percentage points. Chakravarty, Das, and Vaillant (2017) identified several constraints that affect girls' employment in sub-Saharan Africa. These include lower levels of marketable skills, limited access to financing, network constraints stemming from less exposure to the workplace and employers, discrimination, lack of role models, and limited mobility.

Digital Divide

The rise of digital technology in sub-Saharan Africa offers both opportunities and challenges for gender equality. While mobile technology and digital platforms have expanded economic opportunities for women and increased access to services such as education, health care, and information, a significant gender digital divide persists. Harnessing digital technology has the potential to enhance women's participation in the labor market and contribute to fostering inclusive societies and economic growth. Nevertheless, failure to bridge the gender digital divide would deprive women of the full advantages and benefits of digital technology, thereby limiting their participation in the digital economy and restricting their access to resources and opportunities available through digital platforms.

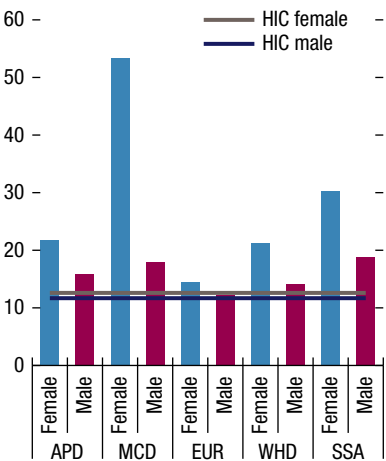
Sub-Saharan Africa presents some of the largest gaps, with women 36 percent less likely than men to use mobile internet, and a 13 percent gender gap in mobile

Figure 1.26. NEET Rates in SSA and High-Income Countries, 2004–21
(Percent of youth population)



Source: World Bank 2021b.
Note: The figure shows the share of young people (ages 15 to 24) who are not in education, employment or training (NEET) in sub-Saharan Africa compared to those in high-income countries. HIC = high-income countries; SSA = sub-Saharan Africa.

Figure 1.27. NEET Rates by Regions, 2021
(Percent of firms)



Source: World Bank 2021b.
Note: The figure shows the share of young people (ages 15 to 24) who are not in education, employment or training (NEET) in sub-Saharan Africa compared to various other regions. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

ownership, equivalent to 95 million women not owning a mobile phone as of 2023 (GSMA 2023). These disparities in access to digital technology perpetuate inequalities, leading to both social and economic costs.²⁷ Social norms can restrict women’s access to and use of digital technologies. Additionally, in several sub-Saharan African countries, taxes imposed on mobile money withdrawals or digital services act as barriers, especially for low-income women (Aranda-Jan and Qasim 2023). Addressing the gender gap in digital technology through improved access, digital literacy programs, and supportive policies is crucial for promoting gender-inclusive economic development.

Entrepreneurship

Sub-Saharan Africa stands out for having the highest rate of women involved in entrepreneurial activity worldwide (World Economic Forum 2022). Several sub-Saharan African countries rank among the world’s top nations with the

²⁷ It is estimated that excluding women from digital platforms has led to a staggering loss of \$1 trillion in GDP (Alliance for Affordable Internet 2021).

highest proportion of female entrepreneurs (Mastercard Index of Women Entrepreneurs 2022). The region's limited employment opportunities often drive both men and women to entrepreneurship out of necessity. This is especially true for women, who face greater constraints in accessing alternative employment opportunities. However, despite their significant presence, female entrepreneurs in Africa face several challenges, including lower profits than do their male counterparts (World Bank 2019). Female entrepreneurs make strategic decisions that differ from those made by male entrepreneurs, including the choice of sector, capital investment, labor utilization, business innovation, and competitiveness. Typically, female entrepreneurs gravitate towards female-dominated industries, such as trade, retail, textile and footwear, pharmaceutical, and perfume products, while male-owned enterprises dominate manufacturing, agriculture, forestry, and fishing (World Bank 2021). Interestingly, female entrepreneurs who cross over to male-dominated sectors tend to perform better than those in female-concentrated sectors (World Bank 2021). Women, however, face numerous obstacles when entering male-dominated sectors, including unconscious biases, societal norms, limited exposure to these fields, and constraints related to time and capital.

Despite their entrepreneurial efforts, female entrepreneurs consistently face significant disparities in business capital compared to their male counterparts.²⁸ They also employ fewer workers and allocate fewer labor hours in their businesses.²⁹ Male entrepreneurs are more likely to adopt advanced business practices and innovations, and they are more likely to engage in competitive practices. Furthermore, legal discrimination against women remains a challenge in several sub-Saharan African countries. While gender gaps in accessing loans from financial institutions are smaller in Africa than in other regions of the world, a World Bank (2019) analysis reveals persistent and substantial gender disparities in the size of outstanding loans for various groups of entrepreneurs.

Furthermore, female farmers in sub-Saharan Africa face significant systemic barriers, leading to substantial gender gaps in agricultural productivity. Since agriculture is the primary employment sector on the continent, addressing these gaps can yield substantial gains for African economies. One major driver of gender inequality in agriculture is the disparity in land ownership between men and women. According to FAO (2018), African women own significantly less agricultural land than men, often due to customary norms discouraging women from claiming ownership or possessing documented evidence of their land property rights. Extensive evidence demonstrates a strong correlation between secure land tenure and increased investment and productivity in agriculture (FAO 2018;

²⁸ When analyzing data from 14 impact-evaluation data sets across 10 African countries, it is evident that male-owned firms typically have more than six times the amount of capital investment, including equipment, inventory, and property, compared with female-owned enterprises (World Bank 2019).

²⁹ The existence of constraints restricting the growth of women-led firms is evidenced by the fact that the share of businesses that are led by women decreases as the size of the firm increases (Ubfal 2023).

World Bank 2020b). Therefore, addressing gender discrimination in land ownership and ensuring secure land tenure for women are critical steps toward promoting their economic empowerment and improving agricultural productivity.³⁰

INEQUALITY OF REPRESENTATION

Inequality of representation of women is a multifaceted and pressing issue that spans both political and corporate realms. In societies around the world, women continue to face significant disparities in their representation within these spheres. In the political sphere, the underrepresentation of women in elected offices and decision-making bodies raises questions about the inclusivity of democratic processes and the fair representation of diverse perspectives. In the corporate world, women continue to encounter barriers that limit their access to executive and managerial roles, emphasizing the broader issue of gender disparities in economic leadership.

Sub-Saharan Africa has made notable strides in promoting women's political participation and decision-making. Many countries in the region have implemented gender quotas and affirmative action policies to increase the representation of women in parliaments and government positions. Despite these efforts, women's political empowerment in the region remains below parity, and women are underrepresented in key decision-making bodies and leadership roles.

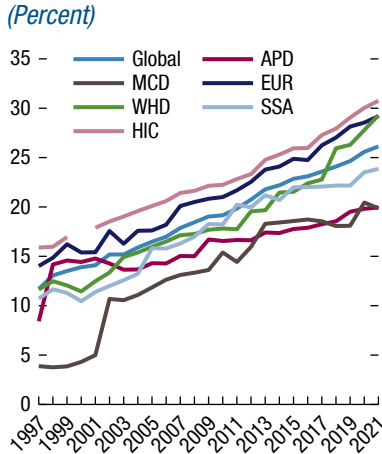
Globally, there has been an increase in the portion of seats held by women in national parliaments (Figure 1.28). In 2021, women held 26 percent of parliamentary seats worldwide, up from 11 percent in 1997. However, as of 2020, no region had achieved gender parity in parliamentary representation. In sub-Saharan Africa, there has been remarkable progress, with the proportion of female parliamentarians increasing from 10 percent in 1997 to 23 percent in 2020. Throughout the continent, the representation of women in parliament varies significantly, ranging from 7 percent to 61 percent.

Similarly, the region made substantial progress in women's representation in ministerial positions. Between 2005 and 2020, the share of women holding ministerial positions increased from 15 percent to 25 percent. This percentage is higher than the global average (Figure 1.29), highlighting the region's positive advancements in gender equality in political leadership.

In terms of women's representation in managerial positions in the private sector, in 2021, 16 percent of firms had female top managers in sub-Saharan Africa, just below the global average (Figure 1.30). According to the Africa Growth Initiative (2022), women accounted for 24.3 percent of company board seats in the region in 2018, with 24.5 percent serving as board chairs. This represented the highest level of any region and was marginally higher than average representation on executive committees at 22 percent (McKinsey 2019). While this figure reflects significant progress in the region, this progress has not been consistent across countries and the region faces challenges in promoting women in middle management positions.

³⁰ For a comprehensive review of the most promising interventions in addressing key constraints to women entrepreneurship see Ubfal 2023.

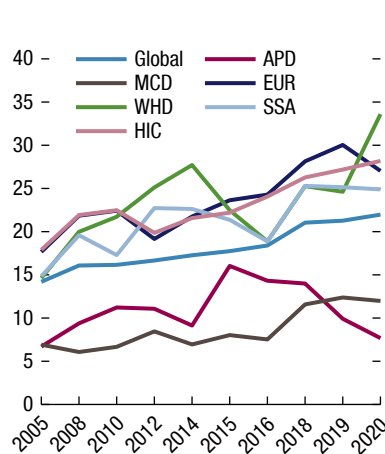
Figure 1.28. Proportion of Seats Held by Women in National Parliament, 1997–2020
(Percent)



Source: World Bank 2021b.

Note: APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

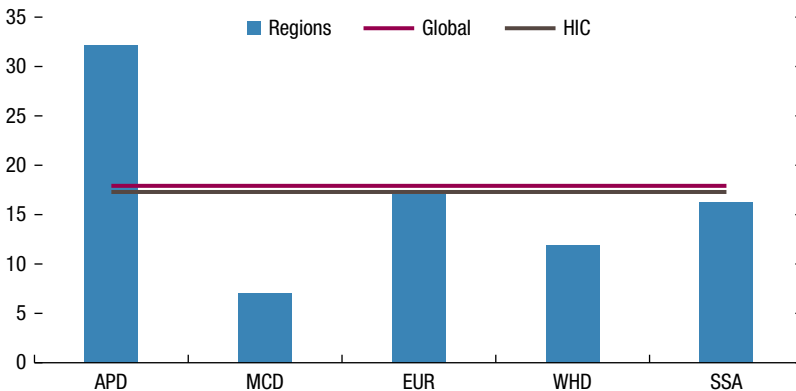
Figure 1.29. Proportion of Women in Ministerial Positions, 2005–2020
(Percent)



Source: World Bank 2021b.

Note: APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Figure 1.30. Firms with Female Top Managers, 2018–2020
(Percent of firms)



Source: Enterprise Surveys, World Bank Survey. Retrieved from the World Bank Gender Data Portal.

Note: The figure shows the percentage of firms in the private sector who have females as top managers. APD = Asia and Pacific; EUR = Europe; HIC = high-income countries; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

CONCLUSION

The analysis of gender inequality in sub-Saharan Africa paints a comprehensive picture of the region's challenges and persistent gaps across various dimensions. While it is evident that gender disparities persist in multiple sectors, hindering progress towards gender equality and sustainable development, it is equally important to recognize the positive developments that have emerged.

Despite the persistence of discriminatory social institutions, societal norms, and harmful traditional practices, sub-Saharan Africa has made strides in enacting positive legal changes that are crucial in promoting women's economic empowerment. There have been improvements in access to education, even though gender disparities remain, particularly at the tertiary level. Efforts to combat maternal mortality rates and adolescent fertility rates are underway, although further progress is needed. Women still face significant barriers to economic empowerment, such as in accessing financial resources, formal employment, and entrepreneurship opportunities, but there have been positive shifts towards increased political representation and entrepreneurship rates.

Addressing gender inequality in sub-Saharan Africa requires a comprehensive and multifaceted approach. This includes promoting legal reforms to protect women's rights, improving access to quality education and healthcare services, combating violence against women, eradicating harmful traditional practices, fostering financial inclusion, bridging the digital gender divide, and promoting women's economic empowerment. Moreover, increasing women's representation in decision-making positions is essential. The approach needs to confront deeply ingrained gender biases and systemic obstacles while exploring innovative strategies to ensure balanced and inclusive representation in both political and corporate domains.

Crucially, collaboration among governments, civil society organizations, international institutions, and communities is pivotal in implementing and sustaining these measures. By prioritizing gender equality as a fundamental development goal and addressing the systemic barriers preventing women to achieve their full potential in sub-Saharan Africa, it will be possible to foster inclusive and sustainable economic growth, improve social well-being, and create a more equitable and just society.

ANNEX 1.1. GENDER EQUALITY INDEXES

Gender Inequality Index (GII)

The Gender Inequality Index is a composite measure of gender inequality defined using three dimensions: reproductive health, empowerment, and the labor market. The reproductive health indicators are the maternal mortality ratio and adolescent birth rate. The empowerment indicators include the percentage of parliamentary seats held by women and the percentage of the population with at least some secondary education by gender. Finally, the labor market indicator corresponds to labor force participation by gender. A low index value suggests low inequality between women and men and vice versa.

African countries continue to report high levels of maternal mortality and adolescent fertility rates; moreover, women are at a disadvantage in educational attainment (Annex Table 1.1.1).

Global Gender Gap Index (GGI)

The Global Gender Index provides an annual assessment of the status and progress of gender inequality across four crucial dimensions: economic participation and opportunity, educational attainment, health and survival, and political empowerment.

Sub-Saharan Africa has shown significant progress in women's economic participation and half of the countries in the region have achieved parity in estimated earned income. However, sub-Saharan Africa ranks lowest globally in educational attainments.

Regarding health and survival, sub-Saharan Africa has closed 97.1 percent of its gender gap, reporting the highest regional score together with Europe and the Western Hemisphere. Eleven African countries have reached full parity in healthy life expectancy, and all countries in the region have achieved gender parity in sex ratio at birth. Political empowerment remains the most significant challenge, although it shows a substantive improvement of 3.3 percentage points from 2021 (Annex Table 1.1.2).

ANNEX TABLE 1.1.1

Regional Performance by IMF Area Department, Gender Inequality Index, 2021

	Maternal Mortality Ratio ¹	Adolescent Birth Rate ²	Share of Seats in Parliament Held by Women	Population with at Least Some Secondary Education (25+)		Labor Force Participation Rate (15+)	
				Female	Male	Female	Male
SSA	457	96	24	30	41	61	72
APD	86	28	20	61	65	51	72
EUR	8	11	31	90	93	52	65
MCD	99	30	19	66	68	32	70
WHD	82	50	29	63	64	52	73

Note: APD = Asia and Pacific; EUR = Europe; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

¹ Deaths per 100,000 live births.

² Births per 1,000 women ages 15–19.

ANNEX TABLE 1.1.2.

Regional Performance, Global Gender Gap Index by Subindex, 2022				
	Economic Participation	Educational Attainment	Health and Survival	Political Empowerment
SSA	0.68	0.88	0.97	0.22
APD	0.66	0.98	0.95	0.19
EUR	0.72	0.99	0.97	0.35
MCD	0.52	0.95	0.95	0.13
WHD	0.69	0.99	0.97	0.30

Source: World Economic Forum 2022.

Note: APD = Asia and Pacific; EUR = Europe; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Social Institutions and Gender Index (SIGI)

SIGI assesses gender discrimination within social institutions across 179 countries. It does so by considering the impact of laws, social norms, and practices that limit women's rights and hinder their access to opportunities and resources. This comprehensive approach allows SIGI to uncover the root causes of gender inequality. Compared to other regions, African countries experienced the highest level of discrimination due to family-related factors and of restrictions in accessing productive and financial resources (Annex Table 1.1.3).

ANNEX TABLE 1.1.3.

Regional Performance, Social Institutions and Gender Index by Components, 2022				
	Discrimination in the Family	Restricted Physical Integrity	Restricted Access to Productive and Financial Resources	Restricted civil liberties
SSA	52.4	29.6	40.9	34.6
APD	50.4	33.7	32.5	38.3
EUR	12.8	18.6	10.2	16.8
MCD	67.9	37.2	37.3	47.3
WHD	21.5	26.1	22.7	17.6

Source: SIGI 2023.

Note: 0 indicates non-discrimination, 100 absolute discrimination. APD = Asia and Pacific; EUR = Europe; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

Human Capital Index (HCI)

The Human Capital Index calculates the contributions of health and education to worker productivity. The final index score ranges from 0 to 1 and it measures the productivity of a future worker of a child born today relative to the benchmark of full health and complete education.³¹

Annex Table 1.1.4 shows the disaggregation of the Human Capital Index by its subcomponents. In the African regions, boys have a higher stunted rate and a

³¹ The Human Capital Index quantitatively illustrates the key stages in a child's human trajectory and their consequences for the productivity of the next generations of workers with three components: (1) survival from birth to school age, measured using under-5 mortality rates; (2) expected years of learning-adjusted school, combining information on the quantity (number of years of school) and quality (harmonized test score) of education; and (3) health (adult survival rates from age 15 to 60 and fraction of children under 5 not stunted; World Bank 2020a).

ANNEX TABLE 1.1.4.

Regional Performance, Human Capital Index by Subcomponents, 2020												
	Probability of Survival to Age 5		Expected Years of School		Harmonized Test Scores		Learning-adjusted Years of School		Percentage of Children under 5 Not Stunted		Adult Survival Rate	
	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male
SSA	0.94	0.93	8.2	8.4	362	362	4.6	4.7	0.71	0.66	0.76	0.70
APD	0.98	0.98	12.1	11.9	438	428	8.6	8.20	0.77	0.76	0.90	0.83
EUR	1.00	0.99	13.2	13.2	494	483	10.5	10.2	0.93	0.92	0.94	0.88
MCD	0.98	0.97	11.2	11.1	412	389	7.5	7.0	0.82	0.79	0.91	0.86
WHD	0.98	0.98	12.1	11.9	421	416	8.2	7.9	0.85	0.83	0.90	0.83

Source: World Bank 2020a.

Note: APD = Asia and Pacific; EUR = Europe; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

lower probability of survival to age 5. Concerning the expected school years, girls are still disadvantaged compared with boys. The gender gap in the Human Capital Index varies quite widely across African economies, ranging from -0.05 to 0.03 .

Gender Development Index (GDI)

The Gender Development Index measures gender inequalities in three basic dimensions of human development: longevity, education, and income per capita.³² Sub-Saharan Africa has the second-lowest gender development index (0.91), followed by Middle East and North Africa and with scores that range from 0.77 to 1 . Sub-Saharan Africa is the only region with girls at a disadvantage compared with boys in terms of expected years of schooling and have the largest gap in terms of mean years of education.

ANNEX TABLE 1.1.5.

Regional Performance, Gender Development Index by Subcomponents (2020)				
	Life Expectancy at Birth	Expected Years of Schooling	Mean Years of Schooling	Estimated Gross National Income per Capita
SSA	1.08	0.95	0.74	0.68
APD	1.07	1.03	0.93	0.60
EUR	1.08	1.05	0.99	0.67
MCD	1.08	1.00	0.94	0.35
WHD	1.09	1.07	1.00	0.60

Source: UNDP 2021a.

Note: APD = Asia and Pacific; EUR = Europe; MCD = Middle East and Central Asia; SSA = sub-Saharan Africa; WHD = Western Hemisphere.

³² The Gender Development Index is calculated as the ratio of female to male Human Development Index values constructed based on three components: life expectancy at birth, expected and mean years of schooling, and the estimated gross national income per capita (United Nations Development Programme 2021).

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Zooming into Women's Financial Access

**Kazuko Shirono, Katia Huayta-Zapata,
Hector Carcel-Villanova, and Esha Chhabra**

The developments of women's financial access in sub-Saharan Africa—including during the height of the COVID-19 pandemic—are depicted in annual data from the IMF's Financial Access Survey (FAS). The FAS results indicate a lower degree of financial access and use via commercial banks in sub-Saharan Africa than in other regions during the prepandemic period, with a sizable and persistent gender gap. During the pandemic years, women in sub-Saharan Africa broadly maintained financial access with no major widening of the gender gap, albeit with a few exceptions, likely supported by countries' policy measures. Nevertheless, given the low and uneven levels of financial access for both men and women in the region, efforts need to continue advancing women's financial access—including improving the availability of gender-disaggregated data on both traditional and digital financial services—to better track the developments and assess policies.

INTRODUCTION

Since the onset of the global COVID-19 pandemic, much has been debated on its potential impact on women's financial inclusion and the need for supportive global policy actions (Azar and Mejía 2020; Schuttenbelt 2020, IFC 2020). Sub-Saharan Africa is no exception given that 350 million of the world's unbanked population live in the region (World Bank 2018). However, evidence is scarce, partly due to a lack of well-curated data in this area. This chapter examines women's financial access in sub-Saharan Africa during the COVID-19 pandemic—drawing on annual supply-side data collected through the IMF's Financial Access Survey (FAS)¹ and publicly available information on relevant gender-sensitive policies to support women's financial access during the pandemic.

This chapter partly draws on Shirono and others (2021), to which Yingjie Fan (Princeton University) also contributed.

¹ The FAS is available on <https://data.imf.org/fas>. Some FAS indicators can also be accessed through third-party websites such as the World Bank's World Development Indicators Databank.

As of March 2023, 29 African countries report gender-disaggregated data to the FAS. While the time coverage and availability of indicators vary across countries, the FAS can offer insight into various aspects of women's financial access in sub-Saharan Africa—such as account adoption and financial service usage—including comparison with other regions.

The FAS gender-disaggregated data suggest that before the pandemic, the degree of financial access in sub-Saharan Africa was lower relative to other regions, with significant gender gaps measured by the share of borrowers and depositors in some cases. During the first year of the pandemic in 2020, the number of female depositors in the region stayed broadly stable or even increased in some countries and continued to rise in 2021, which may reflect the impact of policy measures that countries implemented to maintain financial access. The outstanding value of deposits and loans also stayed relatively stable in the African countries in the sample, albeit with a few exceptions.

The gender gap in financial access remained mostly unchanged in most of sub-Saharan Africa during the COVID-19 pandemic. The gender gap for share of depositors improved in some cases, for example, but some of these improvements were due to a larger decline in the indicators for men than for women. These results indicate the importance of granular data in tracking developments and assessing policy impacts.

These findings should be interpreted cautiously because of the limited sample size and the need to control for omitted variables. As more granular financial access data become available, improved analysis could help better tailor supportive policies. Nevertheless, the data clearly show that efforts need to continue advancing financial access in sub-Saharan Africa.

The rest of the chapter is organized as follows: The next section discusses issues related to gender-disaggregated data on financial access, including key features of the FAS gender-disaggregated data series and indicators. After that is a pre-COVID-19 pandemic snapshot of women's financial access based on FAS gender-disaggregated data, with an emphasis on sub-Saharan Africa. Then there is an overview of policy measures countries have taken to support financial access for women in response to the COVID-19 pandemic. FAS outcome data are presented on women's financial access for sub-Saharan Africa in 2021.

FAS GENDER-DISAGGREGATED DATA

The gender gap in financial inclusion—defined as access to and use of financial services in this chapter²—has important macroeconomic implications through its links to higher economic growth, particularly in countries with lower overall levels of financial inclusion (Sahay and others 2015), income inequality (Aslan

² Financial inclusion is a multifaceted concept covering several dimensions (Espinosa-Vega and others 2020). This chapter focuses on access to and use of financial services as these are key pillars of financial inclusion.

TABLE 2.1.

FAS Gender-Disaggregated Series		
Commercial Banks	Deposit-Taking Microfinance Institutions	Non-Deposit-Taking Microfinance Institutions
Depositors and borrowers	Depositors and borrowers	Borrowers
Deposit and loan accounts	Deposit and loan accounts	Loan accounts
Outstanding loans and deposits	Outstanding loans and deposits	Outstanding loans

Source: IMF Financial Access Survey.

Note: The FAS collects gender-disaggregated data on 15 series. Each cell in the first two columns of the table lists two separate series for deposits and loans.

and others 2017; Sahay and others 2018), and financial stability (Sahay and others 2018; Khera and others 2022). However, a lack of granular data often hampers analysis. Despite progress made in recent years, addressing data issues is often the first step toward closing the gender gap in financial inclusion (Data2X 2019; Oula, Carey, and Kayibanda 2019; FinEquity 2020). This section briefly discusses issues related to gender-disaggregated data on financial access, focusing on FAS supply-side data.

Gender-disaggregated data from the FAS offers a bird's-eye view of the financial inclusion landscape for women before and during the COVID-19 pandemic.³ The FAS is a unique source of annual supply-side data on financial access and use, covering 189 countries. It contains 121 time series, including gender-disaggregated data on 15 series on access to and use of financial services provided by different types of financial institutions (Table 2.1). Data reporting is somewhat uneven with some data gaps—not all countries report the complete set of time series to the FAS, and data coverage varies across countries. Nevertheless, the increasing number of countries reporting these data to the FAS, which more than doubled between 2018 (35 reporters) and 2022 (82 reporters as of March 2023)—including 29 countries in sub-Saharan Africa—reflects the importance of gender-disaggregated information for policy-making. Out of the 82 countries that report gender-disaggregated data to the FAS, about half are low- and lower-middle income countries, some of which report many of the gender-disaggregated data series. For example, Guinea, Pakistan, Zimbabwe (reporting all 15 gender-disaggregated series), and Bangladesh (reporting 10 gender-disaggregated series) were the top gender-disaggregated data reporters in 2022.

The FAS is based on administrative data collected by central banks and financial regulators and thus tends to be more cost-effective than demand-side surveys. FAS data can be collected annually unlike similar statistics from demand-side surveys. The FAS also directly measures financial access outcomes, so the FAS series capture actual outturns, free of estimates or assumptions. The FAS gender-disaggregated series allow for developing several cross-country

³ The FAS gender-disaggregated data have been collected annually since 2018.

comparable indicators to measure different aspects of financial inclusion (see Annex 2.1). As shown in the next section, these indicators offer insights into the state of play before the COVID-19 pandemic.⁴

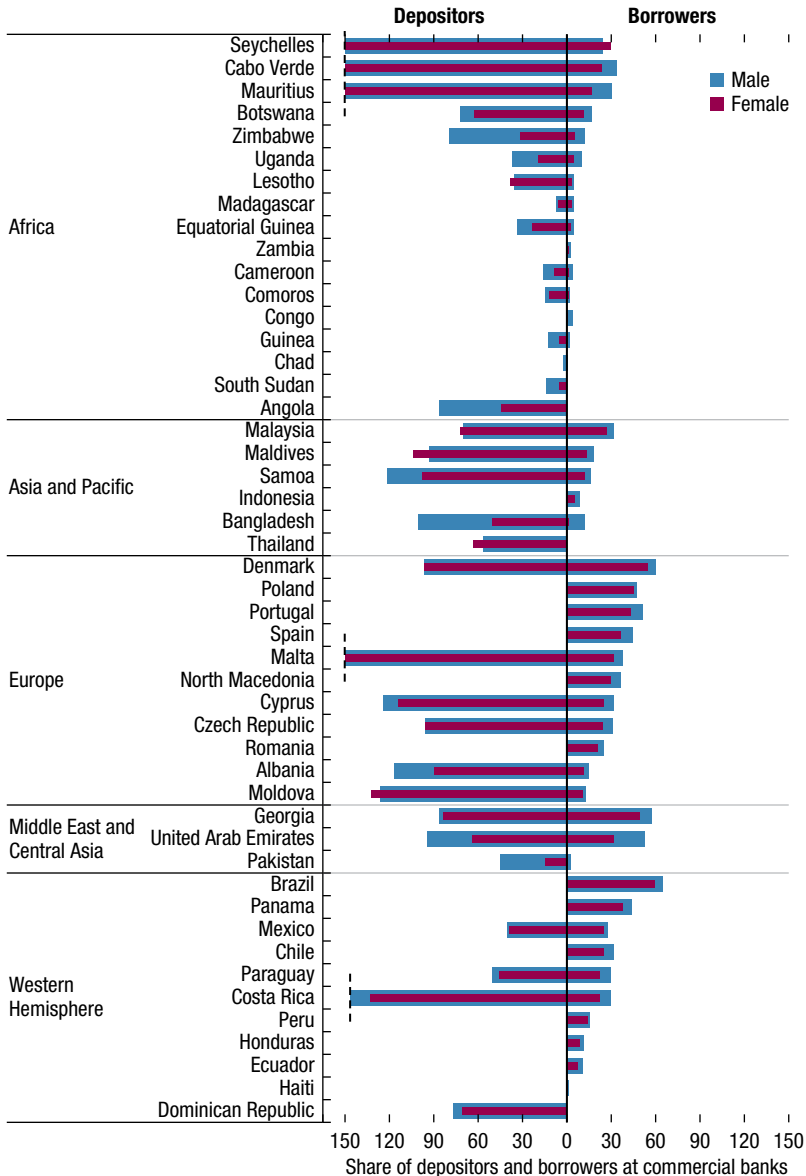
FINANCIAL ACCESS GENDER GAP: A PRE-COVID-19 PANDEMIC SNAPSHOT

Adoption of Bank Accounts

The share of male or female depositors (or borrowers) in the male or female adult population (Annex 2.1) can proxy the size of the banked population. Figure 2.1 shows the size of banked male and female populations for countries across five regions in 2019. Before the COVID-19 pandemic, the size of the banked population and gender gaps therein varied widely across countries, with significant gender gaps in some cases. Multiple factors drive gender gaps in financial access, including structural inequality in income (for example, due to lower female labor market participation) and restricted legal rights (for example, lack of a legal provision to prohibit gender-based discrimination; World Bank 2021). Overall, the global average share of female and male depositors in the adult population in 2019 was 40 and 61 percent, respectively (a 21 percentage point gap). Among sub-Saharan African countries, however, the banked population rate was significantly lower for both women and men—22 and 38 percent, respectively. The financial access gender gap within the region itself varied widely. In some countries, the share of the banked female population was less than half of the banked male population (for example, in Chad, South Sudan, Zimbabwe, and Guinea). Compared with the banked depositors, the gender gap for banked borrowers tends to be smaller globally—in 2019, the average share of female and male borrowers at commercial banks was 19 percent and 24 percent, respectively. For sub-Saharan Africa, the same indicator was 3 percent and 6 percent for women and men, respectively.

Another approach to analyzing gender inequality in financial access is to examine the proportion of women among the banked population. For gender parity, the indicator should be close to 50 percent (see Annex 2.1 for more discussion). The FAS data suggest that the proportion of female borrowers stayed at about 33 percent on average between 2016 and 2019 globally, even though some variation in the pace of progress existed across countries (Figure 2.2). For sub-Saharan African countries, the proportion of female borrowers was 31 percent in 2019, with substantial heterogeneity in the region. In countries such as Republic of

⁴ This chapter mainly uses the data on commercial banks and microfinance institutions when available, given that commercial banks are still the most common financial service providers in many countries. However, some countries may have other types of financial institutions (for example, credit unions) playing a more dominant role. In addition, a more complete analysis requires taking account of digital financial services (see, for example, Comodore 2020, Ayadi and Shaban 2020; Agur, Martinez Peria, and Rochon 2020; and Bazarbash and others 2020).

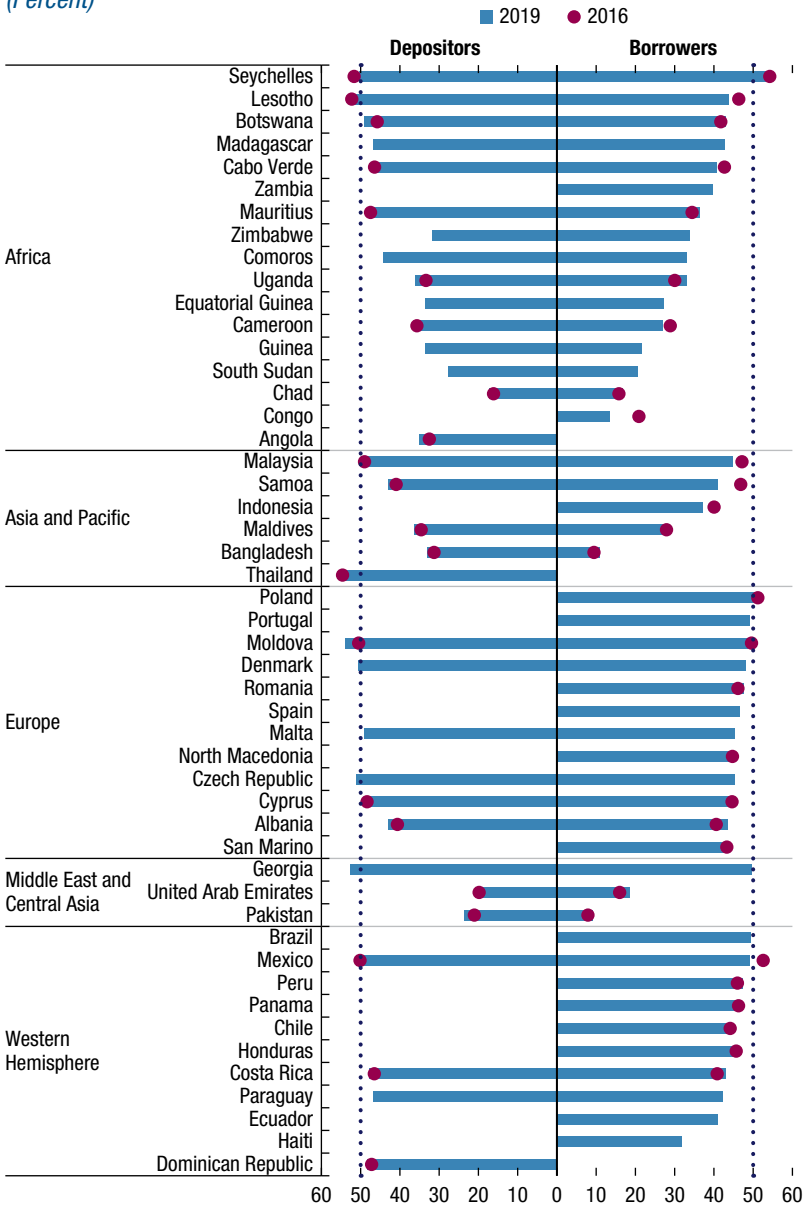
Figure 2.1. Male and Female Share of Banked Population, 2019*(Percent of female or male population)*

Sources: IMF Financial Access Survey; and IMF staff calculations.

Note: This chart shows the share of male and female depositors and borrowers at commercial banks as a percent of the male and female adult population in 2019. The sample was selected based on data availability. Depositors with bank accounts in more than one bank may be counted multiple times in some cases, and thus the indicator can exceed 100 percent of the population. Some countries also include nonresidents and minors (Cabo Verde, Malta, Mauritius, Seychelles), resulting in a significantly higher share of depositors, ranging from 150 to 240 percent of the female or male population.

Figure 2.2. Proportion of Female Depositors and Borrowers at Commercial Banks, 2016 and 2019

(Percent)



Sources: IMF Financial Access Survey; and IMF staff calculations.

Note: This chart shows the proportion of female borrowers to total borrowers at commercial banks for 2016 and 2019. Some countries did not report 2016 data to the FAS. The sample was selected based on data availability.

Congo, Chad, South Sudan, and Guinea, the proportion of female borrowers was less than a quarter of the total banked population, while other countries (for example, Lesotho, Botswana, and Madagascar) had a proportion of female borrowers above 40 percent in 2019. A similar pattern holds for the average proportion of female depositors. The average proportion increased only slightly—from 33 percent to 36 percent—in sub-Saharan Africa during the same period, even though the gender gap measured by this indicator is smaller than the indicator based on total borrowers. Overall, these limited changes over time may suggest persistent structural factors, such as inequality in income and restricted legal rights (Jousse 2021).

The gender gap in financial access may be smaller once microfinance institutions⁵ are considered, as they are known to play a key role in providing credit to women in some low- and middle-income economies (Espinosa-Vega and others 2020). For example, in Guinea, microfinance institutions provide loans to a larger share of the population than commercial banks, accounting for close to 5 percent of both male and female populations with a minimal gender gap. In addition, informal access to finance—such as savings or investment clubs—also plays a role in many developing economies, particularly in rural areas (Singer 2014). In this regard, it is also important to note the growing significance of fintech to help close gender gaps in financial access (Sahay and others 2020).

Financial Service Usage

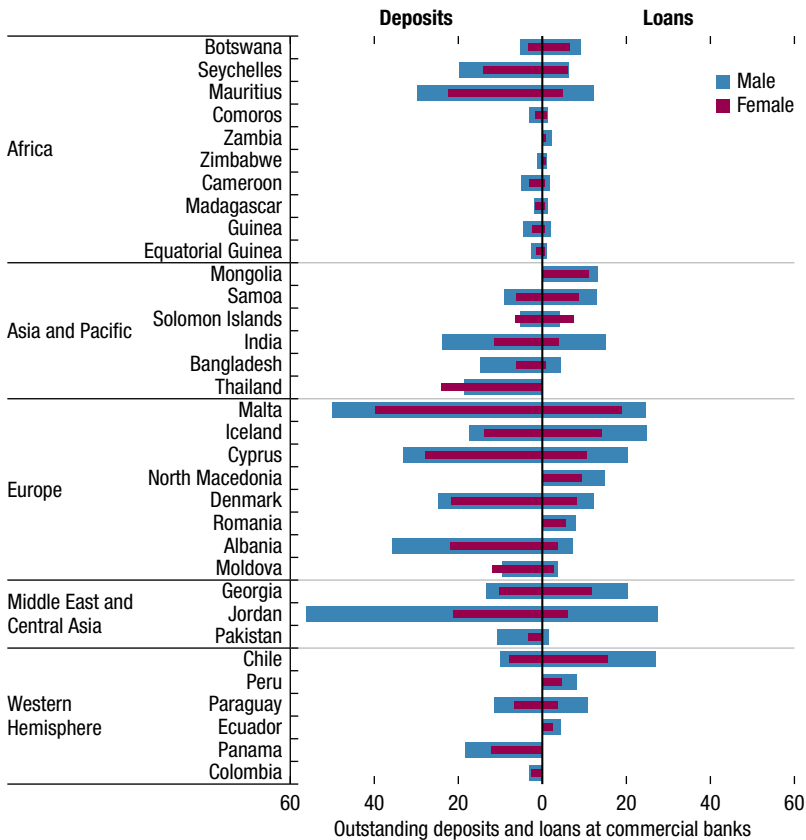
Financial service usage among the banked population can be measured by the value of outstanding deposits or loans. In 2019, the outstanding female-owned deposits at commercial banks were significantly lower than those owned by men in most economies in the sample (Figure 2.3).⁶ In sub-Saharan African countries, the outstanding value of deposits as a share of GDP was significantly lower for both men and women than in other regions with a large gender gap in some cases. The gender gap is smaller for a few countries (for example, Guinea and Zimbabwe) if it is measured by outstanding deposits per depositor. However, these economies also tend to have a large share of the unbanked population.⁷ A similar pattern holds for the outstanding number of female-owned loans at commercial banks.

⁵ Microfinance institutions offer small-scale loans typically to self-employed or informally employed low-income individuals and microenterprises. Microfinance institutions can be either deposit-taking or non-deposit-taking.

⁶ According to the FAS guidelines, gender-disaggregated data on outstanding deposits and loans need to be reported excluding balances in joint accounts. In practice, however, many countries do not exclude joint accounts when disaggregating by gender. See Annex 2.1.

⁷ The gender gap in outstanding deposits or loans as a share of GDP needs to be interpreted with caution as these indicators can be low for women due to lower female income. Outstanding deposits or loans as a share of female income may be a useful measure, but data on female income are not readily available.

Figure 2.3. Female-Owned versus Male-Owned Deposits and Loans, 2019
(Percent of GDP)



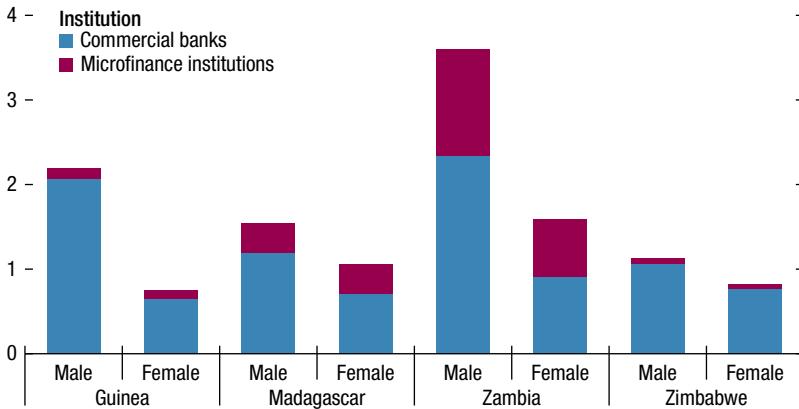
Sources: IMF Financial Access Survey; and IMF staff calculations.

Note: This chart shows the male-owned and female-owned outstanding deposits at commercial banks as a percent of GDP in 2019 or the latest available. The sample was selected based on data availability.

The gender gap in financial service usage seems to persist even when microfinance institutions are considered. Given the relatively high degree of access by women, women's loans from these institutions—measured by the amount of outstanding loans—tend to be close to the level of men's loans (Figure 2.4). However, overall, the gender gap in bank loans dominates in these countries in the sample.

In sum, before the COVID-19 pandemic, the degree of financial access and use via commercial banks was low for both men and women in many economies, particularly in sub-Saharan Africa. In addition, the gender gap in financial access and usage was sizable and persistent. This assessment is incomplete, however, as in many low- and middle-income countries, mobile money plays a key role in

Figure 2.4. Gender Gap for Microfinance Loans versus Bank Loans, 2019
(Percent of GDP)



Sources: IMF Financial Access Survey; and IMF staff calculations.

Note: This chart shows the outstanding loans at microfinance institutions and commercial banks as a percentage of GDP with gender breakdowns for select economies in 2019. The sample was selected based on data availability.

facilitating financial inclusion, including in sub-Saharan Africa (Bazarbash and others 2020; GSMA 2019). A more holistic assessment of financial inclusion will require accounting for the role of digital finance in addition to traditional banking services, but options are limited as supply-side gender-disaggregated data on digital finance are currently scarce.

With this foundational information, the next section examines policy responses adopted during the COVID-19 pandemic to support women's financial access, providing context for understanding financial access outcomes in the first year of the pandemic in 2020.

POLICY RESPONSE TO THE PANDEMIC: A GENDER PERSPECTIVE

Countries took a range of policy actions in response to the COVID-19 pandemic. However, documenting information on policy measures specifically targeted to support women's financial access is not straightforward because such information is qualitative in nature and often scattered across the public domain. In addition, available sources may exclude some measures that have benefitted women but have not necessarily been tailored for them. To bridge this information gap, this section presents policy measures targeting or benefiting women's financial access in sub-Saharan Africa, drawing on the UNDP and UN Women's COVID-19 Global Gender Response Tracker. This tracker contains information on a range of measures announced in response to the pandemic to support women's social and economic security.

TABLE 2.2.

Policy Responses Benefitting or Targeting Women for Selected African Countries	
Gender-Sensitive Policy Measures	Countries
Financial assistance to small and medium enterprises	Benin, Burkina Faso, Cabo Verde, Côte d'Ivoire, Guinea, Lesotho, Niger, Senegal, South Africa
Income support	Burkina Faso, Cote d'Ivoire, Ghana, Liberia, Senegal, Sierra Leone, South Africa, Togo
Loan assistance	Cabo Verde, Cameroon, Niger, Nigeria, Senegal, Uganda
Tax relief	Burkina Faso, Cameroon, Côte d'Ivoire, Madagascar, Mali, Niger, The Gambia

Source: IMF staff.

Note: For more details, see IMF COVID-19 Financial Access Policy Tracker, <https://www.imf.org/-/media/Files/Topics/COVID/FAS-tracker/stafi-2020-fas-covid-19-policy-response-tracker-smes-and-mobile-money-6896899-v22-dmsdr1s.ashx>, which contains information on gender-sensitive policies to support women's financial access according to the classification listed in this figure, drawing on the UNDP COVID-19 Global Gender Response Tracker.

The UN tracker classifies measures into four categories: (1) social protection, (2) labor markets, (3) violence against women, and (4) fiscal and economic policies. The measures under each of these categories are further divided into several types and subtypes. Policies discussed in this section focus on measures intended to help women continue to engage in financial transactions and alleviate credit constraints during the pandemic. These measures were largely drawn from the fourth category of the UN tracker—fiscal and economic policies. Such measures were identified for 69 countries, 19 of which are in sub-Saharan Africa. Some of these measures are specifically geared toward supporting women's financial access ("targeting women"), while others may be for the broader population yet benefit women ("benefitting women").⁸ The results are summarized in IMF's Financial Access COVID-19 Policy Tracker (<https://www.imf.org/en/Topics/imf-and-covid19/Policy-Responses-to-COVID-19>).⁹ Focusing on sub-Saharan African countries, these measures are classified in this section into four categories:

- **Financial assistance for small and medium enterprises (SMEs)** has been implemented in the form of grants and subsidies, directly or indirectly supporting women. In Benin, the authorities announced an XOF 74 billion (around \$124 million) stimulus package including temporary wage subsidies to support formal enterprises, craftspeople, and small service providers in the informal sector, which employs 98 percent of employed women. In Senegal, the government provided subsidies to female entrepreneurs in the informal sector who experienced significant operating losses during the

⁸ These classifications are also included in the gender component of the IMF Financial Access COVID-19 Policy Tracker.

⁹ These 69 countries include 11 low-income, 38 middle-income, and 20 high-income countries. Shirono and others (2021) provide an overview of these policies and show that the most popular gender-sensitive measures for middle- and high-income countries were income support and financial assistance to SMEs, while loan assistance was the most commonly used measure by low-income countries.

pandemic. In Lesotho, the government launched a three-month disbursing grant program of 50 million LSL (around \$3.5 million) for medium, small, and micro businesses, targeting the tourism sector. In Lesotho, women make up about 76 percent of workers in the accommodation and food services.

- **Income support** for women has been provided in the form of direct cash transfers to individuals or wage subsidies to compensate for lost income in certain sectors. For example, in Burkina Faso, the government announced cash transfers for workers in the informal sector, where 98 percent of employed women in the country work. Similarly in Togo, the government set up an unconditional cash transfer program for all informal workers above age 18 who could prove income loss due to the pandemic, and 65 percent of the beneficiaries were women.
- **Loan assistance**, such as loan moratoria, reduced interest rates, and loan guarantees, have been implemented to support sectors where women are overrepresented. In Liberia, as part of the Market Women and Small Informal Petty Traders Bank Loan Program, the government paid the loans owed by women who either work in markets or are petty or small traders in affected regions of the country. The Central Bank of Nigeria pledged N1 trillion (around \$2.5 billion) in loans to boost local manufacturing and production across critical sectors. In Nigeria, manufacturing accounts for 12 percent of women's employment compared with 5 percent of men's employment.
- **Tax relief** includes tax exemptions or delayed filing of returns, aimed at alleviating the financial burden of female entrepreneurs. In Mali, the government introduced tax advantages for sectors particularly affected by the pandemic, including tourism, hotels, and transport. The measure supports women's economic security as accommodation and food service activities employ over two times more women than men in the country.

The most popular gender-sensitive measure among these 19 sub-Saharan African countries was financial assistance to SMEs (adopted by 47 percent of the countries in the sample), followed by income support (42 percent), tax relief (37 percent), and loan assistance (32 percent). Similar patterns were found for other regions; income support and financial assistance to SMEs were the most popular policy responses in Europe, the Middle East and Central Asia, and the Western Hemisphere. Overall, the majority of the countries in the full sample (62 percent) adopted a single gender-sensitive measure, with only three countries implementing all four types.

Income support for individuals and wage subsidies for SMEs (to the extent that they provide support for household income) are likely to help maintain the number of depositors and borrowers as well as the level of outstanding deposits and loans during the COVID-19 pandemic. This support is of great relevance for the sub-Saharan Africa region, where over 90 percent of micro, small, and

medium enterprises (MSMEs) were adversely affected by the pandemic and female-led SMEs were particularly hit hard, with many reporting revenue losses of over 50 percent (IFC 2021). Other policy measures have likely had similar supportive effects on corporate accounts, but such gender-disaggregated data are not readily available. While gender-disaggregated data for households are available in the FAS, further policy analysis, including understanding the effectiveness of these policies, will require more granular data.¹⁰

WOMEN'S FINANCIAL ACCESS DURING THE PANDEMIC

The information on gender-sensitive policies to support financial access provides a useful context in understanding the outcomes during the COVID-19 pandemic based on FAS data. This section sheds light on women's financial access in select sub-Saharan African countries during the pandemic years (2020–21) using the FAS data from the 2021 and 2022 rounds.¹¹

Figure 2.5 shows how the share of depositors and borrowers by gender at commercial banks has changed between 2019 and 2020 or 2019 and 2021, depending on data availability. In some countries, the banked population with deposit accounts increased for both men and women, including some cases where the share of female depositors grew more than that of male depositors—for example, in Angola and Lesotho, the female share increased by 10 and 3 percentage points, respectively, which is more than double the increase for men. This positive change is likely due to the policy measures implemented in response to the pandemic. Like many other countries globally, these economies have taken monetary and fiscal measures to support the economy during the pandemic, including cash transfers and loan assistance (IMF 2021a; IMF 2021b). For example, in Lesotho, the government provided three-month salary subsidies and a grant program in some sectors, including textile, health, and tourism, that mainly benefitted women as they are the predominant workforce in these sectors. These policies are likely to have limited the adverse effects of the pandemic on financial access or even created the need to open bank accounts to receive these government services in some cases.

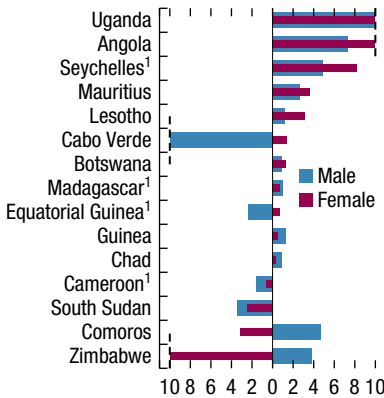
In other countries, however, the banked female population declined, and the degree of reduction varied widely across countries. A similar pattern exists for the share of borrowers, but the magnitude of the change was less pronounced, with a

¹⁰ Other pandemic measures also could have affected financial access and usage indicators. For example, mobility restrictions could have affected spending patterns (for example, lower spending and higher savings), affecting the outstanding amounts of deposits.

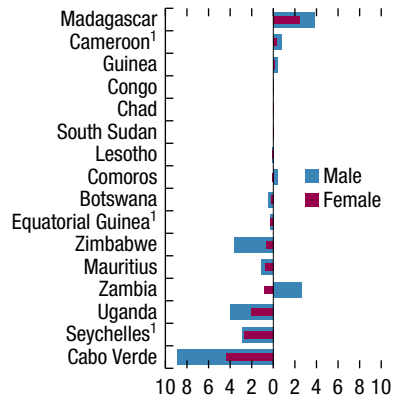
¹¹ The 2022 FAS round collects data for 2021. Newly reported annual data to the FAS are normally disseminated on a rolling basis on the FAS portal starting in June, with an announcement of the main findings in October or November.

Figure 2.5. Variation of Male and Female Depositors and Borrowers in Selected African Countries, 2019–2021
(Percent of male or female population)

1. Variation in Share of Depositors



2. Variation in Share of Borrowers



Sources: IMF Financial Access Survey; and IMF staff calculations.

Note: This figure shows annual changes in the share of male and female depositors and borrowers at commercial banks as a percentage of the male and female adult population in 2021. Positive (negative) numbers indicate an increase (decrease) in the banked population in 2021 compared with 2019. The sample was selected based on data availability. Uganda, Angola, Cabo Verde, and Zimbabwe have a higher variation of share of depositors, ranging from 10 to 35 percentage points of the male or female population. Uganda's variation of share of depositors is 17 percentage points (female) and 34 percentage points (male).

¹Indicates the variation was calculated using 2019 and 2020 data.

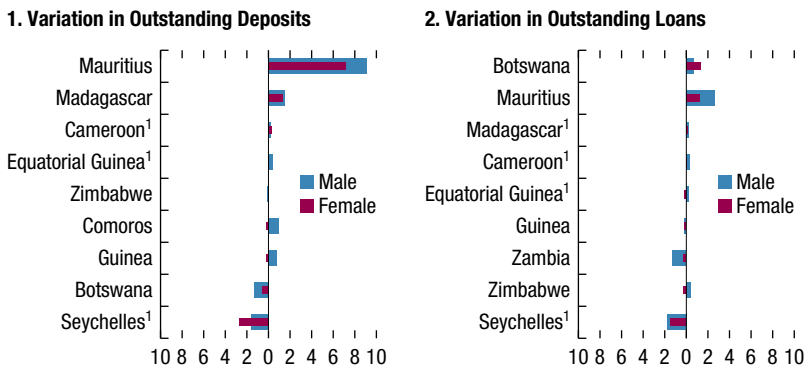
broadly unchanged state in most cases, particularly for 2020. In 2021, the share of borrowers, both men and women, had a greater decline in a few countries.

Figure 2.6 shows the evolution of the outstanding amounts of deposits and loans between 2019 and 2020 or 2019 and 2021, depending on data availability. Most countries in the sample had less than one percentage point change in both deposits and loans in both directions. This outcome likely reflects the fact that these are stock indicators and might not fully reflect the impact stemming from the COVID-19 pandemic, particularly in the 2020 data. Notable exceptions include Mauritius, which experienced a large change in the outstanding amounts of deposits between 2019 and 2021. For Mauritius, this is in part explained by the decline in nominal GDP.¹²

Table 2.3 summarizes the evolution of the gender gap in financial access measured by the indicators used in Figures 2.5 and 2.6 for sub-Saharan African countries. Overall, the gender gap remained broadly unchanged in most countries

¹² Deposits also increased by more than 10 percent for both men and women in nominal terms between 2020 and 2021.

Figure 2.6. Variation of Male- and Female-Owned Deposits and Loans, Selected African Countries, 2019–2021
(Percent of GDP)



Sources: IMF Financial Access Survey; and IMF staff calculations.

Note: This figure shows annual changes in the male-owned and female-owned outstanding deposits and loans at commercial banks as a percent of GDP in 2021. Positive (negative) numbers indicate an increase (decrease) in the banked population in 2021 compared with 2019. The sample was selected based on data availability.

¹Indicates the variation was calculated using 2019 and 2020 data.

between 2019 and 2021, except the gender gap measured by the share of depositors, which shows some narrowing of the gender gap in several countries by varying degrees while in a few cases the gender gap has increased significantly. In some cases, the narrowing of the gender gap was not necessarily due to a relative increase in women's financial access, but rather due to a larger reduction in the level of financial access for men compared with that of women in 2021 (for example, Cameroon for the share of depositors and Uganda for the share of borrowers).

Financial service usage through commercial banks may not provide a complete picture of the impact of the pandemic on vulnerable groups. A closer look at microfinance institutions—a key financial service provider to women in developing economies—reveals that the gender gap also remained broadly unchanged for this segment, albeit based on data for a handful of countries. However, an unchanged gender gap may not necessarily be reassuring given the sizable gender gap combined with low levels of financial access in many of these economies. More substantial efforts will be needed to further advance women's access and narrow the gender gap. In this regard, fintech adoption may play a key role as fintech can provide more efficient and often cheaper financial access to low-income households and SMEs, including those led by women (Sahay and others 2020; Loko and Yang 2022).

Interpreting these findings requires considering several factors. First, countries have taken various policy actions to mitigate the economic fallout from the pandemic, including measures to support women's financial access (IMF 2021a). These policy responses are likely to have limited severe disruptions to financial service

TABLE 2.3.

Evolution of The Gender Gap in Financial Access, 2021 vs 2019
(Percentage change)

Country	Depositors (share of adult population)	Borrowers (share of adult population)	Deposits (percent of GDP)	Loans (percent of GDP)
Angola	▲ -6.8			
Botswana ¹	■ -0.4	■ -0.2	■ -0.7	■ -0.6
Cabo Verde	▲ -30.9	▲ -4.6		
Cameroon ²	▲ -1.1	■ 0.4	■ -0.1	■ 0.3
Chad	■ 0.6	■ 0.0		
Comoros ¹	▼ 7.9	■ 0.5	▼ 1.3	
Equatorial Guinea ²	▲ -3.2	■ 0.0	■ 0.4	■ 0.4
Guinea	■ 0.8	■ 0.3	▼ 1.0	■ 0.0
Lesotho	▲ -2.0	■ -0.1		
Madagascar ¹	■ 0.2	▼ 1.4	■ 0.2	■ 0.1
Mauritius	▲ -1.0	■ -0.3	▼ 1.9	▼ 1.4
Seychelles ²	▲ -3.3	■ -0.2	▼ 1.1	■ -0.3
South Sudan	■ -0.9	■ 0.0		
Uganda	▼ 17.7	▲ -2.0		
Zambia		▼ 3.5		▲ -1.0
Zimbabwe	▼ 15.3	▲ -3.0	■ -0.2	■ 0.7

Sources: IMF Financial Access Survey; and IMF staff calculations.

Note: This table shows changes in the gender gap in 2021 measured by the four indicators used in Figures 2.7 and 2.8.

Positive (negative) numbers indicate increased (decreased) gender gap, grouped into three categories depending on the size of the changes: (1) a green triangle indicates a narrowing of the gender gap in 2021 compared with 2019 by equal to or greater than 1 percentage point; (2) a red reversed triangle indicates an increase in the gender gap by equal to or greater than 1 percentage point; and (3) a yellow rectangle indicates a broadly unchanged gender gap within the range of 1 percentage point. The sample was selected based on data availability.

¹Indicates the data for 2020 were used to calculate the change in the gender gap for one or two of the indicators.

²Indicates that the data for 2020 were used to calculate the change in the gender gap for all four indicators.

usage. Second, the impact of the COVID-19 pandemic might not have been fully evident in the 2020 data, in particular. The timing and the degree of lockdown measures have varied across countries. Due to these timing effects, the impact of the pandemic might have been more pronounced in the second half of 2020, thus not clearly revealed in the annual data. The 2021 data may be subject to a similar effect as conditions might have eased during 2021 in some countries. Similarly, some indicators tend to evolve slowly and may not fully reflect the impact of the pandemic. Finally, these observations are based on a limited set of countries, and more data points and controls will be needed for a complete and formal assessment.

CONCLUSION

The COVID-19 pandemic has amplified the need for better-tailored policies to support women that rely on gender-disaggregated data to track women's financial access. The early outcome data from the FAS presented in this chapter suggest that during the pandemic, women's financial access in sub-Saharan Africa remained unchanged or improved in some countries while others experienced

reversals to varying degrees. Overall, policy measures—income transfers and wage subsidies—undertaken in response to the pandemic seem to have helped support financial access. Nevertheless, given the lower level of financial access compared with other regions, efforts need to continue to further advance women’s financial access in sub-Saharan Africa. In addition, these findings are based on relatively small samples, and a more complete analysis will require more data with better country coverage. Other considerations, such as the role of digital financial services, including mobile money, will also help derive a complete picture.

The FAS gender-disaggregated data series offers a good starting point for countries that currently do not collect such data but are faced with the need to better monitor gender gaps in financial access. For countries that already collect some annual gender-disaggregated data, broadening series coverage (for example, data on female-owned SMEs or digital financial service usage disaggregated by gender) or collecting higher-frequency supply-side data (monthly or quarterly) can provide further insights. These data can also aid in understanding the effect of various policies implemented to support women’s financial access during the pandemic. Nevertheless, gender-disaggregated data from the FAS highlight the need to continue advancing women’s financial access and financial inclusion more generally in sub-Saharan Africa, given the low level of financial service penetration in many of the economies in the region.

ANNEX 2.1. MEASURING WOMEN'S FINANCIAL ACCESS

Financial inclusion is a multifaceted concept, and no single indicator can capture it entirely (Espinosa-Vega and others 2020). The analysis in this chapter is based on several indicators constructed from the FAS gender-disaggregated series to measure the adoption and usage aspects of women's financial inclusion. These indicators have the following analytical features and limitations:

- **Share of male and female depositors (or borrowers) at commercial banks in the male and female adult population:** The share is a proxy for the *size of the banked population among all adults*. The indicator can be higher than 100 percent and may overestimate the level of a banked population if account holders have multiple bank accounts and are not uniquely identified in the data reporting.
- **Proportion of female depositors (or borrowers) in total depositors (or borrowers):** This indicator captures the gender gap *among the banked population*. The closer to 50 percent the female share is, the smaller the gender gap. One caveat for this indicator, however, is that while most countries have the female share of the population at about 50 percent, there are some exceptions where the female share is significantly lower or higher than 50 percent. In that case, this indicator may not necessarily reflect the gender gap.¹³
- **Outstanding value of male- or female-owned deposits (or loans) as a percent of GDP:** This indicator directly measures the relative usage in terms of deposit or loan amounts *among the banked population* as a share of GDP. However, this indicator can be low for women due to lower female income. Using female or male income as a denominator may be more appropriate in this case, but such data are often not readily available.

Understanding the treatment of joint accounts is important when using these indicators. According to the FAS Guidelines and Manual, both joint accounts and joint account holders need to be counted when disaggregating the data by gender—based on the gender of the account holders. The following example illustrates the treatment of joint accounts in FAS reporting (Annex Table 2.1.1).

In this example, there are two deposit accounts—account A is owned by a woman and account B is jointly owned by a woman and a man. In this case, the number of depositors should be reported to the FAS as one male depositor and two female depositors, and the sum of male and female depositors is equal to the total number of depositors ($A + B + B = 3$). The number of accounts should be reported as one male-owned account and two female-owned accounts as gender disaggregation is based on the gender of account holders. In this case, the sum of female- and male-owned accounts ($A + B + B = 3$) is greater than the total number

¹³ See Ritchie and Roser 2019.

ANNEX TABLE 2.1.1.

Treatment of Joint Accounts in the Financial Access Survey			
	Depositors	Accounts	Outstanding Deposits
Total	3 (A+B+B)	2 (A+B)	\$(A+B)
Female	2 (A+B)	2 (A+B)	\$(A)
Male	1 (B)	1 (B)	\$0

Source: IMF staff.

of accounts ($A + B = 2$). Finally, joint accounts are excluded when gender-disaggregating outstanding loans and deposits in the FAS, although in practice, some countries include joint accounts when reporting outstanding loans and deposits in the FAS. Deviations from the FAS guidelines are recorded in the metadata, which can be checked to better understand the treatment of joint accounts by each country.

ANNEX 2.2. SUB-SAHARAN COUNTRIES SUBMITTING GENDER DATA

This annex lists countries with available FAS gender-disaggregated data (Annex Table 2.2.1). While 29 African countries report gender-disaggregated data to the FAS, some were not included in the analysis of this chapter due to insufficient data coverage. Countries covered in the IMF's COVID-19 Financial Access Policy Tracker are indicated by an asterisk in Annex Table 2.2.1. In addition, countries that have information on gender-sensitive policy measures to support women's financial access in the IMF's Financial Access Tracker are highlighted in blue.

ANNEX TABLE 2.2.1.

Sub-Saharan Countries Submitting Gender Data to the FAS							
1	Angola	9	Côte d'Ivoire ¹	17	Madagascar ¹	25	South Sudan
2	Benin ¹	10	Congo, Republic of	18	Malawi ¹	26	Togo ¹
3	Botswana ¹	11	Congo DRC	19	Mali ¹	27	Uganda ¹
4	Burkina Faso ¹	12	Equatorial Guinea ¹	20	Mauritius ¹	28	Zambia ¹
5	Cabo Verde ¹	13	Guinea ¹	21	Namibia ¹	29	Zimbabwe ¹
6	Cameroon ¹	14	Guinea-Bissau ¹	22	Niger ¹		
7	Chad ¹	15	Lesotho ¹	23	Senegal ¹		
8	Comoros	16	Liberia ¹	24	Seychelles ¹		

Source: IMF staff.

¹Countries covered in the IMF's COVID-19 Financial Access Policy Tracker.

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Epidemics, Gender, and Human Capital in Developing Countries

Stefania Fabrizio, Diego B. P. Gomes, Carine Meyimdjui, and Marina M. Tavares

For centuries, epidemics have disrupted lives and had negative consequences on human capital and economics. This chapter investigates the impact of epidemic episodes on school completion in developing countries, examining 338 epidemic episodes across countries from 1970 to 2019. The investigation yielded five key results: (1) estimates show that epidemic episodes are associated with reductions in school completion rates in the same year and one year later; (2) girls are affected more severely than boys; (3) children in primary education suffer more than those in lower secondary education; (4) viral epidemic episodes reduce completion rates more in lower secondary than in primary education, while the opposite occurs for bacterial events; and (5) longer epidemics are more harmful to children in lower secondary education.

INTRODUCTION

Epidemics have been a disruptive force in societies for centuries, causing loss of life and hampering economic activities. Apart from their immediate impact, they can have long-term consequences on growth by reducing children's years of education. This decline can occur directly due to school closures or indirectly due to loss in family income that can lead parents to withdraw their children from school. The COVID-19 pandemic is a stark example of how such events can have global impact. Developing countries are especially vulnerable to adverse human capital effects because children already face significant educational barriers. Primary and secondary education completion rates in these countries still lag behind those in developed nations.¹

Despite their frequency and harmfulness, little is known about the epidemics' impact on school completion rates in developing countries. Estimating the epidemic effect on school completion rates and identifying which students are more vulnerable to these events are critical for accurately measuring the long-term benefits of containing epidemics and improving government policies to

¹ According to the World Development Indicators from the World Bank, the primary and secondary education completion rates in low-income countries are 67 and 39 percent in 2020, respectively. In high-income countries, the completion rates are 99 and 97 percent, respectively.

reach the most vulnerable. This chapter fills in the information gap on the effect of epidemics on education in developing countries by answering three critical questions: (1) What is the impact of an epidemic episode on school completion rates? (2) Is the impact similar for boys and girls and primary and secondary education? and (3) Does the impact depend on the epidemic type or length?

To quantify the epidemics' impact on school completion rates in developing countries, data on historical epidemic episodes from the Emergency Event Database are combined with primary and lower secondary school completion rates from the World Development Indicators from the World Bank. Data were selected from 57 developing countries from 1970 to 2019. Analysis is limited to developing countries because epidemics are more common and more dangerous to children in these countries.

The findings indicate that an epidemic episode significantly reduces completion rates, affecting primary education more than lower secondary education and girls more than boys. Experiencing an epidemic episode is associated with a decline of 2.5 percentage points in the primary education completion rate and a decline of 1.8 percentage points in the lower secondary education completion rate in the epidemic year. Investigating the impact by the children's gender showed that an epidemic episode is associated with a reduction in boys' and girls' primary education completion rates of 1.8 percentage points and 2.8 percentage points, respectively. In lower secondary education, the completion rate falls by 1.8 percentage points for boys and 2.2 percentage points for girls. These results withstand a range of robustness checks, including controlling for government expenditures on education.

The findings also indicate that, regardless of the type of epidemic, an epidemic episode is associated with a decrease in school completion. However, viral events reduce completion rates in lower secondary education more than in primary education, whereas bacterial events have the opposite effect. Finally, findings showed that the length of the epidemic is related to a drop in school completion, and longer epidemics are harmful to students in lower secondary education.

This chapter adds to the growing body of research on the impact of epidemics on children's educational outcomes, discovering that epidemics are associated with a deterioration in years of education. The contribution is twofold: (1) a cross-country approach allows a comparison of the impact of different epidemic types and durations on school completion rates in a way that a single-country study cannot and (2) a study of epidemics across a broader range of countries allows an assessment of the relationship between epidemics and school completion rates while controlling for key factors such as country fixed effects and time fixed effects. Most papers in the literature focus on a single epidemic episode.

The literature on the impact of epidemics on children's outcomes is vast, with several studies focusing on specific epidemic episodes. Archibong and Annan (2017, 2021) examined the impact of the meningitis epidemic in Niger. Their findings indicate that the epidemic adversely affected school attendance,

educational outcomes, and labor market outcomes for affected individuals. Furthermore, the impact persisted for several years after the epidemic had ended. Bandiera and others (2019) conducted a quasi-random experiment to estimate the impact of the Ebola epidemic in Sierra Leone. The study found that younger girls tend to spend more time with men during a crisis, leading to an increase in out-of-wedlock pregnancies and a significant drop in school enrollment. However, the adverse effects can be reversed with interventions that allow girls to allocate their time differently, preventing unplanned pregnancies and enabling them to re-enroll in school after the crisis.

Other studies have focused on the impact of HIV on children's education outcomes. Guo, Li, and Sherr (2012) critically reviewed the global literature on the impact of HIV/AIDS on children's education outcomes. The study found that children affected by HIV/AIDS face multiple educational barriers, including stigma and discrimination, increased caregiving responsibilities, and limited access to health care and social services. More recently, de Azevedo and others (2021) and United Nations Children's Fund (2021) documented the impact of COVID-19 on children's education outcomes. The studies found that school closures, remote learning challenges, and economic hardships due to the pandemic have significantly disrupted children's education. These disruptions have disproportionately affected disadvantaged children, exacerbating preexisting inequalities in access to education.

This chapter also contributes to recent studies that examine the impact of epidemics on individual beliefs and attitudes. Eichengreen, Saka, and Aksoy (2022) examined the effects of epidemic exposure during an individual's "impressionable years" (ages 18 to 25) on their confidence in political institutions and leaders, while Eichengreen, Aksoy, and Saka (2021) measured the impact of such exposure on views about science, trust in scientists, and the perceived benefits of their work.

The rest of the chapter includes the following sections: data description, empirical strategy, effects of epidemic events, government expenditures on education, and conclusion.

DATA DESCRIPTION

The data on epidemic events come from the Emergency Events Database. The Centre for Research on the Epidemiology of Disasters (CRED) launched the database in 1988 with the initial support of the World Health Organization (WHO). The primary goal of the database is to support humanitarian efforts on a national and international scale. The initiative aims to provide an objective foundation for vulnerability assessment and priority setting and to rationalize decision-making for disaster preparedness. It contains essential core data on the occurrence and effects of more than 22,000 mass disasters worldwide from 1900 to the present day. The database comprises various sources, including UN agencies, nongovernmental organizations, insurance companies, research institutes, and press agencies.

Among the epidemic events, the data set includes all the episodes of viral, bacterial, parasitic, fungal, or prion diseases that the WHO declared as an epidemic—defined as a widespread occurrence of an infectious disease in a geographical area at a particular time. From the universe of epidemic events, only those that happened in developing countries were selected, resulting in a panel of 623 unique country-year observations from 1970 to 2019.

This data set is then combined with primary and lower secondary school completion rates and macroeconomic data from the World Bank's World Development Indicators (WDI). Annex Table 3.1.1 summarizes all the data used in the chapter. School completion rates are measured as the number of children completing the final grade of primary or lower secondary education relative to the number of children in the relevant age group. A limitation of this measure is that it does not allow for the identification of whether the decline in completion rates is due to a decline in enrollment or an increase in dropout rates that could affect completion years after the epidemic.

The merged data set maintains only country-year observations with information on school completion rates. Due to the irregularity of school completion rates data in the WDI, the final sample had 338 unique country-year epidemic events, comprising 57 developing countries from 1970 to 2019. Missing data on school completion rates are well distributed among epidemic and nonepidemic years, indicating that epidemics' harm did not cause an unprecedented impact on data collection that could bias the results (see Annex Table 3.1.3).

Table 3.1 presents the distribution of events by type of epidemics. Epidemics are divided between bacterial, viral, and other. Viral epidemics include Ebola, dengue, and viral meningitis, while bacterial epidemics include cholera, bacterial meningitis, and typhoid. The sample has a high concentration of bacterial (65 percent) and viral (29 percent) episodes. Other types of epidemics account for only 6 percent of the episodes. Table 3.2 documents the distribution of

TABLE 3.1.

Distribution of Events by Type of Epidemics				
	Total	Viral	Bacterial	Other
Number of Events	338	98	221	19
Share of Events	100%	29%	65%	6%

Source: Emergency Events Database.

TABLE 3.2.

Duration in Months of Epidemic Events						
	Mean	P25	P50	P75	P90	P95
Months	3.4	1	2	4	8	11

Source: Emergency Events Database.

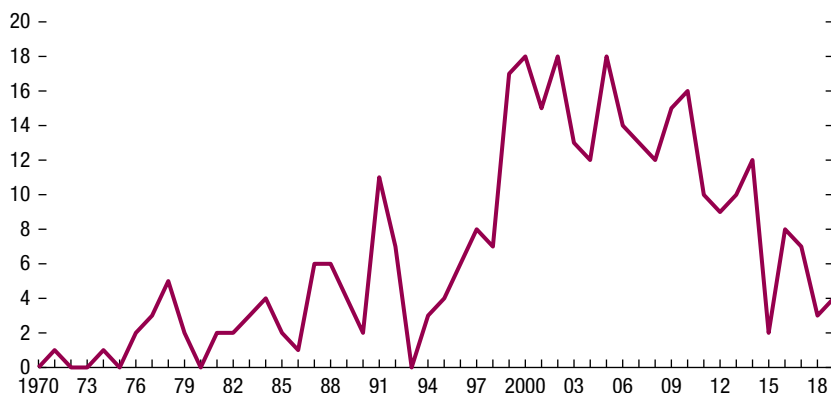
Note: This table summarizes duration information for the 237 epidemic events for which data exist on both the start and end dates (years and months). P25 refers to the 25th percentile, P50 refers to the 50th percentile (median), and so on.

epidemic duration in the sample. The mean and median (P50) durations equal 3.4 and 2 months, respectively, indicating that the distribution of duration is right skewed. The 75th duration percentile (P75) equals four months, showing that all events up to the third quartile of the distribution last less than six months. The 95th percentile (P95) is less than 12 months, showing that epidemics lasting more than one year in the sample are rare, representing less than 5 percent of events.

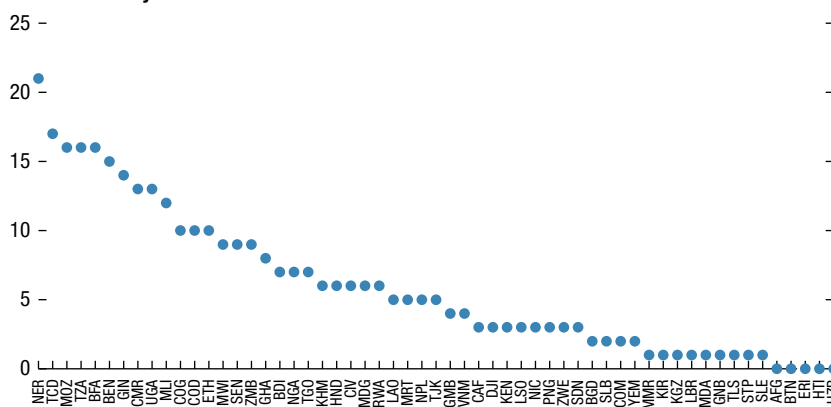
Figure 3.1 presents the historical and cross-country view of the epidemic events in the final sample. Panel 1 shows the number of epidemics reached an all-time high during the 2000s, with 18 episodes recorded in 2000, 2002, and 2005.

Figure 3.1. Number of Epidemic Events

1. Time Series



2. Cross-Country



Source: Emergency Events Database.

Note: Figure uses International Organization for Standardization (ISO) country codes. Panel 2 encompasses data from 1970 to 2019.

Since then, the number of episodes by year has dropped, with four episodes in 2019.² Panel 2 reveals considerable heterogeneity among countries. In the upper tail of the distribution, Niger (NER) is the country with the highest number of cases, 21 in total, followed by Chad (TCD) with 17 cases. In the lower tail of the distribution, five countries have 0 epidemic events: Afghanistan (AFG), Bhutan (BTN), Eritrea (ERI), Haiti (HTI), and Uzbekistan (UZB). Overall, the median and the average number of events per country are 5 and 6, respectively.

Previously stated facts reveal salient patterns about the distribution of epidemic events in the sample: (1) some types of epidemics are more likely than others, (2) long-lasting epidemics are rare events, (3) different periods in history have more events than others, and (4) certain countries suffer more from epidemics than others. The next sections outline an empirical strategy combined with robustness exercises that take these patterns into account.

EMPIRICAL STRATEGY

Fixed-effects panel regression models are used to estimate the impact of epidemics on school completion rates, allowing for the control of unobservable and unmeasurable factors that vary across countries but not over time. The main independent variable of interest is an epidemic indicator created from the Emergency Events Database. This indicator is set to 1 if a country experienced at least one epidemic event in the given year and 0 otherwise. Since epidemics can start at different times of the year, their consequences on school completion rates might materialize in the following year. Therefore, this analysis considers both the current and the one-year-lagged epidemic indicators. The effects of viral and bacterial events—the two most common epidemics in the sample—and the duration of epidemics are also explored.

Countries are indexed by i and years by t . Let $y_{i,t}$ be the school completion rate of interest, $E_{i,t}$ the epidemic indicator, $Y_{i,t}$ the log of real GDP per capita, λ_i country fixed effects, and θ_t year fixed effects. Note that the completion rates variable $y_{i,t}$ refers to any of the six series available from the WDI, where each series concerns a combination of education level (primary or lower secondary) and gender (girls, boys, or all). Then, for each type of completion rates series, the main regression specification can be stated as:

$$y_{i,t} = \beta_0 + \beta_1 E_{i,t} + \beta_2 E_{i,t-1} + \beta_3 Y_{i,t} + \lambda_i + \theta_t + \varepsilon_{i,t}, \quad (3.1)$$

where $\varepsilon_{i,t}$ is the error term. Because the regressions are estimated independently for each series of an education–gender combination, the estimated coefficients are gender and education-level specific. The coefficients β_1 and β_2 measure the current and lagged average marginal effects of epidemics on school completion

² The increasing trend until the 2000s stems from the discovery of new microorganisms, the proliferation of slums resulting from urbanization, and trade openness (Lindahl and Grace 2015; Oliveira and others 2020).

rates in the year of the epidemics. Negative and significant estimates for β_1 and β_2 imply that epidemics are associated with reductions in school completion. Possible confounding factors are controlled for through (1) λ_i country fixed effects to capture unobservable factors that vary across countries but are fixed over time, (2) θ_t year fixed effects to account for possible global trends that may affect the dependent variable, and (3) $Y_{i,t}$ the log of real GDP per capita to capture the country's income level and the fact that richer countries may have more resources to contain epidemic episodes and assist the population and schools.³

Given the high concentration of viral and bacterial epidemic events in the sample, the impacts of these specific types are also tested separately. Following the same approach as the main epidemic indicator, three new specific indicators were created to capture the incidence of viral, bacterial, and other types of epidemic events. Let $V_{i,t}$ be the indicator for viral events, $B_{i,t}$ the indicator for bacterial events, and $O_{i,t}$ the indicator for other types of epidemics. Then, the specification for a particular epidemic event is given by:

$$y_{i,t} = \beta_0 + \beta_1 V_{i,t} + \beta_2 V_{i,t-1} + \beta_3 B_{i,t} + \beta_4 B_{i,t-1} + \beta_5 O_{i,t} + \beta_6 O_{i,t-1} + \beta_7 Y_{i,t} + \lambda_i + \theta_t + \varepsilon_{i,t} \quad (3.2)$$

where $\varepsilon_{i,t}$ is again the error term. The interpretation of the coefficients β_1 to β_6 is the same as before, that is, negative values mean that specific epidemic events are associated with a drop in school completion rates. The controls are the same as in the main specification.

A further investigation explores how the duration of epidemics is associated with school completion rates. Two dummy variables are used to measure a short and long epidemic. An epidemic is short when it is less than or equal to two months (the median) and long when it is more than two months. The duration is calculated using the start and end year or month information contained in EM-DAT. Let $D_{i,t}^s$ indicate a short epidemic and $D_{i,t}^l$ indicate a long epidemic. The specification assessing the role of duration is then given by:

$$y_{i,t} = \beta_0 + \beta_1 D_{i,t}^s + \beta_2 D_{i,t}^l + \beta_3 Y_{i,t} + \lambda_i + \theta_t + \varepsilon_{i,t} \quad (3.3)$$

where the controls are the same as before and $\varepsilon_{i,t}$ is the error term. The interpretation of the coefficients β_1 to β_2 is the same as before, that is, negative values mean that specific epidemic events are associated with a decline in school completion rates. The controls are the same as in the main specification.

Since a high concentration of epidemic events occurs in certain countries and years, robustness exercises were performed to ensure that no specific country or year influenced the results. The primary model was re-estimated (equation 3.1) excluding each country and year.

³ A robustness exercise also controls for government expenditures on education per pupil as a percentage of GDP per capita. Adding this control reduces the sample size by over three-fourths, making comparisons with the previous estimates difficult.

THE EFFECTS OF EPIDEMIC EVENTS

Table 3.3 shows the results of the main regression exercises (equation 3.1) for school completion rates both in primary and lower secondary education. The results are presented for all children and separately by gender (boys and girls). The estimates reveal three principal facts. First, all estimates for current and lagged epidemic indicators are negative, showing that epidemic events are associated with reductions in school completion rates both in the same year and one year later. Second, girls are more severely affected than boys—that is, given the level of education, estimates for girls are always more negative than for boys. Third, primary education suffers more than lower secondary education—that is, conditional on the child's gender (the same holds for all children), estimates for primary education are always more negative than for lower secondary education.

A closer examination of the estimates shows that, for primary education, the fall in completion rates for all children due to epidemic shocks (current and lagged) is significantly greater than 2.0 percentage points on average. Girls' rates drop 2.8 percentage points in the current year and 2.2 percentage points a year later, while boys' rates decrease by 1.8 and 2.0 percentage points, respectively. For lower secondary education, the fall in completion rates varies between 1.8 percentage points (current) and 1.3 percentage points (lagged) on average when considering all children. In the current year, the rates for girls and boys fall by an average of 2.2 and 1.8 percentage points, respectively. A year later, the drop equals 1.9 percentage points for girls and 1.3 percentage points for boys.

The results should be interpreted as a lower bound since only the impact of epidemics on completion rates of the last grades of primary and secondary education are captured. It is reasonable to believe that the adverse effects also occur in the rest of the grades, not only in the last ones.

Many possible reasons could explain why girls may be more vulnerable to failing to complete school during epidemics than boys. First, parents may value boys' education more than girls', leading them to sacrifice a daughter's school

TABLE 3.3.

Effects of Epidemic Events on School Completion Rates						
	Primary Education			Lower Secondary Education		
	All	Boys	Girls	All	Boys	Girls
Epidemics, Current	-2.495*** (0.825)	-1.769* (0.981)	-2.789*** (0.875)	-1.756** (0.768)	-1.767** (0.876)	-2.232*** (0.819)
Epidemics, Lagged	-2.048*** (0.670)	-2.039*** (0.731)	-2.156*** (0.739)	-1.299 (0.794)	-1.295 (0.786)	-1.887** (0.901)
N	1,375	1,271	1,271	1,076	948	948
R ²	0.665	0.554	0.722	0.684	0.607	0.702

Source: Authors' calculations.

Notes: Standard errors in parentheses are robust and clustered at the country level. All specifications control for the log of real GDP per capita and include country and year fixed effects.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

attendance when families suffer a negative income shock (Oster 2009; Garg and Morduch 1998). Second, parents might also value girls' contribution to home production more than boys' during crisis time, leading them to take girls out of school or not enroll them in school (Archibong and Annan 2021; Bjorkman-Nyqvist 2013). Third, parents may also be more likely to support girls' marriage during epidemics because they benefit financially through wedding gifts that are highly valued during periods of hardship, such as during epidemics (Archibong and Annan 2021; Corno and Voena 2016; Corno, Hildebrandt, and Voena 2020; Ashraf and others 2020). Fourth, girls may lose access to safe spaces and become more vulnerable to rapes and unplanned pregnancies that can lead them to drop out of school (Bandiera and others 2019; Denney, Gordon, and Ibrahim 2015; Elston and others 2016; Menzel 2019). Last, girls are more prone to illness during epidemics due to their responsibilities at home, which include caring for younger siblings and sick family members, fetching and treating water (particularly dangerous in the case of cholera), and preparing food (Das 2017; Bjorkman-Nyqvist 2013; Malta and others 2019).

Table 3.4 explores the epidemics' impact depending on the epidemic type (equation 3.2). The disaggregated estimates confirm two of the previously mentioned findings, regardless of the type of epidemic event. Viral, bacterial, and other types of epidemics are all associated with drops in school completion rates for girls and boys in the same year and a year later. Conditional on the level of education, all types of epidemics are more harmful to girls than to boys. The only exception is the bacterial lagged effect for primary education. In that case, the estimates for boys are slightly more negative than for girls.

TABLE 3.4.

The Role of Viral, Bacterial, and Other Types of Epidemic Events						
	Primary Education			Lower Secondary Education		
	All	Boys	Girls	All	Boys	Girls
Viral, Current	-3.048** (1.460)	-2.138 (1.727)	-3.353** (1.361)	-3.474*** (1.032)	-2.701** (1.304)	-3.795*** (1.056)
Viral, Lagged	-0.938 (1.119)	-0.388 (1.239)	-1.073 (1.175)	-1.949** (0.967)	-1.778 (1.089)	-2.839** (1.215)
Bacterial, Current	-2.152** (0.926)	-1.559 (1.083)	-2.482** (0.952)	-0.777 (0.791)	-1.203 (0.835)	-1.254 (0.825)
Bacterial, Lagged	-2.287** (0.876)	-2.600** (0.995)	-2.402** (0.921)	-0.791 (0.898)	-0.946 (0.866)	-1.074 (0.905)
Other, Current	-2.923 (2.483)	-1.513 (2.767)	-2.416 (2.782)	-2.156 (3.293)	-1.928 (2.164)	-2.704 (3.829)
Other, Lagged	-6.559*** (2.027)	-6.393** (2.444)	-6.422*** (2.175)	-1.789 (2.324)	-1.120 (2.507)	-2.810 (3.106)
N	1,375	1,271	1,271	1,076	948	948
R ²	0.666	0.556	0.723	0.686	0.608	0.704

Source: Authors' calculations.

Notes: Standard errors in parentheses are robust and clustered at the country level. All specifications control for the log of real GDP per capita and include country and year fixed effects.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

This exercise reveals that certain epidemic events are more harmful than others, depending on the level of education. From the perspective of all children, for instance, viral episodes reduce completion rates more in lower secondary than in primary education. The opposite happens during bacterial events. These results hold when looking at girls and boys separately. Other types of epidemics are more negative in primary than in lower secondary education for all children combined. However, this last result becomes more mixed when boys and girls are looked at separately, depending on if the effect is in the same year or a year later.

Many possible reasons may account for the differences between the impact of viral and bacterial epidemics on children's completion rates, but data limitations do not allow for the identification of these channels. Differences in the transmission and mortality rates of these two types of epidemics by age group may help to explain the differences in impact. On the one hand, bacterial infections, particularly cholera, are known to be more harmful to young children (Ali and others 2012), which can help explain why young children in primary school are more negatively affected by bacterial epidemics than older children in secondary education. On the other hand, viral epidemics, including dengue⁴ and Ebola,⁵ have higher transmission rates among teenagers and adults than among young children, which can help explain why viral epidemics more negatively impact completion rates of children in secondary education. Another possible reason is that Ebola, a viral epidemic, has an extremely high fatality rate among adults, which can lead to an increase in the number of orphans and loss of income due to deceased parents. This effect is more likely to impact older children's completion rates since these children are old enough to work and have less access to orphanages due to their advanced age.

The importance of the epidemic length is examined next. Table 3.5 presents the results for the regressions that quantify the effect of epidemic duration (equation 3.3).⁶ All estimates are negative, showing that long and short epidemics are associated with a drop in school completion rates. Short epidemics are associated with more significant reductions in school completion in primary education. Girls' completion rates drop by 2.95 percentage points, and boys' fall by 2.56 percentage points. In lower secondary education, longer epidemics are associated with much larger declines in completion rates than in short epidemics. A long epidemic is associated with a drop of 3.69 percentage points in girls' completion rates, while boys' completion rates drop by 2.92 percentage points.

⁴ Dengue is primarily spread by the bite of infected *Aedes* mosquitoes, including *Aedes aegypti* and *Aedes albopictus* mosquitoes. A mosquito becomes infected when it bites a person whose blood contains dengue. See Thai and others (2015) for a discussion about dengue transmission by age.

⁵ Ebola is spread by direct contact with blood or other body fluids (such as vomit, diarrhea, urine, breast milk, sweat, and semen) of an infected person who has Ebola or recently died from Ebola—a possible reason for the lower transmission of Ebola to children since they can be easily isolated from the sick and do not participate in funerals. See Glynn (2015) and WHO (2015).

⁶ The sample sizes of the duration exercises are slightly smaller than those of previous exercises, as monthly durations could be calculated for only a subset of the epidemics in the sample (237 of 338 events) due to missing information on the start month, end month, and year for all epidemics.

TABLE 3.5.

Effects of the Duration of Epidemics on School Completion Rates						
	Primary Education			Lower Secondary Education		
	All	Boys	Girls	All	Boys	Girls
Short	-2.680**	-2.566**	-2.953**	-0.435	-1.036	-0.1370
Epidemics	(1.054)	(1.265)	(1.144)	(1.053)	(1.075)	(1.190)
Long	-2.271*	-1.409	-2.622*	-3.539***	-2.922**	-3.690***
Epidemics	(1.310)	(1.448)	(1.314)	(1.184)	(1.253)	(1.156)
N	1,280	1,174	1,174	926	811	811
R ²	0.668	0.558	0.723	0.670	0.585	0.688

Source: Authors' calculations.

Note: Standard errors in parentheses are robust and clustered at the country level. All specifications control for the log of real GDP per capita and include country and year fixed effects.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

The robustness of the findings is assessed by considering lagged effects of more than a year and country/year importance. To do so, the main model (equation 3.1) was re-estimated first by including additional epidemic lags ranging from two to four years and second by excluding each country and year. The three principal conclusions from the main exercise are still valid. Furthermore, all extra lag estimates are negative, suggesting that epidemics have a long-term negative association with completion rates. These estimates are highly statistically significant for girls in lower secondary education but less for the other groups.

GOVERNMENT EXPENDITURES ON EDUCATION

In this section, an additional robustness exercise explores the importance of government expenditures in education to reduce the epidemic's impact on school completion. Government expenditures on education can significantly reduce an epidemic's negative impact on school completion. Well-funded schools may support students during epidemics, reducing dropout rates. Also, well-funded schools may have better sanitation and water supplies, reducing the need for school closure during epidemics.

To assess its importance, a control was included in the main regression specification (equation 3.1) for the government expenditures on education per pupil as a percentage of GDP. Let $G_{i,t}$ denote the government expenditures variable. Then, the specification now estimated can be stated as:

$$y_{i,t} = \beta_0 + \beta_1 E_{i,t} + \beta_2 E_{i,t-1} + \beta_3 Y_{i,t} + \beta_4 G_{i,t} + \lambda_i + \theta_t + \varepsilon_{i,t}, \quad (3.4)$$

where the rest of the notation is the same as the main regression (equation 3.1). For primary school completion rates, government expenditures are included per pupil in primary education only. Similarly, for lower secondary school completion rates, only government expenditures per pupil on secondary school are considered.

Table 3.6 summarizes the results. Note that adding this control significantly reduces sample size by over three-fourths, making comparisons with the previous

TABLE 3.6.

Results when Including Government Expenditures on Education						
	Primary Education					
	All	All	Boys	Boys	Girls	Girls
Epidemics, Current	0.213 (1.121)	0.361 (1.023)	0.971 (1.172)	1.142 (1.077)	-0.591 (1.179)	-0.501 (1.085)
Epidemics, Lagged	-2.857** (1.118)	-2.918*** (1.080)	-3.019** (1.185)	-3.098** (1.151)	-3.047** (1.289)	-3.088** (1.256)
Government Expenditure	No	Yes	No	Yes	No	Yes
N	247	247	242	242	242	242
R ²	0.740	0.745	0.640	0.65	0.771	0.771
	Lower Secondary Education					
	All	All	Boys	Boys	Girls	Girls
Epidemics, Current	-0.015 (0.862)	-0.011 (0.871)	0.104 (0.943)	0.103 (0.946)	-0.130 (0.866)	-0.121 (0.879)
Epidemics, Lagged	-0.657 (1.115)	-0.657 (1.077)	-0.327 (1.015)	-0.335 (0.992)	-1.059 (1.308)	-0.991 (1.242)
Government Expenditure	No	Yes	No	Yes	No	Yes
N	155	155	155	155	155	155
R ²	0.719	0.719	0.703	0.71	0.703	0.703

Source: Authors' calculations.

Note: Standard errors in parentheses are robust and clustered at the country level. All specifications control for the log of real GDP per capita and include country and year fixed effects. For primary school completion rates, government expenditures per pupil are used on primary education only. For lower secondary school completion rates, government expenditures per pupil are used on secondary school only.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

estimates difficult.⁷ Regardless, the results still display negative estimates overall, especially for the lagged epidemic indicator. When comparing the exercise with and without the government expenditure controls, restricting to the sample of episodes where government expenditure data are available, it was found that this control does not affect the results significantly, indicating that government expenditures in education may not be an important margin that affects the epidemic impact on school completion. Due to the important reduction of the sample size, the uncertainty around the estimates increases, especially for lower secondary education.

CONCLUSION

This chapter offered an analysis of the impact of past epidemics on school completion rates. The results suggest that past epidemics reduced completion rates by an average of 2.5 percentage points in primary education and 1.8 percentage points in lower secondary education. Furthermore, girls were more severely affected than boys. In primary education, completion rates decrease by 1.8 percentage points for boys and 2.8 percentage points for girls. In lower secondary education,

⁷ Annex Table 3.1.2 lists the set of countries used in this exercise.

rates fall by about 1.8 percentage points for boys and 2.2 percentage points for girls. These findings are consistent with those in previous studies that examined single epidemic episodes, finding that epidemics are associated with a decline in school completion rates and that effects are more significant for girls. For example, Archibong and Annan (2021) found a significant reduction in years of education for school-aged girls relative to boys after a meningitis epidemic in Niger. Bandiera and others (2019) found that young girls in highly disrupted villages during the 2014 Ebola outbreak in Sierra Leone had a persistent drop in school enrollment postcrisis. All types of epidemic events were found to reduce completion rates. However, viral epidemic episodes reduce completion rates more in lower secondary than in primary education, while the opposite happens for bacterial events. In addition, longer epidemic episodes are associated with more harmful effects in lower secondary education.

This study highlighted the urgent need for preparation and targeted policies to address the negative impact of epidemics on education outcomes in developing countries, particularly on the most vulnerable groups. The cost of epidemics on school completion rates is substantial, affecting children for many years afterward. More research could identify effective policy interventions to reduce the impact of outbreaks on education outcomes, particularly for girls, and to understand the main channel through which epidemics reduce completion rates.

ANNEX 3.1. DATA DETAILS

ANNEX TABLE 3.1.1.

Data Sources	
Data	Source
Epidemics (All Epidemics, Viral, Bacterial, and Other)	Emergency Events Database (EM-DAT)
School Completion Rate (All, Boys, Girls, Primary, and Secondary)	World Bank's World Development Indicators (WDI)
GDP per Capita	World Bank's World Development Indicators (WDI)
Government Spending on Education per Pupil as a Share of GDP	World Bank's World Development Indicators (WDI)

ANNEX TABLE 3.1.2.

List of Countries in the Sample			
ISO	Country Name	ISO	Country Name
AFG	Afghanistan ¹	MDG	Madagascar ¹
BDI	Burundi ¹	MLI	Mali ¹
BEN	Benin ¹	MMR	Myanmar
BFA	Burkina Faso ¹	MOZ	Mozambique ¹
BGD	Bangladesh ¹	MRT	Mauritania ¹
BTN	Bhutan ¹	MWI	Malawi ¹
CAF	Central African Republic ¹	NER	Niger ¹
CIV	Cote d'Ivoire ¹	NGA	Nigeria
CMR	Cameroon ¹	NIC	Nicaragua ¹
COD	Democratic Republic of the Congo ¹	NPL	Nepal ¹
COG	Republic of the Congo ¹	PNG	Papua New Guinea
COM	Comoros ¹	RWA	Rwanda ¹
DJI	Djibouti	SDN	Sudan
ERI	Eritrea ¹	SEN	Senegal ¹
ETH	Ethiopia ¹	SLB	Solomon Islands
GHA	Ghana ¹	SLE	Sierra Leone ¹
GIN	Guinea ¹	STP	São Tomé and Príncipe ¹
GMB	The Gambia ¹	TCD	Chad
GNB	Guinea-Bissau	TGO	Togo ¹
HND	Honduras ¹	TJK	Tajikistan
HTI	Haiti	TLS	Timor-Leste ¹
KEN	Kenya ¹	TZA	Tanzania ¹
KGZ	Kyrgyz Republic	UGA	Uganda ¹
KHM	Cambodia ¹	UZB	Uzbekistan
KIR	Kiribati ¹	VNM	Vietnam ¹
LAO	Lao PDR ¹	YEM	Yemen
LBR	Liberia	ZMB	Zambia ¹
LSO	Lesotho ¹	ZWE	Zimbabwe ¹
MDA	Moldova ¹		

¹ Countries included in the section titled Government Expenditures on Education.

Note: ISO refers to International Organization for Standardization.

ANNEX TABLE 3.1.3.

Missing Data		Primary School Completion	Low Secondary School Completion
Epidemic Year	Missing	46%	60%
	Nonmissing	54%	40%
Nonepidemic Year	Missing	51%	61%
	Nonmissing	49%	39%

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COVID-19 and Gender Inequality: Impact in Southern Africa

Fatou Kiné Thioune, Giorgia Albertin, Romina Kazandjian, and Tianyuan Wang

This chapter analyzes the impact of the COVID-19 pandemic on gender disparities in the southern African region. It explores the extent and channels through which the COVID-19 pandemic has affected gender inequality. The analysis provides significant evidence for a “she-cession”: the pandemic economically affected women disproportionately relative to men. In South Africa, employment, hours worked, and incomes for women were more negatively affected than for men. In Namibia and Eswatini, women were less likely to be employed during the pandemic and were more likely to have lost employment due to the pandemic. In Lesotho, women were less likely to be formally employed than men during the pandemic. This chapter provides evidence supporting women’s heavier unpaid work burden and occupational gender segregation as transmission channels and that business employment acted as a countercyclical shock absorber for women in South Africa and Lesotho.

INTRODUCTION

The COVID-19 pandemic has been disproportionately impacting women worldwide, and likely even more severely in developing countries, due to heavier unpaid care work burdens, occupational gender segregation in affected sectors, increased gender-based violence, and possibly permanent loss of human capital. Unlike in previous recessions, which mostly affected traditionally male occupations in construction and manufacturing, the current crisis is referred to as a “she-cession,” having damaged the services industry, that is, retail, tourism, and hospitality, which primarily employs females (Alon and others 2020). The United Nations has warned that the negative effects on women are even worse in developing countries, where 70 percent of female employment is in the informal sector, with low wages and high job insecurity (United Nations 2020). Moreover, school and daycare closures have increased the already higher burden of unpaid care work that women perform in the household and have led to missed years of schooling as millions of girls worldwide were forced to permanently drop out of school (United Nations 2020). As a result of restrictions on movement due to lockdown measures, gender-based violence also “increased exponentially” globally (United Nations 2020). Thus, the pandemic and associated lockdown measures exposed and exacerbated pre-existing macro-critical

gender inequalities. Addressing these gaps in the labor market, education, health, and finance will be key to supporting the postpandemic recovery.

This chapter analyzes gender disparities and empirically quantifies the disproportionate impact of the COVID-19 pandemic on women in the southern African region. Using macro-level data and micro-level data from surveys on individuals, stylized facts are derived on pre-existing gender disparities in the region. Pre- and post-COVID-19 pandemic survey data are then used to empirically estimate the disproportionate effect of the pandemic on women's economic outcomes, specifically in South Africa, Eswatini, Lesotho, and Namibia. The chapter draws on an IMF working paper (Thiouné and others, forthcoming) that empirically investigates the impact of the COVID-19 pandemic on gender disparities in the southern African region and related transmission channels.¹ The data show that significant gender inequalities still exist in the Southern Africa Customs Union (SACU) region, especially in employment, political representation, and health, which the COVID-19 pandemic exacerbated. Evidence suggests that the COVID-19 pandemic has widened the gap between male and female economic outcomes in South Africa, Eswatini, Lesotho, and Namibia, as it has decreased women's employment and income more relative to men and has disproportionately increased the unpaid work burden on women. The data also show that businesses—mostly informal—acted as a shock absorber. The disproportionate negative effects of the pandemic on women are due to a heavier unpaid work burden and occupational gender segregation.

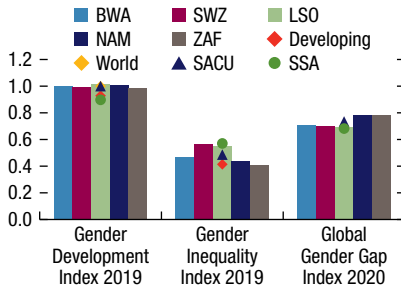
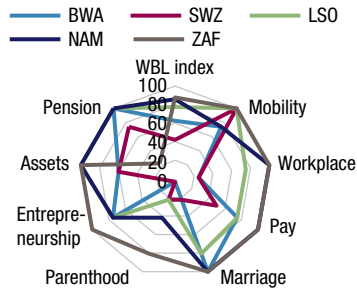
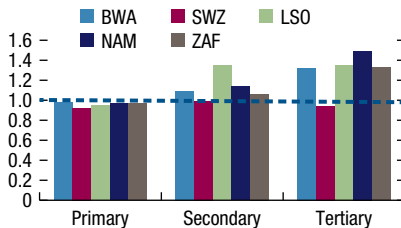
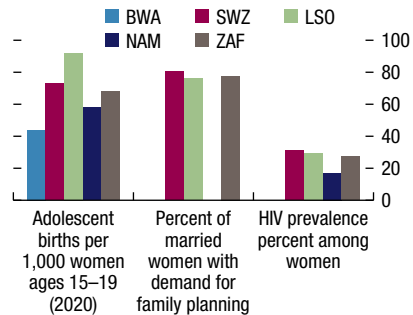
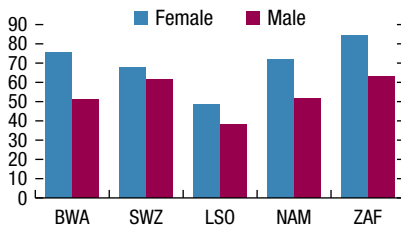
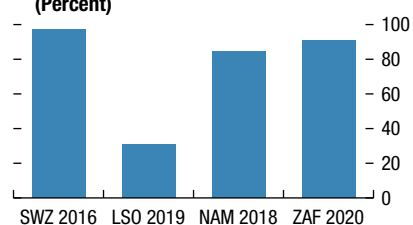
The rest of the chapter presents stylized facts about gender disparities in the region and highlights the main results from the analyses. Some policy recommendations and conclusions are then offered.

STYLIZED FACTS: GENDER DISPARITIES IN SOUTHERN AFRICA BEFORE THE COVID-19 PANDEMIC

While overall gender inequality may not seem too striking in the SACU region, aggregate indices mask significant gender inequalities in several dimensions, such as health, labor markets, and legal empowerment. At the aggregate level, the SACU countries do relatively well with regard to gender parity: Gender Development Indices² are close to 1 for all of them and above the sub-Saharan Africa and Emerging Markets and Developing Economies averages (see Figure 4.1). Their

¹ Bluedorn and others (2021) explore gender gaps during the pandemic in South Africa but do not empirically estimate the impact on gender disparities. Some evidence has also been provided in Lesotho on a “she-cession,” although covering statistical evidence without empirical estimations (Government of Lesotho 2020). Research also exists on COVID-19 and gender in other African countries, such as Ethiopia and Nigeria (Chapter 6). Less research has been done on southern African countries, and this chapter fills this gap in the literature.

² The Gender Development Index is computed as the female-to-male Human Development Index ratio; the Human Development Index captures health measured as life expectancy, education, and gross national income.

Figure 4.1. SACU Countries: An Overview of Gender Inequality**1. Gender Indicators^{1,2}**
(Index, 2019)**2. Women, Business and the Law**
(2019)**3. Gender Parity Index for School Gross Enrollment**
(Girls-to-boys ratio, to the latest available year before 2020)**4. Health Indicators**
(Percent, to the latest available)**5. Employment in Services**
(Percent of female and male employment, modeled ILO estimate, 2019)**6. Female Share of Total Unemployed Population Not Available for Work Due to Unpaid Domestic Work**
(Percent)

Sources: Eswatini CSO; Lesotho BOS; Namibia Statistics Agency; StatsSA; UNDP; UNFPA; World Bank Gender Database; World Economic Forum; World Bank Women, Business and the Law; and World Bank World Development Indicators.

Note: In panel 3, due to the data limitation, the latest available data varies for the following: Botswana (2015), Eswatini (2019), Lesotho (2017), Namibia (2018), South Africa (2019). BWA = Botswana; Eswatini CSO = Eswatini Central Statistical Office; Lesotho BOS = Lesotho Bureau of Statistics; LSO = Lesotho; NAM = Namibia; SACU = Southern Africa Countries Union; SSA = sub-Saharan Africa; StatsSA = Statistics South Africa; SWZ = Eswatini; UNFPA = United Nations Fund for Population Activities; UNDP = United Nations Development Programme; WBI = Women, Business and the Law; ZAF = South Africa.

¹Higher scores for the Gender Development Index and the Global Gender Gap mean better gender equality, while for the Gender Inequality Index they represent worse gender disparities. The Global Gender Gap index is based on Global Gender Gap Reports, which reflect values collected in the previous year ($t-1$).

²The SACU index is calculated in simple average of all five SACU countries. The Index of World, sub-Saharan Africa, and developing countries are referred directly from the data source.

Global Gender Gap³ scores are lower but still relatively high compared to other Emerging Markets and Developing Economies. Yet, the SACU region does worse with regard to the Gender Inequality Index⁴ compared to the Emerging Markets and Developing Economies. Education gaps at the primary, secondary, and tertiary levels have been largely closed, with girls' enrollment even higher than boys' in some countries. Although these indices suggest fairly gender-equal societies in southern Africa, gender gaps persist in the labor market, including labor force participation, employment, and opportunities, especially in legal rights. The gender gap in access to finance persists in most countries and women face more difficulties in running profitable businesses. Although women are as likely as men to have bank or mobile money accounts overall, fewer women borrow money compared to men.

Indicators of women's health still paint a particularly worrying picture. For all five countries, about 60 percent or more of people over the age of 15 years who live with HIV are women. The maternal mortality rate is also above regional and economic peers. Gender-based violence is prevalent in all SACU countries, and as a result of restrictions on movement, it is exponentially on the rise globally during the pandemic (United Nations 2020). According to the United Nations Populations Fund, in Botswana, 67 percent of women have experienced abuse (UNFPA). The Violence Against Children Survey conducted in 2007 reveals that in Eswatini, 48 percent of women aged 13-24 have experienced sexual violence in their lifetime. In Namibia, 33 percent of married women aged 15-49 years have experienced violence from their partner, according to the 2013 Namibia Demographic and Health Survey. In Lesotho, 33 percent of women have been abused by an intimate partner (UNAIDS 2019). SaferSpaces reports that in South Africa, 25 to 40 percent of women have experienced violence from their partners.

In the labor market, gender gaps also remain prevalent. The female labor force participation rate is lower than for males in all countries, and female unemployment is higher in all SACU countries except Namibia. Women earn less and are more likely economically poor. Political representation and empowerment lag, in part due to legal restrictions, except for Namibia and South Africa. Research has shown that legal restrictions in various dimensions drive gender gaps (Gonzales and others 2015). The services, the informal, and the contact-intensive sectors most negatively affected by lockdown policies across the region predominantly employ women. Women's share of employment in services is higher than men's in all SACU countries and is particularly high in South Africa at 84 percent.

³ The Global Gender Gap captures national gender gaps in economic participation, educational attainment, health, and political criteria.

⁴ The Gender Inequality Index captures gender inequality across areas of health (maternal mortality ratios and adolescent fertility rates), empowerment (share of parliamentary seats and education attainment at the secondary level for both males and females), and labor force participation (rates by sex). While the Gender Inequality Index has drawbacks (such as a complicated functional form and a combination of indicators that compare men and women with indicators that pertain only to women), it is preferable to alternatives such as the Gender Development Index, in which one of the main components is not observed and is imputed.

Informal employment is also significant. In all countries except South Africa, the informal sector employs more than 50 percent of women. Most of the unpaid domestic work burden falls on women, keeping them out of the labor force. From micro-level data in all the sample countries except for Botswana, out of the unemployed respondents whose primary reason for not working is unpaid domestic work, the overwhelming majority—over 70 percent—are women.

The COVID-19 pandemic has exacerbated all of these challenges, as described in the next section.

MAIN RESULTS

The previous section provided a short overview of gender equality in the region and initial statistics on how the pandemic has impacted the overall picture. The next subsections present the results on the impact of the pandemic on women through different indicators and for each country.

South Africa

The analysis for South Africa shows that the COVID-19 pandemic affected more women than men and those women experienced a larger loss in employment, reduction in weekly hours worked, and loss in income (Table 4.2.1). These disproportionate effects are due to sectoral and occupational gender differences and childcare and unpaid work burdens mostly falling on women. That is, evidence suggests a “she-cession” in South Africa. However, some evidence also suggests that business employment, especially in informal businesses, acted as a counter-cyclical shock absorber for women. Generally, women in low- and medium-skilled occupations and women with children were particularly affected. However, education reduces the disproportionate negative effect of the pandemic on women.

Two data sources—the National Income Dynamics Survey–Coronavirus Rapid Mobile Survey and the Quarterly Labor Force Survey—were used to conduct the analysis and derive these results. Linear probability panel regressions were run to empirically estimate the effect the COVID-19 pandemic had on women and various outcome variables (Thioune and others forthcoming).

Impact of the COVID-19 Pandemic on Women’s Employment

The analysis shows that the COVID-19 pandemic reduced employment for women significantly more than for men. The pandemic decreased the probability of employment by 12.7 percentage points for men and by 14.6 percentage points for women, decreasing the probability of employment for women by 1.94 percentage points more than men.⁵ Controlling for the transmission channels shows that the type of employment, occupation, children, and education explain the disproportionate negative effect of the pandemic on women.

⁵ The analysis is based on the National Income Dynamics Survey–Coronavirus Rapid Mobile Survey.

Notably, the disproportionately negative effect of the pandemic on women's employment is largest among the self-employed.⁶ Women with regular jobs and casual work and self-employed women were more negatively affected than their male counterparts. No evidence indicates that women who were running businesses were less or more affected relative to their male counterparts. Furthermore, women with regular jobs in low-skilled and medium-skilled occupations were more affected than their male counterparts. Additionally, women with children below seven years old were also more affected than men in the same situations. Similarly, more educated women were more affected than more educated men.

Evidence indicates that the kind and sector of employment and occupation are transmission channels. In all types of employment (except unpaid work), the pandemic disproportionately affected women's employment.⁷ The analysis suggests that the pandemic disproportionately affected more women's employment than men's in all sectors except mining. Women working in agriculture were most disproportionately affected by the pandemic relative to women in manufacturing and services. Regarding occupation type, the pandemic disproportionately affected more women in jobs at all skill levels, with the negative effect being the largest in low- and then in medium-skilled occupations.

Finally, the analysis indicates that the pandemic disproportionately affected women who spend more time fetching water and wood compared to their male counterparts. However, the disproportionate negative impact of the pandemic on women relative to men is larger for women who do not spend time fetching water and wood. The fact that women who spend more time fetching water and wood are less likely to be employed even before the pandemic explains the latter result; therefore, they are less likely to lose employment because of the pandemic.

Analysis using the National Income Dynamics Survey–Coronavirus Rapid Mobile Survey suggests that women running businesses were not disproportionately affected by COVID-19, even after controlling for different transmission channels. These businesses are mostly informal and the associated day-to-day activities, although vulnerable to the pandemic shock, can still be carried out despite COVID-19 related restrictions due to their informal nature. For instance, women selling vegetables at the market can still carry out these activities despite the lockdown, although they might incur a decrease in hours worked or income from these activities due to reduced mobility and economic activity.

The results obtained from the Labor Force Survey are similar to those from the National Income Dynamics Survey as the pandemic did not affect women

⁶ Also included are the respondents' different types of employment, namely regular job, casual work, self-employment, and business employment. Regarding occupation (in the case of a regular job) and business occupation (in the case of a business job or self-employment), the classification follows the International Standard Classification of Occupation, and the different categories are classified based on skill level. Managers, professionals, technicians, and associate professionals constitute the high-skilled occupations; clerical support workers, service and sales workers, craft and related trades workers, armed forces occupations, plant and machine operators, and assembly, and skilled agricultural, forestry and fishing workers are medium-skilled; and elementary occupations are low-skilled occupations.

⁷ The analysis is based on the Labor Force Survey for South Africa.

running businesses more than men. However, the sector of business employment and occupation are channels through which the pandemic affected women's businesses more relative to those owned by men. Controlling for sector and occupation reveals that the pandemic disproportionately affected women running businesses in manufacturing and other sectors and those in low-skilled occupations more than their peers in other sectors.

A more detailed analysis points to informal businesses acting as a shock absorber for the impact of the pandemic on women's employment. To paint a more complete picture, the analysis included the disproportionate impact of the pandemic on women running formal compared to informal businesses. The National Income Dynamics Survey–Coronavirus Rapid Mobile Survey included 16 formal businesses and 398 informal ones, while the Labor Force Survey included 1,759 formal and 7,250 informal ones. The baseline specification was run separately for informal and formal businesses, where the binary business employment variable equals 1 if the person runs an informal and formal business, respectively, and 0 otherwise. From the National Income Dynamics Survey data, significant evidence indicates that the pandemic disproportionately affected women running formal businesses relative to men running formal businesses. However, no disproportionate impact of the pandemic exists for women running informal businesses.

The Labor Force Survey data do not offer significant evidence of the disproportionate effects of the pandemic on either women-run formal or informal businesses. However, results from the National Income Dynamics Survey provide some evidence to suggest that informal businesses served as a cushion for women against the pandemic shock.

This evidence suggests that informal (but not formal) businesses served as a cushion for the pandemic shock, particularly for women. This finding does not necessarily contradict recent research on Northern Africa suggesting that informal employment did not increase and therefore was not countercyclical during the COVID-19 pandemic (IMF 2021a). The findings are specifically on women's business employment relative to men's, which did not decrease in the case of informal businesses relative to men's. The results support the finding of Lambert, Pescatori, and Toscani (2020) that informality plays a key role as a shock absorber, limiting unemployment, and demonstrating that the informality effect is stronger for women than men in South Africa.

Impact of the COVID-19 Pandemic on Weekly Hours Worked and Monthly Income

An investigation of the impact of the COVID-19 pandemic on women's hours worked, which examined the intensive margin of the pandemic's effects on employment, found that the pandemic decreased women's weekly hours worked relative to men's by 1.2 hours. Therefore, not only did the pandemic drive women out of work to a larger extent compared to men, it also disproportionately decreased their aggregate hours worked relative to men. Two main channels, occupation and children, drive the disproportionate effect. Indeed, women in low-skilled occupations were more affected than their male counterparts, and

women with children under seven years old were more affected than men in the same circumstance. This finding suggests that the occupational gender differences and the childcare burden that falls mostly on women explain why women worked fewer hours due to the pandemic relative to men.

Women's monthly net income decreased by 226.3 Rands (about \$13) beyond men's because of the pandemic. This relative decrease might not seem sizable; however, given that the average monthly income for women in the sample is R3780 (about \$239), this drop is equivalent to an additional decrease of 5 percent of their income beyond that of men. The occupations drive the overall negative effect, as women in low-skilled occupations were more affected than their male counterparts. Similarly, women with children under seven years old lost more income than men in a similar circumstance. The number of children in school accentuates the disproportionate effect of the pandemic on women, as having more children attending school increases the disproportionately negative effect of the pandemic on women's income.

Similarly, the Labor Force Survey provides evidence that the pandemic affected women's hours worked more than men's. Data from the Labor Force Survey provide more evidence on the channels through which the pandemic affected women's hours worked, with the type and sector of employment, occupation, domestic work, and education as the significant ones. When the type of employment is considered, the pandemic disproportionately affected women doing regular jobs. Another expected result is that the pandemic disproportionately affected women in services more than their male counterparts. The pandemic also affected women in medium- and low-skilled occupations disproportionately more than men, and the disproportionately negative effect on women is largest in medium-skilled occupations, which is consistent with findings using the National Income Dynamics Survey. Consistent with findings regarding employment, women who spent more time fetching water and wood were more affected than men in the same situation. The pandemic disproportionately affected more educated women less than women who were less educated, as higher levels of education decrease the significant disproportionately negative effect of the pandemic on women.

Eswatini: Women's Unpaid Work Burden Increased Due to the Pandemic

The analysis in Eswatini is based on probit models on binary response questions data, and results are obtained on the probabilities that gender may have had a differential effect on the outcome of interest. The results suggest that the COVID-19 pandemic disproportionately affected women's employment relative to men in Eswatini. However, the pandemic impacted men more significantly in terms of income loss. Specifically, the odds ratios imply that during the pandemic, men were 1.1 times⁸ more likely to be employed, while women were 1.3 times

⁸ When the odds ratio is <1, the reciprocal is taken—so in this case, $1/0.915 = 1.09$ —and interpreted as a lesser probability of women having the outcome in question. When the odds ratio is >1, the odds ratio is taken as is, implying that women are more likely to have the outcome in question.

more likely to have lost their jobs. However, the pandemic caused men in Eswatini to become 1.1 times more likely to lose income. Since women in Eswatini are more likely to be unemployed or out of the labor force relative to men, these findings imply that the pandemic further exacerbated these pre-existing labor market inequalities. However, the COVID-19 pandemic may have also had a positive mechanical effect on the pre-existing gender inequality in incomes, whereby men were more likely negatively impacted due to the pandemic.

Second, the effect of the pandemic on outcomes of domestic and care work is estimated. This work is one of the main hypothesized channels of the disproportionate effect of the pandemic on women, indicating the availability of these data is a major advantage of this survey. The dependent variables are dummies for whether the respondent spent more time on domestic work, childcare, and adult care due to the pandemic. As expected, the results suggest that the pandemic resulted in women spending increased time—specifically, about 1.1 times more time—doing domestic, childcare, and adult-care work relative to men. These results confirm the hypothesis that one way the pandemic disproportionately affected women in Eswatini is by increasing their unpaid care work burden, including domestic and care work.

Lesotho: Women's Businesses Were Resilient to the Pandemic Shock

As with the empirical analysis conducted on the Eswatini data, simple probit models are estimated using the 2020 Rapid Impact COVID-19 survey data for Lesotho. The results suggest that during the pandemic, women were less likely to be formally employed than men, more likely to run a business, and less likely to have stopped working due to the pandemic. Specifically, men were 1.5 times more likely to be formally employed but also 1.3 times more likely to have stopped working due to COVID-19. On the other hand, women were 2.2 times more likely to run a business.

Namibia: The Pandemic Drove More Women Out of the Labor Market

Both panel and probit models are estimated for Namibia using its COVID-19 Household and Jobs Tracker survey data. The results suggest that women were more likely to lose their employment due to the pandemic and less likely to work during the pandemic. Specifically, women were 6.7 percentage points more likely to lose their employment and 6.9 percentage points less likely to work. Women were not significantly more likely than men to have stopped working or changed jobs due to the pandemic.⁹

⁹ Probit models that were run for South Africa establish comparability among all countries in this analysis, and the results are consistent with the previous findings from the data on Eswatini, Lesotho, and Namibia.

POLICY RECOMMENDATIONS

The results from these analyses point toward some policy actions to reduce the gender gap and build more economic resilience in South Africa, Eswatini, Lesotho, and Namibia. Reducing women's occupational segregation and unpaid work burden would strengthen their employment and income resilience to shocks. This chapter provides strong evidence that occupational gender segregation and unequal unpaid work burdens are the two primary channels of the disproportionate impact of the COVID-19 pandemic on women's employment and income relative to men in the southern African region. Country authorities should implement policies that tackle the root causes of gender inequality to strengthen women's resilience to such shocks, catalyzing a stronger, more sustainable, and inclusive postpandemic recovery.

Encouraging girls' education in science, technology, engineering, and mathematics (STEM) fields will reduce occupational gender segregation by ensuring that women have more access to diversified job opportunities and become more resilient to shocks. Disparities in fields of study still exist, with girls more likely to opt-in for non-science fields. As a result, women's employment is more concentrated in the tertiary sector activities that are more vulnerable to shocks like the COVID-19 pandemic, notably hospitality and tourism. Therefore, encouraging women to pursue careers in STEM fields will support their representation in all sectors of the economy, attenuating the disproportionately negative effects of shocks on women due to occupational gender segregation.

Fostering the provision of care services, such as childcare and adult care, and parental leave would give women more flexibility to pursue full-time jobs since many women had to drop out of the labor force to stay at home and care for their children following school closures. Research has shown significant heterogeneity in how the pandemic affected countries' gender gaps, with some countries being more resilient to the shock than others. One of the main reasons the pandemic did not worsen some countries' gender gaps is the availability of affordable childcare (Bluedorn and others 2021). Providing affordable and accessible care services would thus strengthen countries' resilience to such shocks.

In light of the evidence on the importance of business employment, country authorities should complement their efforts to formalize the informal sector by ensuring a robust social safety net. The informal sector is a large component of South Africa's, Eswatini's, Lesotho's, and Namibia's economies. Although the informal sector is vulnerable in terms of pay, job security, and working conditions, it has cushioned against the pandemic shock for informal-sector employees, the majority of whom are women. Recent IMF research (IMF 2021b) suggests that countries aiming to formalize their informal sectors should set up permanent mechanisms to increase the coverage and adequacy of social protection systems and provide adequate incentives for formalization. These mechanisms may include cash transfers for unemployed and informal-sector workers through digital technologies, for instance, sustainable government-to-person mobile transfer programs. In addition to supportive social protection systems, tax policy and administration measures can decrease the incentives for informality and increase the incentives to join the taxpayer registry, for instance, adequate minimal threshold for value-added tax, simpler value-added and corporate

tax systems, low payroll taxes, and simplification of tax payment procedures. Steady and gradual structural policy reforms, including simplifying labor market regulations, competition policy, eliminating excessive regulations and bureaucratic requirements, and digitalization would further support the formalization process.

One way to enhance business employment in these countries is by strengthening financing for micro, small, and medium-sized enterprises (MSMEs). Access to more financing would allow small business owners, the majority of whom are women, to scale up their businesses and become more resilient to economic shocks. Support for MSMEs could include setting up incubators and facilitating business networking as well as increasing financial literacy and inclusion by providing access to formal financial services and preparation of financial accounts, among others. Digitalization and fintech could also play a catalytic role in scaling up MSMEs.

Enabling Environment for Gender Equality

Policies addressing the macro-critical gender inequalities previously discussed require a conducive environment, with legal systems that ensure women's equal economic participation and social freedom. The restrictive legal system in some southern African countries impedes women's agency, limiting their economic participation and contributions. For instance, in Eswatini and Lesotho, the misalignment between and parallel application of civil and customary legal systems allows for persistent discrimination against women. Even though constitutions in these countries promote gender equality *de jure*, customary law remains exempted from these provisions. As a result, women still face discrimination in accessing financial services, such as loans, and in property ownership and inheritance rights, which may obstruct them from owning collateral. This chapter provides strong evidence for the importance of businesses in providing a social safety net for women in these countries. Access to financial services and the ability to acquire assets are crucial in giving women more agency to invest and grow their businesses, thereby enhancing private sector-led growth. Enshrining gender equality in legal frameworks may also contribute to combating gender-based violence and child marriage and may eventually make discriminatory social norms and unequal opportunities obsolete.

Gender-responsive budgeting could support the implementation of these recommendations and help foster an enabling environment conducive to gender equality. Gender-responsive budgeting allows governments to take into account gender inequality concerns when enacting policies and implementing programs in national, state, and local budgets. Given that women's participation is lagging in many aspects of the economy, including the labor market, as well as in health, education, and political representation, gender equality goals should be incorporated into laws, policies, and budgets that aim to promote the well-being of the population.

CONCLUSION

This chapter explores whether the COVID-19 pandemic disproportionately affected women in southern Africa and finds significant evidence for a “she-cession” in South Africa, Eswatini, Lesotho, and Namibia. Specifically, women's

employment, hours worked, and incomes were disproportionately more negatively affected during the pandemic than those of men in South Africa. Likewise, in Namibia, women were more likely than men to have lost employment due to the pandemic. Furthermore, analysis of the dynamic effects of the pandemic suggests that women's employment, hours worked, and income also took longer to recover after the initial shock of the COVID-19 pandemic in South Africa. Findings are complemented with results for Eswatini, Lesotho, and Namibia. In Eswatini, women were less likely than men to be employed and more likely to have lost their jobs due to the pandemic. In Lesotho, women were less likely to be formally employed than men. In Namibia, women were less likely than men to work during the pandemic. However, the findings also suggest that men in Eswatini were relatively more likely to have lost income due to COVID-19, while men in Lesotho were more likely to have stopped working due to the pandemic.

These findings provide evidence supporting the role of women's heavier unpaid work burden and occupational gender segregation as transmission channels of the COVID-19 pandemic shock on gender inequality. Having young children and children in school and spending more time fetching water and wood contributed to the disproportionately negative impact of the pandemic on women's employment, hours worked, and income in South Africa. Furthermore, the results from the analysis in Eswatini further suggest that the pandemic exacerbated women's already disproportionate unpaid domestic, childcare, and adult care work burdens. Moreover, women employed in low- and medium-skilled occupations and certain sectors, such as services, were particularly impacted by the pandemic with regard to employment, hours worked, and income, while more education counterbalanced some of this negative effect.

The evidence also showed that business employment, especially in the informal sector, acted as a countercyclical shock absorber for women in South Africa and Lesotho. Unlike other forms of employment, women's relative to men's business employment was not disproportionately affected by the pandemic. Specifically, women running informal (but not formal) businesses were less negatively affected by the COVID-19 pandemic shock than their male counterparts. Furthermore, women in Lesotho were more likely than men to run a business during the pandemic. This evidence suggests that women in these countries may have used informal business employment to cushion the negative impact of the pandemic.

The observed trends may be operating in the broader southern and even sub-Saharan African region. Due to limited data availability, this baseline model cannot be applied to all countries in the region. The analysis of data from Eswatini, Lesotho, and Namibia, albeit not as granular, confirms some of the findings about the disproportionate impact of the COVID-19 pandemic on women in the region. For cross-country comparability reasons, this methodology was also applied to the data from South Africa, broadly confirming the results from the other countries in this study. More detailed, gender-disaggregated, and uniform-across-countries data would greatly facilitate future research aiming to apply this empirical analysis to the broader southern and even sub-Saharan African region.

ANNEX 4.1. EMPIRICAL STRATEGY: ASSESSING THE IMPACT OF COVID-19 ON GENDER INEQUALITY

Empirical analysis is modeled following Alon and others (2021) as follows:

$$y_{it} = \beta_0 + \beta_1 F_i + \beta_2 D_t + \beta_3 F_i \times D_t + \beta_4 X_{it} + \varepsilon_{it} \quad (4.1)$$

for individual i at time t , where y is the outcome variable of interest, namely a binary employment indicator, weekly hours worked, and nominal income in local currency; F is a female dummy; D is a treatment dummy for the post-COVID time period and equals 0 before COVID (Q1 2020) and 1 after the onset of the pandemic; and X is vector of controls: age, marital status, education, area of residence (rural versus urban), province, and ethnicity. Standard errors are clustered at the individual level. The female dummy equals 1 if the individual is female and 0 if a male. The control population is the male population. The impact of the COVID-19 pandemic on any outcome variable of interest for the male population is captured by β_2 . For the female population, it is captured by the sum of β_2 and β_3 . So, β_3 captures the disproportionate impact of the COVID-19 pandemic on women relative to men and is therefore the main parameter of interest. This empirical analysis can be done only in countries with pre- and post-COVID-19 pandemic data.

Additional specifications capture the transmission channels of the COVID-19 shock. The analysis controls for the industry in which individual i is employed and their occupation to assess the role of occupational gender segregation in the differential effects of the COVID-19 pandemic on women compared to men. The industries and occupations are classified into different major categories and include a dummy for each group of industry or occupation, holding one group constant. The specification controlling for the employment industries is as follows:

$$\begin{aligned} y_{it} = & \gamma_0 + \gamma_1 F_i + \gamma_2 D_t + \gamma_3 F_i \times D_t + \gamma_4 \text{industry1}_{it} + \gamma_5 F_i \times \text{industry1}_{it} \\ & + \gamma_6 \text{industry1}_{it} \times D_t + \gamma_7 F_i \times D_t \times \text{industry1}_{it} + \gamma_8 \text{industry2}_{it} \\ & + \gamma_9 F_i \times \text{industry2}_{it} + \gamma_{10} \text{industry2}_{it} \times D_t + \gamma_{11} F_i \times D_t \times \text{industry2}_{it} \\ & + \gamma_{12} X_{it} + \mu_{it} \end{aligned} \quad (4.2)$$

where industry0 is the control industry group. Similarly, the specification for occupation is as follows:

$$\begin{aligned} y_{it} = & \delta_0 + \delta_1 F_i + \delta_2 D_t + \delta_3 F_i \times D_t + \delta_4 \text{occupation1}_{it} \\ & + \delta_5 F_i \times \text{occupation1}_{it} + \delta_6 \text{occupation1}_{it} \times D_t \\ & + \delta_7 F_i \times D_t \times \text{occupation1}_{it} + \delta_8 \text{occupation2}_{it} \\ & + \delta_9 F_i \times \text{occupation2}_{it} + \delta_{10} \text{occupation2}_{it} \times D_t \\ & + \delta_{11} F_i \times D_t \times \text{occupation2}_{it} + \delta_{12} X_{it} \end{aligned} \quad (4.3)$$

where occupation0 is the control occupation group. Other factors controlled for are the kind of employment respondents have, specifically whether they are own-account workers, working for a wage, or are employers, as follows:

$$\begin{aligned} y_{it} = & \eta_0 + \eta_1 F_i + \eta_2 D_t + \eta_3 F_i \times D_t + \eta_4 \text{jobkind1}_{it} + \eta_5 F_i \times \text{jobkind1}_{it} \\ & + \eta_6 \text{jobkind1}_{it} \times D_t + \eta_7 F_i \times D_t \times \text{jobkind1}_{it} + \eta_8 \text{jobkind2}_{it} \\ & + \eta_9 F_i \times \text{jobkind2}_{it} + \eta_{10} \text{jobkind2}_{it} \times D_t + \eta_{11} F_i \times D_t \times \text{jobkind2}_{it} \\ & + \eta_{12} X_{it} + \mu_{it} \end{aligned} \quad (4.4)$$

To estimate the effect that school closures and increased childcare burdens had on the differential impact of the pandemic on women, the analysis controls for the number of children under seven years old that the individual has and the number of children attending school living in the household, as shown in the following equation (these data are available only in the National Income Dynamics Survey data set):

$$y_{it} = \theta_0 + \theta_1 F_i + \theta_2 D_t + \theta_3 children_{it} + \theta_4 F_i \times D_t \times \theta_5 F_i \times children_{it} + \theta_6 children_{it} \times D_t + \theta_7 F_i \times D_t \times children_{it} + \theta_8 X_{it} + \mu_{it} \quad (4.5)$$

Finally, the analysis controls for the increased household work burden by including time spent fetching wood and water, which is available in the Labor Force Survey data:

$$y_{it} = \lambda_0 + \lambda_1 F_i + \lambda_2 D_t + \lambda_3 time_{it} + \lambda_4 F_i \times D_t + \lambda_5 F_i \times time_{it} + \lambda_6 time_{it} \times D_t + \lambda_7 F_i \times D_t \times time_{it} + \lambda_8 X_{it} + \mu_{it} \quad (4.6)$$

Panel regressions are run following equation (4.1) for South Africa and Namibia, and the analysis is complemented with probit models for Eswatini, Lesotho, and Namibia due to data limitations. Specifically, for Eswatini, Lesotho, and Namibia, the following specification is run:

$$Y_i = Pr(Y_i = 1 | X_i) = \phi(X_i \beta) \quad (4.7)$$

for each individual i , where Y_i is a binary outcome variable, such as whether the respondent is employed or not, whether they lost some income or their job, or whether they spent more time doing unpaid work; X_i is a vector of explanatory variables, the main one being gender, but also including controls such as age and marital status. β measures the effects of the corresponding explanatory variables on Y_i .

ANNEX 4.2. RESULTS

ANNEX TABLE 4.2.1.

South Africa NIDS: Clustering Standard Errors at the Province Level				
Variables	(1) Employment All Forms	(2) Business Employment	(3) Monthly Income	(4) Weekly Hours Worked
COVID-19 Dummy	-0.127*** (0.0159)	-0.0237*** (0.00513)	10.65 (114.4)	-5.203*** (0.521)
Gender × COVID-19 Dummy	-0.0194 (0.0115)	0.00471 (0.00620)	-226.3** (101.2)	-1.226 (0.704)
Education level	0.00866** (0.00328)	-1.04e-05 (0.00281)	-13.18 (53.89)	0.383 (0.217)
Age	0.00576 (0.00377)	0.00444*** (0.00120)	148.3 (96.82)	0.284 (0.284)
Observations	33,076	20,110	8,766	11,772
R ²	0.028	0.004	0.013	0.041
Controls	Yes	Yes	Yes	Yes
FE	Yes	Yes	Yes	Yes

** $p < 0.05$; *** $p < 0.01$.

Source: Authors' analysis using the National Income Dynamics Survey data set.

Notes: Robust standard errors in parentheses. Panel regressions with individual fixed effects. Standard errors clustered at the province level. Columns (1)–(4) dependent variables: employment of all forms, business employment, monthly income, and weekly hours worked. Gender = 0 if male and 1 if female. Controls are age, age squared, education level, marital status, and province. R^2 is the within R^2 . Robust standard errors in parentheses.

ANNEX TABLE 4.2.2.

South Africa LFS: Clustering Standard Errors at the Province Level			
Variables	(1) Employment All Forms	(2) Business Employment	(3) Weekly Hours Worked
COVID dummy	-0.0541*** (0.00455)	-0.0110*** (0.00239)	-3.186*** (0.147)
Gender × COVID-19 dummy	-0.00481 (0.00376)	0.00106 (0.00207)	-1.029*** (0.294)
Age	1.05e-05 (0.000796)	0.000577* (0.000256)	0.222* (0.108)
Education level	0.00395*** (0.000191)	0.000216 (0.000408)	-0.0309 (0.0300)
Observations	176,845	176,845	58,919
R ²	0.015	0.002	0.022
Controls	Yes	Yes	Yes
FE	Yes	Yes	Yes

* $p < 0.1$; *** $p < 0.01$.

Source: Authors' analysis using the labor force survey data set.

Notes: Robust standard errors in parentheses. Panel regressions with individual fixed effects. Standard errors clustered at the province level. Columns (1)–(3) dependent variables: employment of all forms, business employment, and weekly hours worked. Gender = 0 if male and 1 if female. Controls are age, age squared, education level, marital status, and ethnicity. R^2 is the within R^2 . Robust standard errors in parentheses.

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COVID-19 and Gender Gaps in the Labor Market: Evidence from Nigeria and Ethiopia

Chie Aoyagi

A high degree of informal self-employment in the rural agricultural sector characterizes the labor structure in sub-Saharan Africa. The impact of the COVID-19 pandemic on female employment may not appear large because the share of informal self-employment is particularly high among women. Nevertheless, widespread income reduction was observed in rural and urban households. This reduction could limit economic opportunities for women as husbands' control over the household resources is the norm. The chapter also finds that rural children struggled to continue learning during school closures. These countries need gender-sensitive policies to narrow the gap during and postpandemic.

INTRODUCTION

Globally, the COVID-19 pandemic disproportionately impacted women's employment and income. Additional data clarifies the nuances of how women have fared worse than men. In sub-Saharan Africa, about 80 percent of women were considered own-account workers (that is, self-employed workers without employees) or contributing family workers (also known as unpaid family workers) before the pandemic. However, "self-employment" status could mask the true economic impact of the crisis on women. Although not officially recognized as unemployed, their economic activities were often interrupted, reduced, or stopped. The divide between women in urban and rural settings also differentiates the pandemic impacts on them. While women in cities may be more prone to losing their jobs, women in rural areas are more likely self-employed and face greater challenges in maintaining their children's educational progress when schools are closed.

Regardless of where a woman lives and works, the shocks to household income are widespread. Even before the pandemic, women in sub-Saharan Africa were more likely than men to exist under the threat of poverty. According to International Labour Organization (ILO) statistics, 81 percent of female workers in sub-Saharan Africa—compared with 77 percent of male workers—were either extremely, moderately, or close to poor in 2019. The pre-existing gender

inequities threatened women's lives as the public health crisis grew, reversing the progress toward gender equality.

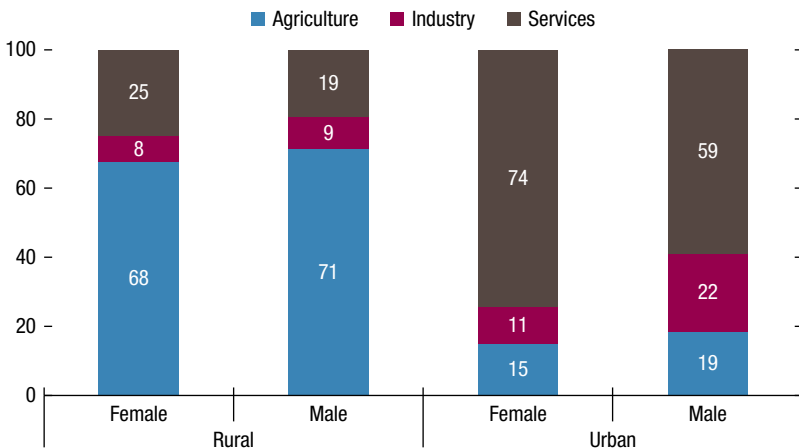
While the gender impact of the COVID-19 pandemic spread to multiple dimensions in sub-Saharan Africa, this chapter focuses on gender differences in employment, household income, and children's education during the early stages of the lockdowns. This study contributes to the growing literature on the household level impacts of the COVID-19 shock, focusing on two sub-Saharan African countries—Nigeria and Ethiopia.

PREPANDEMIC EMPLOYMENT STRUCTURE IN SUB-SAHARAN AFRICA

Figure 5.1 illustrates the regional and sectoral employment distribution in sub-Saharan Africa. As in other parts of the world, more women in sub-Saharan Africa worked pre-pandemic in the services sector, which has been hit harder than agriculture or industry because of reduced demand caused by pandemic-related lockdowns and social distancing. As in other regions, the share of the services sector is higher in urban than rural areas.

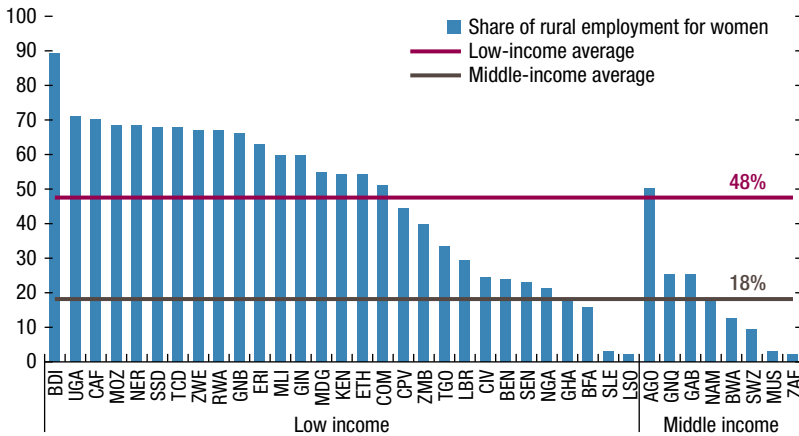
However, sub-Saharan Africa stands out from the rest of the world due to the low urbanization and high employment share in agriculture. The pre-existing divide between women in urban and rural settings is key to understanding how the pandemic has affected overall female employment. The aggregate statistics hide diverse experiences for labor market structure and inequality. The country-by-country data in Figures 5.2 and 5.3 show a strong link between income per capita and the level of urbanization and the share of agriculture in total employment—that is,

Figure 5.1. Employment in Sub-Saharan Africa by Sector
(Percent of total employment)



Sources: International Labour Organization database; and IMF staff calculations.

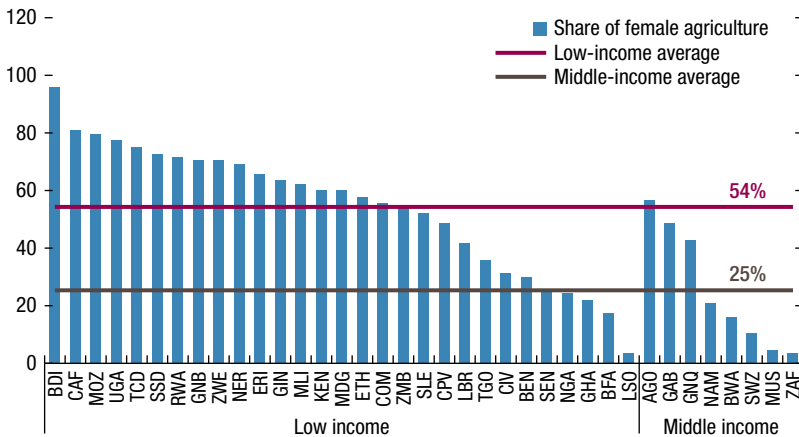
Figure 5.2. Percentage of Rural Female Employment as a Share of Total Female Employment



Sources: International Labour Organization database; and IMF staff calculations.

Note: The figure uses International Organization for Standardization (ISO) country codes. The country classification is from April 2021 WEO.

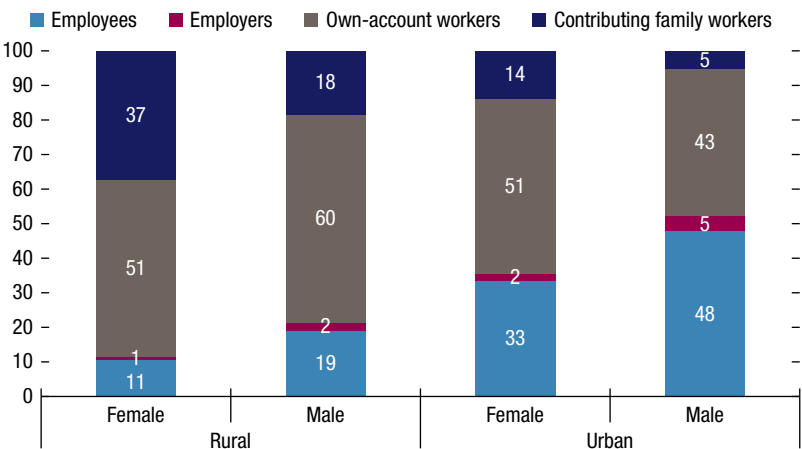
Figure 5.3. Percentage of Female Employment in Agriculture as a Share of Total Female Employment



Sources: International Labour Organization database; and IMF staff calculations.

Note: The figure uses International Organization for Standardization (ISO) country codes. The country classification is from April 2021 WEO.

Figure 5.4. Employment in Sub-Saharan Africa by Employment Status
(Percent)



Sources: International Labour Organization database; and IMF staff calculations.

female employment in low-income countries in sub-Saharan Africa is more likely rural and agricultural compared to middle-income countries in the region.

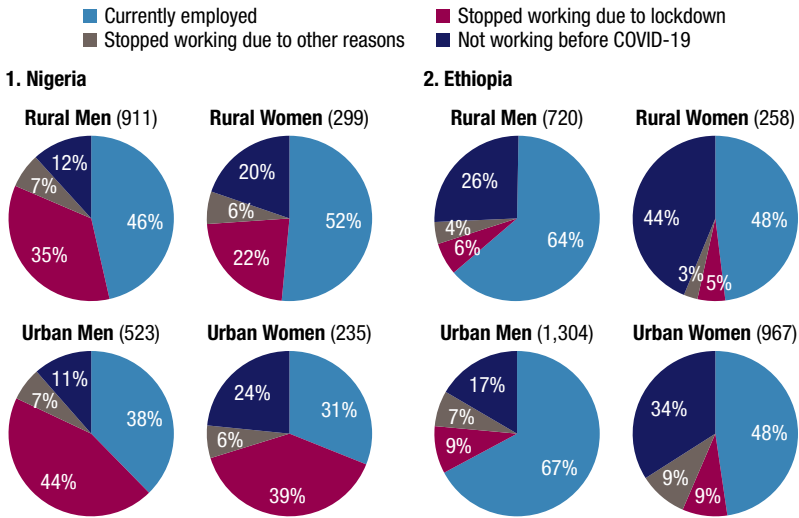
Another feature of sub-Saharan Africa’s labor market structure is the high share of informality—a key determinant of the pandemic’s impact on the overall labor market. Most workers in sub-Saharan Africa are self-employed, especially in rural areas, and often work in the informal sector. According to a paper by the ILO (2018), about 90 percent of employment in Africa was informal. As shown in Figure 5.4, own-account workers (mostly in the informal sector) and contributing family workers (all in the informal sector by definition) together accounted for 88 percent of women and 78 percent of men in rural employment in sub-Saharan Africa. Women were twice as likely as men to be contributing family workers.

The country-by-country outcomes depicted in Figure 5.5 show a strong link between income per capita and the share of own-account and contributing family workers in total female employment, suggesting that informality in employment is particularly high among women in low-income countries in sub-Saharan Africa.

CASE STUDIES OF THE EARLY STAGES OF LOCKDOWN

The discussion on the overall labor market structure for the region in the previous section identified unique features of sub-Saharan Africa while also noting the heterogeneity across the diverse set of countries. This section, however, focuses on Nigeria and Ethiopia, using data from the first round of the World Bank Living Standards Measurement Study–Integrated Surveys on Agriculture Program for High-Frequency Phone Surveys on COVID-19. While these two large countries

Figure 5.6. Survey Results on Employment Situation for Nigerian and Ethiopian Adults



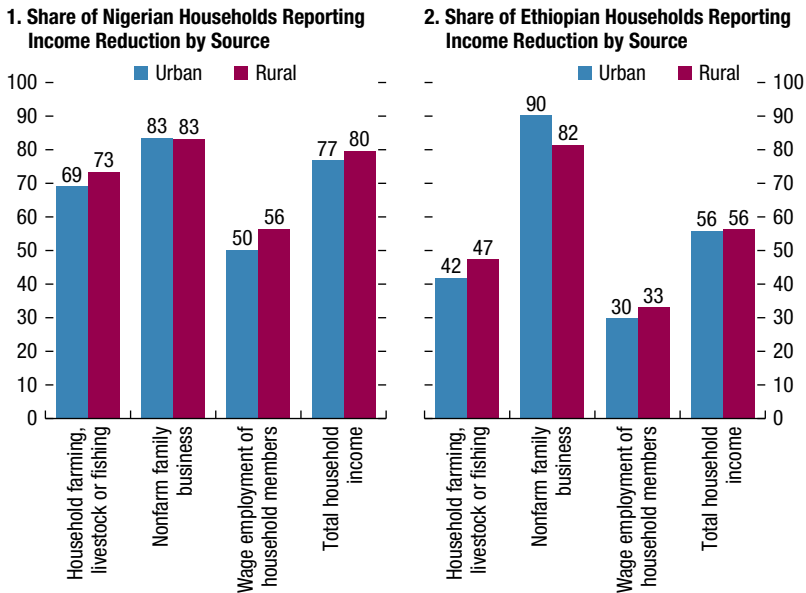
Sources: World Bank COVID-19 phone survey; and IMF staff calculations.

Note: Numbers in parentheses are the number of observations.

On the other hand, for rural and urban areas, the share of men who lost their employment due to the lockdown was greater than women—meaning that rural women were the least affected by employment losses in both countries. However, this finding might be due to widespread informal employment in agriculture among women, especially rural women, for which an activity loss would not appear as lost employment. Bundervoet, Davalos, and Garcia (2021) support this explanation as they found that an employment structure dominated by agriculture and own-account work explains the lower reported job losses in low-income countries worldwide during the early stages of the lockdowns.

Even though employment loss was less severe in rural than urban areas, income reductions were widespread (Figure 5.7). In both Nigeria and Ethiopia, the share of households that reported reduced total household income since the outbreak is equally high between rural and urban areas—about 80 percent in Nigeria and about 60 percent in Ethiopia. Among the top three most common income sources, household farming and nonfarm family businesses are more likely to face reduced income than wage employment, implying a vulnerability in the informal sector and for own-account and contributing family workers. Women may be adversely affected in particular because of their dependence on family farming and family businesses and their initially lower income and higher poverty rate compared to men. Furthermore, because husbands' control over the allocation of household resources is the norm in most parts of West and East Africa (Verschoor and others 2019), reduced household income might further lessen economic opportunities for women in the region.

Figure 5.7. Share of Households Reporting Income Reduction since Mid-March by Income Source
(Percent)

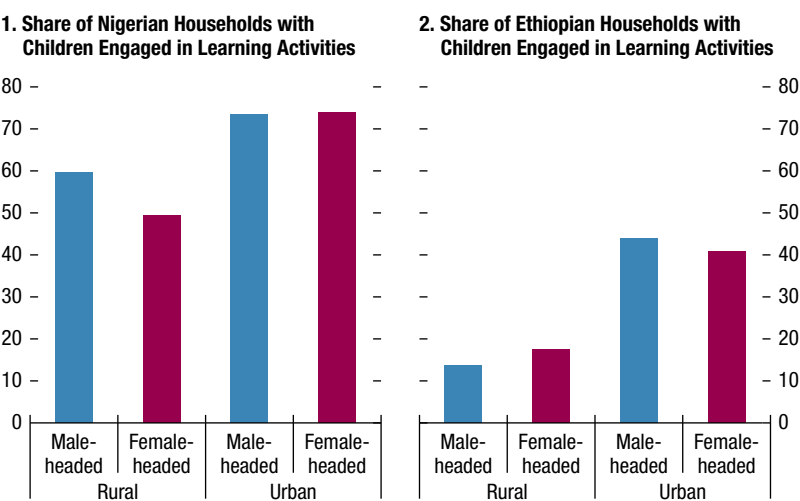


Sources: World Bank COVID-19 phone survey; and IMF staff calculations.

The pandemic has also affected the education of the future work force, possibly leaving long-lasting scars on the economy. The same World Bank surveys were used to study the impact of school closures and families' adaptation to continuous learning for children. On March 26, 2020, Nigeria closed schools nationwide. Likewise, on March 16, 2020, Ethiopia closed all primary and secondary schools. The school closures for both countries continued for about six months until Nigeria and Ethiopia reopened schools on October 12, 2020, and October 19, 2020, respectively. The surveys coincided with the school closures and asked those with primary or secondary school children at home whether the children engaged in any educational activities after the schools closed.

Analysis of responses from over 1,000 Nigerian households and approximately 2,000 Ethiopian households indicates that urban households are more likely than rural households to have children engaged in learning activities at home during school closures (Figure 5.8). Moreover, rural children in female-headed households were the least likely to continue learning during school closures. The educational gap between rural and urban areas had been observed even before the pandemic. For example, according to the 2018 Demographic and Health Survey, in Nigeria, the net school attendance ratio for primary school was 70 percent for urban children and 50 percent for rural children. Yet these results imply that the urban–rural educational gap was widening during the pandemic.

Figure 5.8. Share of Households with Children Engaged in Learning Activities
(Percent)



Sources: World Bank COVID-19 phone survey; and IMF staff calculations.

EMPIRICAL ANALYSIS OF EMPLOYMENT OUTCOMES

The previous section showed employment loss due to lockdowns and learning activities during school closures across different gender and regional groups in Nigeria and Ethiopia. This section quantifies the differences in the employment loss and learning probabilities across those groups using the Nigerian COVID-19 pandemic survey together with the pre-COVID baseline survey—the 2018–19 round of the Nigeria General Household Survey—to supplement household and the respondents’ background information.

The combined surveys identified background information for close to 2,000 respondents, such as gender, age, place of residence, educational attainment, and employment sector, to explain the individual’s probability of losing employment due to lockdowns.

First, the regression results show that being a woman is associated with lower probabilities of employment loss due to lockdowns. However, as discussed in the previous section, this employment loss may reflect the fact that women are more likely to be informal, self-employed agricultural workers and thus would not report losing a job or closing a business even though their return to work might be significantly reduced.

Second, the regression results confirm that living in urban areas and working in the nonagricultural sector are both associated with higher probabilities of lockdown-induced employment loss.

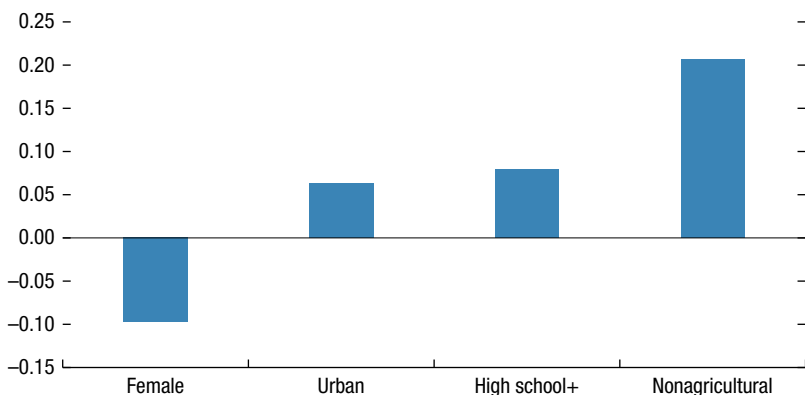
Third, and surprisingly, graduating from high-school or having a higher educational background is a characteristic of an individual with a higher probability of employment loss—which is seemingly contradictory to the finding that global epidemics hurt employment prospects of those with only a basic education. However, the results show that the lockdowns, at least initially, led to higher job losses for people with higher educational levels.

Finally, the respondent's age has a nonlinear effect on the person's probability of employment loss—meaning that until a certain age (estimated around 40 years old), the probability that a person loses a job due to lockdowns increases as the age increases by one year. For individuals older than 40 years, age would have the opposite effect.

To illustrate the magnitude of the effect of those individual characteristics associated with the person's employment-loss probability, Figure 5.9 presents the average marginal effects at age 40. The age of the individual is fixed considering the nonlinear age effects, which are the smallest and almost zero at age 40. While the average marginal effects shown in the chart are particular for the 40-year-old group, the relative magnitude and the sign (positive or negative) of the marginal effects of all other variables remain the same across all age levels. The interpretation of the average marginal effects shown in Figure 5.9 is as follows:

- On average, being a woman is associated with a reduced probability of losing employment by 0.10 point at age 40.
- On the other hand, living in an urban area or having a high school or higher educational background is associated with increased probabilities of employment loss due to lockdowns by 0.06 point and 0.08 point, respectively.
- Overall, the average marginal effects of working in the nonagricultural sector at the onset of the pandemic are the largest and associated with 0.21 point higher probability of an employment loss due to lockdowns.

Figure 5.9. Average Marginal Effects to Employment-Loss Probabilities



Source: IMF staff calculations.

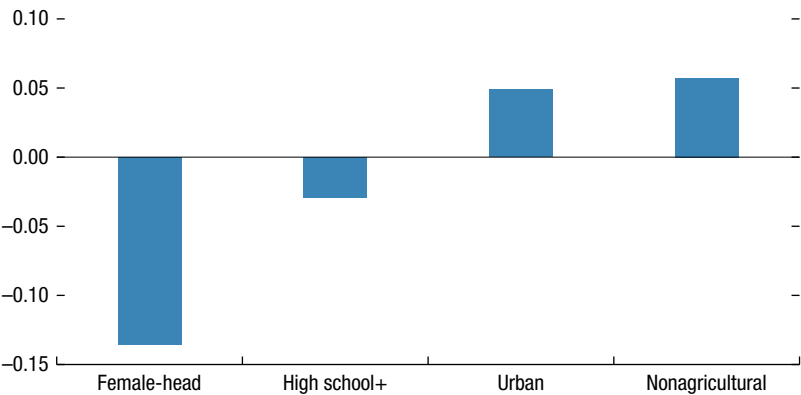
EMPIRICAL ANALYSIS OF EDUCATIONAL OUTCOMES

Similarly, both surveys were used to study how children’s engagement in learning activities during school closures differ across households. The regression results show that children living (1) in a female-headed household, (2) in rural areas, or (3) with an adult agricultural worker are, on average, less likely to continue learning during school closures.

Predicted lower learning probabilities of children in female-headed households are consistent with the fact that female-headed families often face special circumstances. For instance, a single mother could be living alone with her children, thus facing greater challenges in ensuring home-based learning for the children. According to the Nigerian 2018–19 baseline survey, about 80 percent of female heads of household surveyed were widows, 9 percent were separated, and 2 percent were divorced. The predicted lower probabilities for children in rural areas imply that the lack of infrastructure or support necessary to continue learning in rural areas during school closures would widen the pre-existing urban–rural educational gaps. The positive effects of nonagricultural employees in the household suggest that while agricultural employment seemed more resilient against the pandemic shock, children in the household might have to help their family on the farm when schools are closed and face greater challenges engaging in learning activities. On the other hand, whether the respondent has a high school or higher level of education does not have significant effects on the overall learning probabilities of children in the household ($p \geq 0.1$), while the respondent’s education has positive and significant effects on the probabilities of parents teaching children at home during school closures (Aoyagi 2021).

Moreover, to illustrate the magnitude of the effects those individual characteristics have on the learning probability of children in their household, Figure 5.10 presents the average marginal effects for respondents at age 40. As in the case of the employment-loss probability regression discussed previously, age 40 was

Figure 5.10. Average Marginal Effects to Children’s Learning Probabilities



Source: IMF staff calculations.

selected as the fixed point because of the nonlinear age effects, which are almost zero at age 40. The interpretation of the average marginal effects shown in Figure 5.10 is as follows:

- On average, having a female head of household is associated with children's reduced probability of engaging in learning activities during school closures by 0.14 point.
- On the other hand, urban households and households with a nonagricultural worker have on average 0.05 and 0.06 point higher probabilities for children's continuous learning during school closures, respectively.
- The average marginal effects of the respondent's educational background (high school+) on the overall learning activities for children in their household are not significant at 10 percent ($p \geq 0.1$).

Interestingly, the urban–rural gap found by this quantitative analysis of Nigeria is close to the average of low-income and middle-income countries analyzed in Bundervoet, Davalos and Garcia (2021).

CONCLUSION

This chapter studies the initial impacts of the COVID-19 pandemic on employment and children's learning in sub-Saharan Africa by examining the regions' prepandemic labor market structure and using the case study of two low-income African countries—Nigeria and Ethiopia—from survey results that became available in early summer 2020. The World Bank high-frequency phone surveys conducted in Nigeria and Ethiopia during the early stages of the nationwide lockdowns and school closures were used to analyze the differential impacts of the pandemic on individuals and children in their households with various characteristics. In particular, the study focuses on the gender, place of residence, and employment sector of nationally representative respondents to the surveys.

The study finds that a job loss or business closure due to the lockdowns in Nigeria and Ethiopia was more common in urban than rural areas and among male rather than female workers. One caveat to the result is that employment loss probabilities for rural women may not appear large because the majority of these women were informal self-employed and did not “fire themselves.” Nevertheless, income losses were widespread and rural women were one of the most vulnerable groups with a lack of decent work options and a high poverty rate.

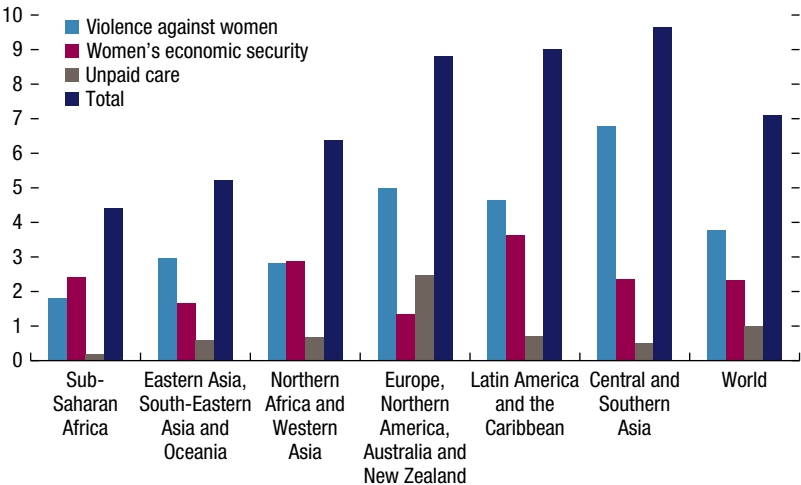
The costs of the pandemic for women are not limited to lost income but also the struggle to maintain children's educational progress. The study finds that, in Nigeria, rural households headed by women are far less likely than urban households to have children engaged in learning activities during school closures. Female-headed households are vulnerable and different in many ways from the average household, not because women head the households but because of the circumstances that lead women to becoming heads of households. For example, in Nigeria, women heads of households are primarily widows. The situation could be particularly troubling in rural areas because of the pre-existing divide.

Compared to children in urban areas, children in rural areas were lagging academically even before the pandemic. Yet these results imply that the urban–rural gap widened during the pandemic—rural children do not have the same internet connectivity, rural schools are less equipped to offer remote learning, and children are more likely to be expected to work to support their households.

While these main survey results are for Nigerian and Ethiopian households, similar geographic and gender divides exist in other countries in sub-Saharan Africa, and a targeted set of women-centered policies can blunt the negative impacts of the crisis and improve prospects for the next generation. As of November 2021, about 90 percent (43 out of 50 countries and territories) of governments in the region have introduced gender-sensitive policies in response to the crisis caused by the pandemic (UNDP-UN Women 2021). For example, the governments of Côte d’Ivoire, Liberia, and Senegal have introduced measures to facilitate access to small business loans during the COVID-19 pandemic; most beneficiaries for these loans are women. The Democratic Republic of the Congo and South Africa initiated a range of fiscal measures to support female-dominated sectors of the economy, such as hospitality and tourism. In Kenya and Rwanda, some in-kind and cash transfer programs prioritize women in beneficiary households or target female-headed households.

These are encouraging signs, but room still exists for accelerating and scaling up gender-sensitive policy measures. Sub-Saharan Africa lags behind other regions in the average number of gender-sensitive measures per country, including those aimed at addressing the surge in violence against women and girls, the unprecedented increase in unpaid care work, and the economic insecurity caused by the large-scale loss of jobs, incomes, and livelihoods (Figure 5.11). In particular,

Figure 5.11. Average Number of Gender-Sensitive Measures by Regions



Sources: United Nations Development Programme-UN Women COVID-19 Global gender response tracker policy measures data set, living database, version 3 (November 11, 2021); and IMF staff calculations.

sub-Saharan Africa has the lowest number of measures to address women's unpaid care burdens, and the region's gender response to the pandemic has been uneven across countries and dimensions.

Stronger policies addressing the unequal impact of the COVID-19 pandemic are warranted to limit widening gender inequality and the urban-rural gap. The pandemic has taken an enormous toll on women's lives and livelihoods, regardless of where they live or how they are employed. Policies should reflect the significant role women often have in their homes and communities and the impact for future generations. Supporting women provides key short-term economic benefits and creates vital building blocks for a more prosperous and inclusive future.

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PART II

Settling on the Basics: Ending Harmful Practices and Developing the Continent

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Eliminating Violence against Women and Boosting the Economy

Rasmané Ouedraogo and David Stenzel

The COVID-19 pandemic and lockdowns led to a rise in gender-based violence. This chapter explores the economic consequences of violence against women in sub-Saharan Africa using demographic and health survey data collected before the pandemic. A two-stage least square method to address endogeneity identified that a 1 percentage point increase in the share of women subject to violence can reduce economic activity, proxied by nightlights per capita by about 9 percent. Nightlights per capita are used as a proxy for economic activity since traditional indicators such as GDP are not available on the subnational level. Economic costs result from a significant drop in female employment. Our findings also show that violence against women is more detrimental to economic development in countries without protective laws against domestic violence, in countries rich in natural resources, in places where women are deprived of decision-making power, and during economic downturns. Beyond the moral imperative, these findings highlight the economic importance of combating violence against women, particularly by reinforcing laws against domestic violence and strengthening women's decision-making power.

INTRODUCTION

One in three women has experienced physical or sexual assault in her life.¹ Worldwide, no place is less safe for a woman than her home. Besides a fundamental human rights violation, a serious public health concern, and devastating for

¹ “Violence against women and girls” refers to: “any act of gender-based violence that results in, or is likely to result in, physical, sexual, or psychological harm or suffering to women, including threats of such acts, coercion, or arbitrary deprivation of liberty, whether occurring in public or in private life” (1993 UN Declaration on the Elimination of Violence against Women, 2). “Intimate partner violence” is one of the most common forms of violence against women and girls but is not identical as it excludes acts of violence not perpetrated by intimate partners and includes violence against men or those with gender-nonconforming identities. The term “gender-based violence” is broader since it includes violence against men, boys, and sexual minorities or those with gender-nonconforming identities. “Violence” also comes in various forms, including sexual, physical, or psychological violence, which this chapter considers violence. This chapter uses the terms “violence against women and girls,” “intimate partner violence,” and “domestic violence” interchangeably for readability. In this chapter, these terms refer to violence against women and girls committed by their intimate partners, including sexual, physical, or psychological violence.

individual well-being, violence against women and girls also has high economic costs. Channels through which violence against women and girls create “economic costs” include lower labor supply, reduced productivity per hour worked, less investment in human capital for both women and children, and possibly lower investment due to higher demand for health and judicial services. Domestic violence adversely affects companies, which incur costs associated with higher rates of absenteeism, staff turnover and reduced productivity at work, and the potential for reputational damage (International Labour Organization and UN Women 2019).

The COVID-19 pandemic led to increased domestic violence against women. Adverse economic and social conditions coupled with lockdown measures exacerbated existing situations of abuse and control. According to the International Growth Centre (Young and Aref-Adib 2020), the number of reported cases of gender-based violence in Nigeria increased from 300 to more than 700 following the introduction of lockdowns in March 2020.

According to UN Women (Puri 2016), the cost (public, private, and social) of violence against women worldwide amounts to \$1.5 trillion. Duvvury and others (2013) found that violence against women can cost up to 3.7 percent of GDP in some countries. Although the sub-Saharan African region has the highest level of domestic violence against women, empirical studies of its potential economic impact are scarce, mainly due to data limitations.

This chapter presents a novel approach to assess the economic impact of violence against women and girls in 18 sub-Saharan countries. The data on the prevalence of violence against women and girls stem from the Integrated Public Use Microdata Series (IPUMS) database compiled by the Demographic and Health Survey Program of the United States Agency for International Development (USAID). This database includes country surveys from the 1980s to the present. The analysis uses several types of violence against women, including physical (hurting, punching, and slapping, among others), emotional (attempts to embarrass, shame, blame, frighten, control, or isolate), and sexual violence. These data were combined with the level of economic activity (approximated by nighttime light data per capita) for 224 districts in sub-Saharan Africa. To address the problem that economic activity could affect the level of violence against women and girls and vice versa (endogeneity), an instrumental variable approach was used to examine the following questions based on cross-district variations:

- How does violence against women and girls affect the level of economic activity in sub-Saharan Africa?
- Is there any evidence that violence against women and girls affects labor supply?
- How do protective laws against domestic violence, natural resources, the decision-making power of women, and the business cycle affect the impact of violence against women and girls in economic activity?

The analysis showed that violence against women and girls has a negative effect on economic activity. The impact is both economically and statistically significant. The results show that an increase by 1 percentage point of women

subject to domestic violence is associated with a drop of per capita nightlights of about 9 percent in a given district. More violence is associated with a significant drop in female employment. Moreover, violence against women is more detrimental to economic development in countries without protective laws against domestic violence, in countries rich in natural resources, in countries where women are deprived of decision-making power, and during economic downturns. The economic cost of violence against women is higher in countries where the gender gap in education between partners is high.

The findings imply that sub-Saharan African countries should strengthen laws to combat violence against women and reinforce women's decision-making power. Furthermore, promoting women's education will be key to reducing the gender gap in education and the influence and control of men. Sub-Saharan African countries should also address the increased levels of domestic violence due to the COVID-19 pandemic to bolster economic recovery.

The rest of the chapter offers a discussion of the main channels through which violence against women and girls affects economic growth; a look at the data and extent of global violence against women and girls with a focus on sub-Saharan Africa; the empirical strategy and baseline results; and a discussion of the transmission channels. The chapter also addresses further variables determining the impact of domestic violence on levels of economic activity and presents concluding remarks.

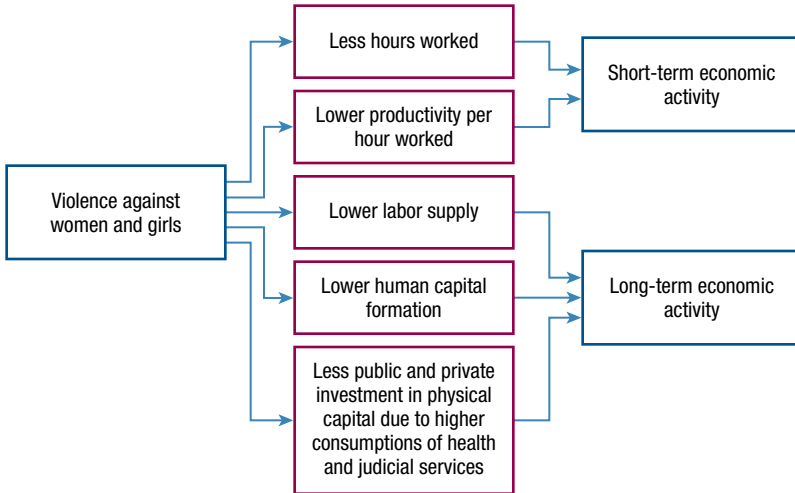
TRANSMISSION CHANNELS, THE DATA, AND STYLIZED FACTS

The main channels through which violence against women and girls potentially affects economic growth are fewer hours worked (absenteeism), reduced productivity per hour worked (presenteeism), lower long-term labor supply (indirect effect due to violence), less investment in human capital formation, and less investment in physical capital due to higher consumption of health and judicial services (see, for instance, Duvvury and others 2013). Figure 6.1 offers a stylized visualization.

The analysis in this chapter is based on 29 integrated Demographic and Health Surveys (DHS) from the USAID's DHS Program from 18 sub-Saharan African countries.² The surveys have been taken in different years, and the University of Minnesota Population Center harmonized them for comparability of variables across countries and over time. The sample covers more than 440,000 women. IPUMS also includes information on current residences so that individuals can be assigned to administrative units, typically districts or provinces, according to the administrative division of each country.

² The data have been extracted from IPUMS Global Health, hosted by the University of Minnesota Population Center. The original data are from USAID's DHS program, and sampling and methodology per country can be found here: <https://dhsprogram.com/>.

Figure 6.1. Violence against Women and Girls and Economic Growth: Transmission Channels



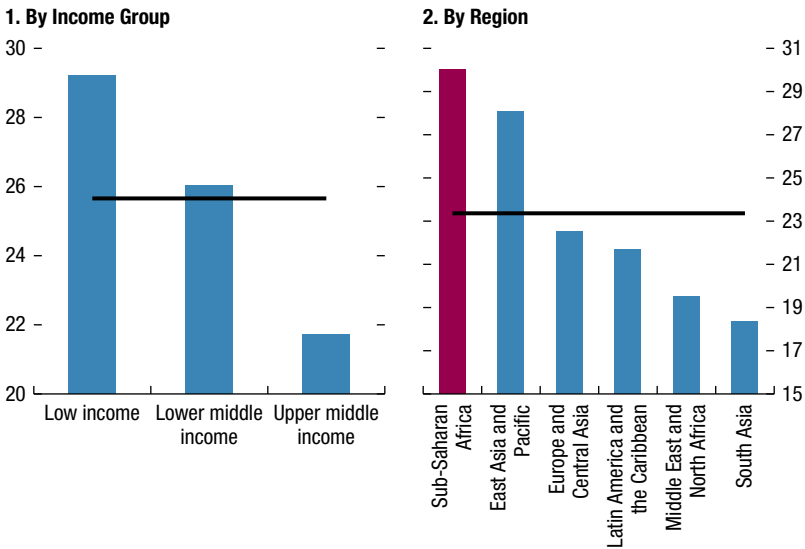
Source: Authors, based on Duvvury and others 2013.

The extent of violence against women and girls by intimate partners is measured by answers to the survey question: “Have you experienced mistreatment by your husband or intimate partner, including punching with a fist or something harmful, slapping, pushing, or throwing something, threatening with harm, emotional violence, and sexual violence?” Drawing on the responses, the share of women subjected to intimate partner violence was calculated at the district level for each variable and construct a composite index of domestic violence, which is the simple average of the previously mentioned variables.

Economic development at the district level is proxied by nighttime lights data provided by National Oceanic and Atmospheric Administration. The most commonly used measure for economic activity, GDP, is not available at the district level. Recent studies have shown that nightlight satellite data are highly correlated with economic activities, with a coefficient of correlation up to 70 percent (Hu and Yao 2019; Pinkovskiy and Sala-i Martin 2016). The data capture cleaned luminosity, after filtering for cloud coverage, other ephemeral lights, and background noise. In this chapter, nightlights per capita are calculated by dividing the sum of all nighttime light pixel values within a district by the district’s population. The population data are from the Gridded Population of the World, Version 4 database provided by the Center for International Earth Science Information Network.

Control variables such as the size of the agriculture sector (proxied by the share of cropland), the urbanization rate (share of respondents living in urban areas),

Figure 6.2. Domestic Violence by Income Group and Region
(Percent of husbands/partners committing physical, sexual, or emotional violence)



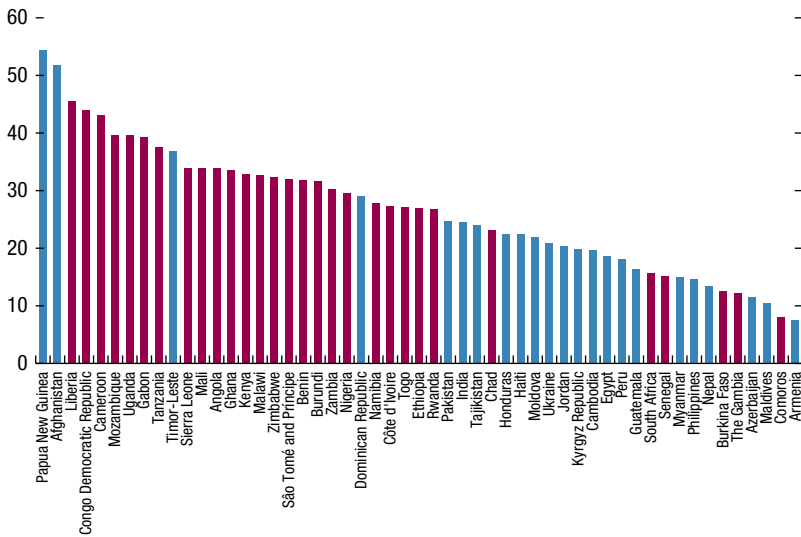
Sources: USAID's DHS Program; and authors' calculations.

Note: Black line indicates average.

infrastructure (time to a water source and total road lengths), education (average number of years of schooling), conflict intensity (number of conflict-related deaths over the population), and religious fractionalization (religious beliefs of respondents) are from the IPUMS data set; road lengths is from the Center for International Earth Science Information Network (2013). These variables are usually considered strong determinants of economic development.

What do the data show about the extent of physical, sexual, and emotional violence against women? Such violence is pervasive, according to the survey data (Figures 6.1–6.3). Moreover, it appears to depend on the country's level of development as the violence is somewhat higher in low-income than middle-income countries.³

³ The sample of countries in the study includes: Afghanistan, Angola, Armenia, Azerbaijan, Bangladesh, Benin, Burkina Faso, Burundi, Cambodia, Cameroon, Chad, Colombia, Comoros, Democratic Republic of the Congo, Côte d'Ivoire, Dominican Republic, Egypt, Ethiopia, Gabon, The Gambia, Ghana, Guatemala, Haiti, Honduras, India, Jordan, Kenya, Kyrgyz Republic, Liberia, Malawi, Maldives, Mali, Moldova, Mozambique, Myanmar, Namibia, Nepal, Nigeria, Pakistan, Papua New Guinea, Peru, Philippines, Rwanda, São Tomé and Príncipe, Senegal, Sierra Leone, South Africa, Tajikistan, Tanzania, Timor-Leste, Togo, Uganda, Ukraine, Zambia, and Zimbabwe. The classification follows the World Bank's classification of countries by income groups: <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups>.

Figure 6.3. Domestic Violence by Country*(Percent of husbands/partners committing physical, sexual, or emotional violence)*

Sources: USAID's DHS Program; and authors' calculations.

Note: Sub-Saharan African countries are marked in red.

The IPUMS data also show that sub-Saharan Africa has the highest rate of domestic violence globally, with more than 30 percent of women having experienced intimate partner violence, followed by the East Asia and Pacific regions. In Afghanistan, Papua New Guinea, Liberia, and Democratic Republic of the Congo, nearly half of all women experienced physical, sexual, or emotional violence committed by a husband or partner.⁴

The percentage of women across sub-Saharan African countries who reported experiencing physical, emotional, or sexual violence by their current husband or partner ranges widely. For example, about 1 percent of women in Comoros reported that a husband or partner punched them with a fist or something harmful, compared with 18 percent of women in Gabon. Women reporting sexual violence ranges from about 2 percent in Burkina Faso to 27 percent in the Democratic Republic of the Congo; emotional violence ranges from about 6 percent in Comoros to 47 percent in Gabon. Slapping occurs frequently, with more than one-third of women reporting being slapped by a husband or partner in Gabon, Democratic Republic of the Congo, Sierra Leone, and Zambia. The threat of harm is also common since nearly 20 percent of women reported they have been threatened with harm in Cameroon, Democratic Republic of the Congo, and Uganda (Table 6.1).

⁴ It is worth noting that these numbers can differ from those from more recent country surveys.

TABLE 6.1.

Type of Domestic Violence							
Country	Did she present any physical results of husband's actions? (%)	Was she ever punched with fist or something harmful? (%)	Was she ever slapped? (%)	Was she ever pushed, shaken, or had something thrown at her? (%)	Was she ever threatened with harm? (%)	Did she ever experience any emotional violence? (%)	Did she ever experience any sexual violence from partner? (%)
Angola	11.8	11.1	29.4	10.6	7.2	25.0	7.8
Burkina Faso	3.6	5.8	8.7	4.2	6.5	9.8	1.5
Burundi	18.8	10.9	36.5	14.1	7.0	25.5	25.6
Cameroon	17.8	16.9	36.4	21.9	19.7	36.9	15.1
Comoros	...	1.1	4.0	3.0	0.4	5.5	1.7
DR Congo	23.4	14.3	40.6	26.0	19.3	38.3	27.1
Côte d'Ivoire	8.1	13.0	19.8	10.6	7.5	18.0	5.1
Ethiopia	6.1	6.7	15.9	11.5	7.2	22.3	7.6
Gabon	...	17.8	42.0	24.9	3.6	47.0	15.5
The Gambia	...	2.9	14.9	5.1	0.6	17.3	2.9
Ghana	8.7	5.0	16.6	9.4	9.0	33.0	6.7
Kenya	13.3	13.3	31.2	19.7	14.4	27.7	13.0
Malawi	9.1	8.7	17.9	8.1	9.0	20.7	15.8
Mali	5.0	8.6	15.5	6.2	6.2	16.1	6.5
Mozambique	6.3	10.7	26.9	9.2	5.4	34.9	7.8
Namibia	8.4	13.0	18.8	15.6	12.0	25.3	7.3
Nigeria	5.0	3.8	14.7	6.3	6.1	21.2	4.8
Rwanda	11.3	11.5	28.3	15.4	9.0	18.8	15.0
Senegal	5.8	7.9	15.2	5.5	4.9	13.4	6.3
Sierra Leone	...	8.8	37.4	21.1	1.3	40.0	6.6
South Africa	5.1	...	...	12.8	6.2	18.7	3.5
Tanzania	25.2	16.8	31.8	15.4	8.2	31.8	12.6
Chad	...	7.7	20.6	10.3	1.3	22.4	8.9
Togo	...	9.4	19.6	10.0	0.8	21.6	8.1
Uganda	19.4	17.1	36.5	21.1	19.4	42.6	24.7
Zambia	14.6	13.0	36.8	14.6	9.3	25.0	17.0
Zimbabwe	11.8	10.5	25.1	10.6	10.4	29.1	13.3

Sources: Integrated Public Use Microdata Series database; and authors' calculations.

EMPIRICAL STRATEGY AND BASELINE RESULTS

The following estimating equation further investigates the impact of intimate partner violence against women on economic activity, proxied by nightlights per capita:

$$Nightlights_{ij,t} = \alpha + \beta Violence_{ij,t} + \gamma X'_{ij,t} + \pi_j + \phi_t + \mu_{ij,t} \quad (6.1)$$

where, for district i from country j at time t , $Nightlights_{ij,t}$ represents nighttime lights per capita, $Violence_{ij,t}$ is the share of women who experienced domestic violence, Vector $X'_{ij,t}$ includes control variables.⁵ π_j and ϕ_t are country and time fixed

⁵ The approach accounts for the influence of variables on economic activity traditionally considered important determinants of economic development (Mamo, Bhattacharyya, and Moradi 2019; Egert, Kozluk, and Sutherland 2009; Alesina and others 2003; Barro 2001; Alesina and others 1996), including the share of the agriculture sector, the degree of urbanization, infrastructure, education, conflict, and religious fractionalization.

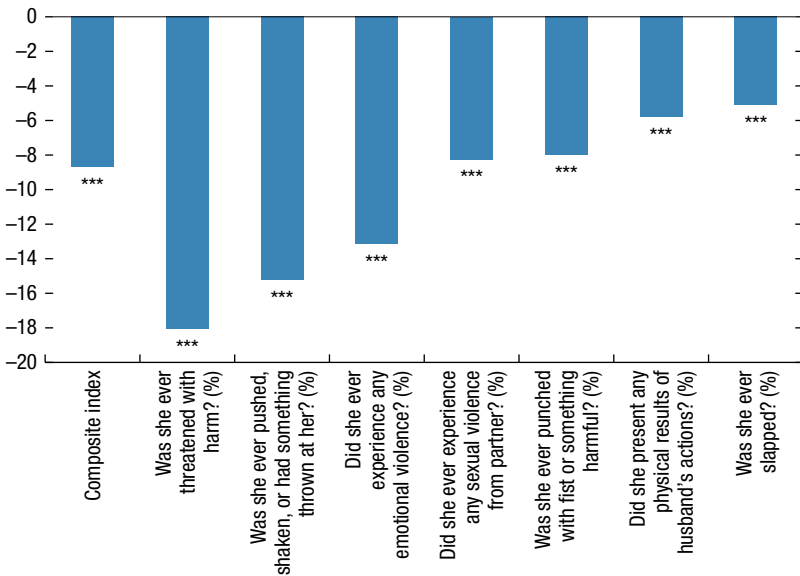
effects, respectively. The inclusion of these fixed effects accounts for country-specific and time-invariant characteristics that may affect nightlight per capita. $\mu_{ij,t}$ is the error term.

The key challenges for estimating the effect of domestic violence on nightlights per capita are reverse causality, omitted and unobservable variables, and measurement errors. First, the level of economic activity could affect the level of domestic violence. For example, during the COVID-19 pandemic, spikes occurred in domestic violence (UN Women 2020). Second, some variables may be subject to measurement error. Data on the prevalence of intimate partner violence collected through surveys could be inaccurate since, generally, victims of domestic violence tend to underreport it, either due to fear of retaliation, privacy concerns, or other societal factors (Felson and others 2006; Krantz 2002). Third, many potential variables affect the level of economic activity at a given point in time and over time. Key control variables and other unobserved variables that could affect both economic activity and domestic violence (for instance, cultural and social norms, and economic inequality) could be missing. Consequently, estimating equation (6.1) could lead to biased results.

To address the first problem of endogeneity, an instrumental variable approach—the two-stage least squares (2SLS) method—was used. The average level of domestic violence in neighboring districts was used as the instrument for domestic violence in a given district. “Neighboring districts” are those that share the same borders. To use the average level of gender-based violence in bordering countries as an instrument, two conditions should be met: (1) the instrument needs to be relevant (meaning that gender-based violence in bordering districts needs to correlate with the level of gender-based violence in a given district) and (2) the instrument should be exogenous (meaning that gender-based violence in bordering districts should be uncorrelated with nightlights per capita in a given district or the error term in estimating equation [6.1]). Several papers have used similar instrumental variables (Acemoglu and others 2019; Caselli and Reynaud 2019; Cherif, Hasanov, and Wang 2018).

Regarding the relevance of the instrument, it seems likely that regional cultural closeness could lead neighboring districts to share some common values and behaviors. Physically close proximity naturally facilitates communication, social interaction, diffusion of cultural values, and learning across different societies; therefore, people may share common behaviors such as the acceptance of domestic violence. It is thus plausible that the level of domestic gender-based violence in neighboring districts is a good predictor of domestic violence in a given district. Regarding the exogeneity of the instrument, as Cherif, Hasanov, and Wang (2018) emphasized, a country could be affected by spillovers from its neighbors mostly, but not exclusively, through trade and finance. To address the spillover issue that could invalidate the exogeneity of the instrument, they controlled for the average real GDP per capita in neighboring countries as a proxy for the spillover effect. This chapter follows this approach and controls for the average nightlights per capita in bordering districts in the second stage. Furthermore, as in Acemoglu and others (2019) and Cherif, Hasanov, and Wang (2018), controlling for time and district

Figure 6.4. The Impact of Violence against Women and Girls on Economic Activity per Violence Type: Baseline Results
(Percentage points)



Source: Authors' calculations.

Note: * $p < .05$; ** $p < .01$; *** $p < .001$.

fixed effects that could affect nightlights per capita allows for parametrical removal of the influence of unobserved fixed characteristics common to districts.

Based on this approach, strong evidence exists that violence against women and girls has a substantial negative impact on economic activity in sub-Saharan Africa. Figure 6.4 shows the estimated coefficients associated with violence against women. The bars represent the size of the impact, and the stars represent the level of significance of the coefficients. Using the composite index of violence against women, the results suggest that an increase of 1 percentage point could reduce nightlights per capita by about 9 percent. The results also indicate that the impact depends on the type of violence. Threatening with harm, punching with a fist or something harmful, and causing emotional violence appear most detrimental to economic activity.

A Transmission Channel

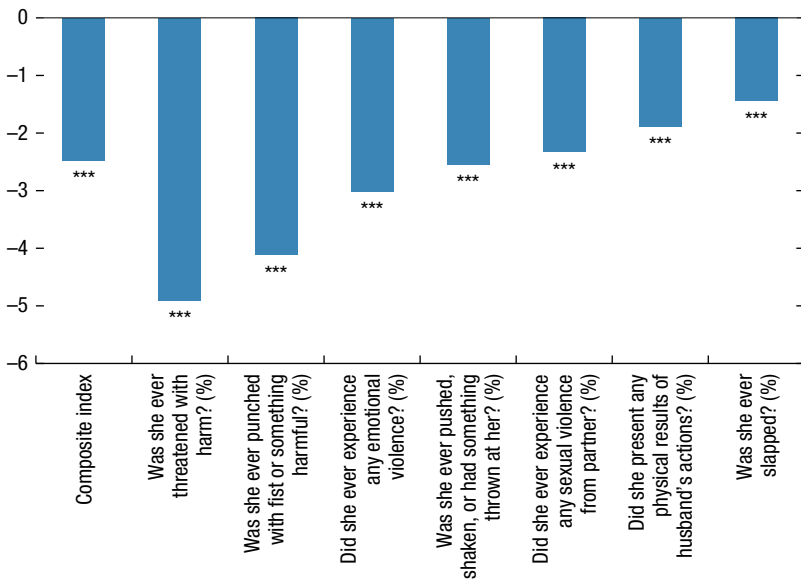
Violence against women and girls reduces economic activity through multiple channels. The evidence shows that a reduced labor supply is important. Extensive literature documents the relationship between employment and domestic violence. Previous studies found that experiencing intimate partner violence is associated with increased absenteeism over the long term and presenteeism in the short term through tardiness, not showing up for work, use of sick days, and

problems with concentration, job performance, and productivity (Reeves and O’Leary-Kelly 2007; Swanberg, Logan, and Macke 2005; Brush 2002). Other studies point out that women who are in abusive relationships tend to experience high rates of job loss and turnover and are frequently forced to quit or be fired (Swanberg and Logan 2005; Bell 2003). Some research papers also emphasize that compared to nonabused women, abused women sustain two to three times more injuries requiring surgery and tend to be socially excluded, making them less likely to be employed (Damonti 2014; Campbell 2002).

Ideally, data on women’s annual hours worked would be needed to assess the effects of intimate partner violence on labor supply. Since these data are unavailable, we use data on women’s participation in the official labor markets (thus excluding agriculture, informal jobs, and unpaid jobs), which the IPUMS data set includes. This approach implicitly assumes that the only way domestic violence affects labor supply is to discourage women from participating in the formal labor market. But domestic violence also reduces labor supply through reduced hours worked per day or missed working days while an affected woman would still be participating in the labor market. Also, the exclusion of agriculture, informal, and unpaid jobs means that the proxy is likely to underestimate the impact of domestic violence on labor supply.

We use the same methodological approach to estimate the impact of violence against women and girls on economic activity. The results presented in Figure 6.5 show a highly negative impact of violence against women and girls on female

Figure 6.5. A Transmission Channel: Female Employment
(Percentage points)



Source: Authors' calculations.

Note: * $p < .05$; ** $p < .01$; *** $p < .001$.

employment, corroborating that women experiencing abuse are less likely to work, which in turn reduces economic activity. However, the evidence also suggests that including female employment in the baseline estimation cannot account for the full impact of violence against women and girls on economic activity and that further channels exist that could include, for example, lower productivity per hour worked due to an inability to focus on work or lower educational attainment associated with violence against women and girls.

MITIGATING AND REINFORCING FACTORS

The data can also be used to shed light on whether certain factors, such as applicable laws on domestic violence, natural resource endowments, economic conditions, and decision-making power, can either mitigate or amplify the effect of violence against women and girls on economic activity. Therefore, an interaction term is included in the baseline estimation between the variable of interest and violence against women and girls. Table 6.2 presents the results.

- **Effects of laws on domestic violence:** The presence of protective laws against domestic violence has proven effective in reducing physical, psychological, and emotional abuse. Laws can not only benefit women seeking to address domestic violence in their relationships but can also help enhance their physical and psychological well-being. Some countries are also adopting protective provisions to help ensure that domestic violence does not affect women's labor force participation. Using data from the World Bank's Women, Business and the Law, it is possible to define a binary variable, taking the value of 1 if the country has adopted protective laws against domestic violence at the time of the survey, and 0 otherwise. Defining a binary variable and adding the interaction term shows that the negative effect of gender-based violence on nightlights per capita diminishes in countries with legal provisions against domestic violence. Legal provisions can help reduce the occurrence of domestic violence by deterring abusive behavior and providing women access to justice for violence-related cases (Dugan 2002), which could enhance their well-being and boost economic participation and contributions.
- **Effects of natural resources:** Previous studies provide evidence that mining zones can be a hostile climate for women in general (International Labour Organization 2007). Opening new mines can trigger a structural shift in employment patterns in Africa, with women leaving the agricultural fields for informal service jobs around the mining sites (for instance, as vendors) or leaving the labor market (Kotsadam, Ostby, and Rustad 2017). Ross (2008, 2012) emphasized that an increase in natural resources leads to lower female employment since revenues tend to push out low-wage and export-oriented factories. Drawing on this employment shift, reduced female economic power, and women's dependence on their partners, Kotsadam, Ostby, and Rustad (2017) concluded that natural resources can

increase women's risk of abuse by their intimate partners. In environments in which women's dependency is high, domestic violence could be elevated, which could also lead to a high adverse economic impact. To test this hypothesis, data on mining sites from the United States Geological Survey (2014) was used to construct a variable that takes the value of 1 if there is a mining site in the district and 0 otherwise. Adding this interaction term shows that the economic costs of domestic violence are higher in resource-rich districts.

- **Economic conditions and commodity price shocks:** Previous studies underscored that a proliferation of intimate partner violence followed the Great Depression (Liker and Elder 1983; Conger and others 1990). Sharp economic downturns could increase abusive behavior in two ways (Schneider, Harknett, and McLanahan 2016): First, the direct experience of job loss and material hardship increases abusive behavior. Men who might otherwise have kept negative behaviors in check may succumb to the stress of challenging economic circumstances and engage in abusive behavior. Second, worsening macroeconomic conditions increase abusive behavior by increasing uncertainty and fear among a broad population segment. The construction of an interaction variable between commodity prices and violence against women and girls allows an exploration of this assumption.⁶ The results show that the economic costs of domestic violence are higher during a commodity price slump, implying that the increase in domestic violence during the pandemic could have detrimental effects on economic development.
- **Decision-making power and gender gap in education:** Women's decision-making power and educational inequality between women and men living in the same household are another important dimension to consider. Many researchers have associated intimate partner violence with unbalanced power and control, and women with limited decision-making power were found to be subject to more physical violence (Rahman and others 2013; Koenig and others 2003). In fact, higher decision-making power could increase women's ability to escape domestic violence, lessen its intensity, or deter violent partners. In addition, education has been shown to help safeguard against abuse (Amegbor and Rosenberg 2019; Garcia-Moreno and others 2005). High educational attainment is negatively associated with being both a victim and a perpetrator of abuse (Cools and Kotsadam 2017; Pierotti 2013; Mocan and Cannonier 2012; Jewkes 2002). The results corroborate these hypotheses. Districts in which women

⁶ Economic conditions at a district level are proxied by a commodity price index based on the main commodities of each district from the IPUMS data set. The commodities include gold, maize, rice, soybean, sugarcane, wheat, sunflower, groundnut, and palm oil. We use the difference in prices between the year prior to the survey and the survey year to build the index with price data from the United Nations Conference on Trade and Development data set.

TABLE 6.2.

Mitigating and Reinforcing Factors					
Dependent variable: nighttime light per capita					
VARIABLES	(1)	(2)	(3)	(4)	(5)
Violence against women	-7.3210*** (2.604)	-14.9031*** (3.395)	-9.4469*** (1.547)	-31.6915*** (5.884)	-20.9654** (7.863)
Violence *law	5.2341** (2.424)				
Law	0.7963* (0.480)				
Violence *natural resources		-10.3803*** (3.071)			
Natural resources		-1.4481*** (0.497)			
Violence *economic conditions			0.7044** (0.224)		
Economic conditions			-0.0273 (0.087)		
Violence *decision power				44.9079*** (8.812)	
Decision power				7.0456*** (1.643)	
Violence *education gap					-21.3012* (12.495)
Education gap					-4.7622** (1.195)
Agriculture	-0.1180 (0.209)	0.1616 (0.250)	0.0417 (0.203)	0.0436 (0.261)	-0.1406 (0.239)
Urban area	0.5708* (0.303)	0.3398 (0.339)	0.3562 (0.311)	0.2835 (0.311)	0.8105*** (0.288)
Time to water source	0.0030 (0.006)	0.0002 (0.008)	-0.0046 (0.005)	-0.0018 (0.005)	-0.0119*** (0.004)
Roads	0.0512*** (0.013)	0.0617*** (0.016)	0.0473*** (0.014)	0.0028 (0.018)	0.0683*** (0.017)
Education	0.6046** (0.297)	0.7957*** (0.272)	0.6616*** (0.191)	1.2004*** (0.315)	0.2835 (0.291)
Conflict	-0.0728** (0.035)	-0.0905** (0.041)	-0.0693** (0.033)	-0.0937*** (0.024)	-0.1072*** (0.028)
Religious fractionalization	0.3103 (0.332)	0.6289 (0.385)	0.2173 (0.326)	0.2694 (0.342)	-0.5366* (0.324)
Neighbors' highlight per capita	0.2941** (0.120)	0.3057** (0.133)	0.2858*** (0.110)	0.3597** (0.147)	0.2235** (0.108)
Constant	-0.3100 (0.406)	0.5240 (0.537)	-0.1097 (0.420)	2.7384*** (0.823)	-2.8206 (1.881)
First stage					
Neighbors' domestic violence	0.7008*** (0.128)	0.4925*** (0.110)	0.8093*** (0.102)	0.2212*** (0.033)	0.1629** (0.070)
Observations	319	337	333	337	337
R ²	0.759	0.584	0.687	0.582	0.802
F-test, p-value	0.00	0.00	0.00	0.00	0.00
Cragg-Donald Wald F statistic	22.03	40.04	69.84	30.70	3.89
Kleibergen-Paap rk LM statistic	18.18	18.61	41.20	29.66	9.16
Anderson-Rubin Wald test	10.82	59.21	44.28	58.92	18.29
District fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes	Yes

Source: Authors' estimates.

Note: robust standard errors in parentheses.

*p < .05; **p < .01; ***p < .001

have more decision-making power or where the level of education is more equal between men and women experience a lower effect of violence against women and girls on economic activity.⁷

CONCLUSION

The research shows that higher violence levels against women and girls are associated with lower economic development. The results suggest that an increase in violence against women by 1 percentage point can reduce nightlights per capita, which is strongly correlated with economic activity, by around 9 percent. Strong evidence suggests that the negative impact of violence against women and girls on the labor supply of victims drives this adverse effect. Furthermore, the results show that the negative economic effects of violence against women is higher in countries without protective laws against domestic violence, countries rich in natural resources countries where women are deprived of decision-making power, and during economic downturns. A significant gender gap in education also contributes to higher adverse effects of violence against women.

The findings have several potential policy implications: First, these findings imply that adopting and reinforcing laws against domestic violence and strengthening women's decision-making power could help lower violence and the economic cost of domestic violence. As previous studies showed, strong laws are crucial to deter violence against women, protect victims of domestic violence, and promote women's participation in the labor markets. Second, the results highlight that sub-Saharan African countries should encourage girls' education to reduce the gender gap in education and attenuate men's influence and control. Third, sub-Saharan African countries should pay attention to the increased levels of domestic violence during the COVID-19 pandemic. As the data show, the economic cost of violence against women is higher during downturns, suggesting that the economic recovery from the current crisis could take longer if domestic violence is not addressed. Finally, sub-Saharan African governments should implement targeted policies to ensure that women benefit equally from the exploitation of natural resources.

⁷ The level of decision-making power is proxied by leveraging an IPUMS survey question about women's ability to make decisions about their health care to construct an indicator of the percentage of women able to make their own decisions regarding health care needs. As for the gender gap in education, the IPUMS data set allows a calculation of the difference between the education levels of husbands and wives at the district level.

ANNEX 6.1

ANNEX TABLE 6.1.1.

List of Country Survey Years in the Sample			
Country	Survey year	Number of districts	Number of individuals
Angola	2015	14	14,379
Burundi	2016	17	17,269
Cameroon	2004	10	10,656
Cameroon	2011	11	15,426
Comoros	2012	3	3,815
Democratic Republic of Congo	2007	9	9,995
Democratic Republic of Congo	2013	9	18,827
Ethiopia	2016	11	15,683
Ghana	2008	12	4,916
Kenya	2003	11	8,195
Kenya	2008	11	8,444
Kenya	2014	11	31,079
Malawi	2010	28	23,020
Malawi	2016	27	24,562
Mali	2006	8	14,583
Mali	2012	6	10,424
Mozambique	2011	10	13,745
Namibia	2013	12	10,018
Nigeria	2008	5	33,385
Nigeria	2013	5	38,948
Senegal	2017	14	16,787
South Africa	2016	7	8,514
Zimbabwe	2005	10	8,907
Zimbabwe	2010	10	9,171
Zimbabwe	2015	10	9,955
Tanzania	2010	24	10,139
Tanzania	2015	27	13,266
Burkina Faso	2010	13	17,087
Zambia	2007	8	7,146
Zambia	2013	8	16,411
Total		224	444,752

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Eliminating Child Marriage and Boosting Growth

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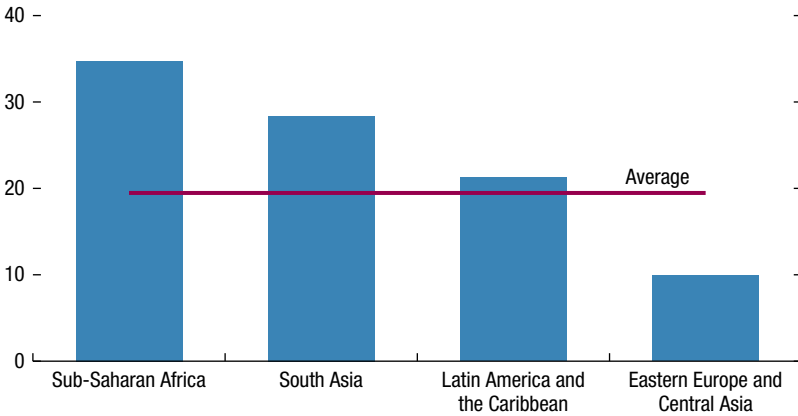
This chapter quantifies the relationship between child marriage and economic growth. It confirms that ending child marriage is critical from a human rights and development perspective, and it also has important macroeconomic implications. The analysis shows that eliminating child marriage would significantly improve economic growth—ending child marriage today would result in a 1.05 percentage point increase in the long-term annual per capita real GDP growth in emerging and developing countries. Given the strong interdependent relationship between education and child marriage, education policies should be prioritized when developing comprehensive strategies to end child marriage. Key complementary policies include addressing health care, social assistance to poor and marginalized children, and excessive school dropout rates for girls due to pregnancy.

INTRODUCTION

Child marriage is a serious human rights issue. As of 2021, 1 out of 5 girls in emerging and developing countries (EMDCs) was married before the age of 18, with the highest rate in sub-Saharan Africa (Figure 7.1).¹ These numbers are likely to have grown dramatically with the onset of the COVID-19 pandemic, where measures to contain the pandemic's spread (such as lockdowns) are likely to have undermined strategies to end child marriage—including through loss of education, reduced access to sexual and reproductive health information and services, and limitations on access to social support networks (United Nations Children's Fund [UNICEF] 2021a). UNICEF (2021b) estimates that the pandemic has put an additional 10 million girls at risk of child marriage by 2030. Overall, the Sustainable Development Goal of putting an end to child marriage by 2030 appears more out of reach.

¹ Child marriage is defined as a marriage or union involving a child under the age of 18 (UNICEF Global Databases 2022).

Figure 7.1. Child Marriages, 2016–21
(Percent)



Source: UNICEF Global Databases 2022.

Note: Percent of women aged 20 to 24 years who were first married or in a union before age 18.

The consequences of child marriage are wide-ranging and inextricably linked with poverty reduction, inequality, and economic growth. The literature has found that child marriage substantially increases fertility and population growth and reduces educational attainment, resulting in lower earnings and productivity. Child marriage also harms the health of mothers, increases intergenerational poverty and inequality, is associated with increased family violence and gender inequality, and reduces women’s decision-making power (Wodon and others 2017a). Through these and other channels, child marriage not only hurts children and families, it also constrains economic growth and development.

Mitra and others (2020) found that if child marriage were to end today, the long-term annual per capita growth in EMDCs would increase by more than 1 percentage point. The paper is the first to quantify the impact of child marriage on economic growth, and this chapter elaborates on the paper’s key findings, including insights on policies—such as education and health—that could be most effective in ending child marriage.

BACKGROUND

Factors contributing to child marriage include social and cultural norms and low levels of income, education, and family and community health (Lloyd 2005; UNICEF 2005; Santhya, Haberland, and Singh 2006; Jain and Kurz 2007; Malhotra and others 2011; United Nations Population Fund 2012; Vogelstein 2013; Klugman and others 2014; UNICEF 2014a, 2014b). For example, social and cultural norms could bias parents’ decision making when assessing the pros and cons of having a child wed at a young age as opposed to, for example, having

the child complete their primary or secondary education. Limited income or low education levels among parents or grandparents could also lead to perceiving early marriage as a way to ensure social and economic stability.

Child marriage has substantial links to poverty and inequality. Wodon and others (2017a) found that if child marriage were to end by 2030, a group of 106 EMDCs would save an average of about \$300 billion per year between 2015 and 2030 from reduced population growth and about \$75 billion per year from improved child health. They also estimated that EMDCs would gain, by 2030, an average of 1 percent of family earnings from improved salaries and productivity. Chaaban and Cunningham (2011) found similarly striking results indicating that allowing girls in EMDCs to complete their education (mainly primary and secondary school) could increase their lifetime earnings by up to 1.5 percent per year. These estimates focus on the impact of child marriage on girls, but child marriage affects boys, too. Boys who marry early are more likely to drop out of school and have lower earnings potential. From a broader perspective, the long-term impact of child marriage on growth will reflect the trade-off between increased labor supply—owing to increased fertility rates—against lower productivity—due to weak education and health outcomes resulting from child marriage.

Besides having negative effects on individuals and communities, child marriage could be associated with unfavorable macroeconomic outcomes—especially muted growth dynamics—when its occurrence is high. However, assessing this direct or indirect impact is difficult because of the feedback loop that exists between child marriage and various socioeconomic factors (Wodon and others 2017a).

The literature on economic growth links per capita economic growth to various macroeconomic variables such as initial human capital, government consumption, public investment, or political stability (Barro 1991). Growth models have been extended to include additional determinants such as social norms (Cole, Mailath, and Postlewaite 1992), corruption (Mauro 1995), income inequality (Berg and Ostry 2011), and economic diversification (Papageorgiou and Spatafora 2012).

The challenge in building a growth model where child marriage is an explanatory variable is addressing the feedback loop previously mentioned (the endogeneity bias). Although child marriage increases poverty and inequality, at the same time, poorer communities (among other characteristics) are also likely to encourage early marriage, especially for girls. While it is challenging to establish the optimal specification of a growth model, we can assume that child marriage is likely correlated with relevant growth determinants such as health or education. Therefore, as Mitra and others (2020) have shown, provided that the endogeneity bias is removed, the link between child marriage and growth could be assessed through the channels of health and education.

Key strategies to prevent or delay child marriage include empowering girls with information, skills, and support networks; educating the community; improving educational quality and access for girls; offering economic incentives to parents; and removing barriers in the legal and policy frameworks (Malhotra

and others 2011; Brown 2012; Kalamar, Lee-Rife, and Hindin 2016; Wodon and others 2018). But no single strategy is enough. For example, adopting laws against child marriage has not ended the practice (Wodon and others 2017b). The World Bank's Sahel Women Empowerment and Demographic regional project and African Union, United Nations Population Fund, and UNICEF initiatives are beginning to drive broader policy changes.² However, comprehensive strategies, including investment in programs and policies to end child marriage, are still lacking.

LINK BETWEEN CHILD MARRIAGE AND GROWTH: A DESCRIPTIVE APPROACH

Child Marriage in Growth Models

A naïve approach to quantifying the link between child marriage and growth is to include child marriage directly in a growth model as an explanatory (or control) variable. Although this approach likely generates a reverse causality bias, it could provide information on some of the interactions at play. For this exercise, we follow the growth model by Sala-i-Martin, Doppelhofer, and Miller (2004), which uses a Bayesian approach to identify the variables that matter most for growth. The following equation is considered:³

$$G_{t,t+T} = \theta_0 + \theta_1 \ln(y_t) + \theta_2 CM_t + \gamma X_t + \varepsilon_t \quad (7.1)$$

where T is the growth horizon, y_t the per capita real GDP in year t , $G_{t,t+T}$ the growth rate of the per capita real GDP between years t and $t + T$, CM_t the incidence of child marriage at t , X_t the vector of the other explanatory variables at t , and ε_t the error term.

Data

Child marriage is proxied by the percentage of women aged 20 to 24 years who were first married or in a union before the age of 18. Data for this variable are often not available as child marriage surveys are not routinely performed (or performed on average every three to five years at the earliest in countries with more than one survey, as shown in the UNICEF Global Database on child marriage). However, given the stability of the distribution of child marriage across countries and over time (Mitra and others 2020), we include all countries with at least one data point. Table 7.1 presents the variables that we use for the analysis. All of the variables are based on the selection process of Sala-i-Martin,

² The World Bank's Sahel Women's Empowerment and Demographic Project was approved in 2014 and has granted housing and academic support to more than 100,000 girls in various sub-Saharan African countries. UNICEF and United Nations Population Fund in 2016 launched the Global Programme to End Child Marriage, which has provided sex education and school attendance support to more than 14 million adolescent girls as of June 2022.

³ The model from which the final results of the study are deduced is presented later.

TABLE 7.1.

Explanatory Variables of Growth	
Variable	Description (Source)
Log of the per capita GDP	Real GDP per capita in PPP terms (IMF)
Child marriage	Percentage of women aged 20 to 24 years, married or in union before age 18 (UNICEF Global Database 2022)
Human capital index	Based on years of schooling and returns to education (Penn World Table 9)
Life expectancy at birth	Total (World Bank World Development Indicators)
East Asia dummy	1 for East Asia countries and 0 otherwise
Share of extractable commodities in total exports	Percent (Updated from Chapter 2, IMF World Economic Outlook, October 2015)
Fraction of tropical area	Percent (Gallup, Mellinger, and Sachs 2010)
Trade openness	Import+Export in percent of GDP (IMF)
Relative price of investment	Ratio price level of capital formation - price level of household consumption (Penn World Table 9)
Share of government consumption	Percentage of per capita GDP (in PPP terms) related to government consumption

Source: Authors' calculations.

Note: UNICEF (United Nations Children's Fund); PPP (Parity Purchasing Power).

Doppelhofer, and Miller (2004), except child marriage. The final database consists of 112 countries, mostly low-income countries,⁴ with variables observed between 1997 and 2017.

Results

We estimate four growth models based on the additional dependent variables that we use besides the initial income level. Model M1 includes only child marriage; model M2 includes child marriage and its squared value (to test the nonlinearity of the potential effects); model M3 includes only the controls from Sala-i-Martin, Doppelhofer, and Miller (2004); and model M4 adds child marriage to M3.

Models M1 and M2 confirm that the effect of child marriage on growth is likely negative and linear, as the square of child marriage is not significant. Given the omission of the other macroeconomic variables affecting growth (translated into a small R^2), the consistency of these estimates is debatable. Nonetheless, the two models confirm the negative correlation between child marriage and growth. The results from model M3 are consistent with the growth literature (Sala-i-Martin, Doppelhofer, and Miller 2004 provide a summary) and have an R^2 of 0.45, indicating reasonable goodness of fit. Box 7.1 presents these results in detail. Finally, model M4 confirms that the marginal effect of child marriage on growth is negative. However, unlike in M1, that effect is not significant—suggesting correlations between child marriage and the other explanatory variables need to be accounted for. The signs, the marginal effects, and the standard errors for the other explanatory variables are consistent with M3 (Box 7.1).

⁴ Due to the data attrition for some countries, the number of observations in some regressions decreases to 76 countries.

TABLE 7.2.

Growth Models with Child Marriage				
Variables	M1	M2	M3	M4
Log of the per capita GDP	−0.80***	−0.79***	−1.26***	−1.27***
Child marriage	−0.02***	−0.03*		−0.00
Child marriage (squared)		0.00		
Human Capital Index			0.76***	0.75***
Life expectancy at birth			0.02	0.02
East Asia dummy			1.04***	1.04***
Extractable commodities in total exports			0.00	0.00
Fraction of tropical area			−0.61***	−0.61***
Trade openness			−0.00	−0.00
Relative price of investment			0.01	0.01
Share of government consumption			−0.02*	−0.02*
Intercept	4.45***	4.57***	3.72***	3.79***
P-value	0.02	0.02	0.00	0.00
R ²	0.16	0.16	0.45	0.45

Source: Authors' calculations.

Note: M1 = Child marriage as the only control;

M2 = Child marriage and its squared value as controls;

M3 = Growth model with controls from Sala-i-Martin, Doppelhofer, and Miller (2004);

M4 = Child marriage and growth factors as controls.

* $p < 0.10$; *** $p < 0.01$.

Box 7.1. Results of Growth Model with Controls from Sala-i-Martin, Doppelhofer, and Miller (2004)¹

For the countries included in the sample, growth per capita between 1997 and 2017 is (1) positively impacted by the human capital index and the East Asia dummy and (2) negatively impacted by per capita GDP in 1997, the fraction of tropical areas in the country, and the share of government consumption in GDP. Besides the extreme growth figures for East Asian countries during 1997–2017 (due to the Asian crisis and a V-shaped recovery as well as rapid economic diversification), these results reflect that a more educated population supports higher productivity and growth; wealthier countries are more likely to experience lower long-run growth than poorer countries that have further to catch up; government consumption in EMDCs often crowds out more productive private investment; and most tropical countries are EMDCs.

Notably, the coefficient for life expectancy is not significant, which may be due to its correlation with the human capital index (Pearson correlation of 0.64 in the database). The coefficients for trade openness and the share of extractable commodities in exports, both proxies for economic diversification, are also not significant. The remaining explanatory variables are not significant but are of the correct sign.^{2,3}

¹ See results of model M3 in Table 7.2.

² Wealthier countries are assumed to have intense commercial interactions worldwide, naturally increasing their trade openness, measured in this paper as (Imports + Exports) / GDP.

³ The results are robust to the addition of the terms of trade and credit to the private sector. See Mitra and others 2020 for more details.

Link between Child Marriage and Fundamental Growth Factors

The channels through which child marriage could impact growth are multiple. Studies outline five main channels: health, education, fertility, labor force participation, and decision making.⁵ Given the overlap across these channels, we consolidate them into the channels of health, education, and labor force participation, with a focus on the first two (using labor force participation only for robustness checks). For example, health and fertility are closely linked, as are education and labor force participation. Moreover, Wodon and others (2017a) showed that although child marriage reduces earnings and household welfare, its impact on women's labor force participation is marginal—where the latter is more related to girls' education attainment. Also, health affects decision making (for example, HIV/AIDS) and education.

To understand the interactions between child marriage and growth factors (especially health and education), we regress child marriage on the growth factors from Sala-i-Martin, Doppelhofer, and Miller (2004). We consider the following equation for child marriage in a given year t (model M5):

$$CM_t = \beta_0 + \beta_1 He_{t-h} + \beta_2 Edu_{t-h} + \beta X_t^2 + \varepsilon_t^2, \quad (7.2)$$

where CM_t is the proportion of child marriage at time t , He_{t-h} and Edu_{t-h} represent overall health and education levels, respectively, lagged by h periods relative to t , X_t^2 is the vector of other explanatory variables at t , and ε_t^2 the error term.

In this equation, child marriage is assumed to depend on lagged values of education and health—where past improvements in the levels of education and health of the family and the community are expected to reduce child marriage. The rest of the explanatory variables are contemporaneous macroeconomic variables reflecting the idea that current economic conditions affect a family's decision to marry their children early. Endogeneity should not be an issue in this setting, as it can be assumed that current rates of child marriage would take several years to affect macroeconomic variables. We apply a lag of five years for the education and health variables, allowing the equation to capture the education and health levels when the girls married (our child marriage data are collected on girls aged 20–24 years).

The results indicate that GDP per capita, the East Asia dummy, and lagged human capital are significantly and negatively linked to child marriage, while the fraction of tropical areas is significantly and positively correlated with child marriage (see Table 7.3). These findings support the idea that the education of the child, the family, and the community help reduce child marriage; poorer countries, often located in the tropics, have higher instances of child marriage; and the rapid economic development in East Asia during the 1990s resulted in an exceptional decline in child marriage.⁶

⁵ Note that these channels imply correlation and not necessarily causality.

⁶ These results are robust to changing the lags on the human capital index and life expectancy from five to seven years. They are also robust to the addition of variables such as the terms of trade, credit to private sector, and labor force participation. The lack of significance of life expectancy may be due to its correlation with the human capital index. See Mitra and others (2020) for details.

TABLE 7.3.

Child Marriage Explained by Growth Factors	
Variables	Coeff.
Log GDP per capita	-13.01*
Human Capital Index	-11.27***
Life expectancy at birth	0.12
East Asia dummy	-12.55***
Extractable commodities in total exports	-0.01
Fraction of tropical area	10.81***
Trade openness	-0.00
Relative price of investment	-1.95
Share of government consumption	-0.05
Intercept	85.00***
P-value	0.00
R ²	0.56

Source: Authors' calculations.

Note: This model is later referred to as Model M5.

* $p < 0.10$; *** $p < 0.01$.

A MODEL FOR CHILD MARRIAGE AND GROWTH

Model

The model builds on the interdependence between child marriage, education, and health as previously discussed. Child marriage is assumed to depend on past levels of health and education and education to depend on past levels of health. Therefore the following simultaneous equation model is used:

$$\begin{cases} G_{t,t+T} = \alpha_0 + \alpha_1 He_t + \alpha_2 Edu_t + \alpha_3 CM_t + \alpha_4 \ln(y_t) + \alpha X_t^1 + \epsilon_t^1 & (7.3) \\ CM_t = \beta_0 + \beta_1 He_{t-h} + \beta_2 Edu_{t-h} + \beta X_t^2 + \epsilon_t^2 & (7.4) \\ Edu_{t-h} = \gamma_0 + \gamma_1 He_{t-2h} + \gamma_3 CM_{t-2h} + \epsilon_{t-h}^3 & (7.5) \end{cases}$$

$$\begin{cases} CM_t = \beta_0 + \beta_1 He_{t-h} + \beta_2 Edu_{t-h} + \beta X_t^2 + \epsilon_t^2 & (7.4) \\ Edu_{t-h} = \gamma_0 + \gamma_1 He_{t-2h} + \gamma_3 CM_{t-2h} + \epsilon_{t-h}^3 & (7.5) \end{cases}$$

Where $G_{t,t+T}$ is the growth rate of the per capita GDP between t and $t + T$, CM represents the proportion of child marriage, He and Edu represent overall health and education levels, y the real GDP per capita, X^1 and X^2 the vectors of other explanatory variables in equations (7.3) and (7.4), respectively, ϵ^i for $i = 1, 2, 3$ are the error terms. h is a lag-time horizon, which is chosen to be five years.

Estimation Strategy

The simultaneous equations model represented by equations (7.3)–(7.5) are estimated through three-stage least squares. This approach helps address potential endogeneity issues created by some variables being both explanatory and dependent. The three-stage least squares method is an instrumental variable (IV) method that corrects for potential correlation in the structure of disturbances across equations. Concretely, it estimates all the model coefficients simultaneously and proceeds in three steps: (1) applying a two-by-two IV regression, (2) estimating a consistent covariance matrix of the errors, and (3) applying a multivariate GLS-type regression by using the covariance matrix estimated in the second step.

TABLE 7.4.

Explanatory Variables in Models (7.3) and (7.4)		
Variable	Model (7.3)	Model (7.4)
Child marriage	✓	
Log of the per capita GDP	✓	✓
Human Capital Index	✓	✓
Life expectancy at birth	✓	✓
East Asia dummy	✓	✓
Share of extractable commodities in total exports	✓	
Fraction of tropical area	✓	✓
Trade openness	✓	
Relative price of investment	✓	
Share of government consumption	✓	
Terms of trade	✓	

Source: Authors' calculations.

Given the attrition in child marriage data, the stability of near-term child marriage rates, and the stability of its distribution over the medium term, it is assumed that:

$$CM_t = CM_{t-2b} = CM \quad (7.6)$$

The controls in X^1 are the same used in model M3 previously described. In addition to health and education, they build on the factors selected by Sala-i-Martin, Doppelhofer, and Miller (2004). Likewise, beside health and education, controls in X^2 are the variables that were significant in model M5 (child marriage as left-hand variable). We control for endogeneity issues by using lagged variables (see Mitra and others 2020). Table 7.4 summarizes the variable selection in (7.3) and (7.4) with details.

Results

Estimation of the simultaneous equations model confirms that child marriage has a significant and negative impact on growth (see Table 7.5). A decrease in child marriage by 1 percentage point can increase per capita growth by 0.04 percentage point per year. This result implies that if child marriage were to end today (proxied by the average rate of child marriage declining to 0), the long-term annual per capita growth in EMDCs would increase by 1.05 percentage points (given the linear relationship between child marriage and growth, as previously highlighted).⁷ This result is consistent with the findings of Wodon and others (2017a) and helps further quantify the increases in per capita growth implied by their results.⁸ The findings also highlight the intergenerational effects

⁷ This result is obtained by assuming that end of child marriage corresponds to a decline of its incidence by its average value in the sample (that is, by 26 percentage points).

⁸ From a sample of 106 countries, Wodon and others (2017a) estimate that the annual benefit from ending child marriage in 2015 would have increased from \$22 billion in 2015 and would be as much as \$566 billion in 2030. This increase, coupled with the leverage effect from the reduction in population growth, would naturally lead to important increases in per capita growth.

TABLE 7.5.

Results of Model with Simultaneous Equations					
Growth ($t, t+T$)		Child marriage (t)		Human Capital ($t-h$)	
Child marriage (t)	-0.04**			Child marriage ($t-2h$)	-0.02***
Log of the per capita GDP (t)	-1.63***	Log of the per capita GDP (t)	-7.89**		
Human Capital Index (t)	0.07	Human Capital Index ($t-h$)	-26.80***		
Life expectancy at birth (t)	0.02	Life expectancy at birth ($t-h$)	0.46**	Life expectancy at birth ($t-2h$)	0.02***
East Asia dummy	0.42	East Asia dummy	-5.60		
Extractable commodities in exports	0.00				
Fraction of tropical area	-0.36	Fraction of tropical area	6.18**		
Trade openness (t)	-0.00				
Relative price of investment (t)	-0.08				
Share of government consumption (t)	-0.02**				
Credit to private sector (t)					
Intercept	7.30***	Intercept	73.23***	Intercept	1.14**
P-value	0.00	P-value	0.00	P-value	0.00

Source: Authors' calculations.

Note: The columns give the results of regressions (7.3), (7.4), and (7.5) of the simultaneous equation model (from left to right).

For a given variable X , $X(t)$ is the value at t and $G(t, t+T)$ is the PPP growth rate between t and $t+T$. h is the lag horizon.

** $p < 0.05$; *** $p < 0.01$.

of reducing child marriage. Lagged by 10 years, child marriage has an impact on growth through its effect on education five years later. A 1 percent reduction in 10-year lagged child marriage improves the human capital index by 0.02 percent.⁹ This result then affects child marriage after another five years and ultimately improves per capita growth by 0.02 percentage point.

Education has a substantial indirect impact on growth through its effect on child marriage. Lagged education has a significant and negative coefficient in the estimation of child marriage. A 0.1 deviation improvement in the five-year lagged human capital index reduces child marriage by 2.7 percentage points. This reduction, in turn, increases per capita growth by 0.1 percentage point per year. Considering the standard deviations of both the education proxy and child marriage data, this suggests that lagged education accounts for almost 80 percent of the effect of child marriage on growth.¹⁰

The impact of health on growth is smaller than that of education and is mainly channeled through the effect of health on education. Lagged health has

⁹ This result represents an average increase of about 1 percentage point, given the average value of the human capital index in our sample.

¹⁰ The 80 percent figure is obtained by estimating the effects on growth associated with 1 standard deviation in each variable of interest.

a significant and positive coefficient in the estimation of education. A one-year improvement in the 10-year lagged life expectancy at birth increases the human capital index by 0.02 percent. This increase feeds through to a reduction in child marriage by 0.6 percentage point (this channel accounts for about 40 percent of the effect of the human capital index on child marriage). The improvement in child marriage then increases per capita growth by 0.02 percent per year. Notably, the effect of the five-year lagged life expectancy on child marriage is significant and positive. This result could be interpreted as better survival rates of children resulting in more children being married, absent policy changes to improve the income levels, education, and cultural practices.

Mitra and others (2020) showed that the model is robust to changes for various assumptions—including income inequality and governance. The following alternative specifications also do not materially change the results: increasing the lag-horizon h to seven years; considering child marriage by age 15 (instead of age 18); successively adding credit to the private sector, and factoring in labor force participation as additional controls in equation (7.3).

CONCLUSION

Child marriage is a serious human rights issue across many emerging and developing countries with wide-ranging socioeconomic consequences. Its adverse impact on population growth, health, education, and earnings—and more broadly, poverty reduction, and income inequality—have been thoroughly examined.

Mitra and others (2020) is the first study to examine the overall impact of child marriage on economic growth. In that paper, we applied a simultaneous equations model to account for the interdependent relationships among child marriage, education, and health and the multiple channels through which they impact economic growth.

While ending child marriage is an important development goal in itself, Mitra and others (2020) found that reducing child marriage could significantly improve economic growth. For example, over the long term, a reduction in child marriage by 1 percentage point could increase annual per capita real GDP growth by 0.04 percentage point. This result means that if child marriage were to end today, the long-term annual per capita growth in EMDCs would increase by 1.05 percentage points.

While some margin of error always exists around these types of cross-country econometric estimates, the results of this analysis support the existence of a strong relationship between child marriage and economic growth. These findings further strengthen the case for EMDC policymakers to allocate more effort and resources toward ending child marriage.

As countries develop comprehensive strategies toward ending child marriage, the results of Mitra and others (2020) can also provide insights on

policy prioritization in developing strategies to end child marriage. In particular:

- The strong interdependent relationship between education and child marriage suggests that education policies should be prioritized when developing comprehensive strategies to end child marriage. For example, past education accounts for almost 80 percent of the effect of child marriage on growth. Through this channel, an improvement in the human capital index (reflecting years of schooling and returns to education) by 0.1 percent could raise per capita growth by 0.1 percentage point per year.
- Improving education, both the quality and years of schooling, will also require stepping up complementary policies such as social assistance to poor and marginalized children, better infrastructure building for hygiene and transportation, and addressing excessive school dropout rates for girls due to pregnancy—in some countries, official policies expel pregnant girls from school.¹¹
- Health care policies also have an important role, though indirect, in ending child marriage as shown by the importance of better health outcomes on increasing levels of—and returns to—education.
- Stepped-up outreach and education for governments and affected communities (especially parents) on the costs of child marriage will also be critical to ending child marriage.

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¹¹ Human Rights Watch proposes an interactive index that compiles information on laws and policies related to pregnancy in school and adolescent mothers in Africa. The right to education for pregnant students and adolescent mothers was not fully protected in 13 African countries as of November 2022. For more details, see Human Rights Watch (2022).

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PART III

Investing Well: Empowering Women and Boosting Inclusion and Resilience

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Closing Education Gaps and Promoting Inclusive Development—Lessons from West Africa

Diego B. P. Gomes, Lisa Kolovich, Vivian Malta, Monique Newiak, Rasmané Ouedraogo, and Sahar Priano

This chapter demonstrates the critical link between gender equality, education, and macroeconomic outcomes, underscoring the profound economic and social benefits derived from closing gender gaps in education. The chapter summarizes the quantitative impact of education policies in Niger, Nigeria, Senegal, and Sierra Leone. The findings reveal a substantial potential to boost GDP and revenues, and to lower income inequality through policies targeting gender parity in education. The chapter outlines the theoretical framework, presents results across contexts, offers policy insights, and concludes with a summary of findings.

INTRODUCTION

Gender equality is not just a matter of inclusion—it has far-reaching positive implications for macroeconomic outcomes, stability, and governance. Vast theoretical and empirical evidence, including in this book, has consistently shown that gender equality is associated with better macroeconomic outcomes at all levels of development, including higher GDP, greater productivity, lower income inequality, and faster economic growth and convergence (Kochhar, Jain-Chandra, and Newiak 2017; Gonzales and others 2015; Sever 2022).

Narrowing gender gaps in education and promoting female labor force participation unlock a cascade of benefits. Increased educational attainment among women leads to higher diversification of output and exports, bolstering economic resilience (Kazandjian and others 2016). Moreover, women tend to invest more resources in their children, fueling school expenditures and higher school enrollment (Aguirre and others 2012). Increased educational attainment among women translates into higher human capital, a key driver of economic growth and better development outcomes (Barro 2001; Krueger and Lindahl 2001). Conversely, unequal access to education perpetuates income disparities and stifles progress (Galor and Zeira 1993; Gonzales and others 2015).

But just how substantial are the gains from closing gender gaps in education? This chapter outlines a journey to quantify the macroeconomic impact of various education policies in Niger, Nigeria, Senegal, and Sierra Leone. The micro-founded general equilibrium model, which features diverse agents, gauges the macroeconomic and distributional effects of education policies—especially those targeting girls. The results are astounding: in Senegal, implementing a policy ensuring a minimum of five years of education for all children could increase GDP by 8 percent (Kolovich, Malta, and Tavares Mendes 2018). In Nigeria, aligning girls' education levels with boys' levels across income groups could boost GDP by 5 percent (Malta and Newiak 2019). In Niger, closing the gender gap in schooling across income percentiles could ignite an impressive 11 percent long-term GDP surge (Ouedraogo and Gomes 2023). And in Sierra Leone, the benefits of education loom large, the result from a similar reform could amount to 8 percent of GDP in the long run (Malta, Newiak, and Sandy 2020).

The rest of this chapter lays out the theoretical framework underpinning this analysis, then presents the results of the framework's application to different contexts in Niger, Nigeria, Senegal, and Sierra Leone. A policy section synthesizes the insights gleaned from these four case studies, and a conclusion summarizes the results.

LITERATURE REVIEW

Gender inequality in education has wide-ranging implications for macroeconomic performance, and a growing body of research has explored these effects. There are various channels through which gender inequality can exert a negative impact on macroeconomic performance. King and Hill (1993) and Knowles, Lorgelly, and Owen (2002) use a Solow growth framework and discover that gender gaps in education have a substantial and statistically significant negative impact on GDP levels. In addition, Dollar and Gatti (1999), Forbes (2000), Appiah and McMahon (2002), Klasen (2002), Yamarik and Ghosh (2003), Gonzales and others (2015), and Kazandjian and others (2016) investigated the impact of gender disparities on economic growth. Their findings show that gender gaps in education have a negative impact on future economic growth. Furthermore, they challenge the earlier findings of Barro and Lee (1994) that suggested a potential negative relationship between female education and economic growth, demonstrating that such conclusions do not withstand rigorous econometric analysis.

Gendered educational gaps and growth are linked by several channels. First, gender inequality in education translates into lower human capital stock, a key driver of economic growth and better development outcomes (Barro 2001; Krueger and Lindahl 2001). Also, inequality in education perpetuates income inequality and curbs progress (Galor and Zeira 1993; Gonzales and others 2015). In nations with large educational disparities between girls and boys, the limited development of female human capital slows technological uptake and innovation (Barro 2001; Krueger and Lindahl 2001). Furthermore, in cases where diminishing marginal returns to education are observed, the decision to limit girls' education to lower

levels while providing higher levels of education to boys implies that the marginal return on educating girls surpasses that of boys (World Bank 2001; Knowles, Lorgelly, and Owen 2002). Klasen (2002) uses cross-country and panel regressions to show that the costs of educational gaps in the Middle East and North Africa and South Asia respectively amount to 0.9 and 1.8 percentage point differences in growth when compared to East Asia.

Second, closing gender gaps in education benefits future generations. Women tend to invest more resources in their children, resulting in higher school expenditures and child enrollment (Aguirre and others 2012). Encouraging women's education has been established to lower fertility rates and child mortality rates, and to foster education and thus human capital of future generations (Aguirre and others 2012). All these factors help to boost economic growth.

A third argument contends that increased female educational attainment promotes greater diversification of output and exports, thereby strengthening economic resilience (Kazandjian and others 2016). Berge and Wood (1994) support that a well-educated female labor force is critical to driving the growth of manufacturing exports. The diversification of the economy is stimulated through the accumulation of human capital, enabling the growth of skill-intensive industries, the emergence of new technologies, and facilitating the spread of technology among firms. Conversely, gender disparities in education have a direct adverse effect on diversification by limiting the available pool of human capital (Galor and Zeira 1993; Gonzales and others 2015). The results corroborate other studies that show a well-educated workforce is critical not only for achieving economic diversification but also for significantly contributing to the improvement of export quality (IMF 2014).

THE THEORETICAL FRAMEWORK

The analysis in this chapter employs an IMF-developed model-based framework known as the Dynamic General Equilibrium Life-Cycle Model with Heterogeneous Agents to quantify key transmission channels through which gender-responsive policies impact (1) female labor force participation, (2) earnings, (3) economic growth, (4) income inequality and poverty, and (5) public finances. The country-specific model allows for individuals to be different in key aspects, such as (1) gender, (2) stage of life, (3) labor skills, and (4) access to savings. The model also examines gender biases in both the workplace and the household, which create barriers for women to join the labor force. To account for the fact that the informal sector overrepresents women, particularly in low-income countries and emerging markets, the model also allows for workers to decide between participating in formal or informal jobs.

In each life period, households (comprising a man and a woman) make decisions about consumption of goods and services produced in formal and informal sectors while making labor supply decisions separately. Men decide the number of hours worked in formal or informal sectors, while women decide first if they participate in the labor market and, if they do, how many hours to work in the

formal or informal sectors. Households incur a utility cost when women participate in the labor market, which comes from the need to coordinate home production, child or elder care, and other unpaid work, as well as to comply with laws and social norms that create barriers for women to work outside the household. Human capital is determined by initial skills and years of education but also evolves endogenously through on-the-job experience. Formal sector production uses capital and labor inputs, while informal sector production uses only labor. Women face wage discrimination, and households pay taxes on formal sector goods and services, purchases, and earned income. Corporate revenues in the formal sector are also taxed. The government spends revenues on public consumption, public education, and transfers (see Malta, Martínez Leyva, and Tavares 2019 for more details).

RESULTS FROM COUNTRY CALIBRATIONS

Niger¹

Gender development in Niger stands out as one of the lowest in sub-Saharan Africa. The United Nations Gender Development Index measures gender inequalities in achievement in three basic dimensions of human development: (1) *health*, measured by female and male life expectancy at birth; (2) *children's education*, measured by female and male expected years of schooling for children; and (3) *adult education*, measured by female and male mean years of schooling for adults ages 25 years and older. The index shows that, when aggregating these categories, Niger performs worse than the average in sub-Saharan Africa, West African Economic and Monetary Union (WAEMU) countries, or peers in the Sahel region.

This underperformance is due to lower education attainment in Niger. The average years of schooling is only 1.7 and 2.8 for females and males in Niger, respectively, compared to 5.1 and 6.9 years in sub-Saharan Africa and 2.5 and 4.6 years in the WAEMU region. The disparities are also large when it comes to the expected years of schooling, which measures the number of years of schooling that a child of school entrance age can expect to receive if the current age-specific enrollment rates persist throughout the child's life.

Girls' elevated school dropout and child marriage rates significantly constrain gender equality. Government spending on education—3.8 percent of GDP in 2020—is commendable. While slightly below the WAEMU average of about 4 percent of GDP, Niger's education spending is above the average in sub-Saharan Africa (3.4 percent of GDP) and the Sahel region (3.5 percent of GDP). However, this relatively high education spending poorly translates into better education outcomes for women due to elevated school dropout and early child marriage rates. About 45 percent of girls drop out of primary school in Niger,

¹ This section draws directly from Ouedraogo and Gomes 2023.

compared to 20 percent in the WAEMU region and 22 percent in sub-Saharan Africa. In addition, more than 75 percent of women get married by age 18 in Niger, compared to 35 percent for peer countries in the WAEMU region and 41 percent in sub-Saharan Africa.

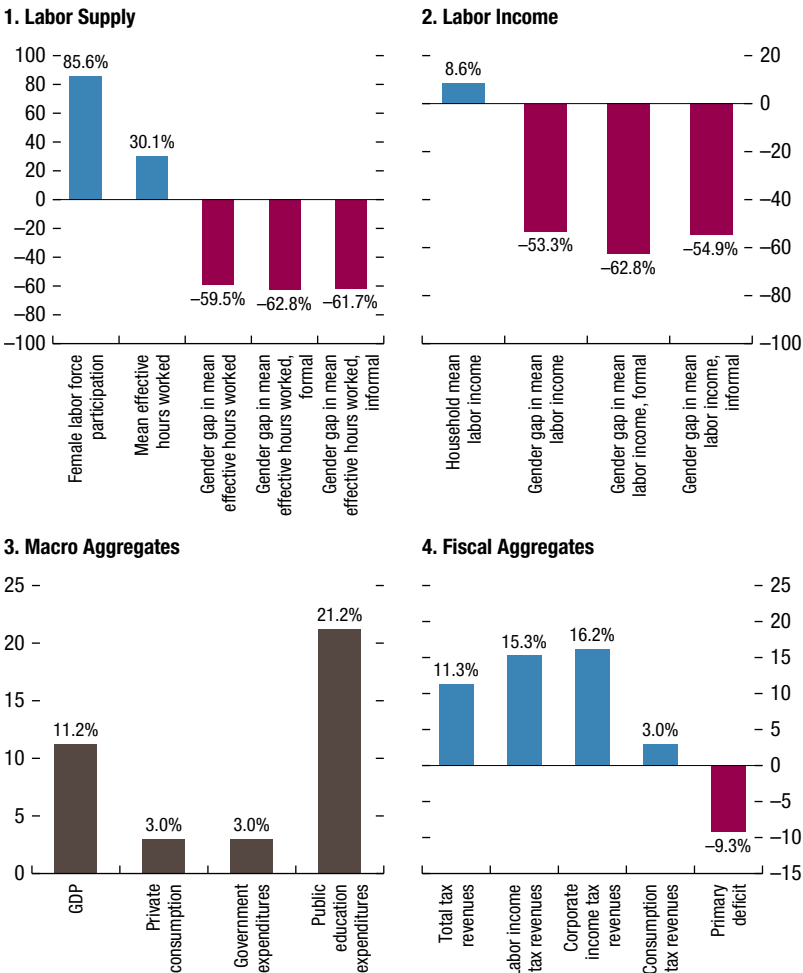
These educational gender disparities undoubtedly pose significant development challenges and have macroeconomic implications. This chapter quantifies the impact of closing education gaps between girls and boys using the framework described in the previous section. The model is calibrated to match key variables of Niger's economy in 2018, such as the formal sector's share of GDP, government expenditures as a percentage of GDP, government expenditures on education as a percentage of GDP, the Gini index of household income, and female labor force participation rate, among others. The benchmark year was considered to be 2018 due to the availability of microdata for Niger, notably from the 2018 edition of the *Enquête Harmonisée sur les Conditions de Vie des Ménages*.

The model simulates a public policy that increases education spending to include more girls in schools to equalize the number of years of education for boys and girls within the same income percentile. This approach is used because in Niger, despite having a low overall level of education, men have more years of education than women at every income percentile. These differences, on average, range from slightly more than 0.3 additional year for men at the lower end of the distribution to approximately 1.4 additional years at the upper end. Also, the gender gap in educational attainment grows monotonically across the income spectrum, suggesting that gender inequality in human capital formation rises with income.

Raising girls' education to the same level as boys' triggers a series of positive effects on the economy (Figure 8.1). Women enter adulthood with more education and thus a larger stock of human capital. As a result, they benefit from higher labor-market returns to education and higher returns to experience if they choose to work. Therefore, the incentives for women to enter the labor force and work more hours increase. According to the model, the new policy will likely increase female labor force participation by 85.6 percent. Working women begin to supply more hours to the labor market, which, combined with their higher level of human capital, increases mean effective hours worked by 30.1 percent. The gender gap in mean effective hours worked, as measured by the male-to-female ratio, falls by 59.5 percent. The same is true for the hours gender gaps in formal and informal labor markets, which have shrunk by 62.8 and 61.7 percent, respectively. Working women earn more in the labor market due to their higher level of human capital and higher supply of work hours, which contributes to increased household income and closing of the income gap between men and women. Household mean labor income goes up by 8.6 percent, and the overall gender gap in mean labor income, as measured by the male-to-female ratio, goes down by 53.3 percent. The same reductions occur individually in the formal and informal labor markets—which have gap reductions of 62.8 and 54.9 percent, respectively.

These labor market developments have macroeconomic and fiscal implications. Total economic output, as measured by GDP, rises by 11.2 percent, owing primarily to the increase in effective hours worked (Figure 8.1, panel 3). Higher labor income enables households to consume more, resulting in a 3 percent increase in aggregate private consumption. On the other hand, to implement the new education policy, the government must increase its public education spending by 21.2 percent, raising total government spending by 3 percent. Such an increase in education spending would imply budget reallocation and efficiency improvement. However, increased labor income, consumption, and production

Figure 8.1. Niger: Simulation Results of Increased Education Expenditure
(Changes from benchmark, percent)



Source: Ouedraogo and Gomes 2023.

result in increased tax collection, which can be used to fund these additional fiscal outlays. Revenues from taxes on labor income, corporate income, and consumption grow by 15.3, 16.2, and 3 percent, respectively (Figure 8.1, panel 4). As a result, total tax revenue increases by 11.3 percent, providing more than enough resources to cover the new education expenditures, implying a 9.3 percent reduction in the primary deficit.

Nigeria²

Education outcomes in Nigeria are among the weakest worldwide and are deteriorating in some parts of the country. Access to education is highly unequal across states and individuals' income and gender. Building human capital to boost work force productivity will be essential to reap Nigeria's demographic dividend. Nigeria's population is growing by more than 2½ percent annually, and its youth account for more than half of the population—presenting tremendous opportunities. However, a larger population will also add pressure on public services and infrastructure, which are already under considerable strain—Nigeria hosts the largest number of out-of-school children worldwide. Experiences from other countries suggest that declining dependency ratios (increasing the ratio of the working-age population to those too young or too old to be in the labor market), provide an opportunity for a demographic dividend. To reap this dividend, however, investments in human capital need to increase to allow workers to enter into the labor market at higher wages and higher productivity employment.

Quality issues in Nigeria's education sector impede the effective accumulation of human capital, and educational attainment varies strongly across states. Using numbers available as of 2018, the average Nigerian spent more than eight years in school, but substantial issues affect in the quality of education, halving the effective level of schooling. For instance, using Nigeria's 2013 demographic and health survey, only 24 and 31 percent of teachers scored at least 80 percent on a fourth-grade test in the language and mathematics curriculum, respectively, and 4 out of 5 persons 15 to 24 years old were not able to read a full sentence (World Bank 2018). Education can either be nearly universal in some states like Lagos or see the majority of children out of school in others like Bauchi.

Access to education also varies widely across individuals' incomes and gender. A 2013 demographic and health survey revealed the following statistics: Among the 20 percent of the poorest in Nigeria, a child's probability of not attending school was more than 70 percent; the probability was 2 percent for the richest quintile, contributing to Nigeria's high number of out-of-school children (a minimum of 10 million children). The probability of completing upper-secondary education was particularly low for poor girls: 1 percent for the bottom quintile and 6 percent for the second quintile.

As in the previous section, the micro-founded general equilibrium model was applied with heterogeneous agents to quantify the gains from narrowing

² This section draws directly from Malta and Newiak (2019).

education gaps. The model is calibrated to match the main features of the Nigerian economy. These features include labor market outcomes (participation, income), income distribution, education (in years of schooling), gender inequality, and the government (see Malta and Newiak for more details).

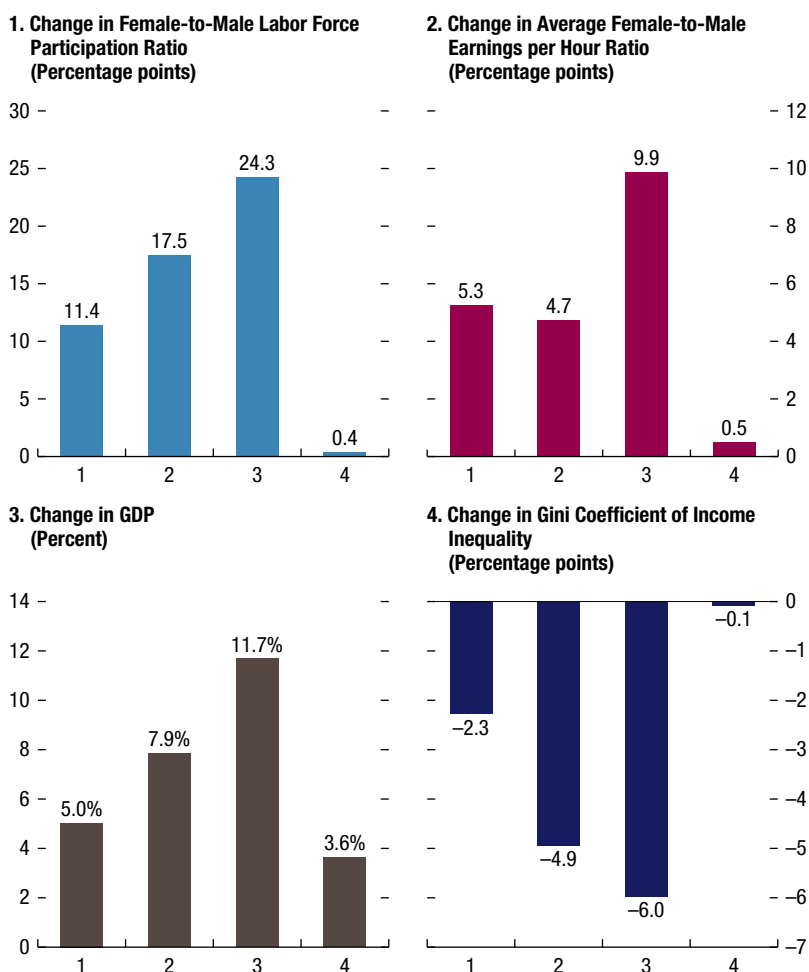
Four scenarios give a gradual picture of possible gains from higher education (Figure 8.2), reaching up to 12 percent of GDP (and higher levels for more ambitious policies not shown here):

- *Scenario 1: Equalize the number of years of education for boys and girls within the same income percentile.* With more girls educated, the female-to-male labor force participation ratio rises from 76 percent to 87 percent, and increased productivity raises women's average hourly wages by more than 8 percent. Men's average hourly wages remain constant and the long-term level of GDP increases by 5 percent. With productivity increasing mainly at the lower end of the income distribution (where gender gaps are highest), income inequality—as measured by the Gini coefficient—decreases 2 percentage points.
- *Scenario 2: Increase education for both boys and girls, so that each child benefits from at least the current median years of education for boys.* This policy increases the ratio of female-to-male labor force participation to 93 percent because the boost in human capital disproportionately impacts girls. Without additional reforms in the tax system, government revenues increase only a little (0.2 percent of GDP), mainly driven by the added activity of women of middle income in the formal sector, as most women will join the informal sector so as not to incur income or corporate tax. Earnings rise for both women (+10.4 percent) and men (+2.7 percent). GDP increases by almost 8 percent, higher than in Scenario 1, because access to education and higher productivity spreads to a wider range of workers. The Gini coefficient decreases by 5 percentage points, because people in the lower half of the income distribution no longer differ in their level of education, and gender gaps in education close.
- *Scenario 3: After increasing education as in Scenario 2, reduce barriers that pose disincentives for women to join the formal and informal labor market.* In this scenario, in addition to increasing education levels for boys and girls, the government addresses discrimination in the formal and informal labor sectors. Examples of such policies are, for example, implementing equal pay for equal work rights, equal access to resources (including land and credit), policies that make it easier to balance work and household responsibilities, and policies that address social norms and negative attitudes toward working women. In this simulation, average education increases to 9.8 years for girls and 10.3 years for boys. With higher returns from labor, gender gaps in labor force participation would completely close, and women's average wages would increase by 18.1 percent (and by 3.2 percent for men). Government revenues would increase marginally (¼ percent of GDP) through higher incomes in the formal sector. GDP rises by 11.7 percent, on the back of higher productivity, including from removing distortions from

the labor market. The Gini coefficient decreases by 6 percentage points as lower-income households become more educated and women join the labor market in an environment with fewer obstacles or discrimination.

- *Scenario 4: Increase the efficiency of education, that is, increase the quality-adjusted years of learning at the given years of schooling, by 16 percent (to the level in sub-Saharan African's lower middle-income countries).* This change boosts GDP by 3.6 percent through higher productivity for all children. The impact on other outcomes discussed in this chapter is smaller since relative education gaps remain the same across individuals at different levels of income and between boys and girls.

Figure 8.2. Nigeria: Impacts from Different Education Scenarios



Source: Malta and Newiak 2019.

Sierra Leone³

Overall human capital outcomes, including in education and health, are challenging in Sierra Leone. In 2019, the World Bank's Human Capital Index—capturing several dimensions of national health and education outcomes globally—ranked Sierra Leone among the seven countries with the largest human capital challenges worldwide, driven by weak health indicators, including child mortality and stunting. Education outcomes remain challenging. Although children completed an average of nine years of education, the suboptimal quality of education results in learning outcomes on par with receiving only 4½ years of high-quality education. These indicators highlight the challenges in a country once referred to as the “Athens of West Africa” around independence in 1961, particularly as population growth alone will put pressure on the educational system.

The Ebola health crisis was a significant setback for health and education outcomes. The immediate impact of the Ebola crisis on health outcomes was dramatic and affected other health areas as health resources shifted to Ebola-related programs (Save the Children 2016). In addition, UNICEF (2017) reported that the direct consequences of the Ebola outbreak on the education sector were substantial and reversed some of the progress toward the Millennium Development Goal of achieving universal primary education. Specifically, UNICEF reported that: all schools closed and 1.7 million primary and secondary students did not attend school for at least nine months; the crisis left 3,300 Sierra Leonean orphans who were less likely to attend school than their peers; the deaths of teachers increased pupil-teacher ratios; and maintenance of school infrastructure stopped. The UNICEF report estimated the indirect cost of lost future human capital due to the crisis at 8.8 to 9.8 percent of Sierra Leone's GDP in 2016.

When UNICEF conducted the analysis featured in this report, Sierra Leone faced significant challenges in closing disparities in access to and completion of education. The final level of education and completion rates varied considerably depending on individuals' demographic characteristics and their household location. While school enrollment rates increased over the past two decades, these increased rates do not imply that all who attend school will complete schooling.

The report also indicated that the level of education varied widely across income groups. A boy in the bottom 20 percent of the income distribution was more than five times as likely to receive no education than a boy in the richest family. A boy from the richest 20 percent of households was more than 10 times more likely to complete secondary education or achieve higher education than a boy from the poorest 40 percent of households, and large education gender gaps existed across all income groups. The probability of a girl receiving no education was more than 10 percentage points higher than for a boy in every income group.

Secondary or higher school attainment increased disproportionately with income for boys, but showed a large uptick for girls only in the richest income quintile. Hence, while enrollment levels were similar for boys and girls, girls were

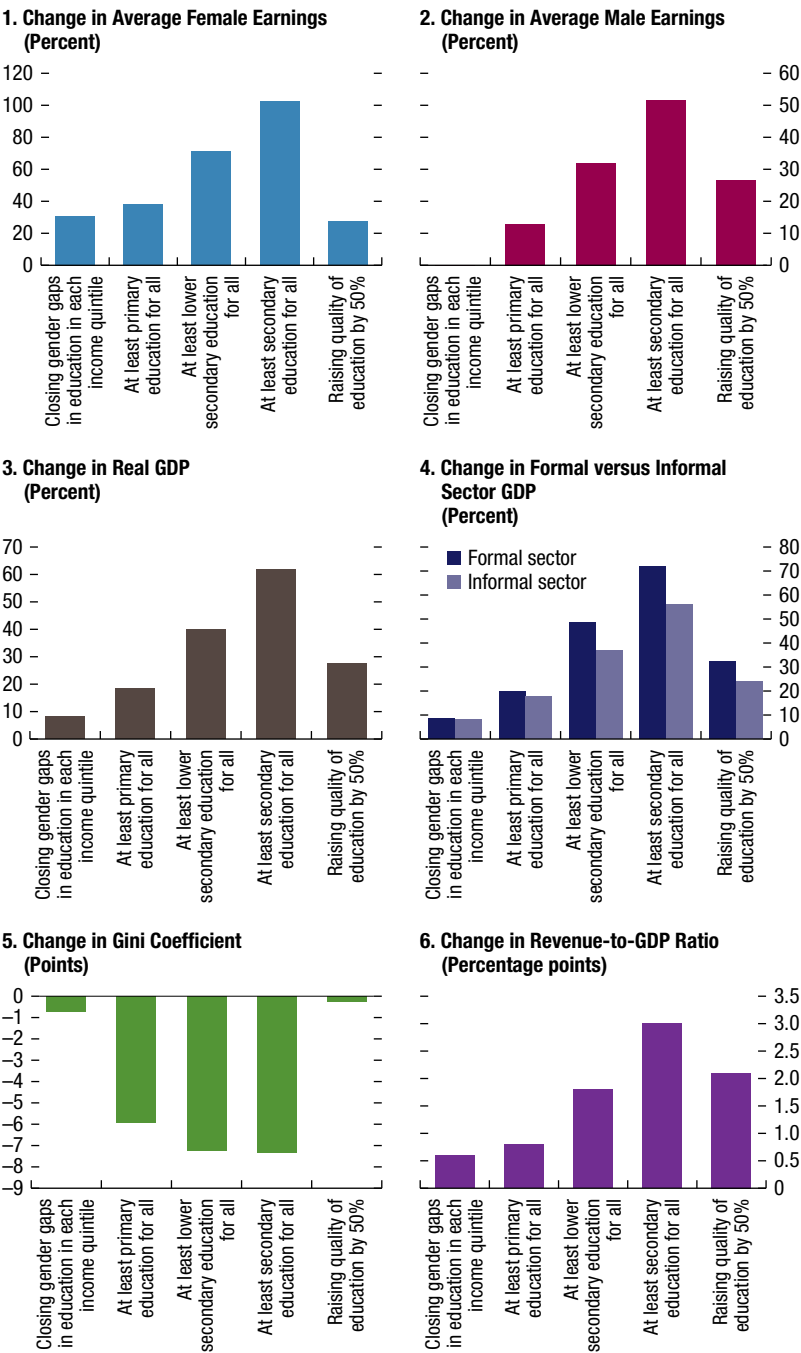
³ This chapter draws directly from Malta, Newiak, and Sandy 2020.

more likely to drop out and not complete their education. The rural-urban divide in education was also substantial. Children in rural areas were two to three times more likely not to receive any education; girls were 16 times more likely, and boys were five times more likely to complete secondary or higher education than boys (girls) in rural areas.

Given the aforementioned statistics and the government's intention to address them, a range of scenarios were tested with the following main results (Figure 8.3):

- *Scenario 1: Close gaps between boys and girls in access to education within each income quintile.* This measure assumes a substantial increase in girls' education levels while not addressing the educational disparities across income groups. The additional education would boost economic activity through higher female human capital and labor force participation, lifting women's purchasing power with higher wages (an increase of 30.9 percent). GDP (composed here of both formal and informal sectors) would increase by 8.2 percent in the long term. With higher skills, more women would join the (more competitive) formal sector, and the formal and informal sectors would grow by 8.6 percent and 8 percent, respectively. Government revenues under current tax policies would be larger by 0.6 percent of GDP in the new steady state. Since the bottom of the income distribution has the widest educational gender gaps, income inequality would decline (1-point drop in the Gini coefficient), and the income ratio of the top 10 percent of the income distribution to the bottom 10 percent would drop by 3.6 percent (from an average income ratio of 9.4 to 9.0).
- *Scenario 2: Ensure all children complete at least primary education.* In this scenario, education increases for all children that currently do not complete primary education, thus raising girls' education disproportionately. The strong boost in human capital and the resulting increase in wages (38 percent for women; 13 percent for men) increases GDP by more than 18 percent in the long term (19.6 percent and 17.7 percent in the formal and informal sectors, respectively). However, assuming the policy is not accompanied by anti-discrimination policies to level the playing field for men and women in the labor market, productivity levels (output per hours worked) in the formal sector would drop by 5.1 percent. In contrast, the informal sector would benefit from increased education at the bottom of the income distribution (28.9 percent increase in productivity). Under current tax policies and formal sector size, government revenues would increase by 0.8 percent of GDP in the new steady state. Since the largest human capital gaps are at the bottom of the income distribution, income inequality would drop substantially, with the Gini coefficient falling by almost 6 points and the income ratio of the top 10 percent of the income distribution to the bottom 10 percent declining by 35.7 percent (from an average income ratio of 9.4 to 6.0).
- *Scenario 3: Ensure all children complete at least lower secondary education.* This scenario magnifies the channels discussed in Scenarios 1 and 2, increasing

Figure 8.3. Sierra Leone: Impact from Reducing Gender Gaps



Source: Malta, Newiak, and Sandy 2020.

GDP by 40 percent (48.4 percent and 36.9 percent for the formal and informal sectors, respectively). Overall productivity in the formal sector would increase by 9.0 percent, while the informal sector would see a 54.8 percent productivity gain, mainly due to increases in education at the lower end of the income distribution where informal labor is more common. Increased consumption and wages would boost government revenues by 1.8 percent of GDP. Income inequality would decline significantly (with the Gini coefficient dropping by more than 7 points) as human capital would increase, in particular for poorer individuals. The income gap between the top and bottom 10 percent of the income distribution would drop by more than 44 percent (from an average income ratio of 9.4 to 5.2).

- *Scenario 4: Ensure all children complete upper secondary education.* This highly ambitious—and thus illustrative—scenario would result in boosting GDP by more than 60 percent (72 percent and 56 percent in the formal and informal sectors, respectively). Labor supply in the informal sector would decline by 12 percent, substituted by more productive work, and productivity in the formal and informal sectors would rise by 28 percent and 77.5 percent, respectively. The revenue-to-GDP ratio could increase by 3 percentage points. The Gini coefficient would drop by more than 7 points, while the income gap between the top and bottom 10 percent of the income distribution would drop by more than 45 percent.
- *Scenario 5. Raise the quality of education by 50 percent (by increasing the number of years of quality-adjusted schooling by 50 percent).* Policies to increase the average quality of education, even in the absence of actions to raise the number of years of schooling, could still boost GDP by 27.3 percent (32.5 percent and 23.8 percent for the formal and informal sectors, respectively), on the back of an increase in human capital across all income groups. Productivity in the formal and informal sectors would increase (by 31.4 percent and 22.1 percent, respectively). Income inequality, however, would remain relatively unaffected as the measure does not change relative endowments of education across the different strata of the society.

Senegal⁴

Senegal has taken important steps to close gender gaps. Gender gaps in both enrollment and completion rates for primary education have closed and even reversed (IMF 2020). According to the United Nations Educational, Scientific and Cultural Organization, from 1999 to 2016, gross enrollment rates in primary education improved from 59 percent to 88 percent for girls, while the rates improved from 71 percent to 78 percent for boys. Primary education completion rates rose from 33 percent for girls and 43 percent for boys in 2000 to 64 percent and 54 percent, respectively, in 2016. However, the average years of education in

⁴ This section draws directly from IMF 2020.

Senegal was only 2.8 in 2015 (according to the United Nations Development Program), lower than the average of WAEMU (3.0 years) and sub-Saharan Africa (5.1 years). More progress is needed as girls' completion rates in secondary education and enrollment in tertiary education are still substantially lower than those of boys. The Demographic and Health Survey program reported that, in 2012, the average female completion rate in secondary education was only 13 percent compared to 21 percent for boys. The female completion rate for tertiary education doubled from 4 percent in 2006 to 8 percent in 2016; however, the male rate—which increased from 8 percent to 13 percent—is still relatively higher.

Household survey data was also used to examine gender gaps in education along the urban–rural divide. Across sub-Saharan Africa, on average, educational attainment in rural areas is significantly lower than in urban areas. In the case of Senegal, the 2011 household survey showed that boys and girls between the ages of 10 and 14 in rural regions in Senegal have on average approximately 1.5 years less education than their urban counterparts. The urban–rural divide increased for boys and girls between ages 15 and 19, with urban students completing approximately twice as many years of education as their rural counterparts. Examining gender gaps in the number of years of education again reveals differences between urban and rural regions. Gender gaps are wider in urban areas than in rural areas (1.3 compared to 0.7 years of education gaps, on average). However, in percentage terms, women in rural areas of Senegal complete much less education than boys. Although the difference in urban areas averages 31 percent, the difference increases to 57 percent in rural areas.

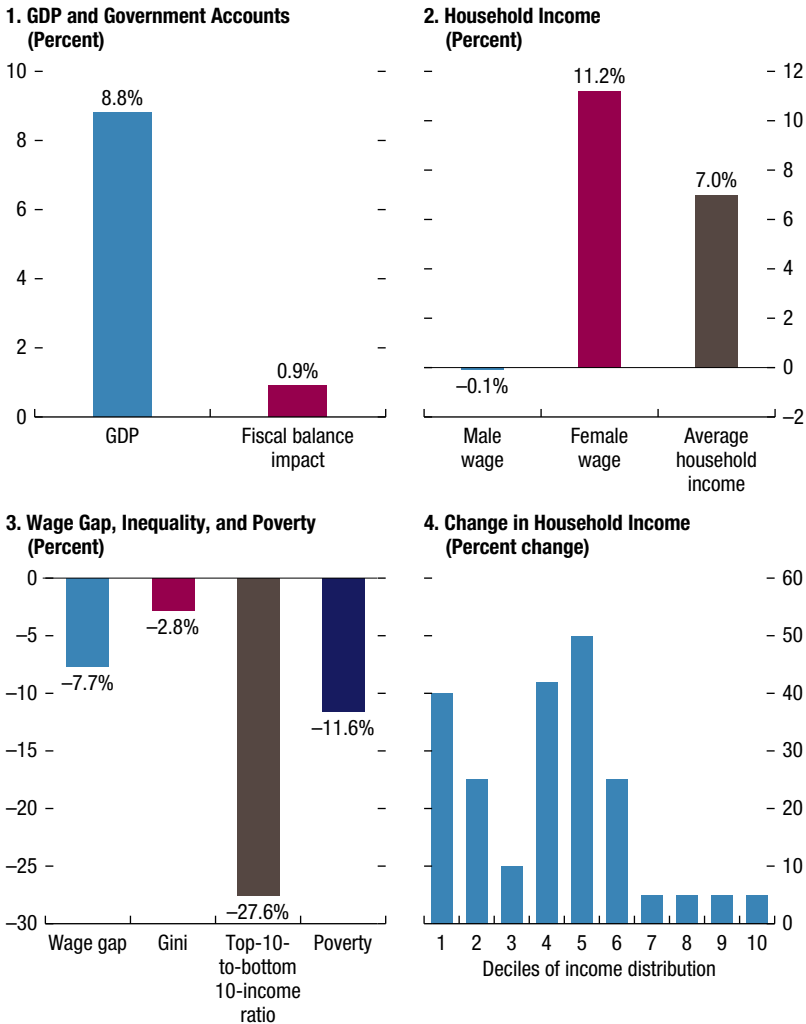
The combination of low education, gender roles, early pregnancy, and early marriage can create a poverty trap for women and girls. Expectations play a prominent role in economic outcomes, which is no different for women in poor employment conditions. Parents expecting lower returns from their daughters in the labor market will have fewer incentives to keep them in school. Girls with less education and fewer professional opportunities may not prioritize improving their skills and may choose to stay out of the labor force or have more flexible jobs allowing them to reconcile the demands of work at home versus outside the home. This cycle can leave them trapped in informal and lower-paying jobs.

The model was applied to Senegal, which also exhibits the characteristics of a typical low-income country, including a sizable gender gap, high informal sector employment, low levels of education, and predominance of the agricultural sector. In addition, as in many other low-income countries, women in Senegal face additional penalties in the labor market, including lower returns from experience and other “unexplained” gender pay gaps.

Closing educational gender gaps in Senegal increases female labor force participation and results in substantial GDP gains (Figure 8.4). In the baseline scenario, working women have 35 percent fewer years of schooling on average than men, and this gap widens for households at the bottom 50 percent of the income distribution. The exercise envisages closing the gap so that men and women at every income level will have the same number of years of education (differences will remain between individuals at different income levels).

Model simulations suggest that closing education gaps boosts female labor force participation by almost 19 percentage points, while output grows by about

Figure 8.4. Senegal: Macroeconomic and Distributional Effects from Closing Gender Gaps in Education



Source: IMF 2020.

9 percent in the long run, as a more educated female labor force increases overall labor productivity. With higher skills, some low- and lower-middle-income women join the labor force in the model, while young female workers continue to work throughout their lifetimes due to the financial returns from experience. Women with less education and experience would join the large informal sector, while middle-income and upper-middle-income women shift more of their working time to the formal sector, which has higher wages and more secure jobs.

Gender wage gaps, income inequality, and poverty rates fall. Because of the rise in female education, the gender pay gap would shrink. Despite higher competition in the labor market, the growth in aggregate demand is enough to offset pressures on men's wages, which decline only marginally. Closing education gaps would also reduce poverty substantially, as poorer and less educated women would gain relatively more years of education. This educational level increase would reduce inequality, shrinking the income gap between the top 10 percent of the income distribution and the bottom 10 percent. In the long run, this measure would bring net fiscal revenue gains, reflecting a higher collection of consumption and labor income tax revenues due to higher economic activity, which could be channeled to finance social and development spending.

PUTTING THE COUNTRY CASES TOGETHER: POLICY RECOMMENDATIONS

The model simulations presented in this chapter provide insight into the macroeconomic and distributional impacts of gender-responsive education reforms. The goal is to support policymakers in making better-informed decisions and considering the most effective and sustainable measures to boost female school enrollment and economic growth while reducing income inequality and poverty.

With limited fiscal space and vast challenges, implementing educational reforms in these countries will require evidence-based prioritizing, sequencing, and targeting of policies. Careful monitoring of education spending and its impacts will help reassess programs and priorities. Enhancing sector governance is vital to ensure that spending aligns well with needs at the various levels. Strengthening systems for a multisector effort to improve education will be critical. Such improvements involve: (1) clarifying mandates and strengthening accountability structures across institutions, policies, and acts; (2) realistic sector planning; (3) significantly reducing discrepancies in resources transferred from the center and those arriving at district levels; and (4) enhancing transparency and accountability in existing systems for allocating resources to schools.

Furthermore, comprehensive reform to improve educational quality and efficiency of services needs to be supported by efforts to improve health outcomes, especially for children, and access to infrastructure. Improving and institutionalizing administrative data collection on the education system will allow for better monitoring and evaluation of education-related reforms. Improving the quality of teachers to raise the efficiency of education would require new recruitment efforts, deployment, and training, along with higher salaries for teachers. School nutrition and health programs may also encourage girls to remain in school.

In some cases, legal reforms can also address underlying factors that reduce female school enrollment and completion. These reforms include laws to enshrine equal opportunities and children's rights and eliminate gender-based violence. Enforcing civil and criminal law when customary law contradicts nondiscriminatory policies is also critical.

CONCLUSION

This chapter summarizes the macroeconomic impacts of gender-responsive education policies in four countries—Niger, Nigeria, Senegal, and Sierra Leone—to provide solid evidence on the macroeconomic benefits and importance of gender equality in education. Ensuring equality of opportunities for girls to attend school brings positive implications for economic growth, productivity, and income equality.

With the Dynamic General Equilibrium Life-Cycle Model with Heterogeneous Agents, it was estimated that GDP gains could range from 5 to 40 percent in the long term. In addition, policies aimed at equalizing access to education support higher female labor force participation, increased household income, and reduced income inequality. Addressing additional barriers to female education, such as access to infrastructure, high adolescent fertility rates, and legal hurdles could further support equality in education. Gender-responsive education policies help level the playing field and promote inclusive and sustainable economic development.

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The Impact of Women's Empowerment on Human Capital of the Next Generation: Looking within the Household

Olivia Goldin

This chapter focuses on the instrumental impact of women's empowerment within the household on children's human capital, particularly health. This study and other evidence indicate that women's economic empowerment, including employment, and women's control of resources in the household improve children's health. Women's empowerment and control of resource allocations are even more important to children's health in households with fewer resources to share among household members, which is particularly pertinent when households face hard times. This work emphasizes the benefits of women's economic empowerment and enabling women to control resources, including through social protection and cash transfer programs.

INTRODUCTION

Women's empowerment has been shown to have a catalytic effect on improving the productive capacity of the next generation (Chapters 1 and 2). This chapter focuses on the impact of women's empowerment within the household on children's health, with an empirical study in South Africa. Strong economic evidence supported by the analysis indicates that children's health is improved by women's economic empowerment, such as employment prospects, and in particular women's control of resources in the household. More generally, women's empowerment is a prerequisite to achieving all of the Sustainable Development Goals (OECD 2021; United Nations 2023; World Bank 2011). The positive impact on the human capital of the next generation constitutes one way that women's empowerment has a multiplier effect, making it "smart economics" (World Bank 2011; Revenga and Shetty 2012).

This study and economic literature additionally indicate that women's empowerment is more important for children's health when households are poor and have few resources to share among household members. This analysis makes these findings even more important when households face hard times. In the wake of the COVID-19 pandemic and the war in Ukraine, people worldwide have suffered

rising poverty and hunger, particularly in sub-Saharan Africa, where 132 million people (roughly 13 percent) were estimated to be acutely food-insecure in 2022 (IMF 2023), with staple food prices in sub-Saharan Africa surging 24 percent from 2020–22, and food making up almost 40 percent of people's consumption baskets (Okou, Spray, and Unsal 2022; IMF 2022). As price-takers for global commodities, sub-Saharan African and other developing countries are particularly susceptible to external shocks, as Sub-Saharan Africa is a large importer of food and energy while commodities dominate exports. Climate change threatens children's health through both income shocks (for instance, threatening agricultural yields) and health shocks (for instance, disease outbreaks). In the face of these crises, optimal allocation of scarce resources within households can improve prospects for the next generation, with potentially minimal cost.

The evidence has policy implications in emphasizing the benefits of women's economic empowerment and the benefits of women's ability to control resources, including social protection and cash transfer programs. Duflo (2003) and numerous other studies have indicated that the efficiency and effectiveness of transfers in improving children's human capital increases when women receive the transfers. Many large-scale cash transfer and social protection programs aimed primarily at improving children's health and education outcomes are therefore directed toward women. Examples include some advanced economies' child benefit programs (including the UK for a long time) and large-scale cash transfer programs such as Oportunidades/Progresá in Mexico and Bolsa Família in Brazil. The evidence that women's empowerment improves children's outcomes may depend on gendered norms of women bearing most of the domestic responsibilities. Some programs, especially conditional cash transfers, that have focused on women's empowerment as only instrumental for benefitting children have unintentionally reinforced existing gender roles and increased women's burdens of responsibilities (Cookson 2018; Orzoco Corona and Gammage 2017; Molyneux and Thomson 2011). To improve children's outcomes and also to truly empower women, interventions must be sensitive to gender equality and seek opportunities to transform gendered norms.

BACKGROUND

Children's Health Links to Economic Development

Children's health and human capital development are critical to well-being, productivity, and economic development. Health is a core dimension of human capital. A child's health is likely to impact local, national, and wider prosperity through important economic outcomes such as the individual's income, educational attainment, cognitive ability, behavioral problems, and health later in life (Almond 2006; Banerjee and others 2010; Carneiro, Meghir, and Paredy 2013; Duc 2011; Mani 2012). Poor nutritional status during childhood and its lasting negative impact on economic growth and health is well documented in the literature (Glover-Amengor and others 2016).

Within households and across society, decisions are made on how much education or health care is provided to individual children. These decisions “have significant implications for the intergenerational transmission of poverty and the potential for upward mobility across generations” (Doss 2013, 10). Low investment in child health therefore has far-reaching consequences for economic growth and welfare (Duflo 2003).

Representing Children's Health

This study analyzes children's health in terms of their nutritional status, specifically anthropometric status. Anthropometric status generally reflects overall health and represents the child health indicator used in most relevant studies (for instance, Thomas 1990; Chen and Li 2009; Abreha and Zereyesus 2020). Nutritional intake is a determinant of children's health, and nutrition is a dominant health problem for children in developing countries, particularly in sub-Saharan Africa, where about 45 percent of all child deaths are linked to malnutrition (WHO 2021).

Representing Women's Empowerment in the Household

Different household members can have different preferences and make different decisions about allocating household resources, including supporting children's health. Through bargaining with each other, household members negotiate to obtain power in making decisions over resources. Each individual's bargaining power determines their options if members do not agree. Norms and rules (for instance, violence, divorce, or support networks) and economic possibilities (through income, employment opportunities, education, and asset ownership) determine people's options outside of agreement. Bargaining power itself is “fundamentally unobservable” (Doss 2013, 11) but can be represented in practice by decision making over resources (an outcome of bargaining) or outside opportunities (that determine bargaining power).

Doss (2013) indicated that the best representation of women's bargaining power is whether a woman can make decisions, including about expenditures, as other proxies may be a means to obtaining control of resources but not a requisite for implementing one's preferences or doing so effectively. This study uses this variable, focusing on decision-making power over day-to-day expenditures, to directly identify control over inputs into children's health.

Women's opportunities outside the household do not necessarily translate to increased capacity to exercise preferences within the household. In theory, outside opportunities increase bargaining power only when they are “gainful” outside options, that is, when a partner's options outside the household have payoffs at least as high as those the partner would have from remaining in the current situation. This scenario gives that partner a credible “threat point” for walking out on a deal (Sen 1992; Handa 1996; Muthoo 1999, 2000, 2002; Browning, Chiappori, and Weiss 2011; Doss 2013). Studies of outside opportunities include employment or earnings (Qian 2008; Duflo and Udry 2004; Lundberg and Polak 1993, 1996; Lundberg, Polak, and Wales 1997), exogenous income transfers (Duflo

2003), women's education (Handa 1996, 1999; Currie and Moretti 2003; Chopra 2003; Wamani and others 2005; Lindelow 2008; Medrano, Rodriguez, and Villa 2008; Chen and Li 2009; Anyamele 2009; Cleland 2010; Lesiapeto and others 2010; Bain and others 2013; Essilfie, Sebu, and Annim 2020), women as a household head (Handa 1999; Gaiha and Kulkarni 2005; Luseno and others 2014), and asset and land ownership (Allendorf 2007; Doss 2006; Beegle, Frankenberg, and Thomas 2001; Quisumbing 1994). Indicators of such outside opportunities can also be associated with factors that may separately impact children's health and confound understanding of a causal pathway. For example, household adults who are working more can support children's health through higher income and better networks through one pathway. Through another pathway, however, increased nondomestic work for men and women has been shown to compromise children's health and nutrition through decreasing time with children in certain settings (Andrabi, Das, and Khwaja 2012; Johnston and others 2018) and time use patterns shape nutritional outcomes (Hull 2013). This relationship could lead to an underestimation of the effect of an individual's employment on children's health.

Women's Empowerment Links to Children's Human Capital

Strong evidence indicates that women's empowerment in the household leads to an improvement in children's health and more human capital accumulation compared to men's (Thomas 1990; Lundberg and Polak 1993, 1996; Lundberg, Polak, and Wales 1997; Duflo 2003; Duflo and Udry 2004; Doss 2006; Qian 2008).

Studies exploiting exogenous income shifts, such as through policy changes, importantly indicated the causal effect of women's empowerment on children's health by ruling out potential endogeneity (Duflo 2003; Duflo and Udry 2004; Qian 2008). In South Africa, Duflo (2003) used a natural experiment showing that young girls who live with a woman eligible for a transfer program gained 0.6 standard deviations of weight for height, but gained statistically insignificant 0.056 standard deviation when living with a man eligible for a transfer, while there was no significant effect for boys.

Studies in developing country settings have indicated that women typically spend a higher proportion of their income and the cash they control on food and health care for children compared to men, while men use more of their income for personal expenditure, and spend more conspicuously than women. Examples of studies that illustrate this finding include in Kenya, Guatemala, Brazil, and the Philippines (Quisumbing 1994; Quisumbing and others 1995), Brazil (Thomas 1990, 1994), and Côte D'Ivoire (Duflo and Udry 2004). The systematic literature review by Santoso and others (2019) points to how money controlled by women is more likely to be used to purchase food and health care for children of the household than money controlled by men, and this results in a positive influence on household-level calorie availability and health outcomes in many studies, including in Rwanda (Von Braun and others 1991) and Mali (Castle 1995). Borges (2007) indicated that on average worldwide, women reinvest 90 percent

of their income in their families and communities, contrasting with men who reinvest just 30–40 percent of their income. Thomas' seminal study in Brazil (1990) indicated that maternal income increased family nutrition four to seven times more than paternal income.

Numerous systematic reviews of recent studies have been conducted, although these studies are almost all cross-sectional, leading to challenges identifying causality. Abreha and Zereyesus (2020) indicated that decision making was the most common measure of women's empowerment among studies reviewed, and most studies in sub-Saharan Africa indicated that women's higher decision-making power is positively associated with better children's health status. Many of these studies examined Demographic Health Surveys (DHSs), including a study across 26 sub-Saharan African countries (Siedu and others 2021) showing that households in which women had greater decision-making power were more likely to seek health care for children. For example, they were more likely to seek treatment for childhood diarrhea and complete child immunizations. For eight sub-Saharan African Demographic Health Surveys (Singh, Bloom, and Brodish 2012), women's decision making is associated with increased likelihood that children will be treated for respiratory illness. Essilfie, Sebu, and Annim (2020) indicated that female decision making in Ghana is associated with improved children's nutritional status. Another systematic review (Santoso and others 2019) found an inconclusive relationship, although all studies reviewed were observational and cross-sectional, with suspected endogeneity issues of factors not accounted for. Several cross-sectional studies in sub-Saharan Africa have indicated a strong association between women's empowerment and positive feeding practices for infants and young children in some African countries studied (Mali, Rwanda, Sierra Leone, and Ghana) although results were mixed in others (Ethiopia, Nigeria, Uganda, and Zimbabwe) (Kobina Dadzie, Amo-Adjei, and Esia-Donkoh 2021; Na and others 2015; Malapit and Quisumbing 2015).

A mother's education also strongly predicts children's health, including when considering effects such as a mother receiving health information (Handa 1996, 1999; Currie and Moretti 2003; Chopra 2003; Wamani and others 2005; Lindelow 2008; Medrano, Rodriguez, and Villa 2008; Chen and Li 2009; Anyamele 2009; Cleland 2010; Lesiapeto and others 2010; Bain and others 2013; Essilfie, Sebu, and Annim 2020). In Uganda, educating mothers is associated with greater adoption of health-promoting behaviors within the household (Nyqvist and Jayachandran 2017). Abu-Ghaida and Klasen (2004) found that in Bangladesh, better educated women are more capable of processing information, using health care facilities, and keeping their living environment clean. A United Nations Educational, Scientific and Cultural Organization (UNESCO) study (2014) claimed that, in low-income countries, 1.7 million (4 percent) fewer children would suffer from stunting if all women had completed primary education. The study also showed that if all women had completed secondary education, 12.2 million (26 percent) fewer children would suffer from stunting, as education leads parents to improve health and hygiene practices and improves nutrition in their children's diet.

Women's household headship is another proxy for women's power in the household. While Handa (1999) provided evidence that household heads can win resources on behalf of their own children, using household head as a proxy for women's empowerment relies on the assumption that women can implement their preferences more efficiently in female-headed households than male-headed households. Mixed evidence exists on the impact of women as a household head, including positive impacts on children's health in India (Gaiha and Kulkarni 2005) and no significant impact in a Malawi cash transfer study (Luseno and others 2014). These findings could be because household headship represents household characteristics that negatively affect children, such as that female heads of household often bear the sole responsibility for earning income (Richards and others 2013) and may have more limited access to resources than the average male-headed household. For instance, in a study in Ghana, female-headed households had lower access to agricultural inputs (Doss and Morris 2001).

Resource allocation when resources are constrained have notably varied by gender, with women receiving less (Sen 1992, 2003), including in cases of extreme weather shock (Anttila-Hughes 2012; Foltz and Han 2013). When women's bargaining power and/or control of resources increased, some empirical studies indicated bigger improvements in girls' health than boys' health in the household (Thomas 1994; Duflo 2003; Qian 2008).

Channels of Impact of Women's Empowerment on Children's Human Capital

The phenomenon of women's intra-household empowerment leading to increased spending on children's health has been attributed to the tendency for women to value children and emphasize children's well-being more than men (Nyqvist and Jayachandran 2017). While women may not inherently value children more than men, there are many socioeconomic reasons that they do. Cultural and social norms often encourage women to prioritize ensuring household members receive sufficient food allocations, particularly in poor rural settings. In poor, rural South Africa, mothers are considered the "most important health worker in the household" (Medrano, Rodriguez, and Villa 2008, 612). Moreover, women often have "gendered indigenous knowledge" of the domestic sphere and matters relevant to health, including agriculture, maintaining water supplies, and food security, facilitating better-informed and more productive decisions on children's health inputs (Terry 2009, 13). As women in these settings tend to be poorer and specialize in nonmarket (domestic) labor, they are likely to rely more on their children for future support, and therefore they may prioritize their children's human capital accumulation to a greater extent than men. Women may also care more about children's daily needs and well-being because they tend to spend more time with children and thus may be more familiar with them than men, who typically specialize in market labor. Given that women tend to care and know more about children than men do, women are expected to make decisions about daily expenditures to support children's health more than men.

Direct experimental evidence on preferences indicates that compared to men, women prefer health outcomes (Lampietti and others 1999; Prabhu 2010) and spending on children (Kusago and Barham 2001). Such direct experimental evidence identifies preferences about outcomes such as children's health through asking men and women what they prefer and comparing. This valuably augments most empirical studies of the impacts of female empowerment, including the empirical analysis in this chapter, which infer the preferences about outcomes through outcomes, inferring how preferences differ between men and women through the improvement in outcomes associated with proxies for women or men having more ability to implement their preferences.

The link between material allocations to children (for instance, food resources and health inputs) to improvements in children's diet and nutritional status is well established (Santoso and others 2019; United Nations Children's Fund [UNICEF] 2021). Women's decision-making power could also affect children's health through other channels, such as women's health and access to maternal health care. Qualitative evidence from a study in Ghana indicated women's lack of decision-making autonomy and wider gender inequality undermined access to maternal health care (Ganle and others 2015).

Evidence also explores the impacts of women's empowerment on children's education. Using exogenous increases in sex-specific incomes, Qian (2008) shows that children's education improves when female income increases but when male income increases, girls' educational attainment decreases and there is no effect for boys. Wider evidence is mixed in the context of a cash program in Morocco, targeting to mothers versus fathers made no difference to children's educational attainment (Benhassine 2015).

IMPORTANCE OF WOMEN'S EMPOWERMENT FOR CHILDREN'S HEALTH WHEN HEALTH INPUTS ARE MORE SCARCE

Resource Scarcity

Health inputs are more important when households are poorer or children's health is more threatened (for instance, by disease). There is a higher return of additional allocations of health inputs for children's health status.

If women's power over resources affects children through the allocation of resources, women's decision making would likely improve children's nutrition where resources are scarce and therefore contested within the household (Heckert, Olney, and Ruel 2019; Santoso and others 2019). This phenomenon amounts to diminishing marginal returns of health-to-health inputs. Reduced household wealth constrains health inputs, including food, medicine, water, and shelter. Poverty is associated with increased stress, mental health problems, reduced cognitive ability, and poor health (Mani and others 2013; Haushofer and Fehr 2014). The effect of women's empowerment in the household is therefore expected to be stronger for poorer households.

Income tends to improve children's nutritional status. Empirical evidence has shown reductions in children's health and educational attainment due to negative

income shocks (Mincer 1958; Subramanian and Deaton 1996; Jacoby and Skoufias 1997; Jensen 2000; Banerjee and Mullainathan 2010), including income shocks from extreme weather crises (Anttila-Hughes 2012; Baez, Caruso, and Niu 2018). Real income losses tend to result in replacing more expensive but potentially more nutritious foods (including meat and fish) with cheaper foods such as cereals (Subramanian and Deaton 1996; Banerjee and Duflo 2007; Anttila-Hughes 2012). Increased resource scarcity and resource competition in a society also increases risk of instability and conflict, with negative impacts on human capital and wealth accumulation.

Climate Shocks

Climate change has been called “the biggest global health threat of the 21st century” (Costello and others 2009, 1693). Countries in sub-Saharan Africa are particularly vulnerable to changes in climate and the impacts of rainfall variability (Niang and others 2014). Extreme weather events are becoming increasingly frequent with climate change. Extreme weather has been consistently associated with reduced children’s nutritional status in the literature (Maccini and Yang 2009; Skoufias and Vinha 2012; Anttila-Hughes and Hsiang 2013; Andalón and others 2016), including in sub-Saharan African and South Africa (Thompson, Matamale, and Kharidza 2012; Foltz and Han 2013; Baez, Caruso, and Niu 2018).

Weather events such as extreme rainfall tend to negatively affect children’s health through health and income shocks. The health shocks include increased disease burden brought about by jeopardizing water, hygiene, and sanitation, which increase water-borne disease (UNICEF 2017). Children living in sub-Saharan Africa are expected to disproportionately bear these risks (Thompson, Matamale, and Kharidza 2012). In particular, floods increase the prevalence of disease-carrying insects such as mosquitos and worms, and extreme flooding can physically harm people, especially children. Extreme weather events also threaten children’s health indirectly by affecting household resources, with subsequent reduced investments in human capital. For poor, rural households, the effect is particularly strong due to the strong impact of extreme rainfall on agricultural production and incomes. Studies have therefore used rainfall shocks as the exogenous instrumental variable for income shocks for poor, rural communities (Angrist and Krueger 2001; Duflo and Udry 2004; Miguel, Satyanath, and Sergenti 2004; Miguel 2005). Extreme flooding also directly causes loss of physical assets (Anttila-Hughes and Hsiang 2013).

This study focuses on the impact of rainfall, as rainfall is considered “the most important source of climate variability for livelihoods” of South Africans (Flato, Muttarak, and Pelser 2017, 48). In accordance with the literature on climate shocks, extreme rainfall shocks will likely negatively affect children’s health.

With the expectation that the impact of women’s empowerment in the household will be stronger when health inputs to children are more constrained, the impact of women’s decision making in the household is likely stronger in the face of extreme weather events. Evidence for this anticipated finding is limited, and insufficient climate data limit potential conclusions of our study. Foltz and Han (2013) indicated that in Mali, the number of women in the household over

childbearing age (over 45 years) improved children's health; however, no significant interaction exists between this effect and extreme positive rainfall shocks. Women's indigenous knowledge of resource management (Terry 2009) will likely give women's empowerment a vital role in climate adaptation (ActionAid 2007).

THEORETICAL FRAMEWORK

Research over the past 25 years, including that in the previous sections of this chapter, has established that preferences differ within households. This research has shown that household decision making over allocation of resources is not the same across potential decision makers, going against the previously dominant theories of unitary preferences within the household (Samuelson 1956; Becker 1981).

We construct a theoretical framework using Besley and Ghatak's (2001) incomplete contracting framework for public goods. The benefits of children's health are public goods (nonrival and nonexcludable) for each potential decision maker (Rasul 2000). Besley and Ghatak demonstrated that decision-making authority should be allocated to the party that values the benefits of the public good the most regardless of involvement in investments for the good. Involvement in investments could be through personal contribution of money. Allocating ownership to the party with the highest valuation improves investment incentives for both parties and results in the highest possible level of joint surplus.

Applying Besley and Ghatak's framework indicates that if women value children's health more than men do, then in order to maximize children's health women should be allocated decision-making power. This allocation produces higher human capital for children (the "joint surpluses" of the investment in children), regardless of either party's investment involvement. Application of the Besley and Ghatak framework assumes that contracts cannot habitually be made on the investment levels or realized human capital of children, leading to an incomplete contracting "holdup" problem and parties underinvesting in children's health compared to the "first best" optimal that would require contracts. Achieving the second-best household investment in children requires allocating control of resources to the party that cares more, regardless of investment involvement, such as in earning the money to invest. The literature has indicated that women tend to be the more caring party. (Annex 9.1 provides further details on the theoretical framework.)

This study empirically tests the hypothesis that women making decisions on daily expenditures is associated with improvements in children's health compared to men making these decisions.

Women's economic opportunities outside the household (such as employment) are expected to increase their bargaining power and decision-making power over household resources in the household, with potential to enable women to implement their preferences more effectively. Women's economic empowerment, such as employment, is not the only factor determining the allocation of decision-making power. Other factors also likely enhance women's effectiveness in implementing their preferences—for instance, gaining support and knowledge through

networks—that also affect their decision-making power. Through separate variables for decision makers’ gender and economic characteristics, this study analyzes whether the impact of women’s decision-making power is dependent on their economic empowerment in terms of employment status. This link could have policy implications, adding to the reasons why women’s economic empowerment enables economic development through supporting children’s health.

Data and Empirical Methodology

Data Origin

This study employs a panel of longitudinal household-level data—the four-wave panel data set from the National Income Dynamics Survey (NIDS) produced by the Southern Africa Labour and Development Research Unit (SALDRU 2017a, 2017b, 2017c, 2017d). The nationally representative panel data set surveyed 28,000–38,000 individuals from roughly 10,000 households in each of the four waves (2008, 2010–11, 2012, 2014–15). Each wave contained seven surveys, and the 1,500 variables provided extensive data on individuals’ and households’ demographic and socioeconomic characteristics, including detailed anthropometric data for children, intra-household decision making, household structure and composition, labor market participation, education, and other economic activity. These data facilitate controlling for—and examining the effects of—relevant health and socioeconomic factors. Inclusion of wider factors helps to minimize endogeneity, while the relatively large sample allows examination of restricted samples. The study uses high-resolution global terrestrial monthly climate data developed by the University of East Anglia Climate Research Unit (UEA CRU 2014; Harris, Jones, and Osborn 2014). For further details on data adjustments see Annex 9.2.

South Africa

The South African context is not typical of sub-Saharan Africa, characterized by one of the world’s most consistently unequal income distributions (World Bank 2023a) with high per capita income levels for a development context but pronounced poverty (World Bank 2023b). The range of wealth in the nationally representative sample facilitates examination of the role of resource scarcity for the impact of climate on children’s health. Located at the southern tip of Africa, the country also hosts a range of climatic zones with rainfall variation across districts. The approach to the rainfall variable reflects this variation.

Sample

The sample includes children aged 2 to 15 years, in line with the NIDS definition of children and evidence on using anthropometric measures (for example, Roemling and Qiam 2013). The main focus and specification sample is resource-scarce households. See Annex 9.3 for details on further sample exclusions and the definition of resource scarcity.

Key Variables

Dependent Variable—Children's Health

There are several ways to measure children's health, including anthropometric measures. For this chapter, children's health is assessed through the anthropometric z -score of BMI (body mass index)-for-age, measuring children's nutritional status and health, in keeping with the literature (for example, Thomas 1997). Robustness is checked against the z -score of weight-for-age and excludes obese children. Annex 9.4 provides further details.

Explanatory Variable—Women's Empowerment

The variable for whether a woman makes decisions in the household uses adults' responses to the question, "Who makes decisions about day-to-day household expenditures? (for example, groceries)." The decision maker was used for daily expenditure as choices and allocation of daily expenditure to determine children's intake of nutrition, given constraints of the household budget.

Explanatory Variable—Extreme Rainfall

Potential impacts of extremely high rainfall (potential flooding) were examined using a dummy variable for whether rainfall in the last completed rainy season in a household's district was in the top 5 percent of the district's rainfall, in terms of deviations from the district-level long-term mean for each southern African rainy season in 1901–2015. This follows the literature (including Maccini and Yang 2009; Blankespoor and others 2010). Using precipitation relative to a long-term mean rather than absolute precipitation improves the exogeneity of the variable, as each district's (long-term) usual rainfall was controlled for. Usual rainfall is likely related to socioeconomic status and accounts for local adaptation and preparedness.

Model

The study employs an ordinary least squares (OLS) panel data model. Unobserved heterogeneity and hence potential endogeneity were minimized by controlling for relevant socioeconomic, environmental, proximal and inherent risk factors for poor health as part of the control variables, in keeping with the literature. Annex 9.5 provides further detail on the empirical methodology. The full results in Annex Table 9.6.1 describes the variables in the model.

RESULTS

Female Empowerment

The empirical analysis indicates a significant positive impact of women's empowerment in the household on children's health. A female making decisions over daily expenditures in the household compared to a male making these decisions improves children's health by 0.077 standard deviation of the z -score of BMI for children of

the same age, for the average 2- to 15-year-old child in a resource-scarce household (Table 9.1, column 1). This result is statistically significant at the 1 percent level—meaning that for 10-year-olds with a BMI at the 25th percentile for their age, women’s decision making is roughly associated with an increase in their BMI from 14.7 to 15.0, moving significantly toward healthy status. Annex 9.6 provides the full table. The result is robust and significant, though, as anticipated, slightly smaller for specifications using decision making over large, unusual expenditures, excluding obese children, or including non-resource-scarce (richer) households.

The role of wider economic empowerment

The study examines the role of women’s economic empowerment versus decision making over household expenditures through variables for decision makers’ characteristics. The variables for decision makers’ characteristics may influence each other; for example, an individual’s employment makes them more likely to become the decision maker. The interaction variable for whether a decision maker is both female and employed indicates that the impact on children’s health of a female making decisions within the household depends on her employment status; having a female household decision maker who is employed is associated with an increase of 0.131 standard deviation of the z -score (Table 9.1, column 2). This analysis indicates no statistically significant impact of an unemployed female decision maker versus an unemployed male decision maker. This finding also suggests there is no significant effect of a male decision maker being employed versus unemployed, all else equal. Employment status may improve female decision makers’ impact on children’s health because they have stronger bargaining power through options outside the household and they are more likely to have stronger control over their own income. Adding further characteristics of the decision maker indicates male decision makers improved children’s health to a greater degree when they were more educated—by 0.01 standard deviation for each extra year of schooling—but schooling is not significant for female decision makers to have an impact, perhaps suggesting the impact is captured through employment (Table 9.1, column 4).

Resource scarcity

The impact of women’s empowerment on health is generally stronger in poorer households, as shown in Figure 9.1, which indicates a trend of higher coefficients for poorer quintiles of income, with a coefficient of 0.088 for the poorest quintile and 0.039 for the highest quintile. While the impact remains positive, the low statistical significance in quintiles is likely due to the low sample size.

The rainfall shock to children’s health

Extremely high rainfall two rainy seasons before the survey year is associated with significant decreases in children’s health of 0.194 standard deviation of the z -score of BMI for age (full results table in Annex 9.6). This finding supports the literature indicating the negative impact of climate shocks on children’s health, which could occur via disease burden and compromising agricultural yields. The interaction with women’s empowerment is insignificant, though better accounting for hydrological systems is needed for the analysis of rain shocks to be credible.

TABLE 9.1.

The Impact of Women's Decision Making and Employment Status on Children's Health

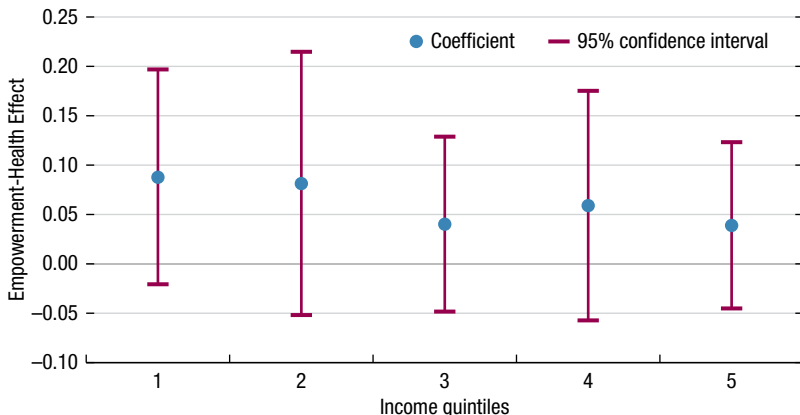
Variables	(1) Main	(2) Interaction of Decision Maker Being Female and Employed	(3) Significant Characteristics of Decision Maker	(4) Decision Maker's Characteristics
Decision Maker Is Female	0.077*** (0.022)	0.029 (0.031)	0.017 (0.031)	0.118 (0.091)
	0.000	0.348	0.586	0.194
Decision Maker's Age				-0.000 (0.001)
				0.832
Age for Female Decision Makers				-0.001 (0.001)
				0.265
Decision Maker's Employment Status		-0.046 (0.044)	-0.070 (0.044)	-0.077 (0.053)
		0.297	0.116	0.144
Decision Maker Is Female and Employed		0.131*** (0.050)	0.139*** (0.051)	0.141** (0.057)
		0.009	0.007	0.013
Decision Maker's Schooling			0.008*** (0.003)	0.010* (0.006)
			0.002	0.085
Decision Maker's Schooling if Female				-0.005 (0.007)
				0.451
Observations	24,535	24,518	24,501	24,501

Source: Author's own analysis using National Income Dynamics Survey (NIDS) produced by the Southern Africa Labour and Development Research Unit (SALDRU 2017a–d) and climate data produced by the University of East Anglia Climate Research Unit (UEA CRU 2014).

Note: Robust standard errors are in parentheses. *P*-values reported below standard errors. Standard errors clustered at district and wave level. FE = clustered at level of Districts and Waves; pid = 14,171 (column 1).

p* < 0.1; *p* < 0.05; ****p* < 0.01.

Figure 9.1. The Coefficient for Women's Decision Making in the Household on Children's Health



Source: Author's own analysis using National Income Dynamics Survey (NIDS) produced by the Southern Africa Labour and Development Research Unit (SALDRU 2017a–d) and climate data produced by the University of East Anglia Climate Research Unit (UEA CRU 2014).

CONCLUSION

Results and Implications

This study's empirical analysis supports the wider evidence indicating that women's empowerment in the household improves children's health, particularly through women's employment. This link indicates the importance of women's economic empowerment outside of the household for household dynamics. The study supports the literature which has shown that when women control household resources, children have better human capital outcomes.

This study also indicates that women's empowerment generally matters more for children's health when households are poorer, making women's empowerment more important for children's health in challenging times. The world faces rising poverty and food insecurity in the aftermath of global shocks. Improving children's health is even more urgent in the face of direct health shocks, such as disease outbreaks, and shocks to household resources, such as through economic downturns experienced worldwide following the Russian invasion into Ukraine and the COVID-19 pandemic. Climate change is expected to accelerate the threats to children's health through household income and health channels.

The results suggest that a multipurpose or holistic policy focusing on children's health and women's empowerment is advantageous. Policy and programs aiming to improve children's health can do so indirectly by increasing women's economic empowerment. Economic empowerment enhances women's bargaining power and control of resources in the household, which is associated with improvements in children's health. Policies that can increase women's economic empowerment include improving employment opportunities (for instance, developing sectors that employ women), enhancing education and skills training for women, and changing rules and norms such as ease of divorce, violence against women, and societal perceptions of women's roles (Agarwal 1997; Browning, Chiappori, and Wiess 2011; Muthoo 1999; Sen 1992).

Social protection and cash transfer programs can usefully target women, and many programs already do. Making women central to cash programs can be key to their success for children's outcomes, as women can most often be relied upon to fulfil responsibilities to their children and spend the money they receive in accordance with children's needs (Molyneux and Thomson 2011). The World Bank and others have developed significant evidence on how to design inclusive and impactful government-to-person transfers to women, including ensuring that payments are deposited directly into women's accounts.

To enable sustainable development and gender equality, policymakers must consider the gains to women of increased control rights over resources. Programs that treat women only as an instrument for benefiting children can risk exacerbating gendered social norms, reinforcing women's caring role, and exacerbating women's time poverty through additional unpaid responsibilities, particularly conditional cash transfer schemes, such as in Peru (Cookson 2018) and in Mexico and Ecuador (Economic Commission for Latin America and the Caribbean 2012; Orozco

Corona and Gammage 2017; Molyneux and Thomson 2011). Women already bear most of the burden of unpaid domestic responsibilities, spending on average three times longer on unpaid care work than men worldwide (World Bank 2023c). To empower women, development actors can seek opportunities to transform social norms, including on caregiving (for example, through the work of Equimundo) and tackling issues in the care economy that this book addresses. The design, implementation, and evaluation of interventions must be sensitive to gender inequalities. Cash transfers to women have the potential to benefit women—through greater financial inclusion, increasing social status, and wider economic empowerment (World Food Programme 2022). Growing evidence indicates cash transfers reduce violence against women (Baranov and others 2020). If gender norms could transform such that the average man invested as much in children as the typical woman, household members' health and well-being would improve.

With human capital of the next generation threatened by climate change and global shocks that affect household resources, it is more critical than ever that policymakers look within households to enable economic development. Enabling women's economic empowerment and women's power to make decisions in the household is smart economics.

Limitations and Further Research

This study augmented the literature through empirical analysis of panel data and analyzing control rights directly and separately to female employment and income.

The evidence that women's empowerment improves children's outcomes may depend on gendered norms of women bearing most domestic responsibilities, which is the dominant culture in most contexts worldwide. Future research could identify whether women's control of household resources matters if gendered norms shifted significantly such that women and men split domestic and caring responsibilities equally.

The reliability and accuracy of the results could be improved by further limiting potential endogeneity of household-level variables, including potential influence of female decision making and other explanatory variables. This improvement could be achieved using an exogenous source of variation in resource control, such as a randomized cash transfer program.

Further research could hone the analysis of how women's empowerment in the household protects children from the negative impacts of extreme weather. Such research would be strengthened by better accounting for hydrological systems, including data for the upstream water catchment, water treatment, and the extent of floodplains in agricultural land. The climate analysis would also be improved by increased location precision of rainfall data through microdata with more precise geographical units within district councils and greater time precision. Further work would assess the interaction effect on children's human capital of women's intra-household decision making with extreme weather, identifying whether women's empowerment protects children from the negative impacts of extreme weather in climate crisis.

ANNEX 9.1. FURTHER DETAILS ON THE THEORETICAL FRAMEWORK

Francesconi and Muthoo (2011) found that the theoretical conjecture holds even if the child's health were a slightly impure public good, that is, excludable or rivalrous to a small extent. It seems plausible that children's health could be slightly impure, for example, due to the potential for sole custody by one party.

Suppose for simplicity of explanation that a woman has a higher valuation of the good, while her partner, a man, is the sole investor. We can plausibly assume that the highest investment level maximizes the joint surplus of their child's health. Therefore, both parties gain larger payoffs from a higher investment than a low one, and the payoffs are higher for whoever cares more about the child (in this scenario, the woman). However, since neither the investment level or health quality is contractible, upfront payments between parties cannot guarantee the high investment level. Following the incomplete contracting literature, we assume the parties bargain over the surplus after the investment is sunk, and investment choices depend upon each party's share of the surplus.

Since resource control affects the parties' payoffs from disagreement, it also affects their share of the surplus and investment incentives. If the man (the party with the lower valuation) has control rights on the determinants of a child's health and he chooses the high investment level, then even in the event a disagreement, he will continue the investment since he receives a higher ex post payoff through the child's health. Hence, the woman (whom, for simplicity, we are assuming is the higher valuation party) cannot be induced to contribute anything ex post if the man has already made a high investment, and he will prefer the low investment level. If the woman has control rights, neither party receives a disagreement payoff, as the investment requires the man's continued participation to generate surplus. Applying the Nash Bargaining Solution, bargaining results in splitting the surplus equally. The man will now choose a higher investment level than if he had the control rights due to the higher payoff. Hence, women's decision making yields the surplus maximizing level of investment: investment incentives increase if control rights are allocated to the more caring party, regardless of who has the money to invest.

ANNEX 9.2. DATA ADJUSTMENTS

Merging Data Sets

The climate data was merged with the micro-data by the geographical center-point of each district council. The district council is the ultimate geographical level for the weather data while each precipitation reading refers to a 2,500 square-kilometer area. This relies on the assumption that rainfall deviations from trends are homogenous within district councils. For time period, rainfall was examined during the rainy season closest to the NIDS wave, when the rainfall impacts are the greatest and the largest rain variation exists (Flato, Muttarak, and Pelsler 2017). The monthly climate data for South Africa for precipitation total

was used, employing the full-time series from 1901–2015 to create the baseline reference of long-term mean and standard deviation by season and year for the rainfall shock variables.

The rainy “southern African monsoon” season occurs in October to March for most of South Africa (Bellprat and others 2015). The climate data were merged with the micro-data using separate geo-mapping data from South Africa’s 2011 Census (Statistics South Africa 2017a). The geographical census data connected each longitude–latitude pair with its corresponding geographical labels, including district and the districts’ center-point coordinates. After adjusting these data to be the same precision as the climate data, the data were merged by center-point coordinates. The micro-data were then merged with the climate data based on South Africa’s 52 District Councils’ Municipal Demarcation Board codes present in both data sets, following the boundaries used in South Africa’s 2011 Census.

Deflation

Since NIDS collected data for each Wave over at least one year, all price or income data were deflated including household income, household wage, and household expenditure by month. The deflators are derived from price indices from the national headline consumer price index published by Statistics South Africa (2017b), as described on the NIDS website (SALDRU 2017a).

ANNEX 9.3. THE SAMPLE

Children

Anthropometric measures such as weight-for-age are less reliable for under two-year-old children due to their different body compositions (Roemling and Qiam 2013). Children who are pregnant or lactating were excluded, as these states would constitute undesired heterogeneity. Sample members are household residents (defined as residing in the household at least four nights a week) such that they are impacted by household phenomena.

Household Resource Scarcity

Household resource-scarcity is defined as insufficient income to provide a minimum of health-improving food to each member of the household, as specified by the World Bank (Haughton and Khandker 2009). Resource-scarcity was therefore calculated using the recommended minimum monthly cost in South Africa of sufficiently feeding one person in the base year for deflation (2015) of R1958 (equivalent to roughly \$150 in 2015) (Numbeo 2015). This figure was then multiplied by each individual’s household size to make an indicator variable for household monthly expenditure under the resource-scarcity line. Thus, 85 percent of the sample members were defined as living in resource-scarce households. This percentage roughly matches the national poverty line (National Treasury & Statistics South Africa 2007).

Movers

To correctly estimate the clustered standard errors using Stata, panels of individuals must be nested within district clusters. Any individuals who moved districts were therefore removed between waves for the main specification.

ANNEX 9.4. MEASURE OF CHILDREN'S HEALTH

BMI-for-age z -score (BAZ) is weighted for age (in days). Accounting for weight, height, and age helps identify malnutrition, which is more sensitive to short-term changes than height-only measures. For robustness, this measure is compared with the weight-for-age z -score (WAZ). Using WAZ for our analysis significantly reduces the sample size and results are not robust for this (p -value of 0.149). If policy aims to avert undernutrition and hunger, overweight children provide useful variation in this analysis. Robustness of the findings are confirmed for excluding obese children, as obese could be considered unhealthy but may be detected as a “health improvement” through BAZ or WAZ. The BAZ measure is derived by the data producers, as described in the NIDS manual (Chinhema and others 2016). The reference populations were the WHO international growth standards (WHO 2006; de Onis and others 2007; WHO 2023).

ANNEX 9.5. FURTHER DETAILS ON EMPIRICAL METHODOLOGY

Accounting for time and space “fixed” effects

A fixed-effects model was used with both district and year fixed effects. The district fixed effects allow for time-invariant unobserved (otherwise omitted) differences across districts, for instance community gender norms, local public services, local preparation, and adaptation for rainfall. The year fixed effects capture unobserved trends, such as sample-wide technological changes in health care or climate adaptation. Furthermore, using fixed effects allows potentially nonrandom attrition to be correlated with—and partially captured by—the unobserved (fixed) effects (Wooldridge 2013), helping to diminish endogeneity.

The estimated standard errors are clustered at the level of the 52 districts, following Bertrand, Duflo, and Mullainathan (2004). In this way, intra-district correlation was allowed for among model errors for each individual, relaxing the assumption of independence between observations (Cameron and Miller 2015).

Rainfall Dynamics

Two lags of rainfall by rainy season year were included, following optimal lag selection best practices. Examining the dynamic impacts of climate shocks on children's health was limited as data for socioeconomic conditions are for the “present” period. Effects may dissipate over time and be conflated with more recent conditions and unobserved “noise.” Furthermore, wave fixed effects in time

will not apply to lags referring to years outside the waves, so rainfall trends are not fully captured through the wave fixed effects.

The dynamic impacts (between time periods) on children's health of floods and droughts will likely differ. Drought may contribute incrementally to low health via sustained water scarcity and poor crop yields that lead to undernutrition in the medium to long term. By contrast, floods may have more of an immediate ("shock") effect. Floods are likely to increase infectious disease prevalence quickly; compromise water, health care, and other infrastructure; and jeopardize the harvest during the rainfall rather than cumulatively over the growing season. Floods are therefore expected to have a more immediate effect on health.

ANNEX 9.6. FULL RESULTS

ANNEX TABLE 9.6.1.

Full Results: Impacts on Children's Health				
Variables	(1) Main	(2) Interaction of Decision Maker Being Female and Employed	(3) Significant Characteristics of Decision Maker	(4) Decision Maker's Characteristics
<i>Household Decision Maker's Characteristics</i>				
Decision Maker Is	0.077***	0.029	0.017	0.118
Female	(0.022)	(0.031)	(0.031)	(0.091)
	0.000	0.348	0.586	0.194
Decision Maker's Age				−0.000
				(0.001)
				0.832
Decision Maker's Age If				−0.001
Decision Maker Is				(0.001)
Female				0.265
Decision Maker Is		−0.046	−0.070	−0.077
Employed		(0.044)	(0.044)	(0.053)
		0.297	0.116	0.144
Decision Maker Is		0.131***	0.139***	0.141**
Female and Employed		(0.050)	(0.051)	(0.057)
		0.009	0.007	0.013
Decision Maker's			0.008***	0.010*
Schooling			(0.003)	(0.006)
			0.002	0.085
Decision Maker's				−0.005
Schooling if Decision				(0.007)
Maker Is Female				0.451
<i>Individual Characteristics</i>				
Female Dummy	0.084***	0.083***	0.081**	0.080**
(Compared to Male)	(0.032)	(0.032)	(0.032)	(0.032)
	0.009	0.010	0.012	0.012
Age in Years	−0.254***	−0.255***	−0.255***	−0.255***
	(0.010)	(0.010)	(0.010)	(0.010)
	0.000	0.000	0.000	0.000
Age Squared	0.012***	0.012***	0.012***	0.012***
	(0.001)	(0.001)	(0.001)	(0.001)
	0.000	0.000	0.000	0.000

(continued)

ANNEX TABLE 9.6.1. Continued

Variables	(1) Main	(2) Interaction of Decision Maker Being Female and Employed	(3) Significant Characteristics of Decision Maker	(4) Decision Maker's Characteristics
Race Dummy: Colored (Compared to Black)	-0.410*** (0.069) 0.000	-0.406*** (0.068) 0.000	-0.409*** (0.067) 0.000	-0.413*** (0.067) 0.000
Race Dummy: Asian (Compared to Black)	-0.739** (0.335) 0.027	-0.722** (0.340) 0.034	-0.753** (0.339) 0.026	-0.758** (0.337) 0.025
Race Dummy: White (Compared to Black)	0.413 (0.279) 0.138	0.425 (0.273) 0.120	0.416 (0.271) 0.126	0.418 (0.274) 0.128
<i>Household Characteristics</i>				
Log Household Income	0.060*** (0.017) 0.001	0.055*** (0.017) 0.001	0.053*** (0.016) 0.001	0.057*** (0.016) 0.000
Mother Is Dead Dummy (Compared to One Parent Being Alive)	-0.083** (0.042) 0.047	-0.078* (0.042) 0.068	-0.073* (0.043) 0.085	-0.070* (0.042) 0.100
Father Is Dead Dummy (Compared to One Parent Being Alive)	-0.028 (0.025) 0.269	-0.025 (0.025) 0.314	-0.023 (0.025) 0.349	-0.022 (0.025) 0.370
Number of Female Siblings	-0.047*** (0.011) 0.000	-0.047*** (0.011) 0.000	-0.045*** (0.011) 0.000	-0.046*** (0.011) 0.000
Number of Male Siblings	-0.020 (0.012) 0.104	-0.019 (0.012) 0.116	-0.017 (0.012) 0.146	-0.018 (0.012) 0.142
Number of Female Siblings if Female	0.013 (0.013) 0.339	0.013 (0.013) 0.326	0.013 (0.014) 0.335	0.013 (0.013) 0.332
Number of Male Siblings if Female	-0.028** (0.012) 0.015	-0.028** (0.012) 0.015	-0.027** (0.012) 0.020	-0.027** (0.012) 0.020
<i>Village Characteristics</i>				
High Rainfall in Last Rainy Season	0.039 (0.061) 0.527	0.040 (0.061) 0.512	0.040 (0.060) 0.506	0.039 (0.060) 0.516
High Rainfall Two Rainy Seasons Ago	-0.194*** (0.070) 0.005	-0.191*** (0.069) 0.006	-0.189*** (0.069) 0.006	-0.190*** (0.069) 0.006
High Rainfall Three Rainy Seasons Ago	0.068 (0.047) 0.155	0.072 (0.048) 0.131	0.073 (0.049) 0.133	0.073 (0.049) 0.133
Urban Dummy (Compared to Rural)	0.073* (0.039) 0.063	0.066* (0.038) 0.087	0.051 (0.039) 0.188	0.051 (0.039) 0.187
Observations	24,535	24,518	24,501	24,501
Number of pid	14,171	14,168	14,159	14,159
District FE	Yes	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes	Yes

Source: Author's own analysis using National Income Dynamics Survey (NIDS) produced by the Southern Africa Labour and Development Research Unit (SALDRU 2017a–d) and climate data produced by the University of East Anglia Climate Research Unit (UEA CRU 2014).

Note: Robust standard errors are in parentheses. *P*-values reported below standard errors. Standard errors clustered at district and wave level. FE = clustered at level of Districts and Waves; pid = 14,171 (column 1).

p* < 0.1; *p* < 0.05; ****p* < 0.01.

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Gender Inequality and Electoral Violence in Africa: Unlocking Women's Peacemaking Potential

Rasmané Ouedraogo and Idrissa Ouedraogo

This chapter uses individual-level data from the sixth round of Afrobarometer surveys to examine the impact of gender equality on electoral violence in Africa. The data cover more than 40,000 individuals from 30 countries. The results highlight that gender equality is associated with lower electoral violence. Countries with higher levels of female labor force participation tend to be less subject to electoral violence due to the peaceful influence of women and improved macroeconomic outcomes. Quantitatively, the empirical estimates show that an increase in female-to-male labor force participation ratio by 1 percentage point is correlated with a reduction of the probability of electoral violence across the continent by around 4.2 percentage points. This chapter's findings support the long-standing view that female empowerment contributes to the reduction of violence and underscore the urgency of addressing gender inequality in Africa.

INTRODUCTION

*If non-violence is the Law of our being, the future is with Women.
Who can make a more effective appeal to the heart than a woman?—
Mahatma Gandhi, "To the Women of India, Oct. 4, 1930"*

Elections constitute one of the most important pillars of democracy, but in most African countries, elections are characterized by the high possibility of election-related violence. During elections, the stability and security of African states hang in the balance. Electoral violence is a major problem on the continent and threatens the development and consolidation of democracy. According to Buchard (2015), around 50 percent of elections are still subject to violence, although there has been substantial annual variation in the frequency of violent episodes in recent years. Violence in Côte d'Ivoire following the 2010 presidential election may have displaced as many as 1 million people and caused more than 3,000 deaths; the Kenyan 2007–08 postelection turmoil killed 1,300 people and displaced some 250,000–600,000; and in Nigeria more than 1,000 people died over the 2011 polls (Atwood 2012).

Yet no study is known to have investigated the empirical impact of gender equality on electoral violence. Electoral violence is carried out to determine, delay, or to otherwise influence an electoral process (Fischer 2002) and take place before, during, or after an election. To explain electoral violence in Africa, scholars have posited several theories, including those related to ethnic and political cleavages (Colombo, D'Aoust, and Sterck 2017; Eifert, Miguel, and Posner 2010), natural resources (Collier, Hoeffler, and Rohner 2008), political competition (Sterck 2017; Collier and Vicente 2012), weak governance (Gutiérrez-Romero 2014), and fragility and past conflicts (Collier and Vicente 2012, Brancati and Snyder 2012). None of these studies empirically explored the impact of gender equality on electoral violence despite the recognition in the literature that closing the gender gap is correlated with a reduction in conflict occurrence (Demeritt, Nichols, and Kelly 2014; Melander 2016; Caprioli 2005). Existing studies associate women with an aversion to violence and an inclination toward peace, and they argue that including women in society (both politics and economics) makes violent conflict less likely (Demeritt, Nichols, and Kelly 2014; Goldstein 2001). Countries with greater gender equality and female participation generally tend to adopt more peaceful preferences, leading to a reduced likelihood of political violence (Goldstein 2001).

This chapter uses individual-level data from the sixth wave of the *Afrobarometer* survey to explore the impact of gender equality on electoral violence in Africa. The study estimates the effect of female labor force participation on the perception of electoral violence and controls for several covariates. The endogeneity problem is addressed by employing an instrumental variable approach in which the female-to-male labor force participation ratio is connected with rainfall growth rate. Rainfall growth rate is positively associated with female labor force participation, because it presents job opportunities for women in the agriculture sector. According to Jayne, Yeboah, and Henry (2017), the agriculture sector remains the single largest source of employment in Africa, and women make up almost 50 percent of the agriculture labor force on the continent (Food and Agriculture Organization 2011).

The results show that gender equality is associated with a reduction in electoral violence in Africa. The findings imply that countries with higher levels of female labor force participation tend to have lower levels of electoral violence. An increase in female-to-male labor force participation ratio by 1 percentage point is correlated with a decrease in the likelihood of electoral violence by 4.2 percent. The chapter underscores the importance of achieving gender equality and women empowerment in Africa and provides a novel contribution to the long-standing debate about the role of gender inequality as a determinant of civil conflicts.

The rest of the chapter discusses the mechanisms through which gender inequality affects conflicts and violence; it then presents the data, provides some descriptive statistics, and describes the empirical strategy. Finally, the chapter reports and analyzes the results and offers some concluding remarks.

MECHANISMS

Gender equality can affect electoral violence through several mechanisms; these include a structural violence channel as well as norms and social roles.

Structural Violence

Coined by Galtung (1969), *structural violence* refers to a form of violence wherein some social structure or social institution may harm people by preventing them from meeting their basic needs. In cases of structural violence, gendered structural hierarchies, which are maintained by norms of violence and oppression, should result in higher levels of intrastate violence by inuring people to aggression and by providing the framework for justifying it (United Nations Educational, Scientific and Cultural Organization 1995).

Often aimed at women, structural violence is maintained through gender socialization, gender stereotyping, and a constant threat of violence—all of which insidiously identify women as inferior, thus influencing their actions at all levels (Caprioli 2005). Structural violence can be expressed in higher rates of female unemployment, unequal access to goods and services, and exploitation. Winter and Leighton (2001) argued that structural violence is the violence of injustice and inequity—embedded in ubiquitous social structures and normalized by stable institutions and regular experience.

Galtung (1975) identified four components of structural violence: (1) exploitation, which is focused on the division of labor, with the benefits being asymmetrically distributed; (2) penetration, which necessitates the control by the exploiters over the consciousness of the exploited, thus resulting in the acquiescence of the oppressed; (3) fragmentation, which means that the exploited are separated from each other; and (4) marginalization, with the exploiters as a privileged class observing their own rules and forms of interaction. The third component refers, for instance, to the exclusion of women from the labor market, which is the subject of this study. It results from women having fewer job opportunities outside the home that would allow for participation and create a sense of efficacy.

Structural violence creates the foundation for structural inequality and conflicts. The term *structural inequality* refers to the system of privilege and inequality created and maintained by interlocking societal institutions. Women are subjected to structural inequality, which results from male domination; gender stereotypes; sexism; lack of opportunities, decision-making power, and adequate education; barriers to accessing social or economic resources; and all other aspects resulting from non- or under-participation of women. Caprioli (2005) argued that gender inequality should have a substantial impact on intrastate conflict based on the direct impact of structural inequality, with its inherent norms of discrimination and violence and the role that it plays in facilitating ethnic rebellion and conflict. Moreover, gender discrimination and structural violence are important aspects in mobilizing groups and in legitimizing violence. Inequality can create grievances between groups that, especially when exploited by politicians, deepen animosity

between them and increase the likelihood of bloodshed (Atwood 2012). In the same vein, Midlarsky (1999) highlighted that inequality, when extreme and systematic, leads to political violence.

Norms and Social Roles

The second mechanism is about the norms and social roles of men and women. In many societies, the tasks and responsibilities imposed on people are divided into male and female gender roles. Under this division of labor, it is argued that gender equality as a norm is associated with a low occurrence of violence.

In general, women are more violence averse. Hudson and others (2008–09) argued that societies dominated and controlled by men are subject to an environment where violence and aggression are considered the norm. Thus, societies with a very high level of male dominance in politics tend to be controlled by hyper-masculine political cultures, which can probably fuel electoral violence. This norm prescribes violence as a means to resolve conflict also on the highest decision-making levels (Forsberg and Olsson 2016). This is typically the case for electoral violence in which some groups do not accept the poll results or adhere to the democratic process; they then resort to violence to achieve their ends. Wolfgang and Ferracuti (1967) used the “subculture of violence” perspective to emphasize that social groups exhibit high rates of violent behavior because of group members’ adherence to values and norms that support, legitimize, and encourage violent behavior, and these values and norms often emphasize an ideal of masculinity.

In this regard, gender equality is associated with lower grievances and conflicts. Societies characterized by gender equality are ingrained with norms that men and woman treat each other with respect (Melander 2005). These norms then transfer to other societal relations, such as between ethnic groups and political parties, and can explain the relative peacefulness of such states. Demeritt, Nichols, and Kelly (2014) argued that gender equality as a norm prescribes respect and resolution of conflict without violence, and hence equality norms may prevent grievances from escalating to violent conflict in the first place. Incorporating women in society reflects changes in gender construction and particularly reflect the relaxation of the norm of men-as-warriors. From this perspective, states that emphasize gender equality and female participation in traditionally male society also tend to adopt more generally peaceful preferences, leading to a reduced likelihood of political violence (Demeritt, Nichols, and Kelly 2014; Goldstein 2001). Accordingly, the inclusion of women in societal decision making could lead to them increasingly exerting the inherent pacific influence and/or redefining traditional gender roles in ways that privilege nonviolence—in the way constructivists anticipate.

Some authors rely on the biological and reproductive role of women to explain these peaceful attitudes. This role leads to an inclination to give life, not take it (Demeritt, Nichols, and Kelly 2014). With this natural preference for peace, women would prefer to prevent societal problems from escalating to conflict and attempt to de-escalate armed conflicts when they do occur (O’Mahoney 2012).

DATA SOURCES AND DESCRIPTIVE STATISTICS

Data Sources

This chapter's primary data source is round six of the *Afrobarometer* surveys. The *Afrobarometer* is an independent, nonpartisan research project that measures the social, political, and economic conditions in Africa. *Afrobarometer* began in 1999 with only 12 countries in the first round; surveys covered 36 countries in the sixth round, conducted in 2014–15. Nationally representative samples of individuals who are older than 18 years are selected both in rural and urban areas of the participating countries. *Afrobarometer* uses multilevel random selection methods to generate the samples, which constitute a representative cross-section of the population. In the sixth wave, more than 53,000 people were surveyed, with sample sizes ranging between 1,200 and 2,400 people depending on each country's population size.

Measurement of Electoral Violence

The survey data was used to measure electoral violence; the main issue that arose was the definition of the term. The United Nations Development Program (UNDP) defines *electoral violence* as “acts or threats of coercion, intimidation, or physical harm perpetrated to affect an electoral process or that arise in the context of electoral competition. When perpetrated to affect an electoral process, violence may be employed to influence the process of elections—such as efforts to delay, disrupt, or derail a poll—and to influence the outcomes: the determining of winners in competitive races for political office or to secure approval or disapproval of referendum questions” (UNDP 2009, 4).

Similarly, Fischer (2002) defined electoral violence as any random or organized act or threat to intimidate, physically harm, blackmail, or abuse a political stakeholder in seeking to determine, delay, or to otherwise influence an electoral process. As such, electoral violence includes acts—such as assassination of opponents or spontaneous fistcuffs between rival groups of supporters—and threats, coercion, and intimidation of opponents, voters, or election officials (UNDP 2009).

The measurement of electoral violence was based on question Q49 from the sixth round of *Afrobarometer* surveys, which provides an assessment about the respondent's personal experience with or perception of political intimidation and violence. In question Q49, the respondents are asked: “During election campaigns in this country, how much do you personally fear becoming a victim of political intimidation or violence?” Response choices include “A lot,” “Somewhat,” “A little bit,” and “Not at all.” The three first answers were merged and coded as 1, which indicates that the respondent feared becoming a victim of political intimidation or violence, and the last answer, “Not at all,” was coded as 0.¹

¹ The results remain consistent even if the intensity of electoral violence is considered.

Measurement of Gender Equality

In this chapter, the female-to-male labor force participation ratio is used as a measure of gender equality; the data are available from the sixth round of *Afrobarometer* surveys. The female-to-male labor force participation ratio is the ratio between the female labor force participation rate and the male labor force participation rate. Question Q95 asks: “Do you have a job that pays a cash income?” The survey also includes the gender of the respondent (see question Q101), which allows the calculation of the female labor force participation rate and the male labor force participation rate for each administrative region; calculating the ratio between the two yields the female-to-male labor force participation ratio.

Control Variables

To alleviate the problems of confounding caused by omitted variables, several are included:

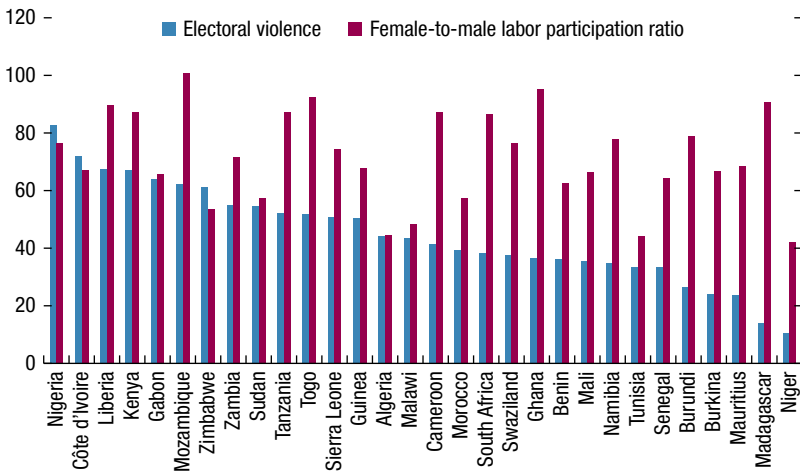
- Sociodemographic variables: age, level of education, residence, and employment status.
- Political affiliation and election-related variables: political party membership, interests in political affairs, views about the elections, and media coverage and participation of political parties.
- Trust in institutions: trust in electoral commission, government, ruling party, police, army, and courts of law.
- Media: access to radio, television, newspapers, and social media.
- Socioeconomics: access to food, water, medical care, cash income, transportation, and electricity.

Annex Table 10.1 lists the control variables.

Descriptive Statistics

The sample covers 30 countries and consists of more than 40,000 respondents. Country sample sizes range from 630 people in Liberia to 2,395 in Malawi. Annex Table 10.2 displays descriptive statistics. On average, 46 percent of total respondents reported being a victim of electoral violence. Furthermore, the female-to-male labor force participation ratio is, on average, 73 percent; thus, around 73 women of the sample are working per every group of 100 working men. This share differs widely among countries. Figure 10.1 presents, by country, the average values of the female-to-male labor force participation ratio and electoral violence. It shows that electoral violence is widespread in Nigeria, Cote d'Ivoire, Liberia, and Kenya, where more than two-thirds of respondents reported being victims of political intimidation and violence. On the contrary, the share of respondents who have been targeted by electoral violence is low in Niger, Madagascar, Mauritius, and Burkina Faso. In these countries, less than one-fourth of respondents reported being subjects of electoral violence.

Figure 10.1. Gender Equality and Electoral Violence per Country
(Percent, average values)



Sources: Authors' calculations; and *Afrobarometer* data.

The female-to-male labor force participation ratio is high in Mozambique, Ghana, Togo, and Madagascar (above 90 percent); it is low in Niger, Burundi, Tunisia, Algeria, and Malawi (below 50 percent).

Empirical Strategy

The objective was to estimate the impact of gender equality on electoral violence. The model is as follows:

$$\text{Electoral Violence}_{ij} = \alpha + \beta \text{Gender Equality}_{ij} + \gamma X'_{ij} + \pi_j + \mu_{ij} \quad (10.1)$$

Where, for individual i from country j , $\text{Electoral Violence}_{ij}$ is a binary variable taking the value of 1 if the respondent reports being a victim of electoral violence, $\text{Gender Equality}_{ij}$ represents the female-to-male labor force participation ratio. Vector X'_{ij} includes socioeconomic, demographic, political, and election-related variables. π_j are country fixed effects. The inclusion of country fixed effects will account for observable and unobservable country-specific characteristics that may explain electoral violence. The error term is μ_{ij} .

Attempts to estimate the impact of gender equality on electoral violence will suffer from the common error of mistaking cause for effect and vice versa—known as endogeneity issues—due to measurement errors, the omission of variables, and reverse causality. The direction and magnitude of this bias will depend on the relative and potentially offsetting effects of reverse causality and measurement error. The most problematic issue arises from reverse causality, given that election-related violence may affect women's participation in the labor market. The direction of this bias is most likely negative, given that electoral violence

might negatively affect the female-to-male labor force participation ratio. The second bias, related to measurement error, may come from any of the control variables, including gender equality. The omitted variables constitute another source of endogeneity; consequently, ordinary least squares regressions are likely to be biased.

To address this endogeneity issue, an instrumental variable probit model was used to estimate equation (10.1). The exogenous variations in rainfalls were used as instrument for female-to-male labor force participation ratio. The idea is that an economic shock emerging from extreme rainfalls will lead to a loss of jobs for women, thus affecting the female-to-male labor force participation ratio and, consequently, triggering electoral violence. An extreme economic shock stemming from a decline in rainfall may disproportionately affect women, as men could have more coping strategies than women do. Several studies have shown that women's job mobility and transition tend to be low compared to those of men (Looze 2017). Royalty (1998) found that women are more likely than men to experience job-to-nonemployment rather than job-to-job transitions. This finding, in turn, implies that the unemployment rate for women will be higher than that of men when an adverse shock in rainfall materializes. Weather shocks are plausible instruments for developments in the labor markets in economies that largely rely on rainfed agriculture, as do those in Africa (Miguel, Satyanath, and Sergenti 2004).

RESULTS

Table 10.1 reports the results. The simple probit model is used in columns 1 through 5, and the instrumental variable probit model is used in columns 6 through 10. From the outset, one can observe that the coefficient associated with gender equality is negative and significant in all columns, suggesting that gender equality is negatively associated with electoral violence. This analysis focuses on the instrumental variable probit model.

Column 6 presents the results of the estimates without any control variables except the variable of interest. Country fixed effects are included in all regressions with the purpose of controlling for all co-determinants of electoral violence and gender equality. The coefficient associated with the female-to-male labor force participation ratio, the coefficient of interest, is negative and significant at the 1 percent level. Thus, an increase in gender equality reduces the likelihood of electoral violence. Quantitatively, an increase in the female-to-male labor force participation ratio by 1 percentage point is correlated with a decrease in the likelihood of electoral violence by 4.2 percentage points.² The findings are consistent with some previous studies that highlighted that gender equality is negatively associated with the occurrence of conflicts (Demeritt, Nichols, and Kelly 2014; Caprioli 2000, 2005).

² Marginal effect calculated at mean values.

TABLE 10.1.

Impact of Gender Equality on Electoral Violence										
VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Probit					IV probit				
Gender Equality	−0.073*** (0.020)	−0.075*** (0.020)	−0.084*** (0.029)	−0.077*** (0.029)	−0.076*** (0.029)	−1.313** (0.610)	−1.233* (0.664)	−1.394*** (0.511)	−1.470*** (0.531)	−1.661*** (0.480)
Sociodemographic		Yes	Yes	Yes	Yes		Yes	Yes	Yes	Yes
Political Affiliation and Trust in Institutions			Yes	Yes	Yes			Yes	Yes	Yes
Media				Yes	Yes				Yes	Yes
Socioeconomic Conditions					Yes					Yes
Constant						0.940*** (0.306)	1.111*** (0.309)	1.533*** (0.226)	1.474*** (0.230)	1.757*** (0.209)
<i>First stage</i>										
Rainfall Growth Rate						0.035*** (0.006)	0.033*** (0.006)	0.061*** (0.008)	0.057*** (0.008)	0.059*** (0.008)
Observations	40,034	39,588	20,880	20,634	20,500	40,034	39,588	20,880	20,634	20,500
Log Likelihood	−44831	−44171	−23413	−23107	−22882	−32270	−31597	−15964	−15696	−15457
Wald chi-square	5313	5408	3967	3954	4032	5641	5602	4399	4500	4940
Wald <i>p</i> -value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Wald Chi-square Test of Exogeneity						4.30	3.31	6.68	6.55	9.25
Exogeneity Test Wald <i>p</i> -value						0.04	0.07	0.01	0.01	0.00
Wald Test of Weak Instrument						3.31*	2.97*	5.11**	5.07**	7.01***
Anderson-Rubin Test of Weak Instrument						3.71*	2.84*	5.77**	5.78**	8.32***

Source: Authors' calculations.

Note: The results in columns 6 through 10 suggest that our instrument performs very well in the first stage. Rainfall growth rate is a positive and significant predictor of female labor force participation in Africa.

Statistical tests do not invalidate the econometric method. At the bottom of the table is Wald $\chi^2(2)$ score of the exogeneity of the instrumented variable. The null hypothesis of no endogeneity is rejected. The

Wald test and the Anderson-Rubin test regarding the strength of the instrument are strongly significant; therefore, the hypothesis that the instrument is weak is rejected. Standard errors in parentheses.

* $p < 0.1$; ** $p < .05$; *** $p < .01$.

Column 7 includes sociodemographic variables, such as respondents' level of education, age, living area, and employment status. As shown by the coefficient associated with the gender equality index, a negative correlation exists between gender equality and electoral violence. Moreover, the coefficients associated with education and with age are negative and strongly significant, suggesting that older, highly educated people are less likely to report electoral violence.

Column 8 includes the political affiliation of the respondent as well as some election-related matters. These variables capture whether respondents report being a member of a political party, their interests in public affairs, their participation in the last election and judgment about it—that is, fairness of votes and participation of opposition parties, media coverage, and the counting of votes. In fact, the outbreak of electoral violence is often related to voters' misjudgment of the election process. The exclusion of opposition parties can also spur electoral violence. The trust of respondents in the national electoral commission, local government council, ruling party, police, army, and courts of law are also controlled for. A lack of trust in the national election commission, the ruling party, police, and army can lead to protests and electoral violence, particularly when voters do not trust the courts of law to reconsider irregularities. Even controlling for these important covariates, the results still show that the coefficient associated with the variable of interest—gender equality—is negative and significant at the 1 percent level; the coefficient is higher than that of column 6. Not surprisingly, the results show that respondents who reported that the election was fair and free, and that the votes were counted fairly, are less likely to be involved in electoral violence. Trust in the election commission, ruling party, and army are also associated with lower electoral violence, suggesting that more accountable, transparent, and effective ruling parties could help prevent electoral violence in Africa. In contrast, those respondents who think that the opposition parties were prevented from participating in the election are more likely to report electoral violence.

Given the increasing role of the media during election periods and daily life, the respondent's media channel is controlled for in column 9. The media play an important role during elections and can act as a watchdog to ensure plurality of views, opinions, and transparent political processes as well as offer a platform for candidates and voters to discuss important issues. However, the media can also amplify or even incite national prejudices or tensions during contentious elections. The following media channels were included: radio, television, newspapers, and social media. The results in column 9 show that controlling for respondents' media channel does not change the core results. The coefficient associated with gender equality remains negative and strongly significant at the 1 percent level. Furthermore, the coefficients associated with some media channels are positive and significant. This finding is in line with that of Abdi and Deane (2008), who discovered that the media had undermined democracy during the Kenya's post-election violence of 2008.

Finally, respondents' access to basic needs and assets including food, water, medical care, cash income, transportation, and electricity are included in column 10. Poverty rates are very high in Africa, and this desperate situation may lead to electoral

violence; the literature has shown that some politicians can exploit the impoverished conditions of the populations to mobilize and lead them to rebellion (Goodhand 2001). Poverty compounds vulnerability to insurgency at the individual and community level by lowering the opportunity cost of mobilizing for violence (Humphreys and Weinstein 2008). As column 10 shows, the impact of gender equality on electoral violence remains negative and significant at the 1 percent level. The results also indicate that respondents who have never been without food, medical care, transportation, and electricity are less likely to report electoral violence.

CONCLUSION

This chapter provided an empirical analysis of the impact of gender equality on electoral violence in Africa. It used individual-level data from the sixth wave of *Afrobarometer* surveys, which allowed an in-depth exploration of the role of gender in election-related violence. The sample covers more than 40,000 individuals from 30 African countries. The results show a strongly significant and negative correlation between gender equality and electoral violence in Africa. Based on the estimates, an increase in female-to-male labor force participation ratio by 1 percentage point is associated with a decline of election-related violence by around 4.2 percentage points.

The findings imply that promoting gender equality and equal opportunity could help reduce the problem of recurrent electoral violence in Africa. With many conflicts and civil wars on the continent originating from electoral violence, the results of this chapter underscore an important way of fostering peace and boosting democracy. African countries should promote gender equality to support women's economic empowerment. Improving women's participation in the labor market should be a priority not simply for equity's sake but also for the positive economic impact (Chapter 2; Ostry and others 2018; IMF 2018) and the potential to bring peace. The quest for peace in Africa must involve fighting all forms of inequality and discrimination against women.

ANNEX 10.1.

ANNEX TABLE 10.1.1.

List and Definitions of Control Variables

The sociodemographic variables include:

- Age: continuous variable (in years)
- Education: dichotomous variable taking the value of 1 if the respondent has a tertiary education degree, and 0 otherwise
- Urban: dichotomous variable taking the value of 1 if the respondent resides in urban area, and 0 otherwise
- Employed: dichotomous variable taking the value of 1 if the respondent is employed, and 0 otherwise

Political affiliation and election-related variables include:

- Political party member: dichotomous variable taking the value of 1 if the respondent is affiliated to a political party, and 0 otherwise
- Public affairs interests: dichotomous variable taking the value of 1 if the respondent is interested in public affairs, and 0 otherwise.
- Voted: dichotomous variable taking the value of 1 if the respondent voted during the last election, and 0 otherwise
- Fair election: dichotomous variable taking the value of 1 if the respondent thinks that the last election was completely free and fair, and 0 otherwise.
- Voters' views reflected: dichotomous variable taking the value of 1 if the respondent thinks that elections ensure that voters' views are reflected, and 0 otherwise
- Votes counted fairly: dichotomous variable taking the value of 1 if the respondent thinks that votes are always counted fairly, and 0 otherwise
- Media fair coverage: dichotomous variable taking the value of 1 if the respondent thinks that the media provides fair coverage of all candidates during the last election, and 0 otherwise
- Nonparticipation of opposition parties: dichotomous variable taking the value of 1 if the respondent thinks that opposition has been prevented from running, and 0 otherwise

Trust in institutions and people involved in election organization variables include:

- Trust electoral commission: dichotomous variable taking the value of 1 if the respondent trusts the national electoral commission, and 0 otherwise
- Trust government: dichotomous variable taking the value of 1 if the respondent trusts the elected local government council, and 0 otherwise
- Trust ruling party: dichotomous variable taking the value of 1 if the respondent trusts the ruling party, and 0 otherwise
- Trust police: dichotomous variable taking the value of 1 if the respondent trusts the police, and 0 otherwise
- Trust army: dichotomous variable taking the value of 1 if the respondent trusts the army, and 0 otherwise
- Trust courts of law: dichotomous variable taking the value of 1 if the respondent trusts the courts of law, and 0 otherwise

Media-related variables include:

- Radio: dichotomous variable taking the value of 1 if the respondent gets news from the radio, and 0 otherwise
- TV: dichotomous variable taking the value of 1 if the respondent gets news from television, and 0 otherwise
- Newspapers: dichotomous variable taking the value of 1 if the respondent gets news from the newspapers, and 0 otherwise
- Social media: dichotomous variable taking the value of 1 if the respondent gets news from the social media, and 0 otherwise

Socioeconomic variables include:

- Access to food: dichotomous variable taking the value of 1 if the respondent has never been without food, and 0 otherwise
- Access to water: dichotomous variable taking the value of 1 if the respondent has never been without water, and 0 otherwise
- Access to medical care: dichotomous variable taking the value of 1 if the respondent has never been without medical care, and 0 otherwise
- Access to cash income: dichotomous variable taking the value of 1 if the respondent has never been without cash income, and 0 otherwise
- Access to transportation: dichotomous variable taking the value of 1 if the respondent has access to transportation, and 0 otherwise
- Access to electricity: dichotomous variable taking the value of 1 if the respondent has access to electricity, and 0 otherwise

Sources: Afrobarometer surveys; and authors' coding.

ANNEX 10.2

ANNEX TABLE 10.2.1.

Descriptive Statistics					
Variable	Obs	Mean	Std. Dev.	Min	Max
Electoral Violence	40,034	0.46	0.5	0	1
Gender Equality	40,034	0.73	0.3	0	3
Employed	39,875	0.39	0.5	0	1
Education	39,932	0.09	0.3	0	1
Age	39,830	37.31	14.3	18	105
Urban	40,034	0.39	0.5	0	1
Political Party Member	36,708	0.63	0.5	0	1
Voted	37,956	0.70	0.5	0	1
Fair Election	36,611	0.44	0.5	0	1
Voters' Views Reflected	32,899	0.37	0.5	0	1
Votes Counted Fairly	37,377	0.35	0.5	0	1
Media Fair Coverage	36,012	0.25	0.4	0	1
Nonparticipation of Opposition Parties	36,237	0.07	0.3	0	1
Public Affairs Interests	39,763	0.56	0.5	0	1
Trust Electoral Commission	36,910	0.27	0.4	0	1
Trust Government	38,471	0.22	0.4	0	1
Trust Ruling Party	36,639	0.24	0.4	0	1
Trust Police	39,432	0.26	0.4	0	1
Trust Army	38,925	0.41	0.5	0	1
Trust Courts of Law	38,680	0.27	0.4	0	1
Radio	39,985	0.82	0.4	0	1
Television	39,963	0.60	0.5	0	1
Newspapers	39,842	0.40	0.5	0	1
Social Media	39,427	0.25	0.4	0	1
Access to Food	39,974	0.54	0.5	0	1
Access to Water	39,972	0.54	0.5	0	1
Access to Medical Care	39,915	0.50	0.5	0	1
Access to Cash Income	39,888	0.25	0.4	0	1
Access to Transportation	39,885	0.82	0.4	0	1
Access to Electricity	40,026	0.63	0.5	0	1
Rainfall Growth	40,034	0.04	0.3	-0.8	4.2

Sources: Afrobarometer surveys; and authors' calculations.

Note: Max = maximum; Min = minimum; Obs = observations; Std. Dev. = standard deviation.

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A Causal Story: Gender Equality and Economic Growth

Ata Can Bertay, Ljubica Dordevic, and Can Sever

This chapter emphasizes the importance of gender equality as a development goal by focusing on its role in economic growth. Gender equality can foster economic growth through multiple channels; however, endogeneity concerns have made it challenging to identify the causal link between gender equality and economic growth. This chapter lays out the argument, based on Bertay, Dordevic, and Sever (2020), that higher gender equality causes economic growth by enabling a better allocation of female labor. In sub-Saharan Africa, a region severely hit by the COVID-19 shock and with large gender gaps, achieving gender equality offers countries large economic development gains.

INTRODUCTION

Gender equality has been at the forefront of the economic development agenda for many years, and the United Nations (UN) has included it as one of the 17 Sustainable Development Goals.¹ Achieving gender equality is a goal in and of itself. In addition, as women constitute about half of the world's population, and their access to equal opportunities and involvement in economic activities has significant consequences on the global economy. Against the background of slowing productivity growth globally and a more shock-prone world, empowering women is now even more important to set the stage for inclusive, resilient, sustainable, and green growth. This chapter seeks to give an overview of recent trends in gender equality that can contribute to better economic performance. It also presents evidence of gender equality's criticality for macroeconomic performance more generally, with a particular focus on the sub-Saharan Africa region and the COVID-19 crisis.

Pursuing gender equality is a complex issue with many dimensions involving both opportunities (for instance, in the area of law, as well as access to health services, education, and finance) and outcomes (for example, regarding labor force participation, earnings, and women in positions of power). The last several decades have witnessed considerable improvement toward a gender-equal world

¹ The fifth Sustainable Development Goal aims to achieve gender equality and empower all women and girls. See the UN's 2030 Agenda for Sustainable Development (UN General Assembly 2015).

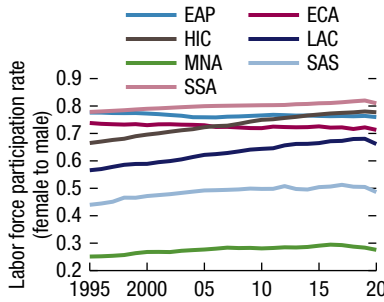
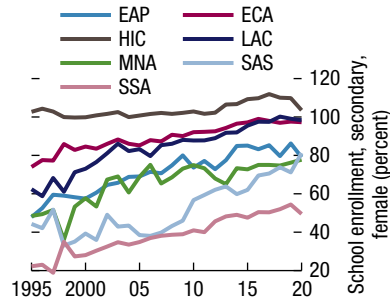
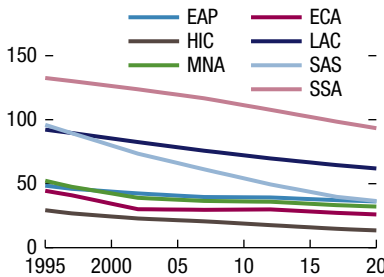
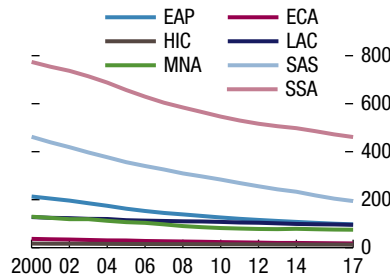
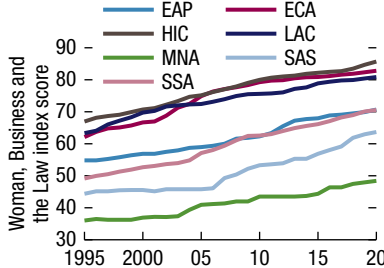
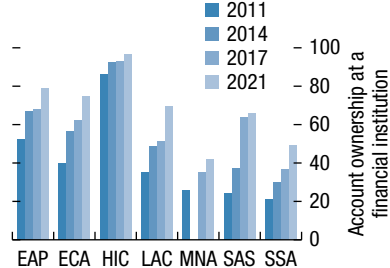
in many of these aspects; however, progress has come at a slower pace in several dimensions (Chapter 1). In addition, gender equality is not an issue specific to developing countries. Many high-income countries also have room for improvement, as macroeconomic measures in these countries may hide gender disparities such as vertical and horizontal segregation and the lack of women in high-level positions (for instance, on corporate boards or in public office).

Progress on gender equality in sub-Saharan Africa has been mixed. Compared to other developing regions and even most high-income countries, sub-Saharan Africa fares relatively well in terms of the gender gap in labor force participation (Figure 11.1, panel 1). Yet, there is a considerable variation within the region, with higher gender gaps in the north than in the south (International Labour Organization 2022), and women are often employed in informal, lower-quality jobs. On education—proxied by female gross secondary school enrollment rates² (panel 2)—and reproductive health (panels 3 and 4)—proxied by adolescent fertility rate and maternal mortality ratio—sub-Saharan Africa has made remarkable progress. Yet overall, the region continues to persistently exhibit the most concerning statistics. Moreover, women’s legal rights, as measured by the World Bank’s Women, Business and the Law Index,³ have been improving in all regions (panel 5), including in sub-Saharan Africa, which has shown significant progress in the realm of law, while legal reforms that support gender equality are still needed to ensure fair treatment of women in the region (see also Chapter 7). Finally, despite steady improvements in women’s access to financial services, as measured by female account ownership at a financial institution or with a mobile-money-service provider, women’s access to financial institutions remains below 50 percent in sub-Saharan Africa (panel 6)—still very far from the 96 percent rate seen in high-income countries (see also Chapter 9).

The COVID-19 pandemic has likely widened these long-standing gender disparities. The pandemic-induced economic contraction has been dubbed a “she-cession” for its disproportionate impact on women worldwide (Alon and others 2021); its effects were likely felt even worse in developing countries (as documented, for instance, in Chapter 15 for the southern African region) due to women’s vulnerable position at work and at home. The International Labour Organization (ILO) reports that the pandemic affected women’s jobs disproportionately: females experienced job losses to a larger extent than did their male counterparts (ILO 2021). The crisis hit women workers particularly hard in sub-Saharan Africa due to their high share in contact-intensive sectors and

² Because gross school enrollment ratios are calculated as the ratio of the number of enrolled students (regardless of their age) to the population of the age group that officially corresponds to the level of education shown, the ratios can be larger than 100 percent.

³ This index focuses on the legal impediments to the economic inclusion of women starting from the time they enter the labor force and continuing through retirement. In particular, the index includes eight aspects regarding women’s legal rights—namely, gender-based discrimination in laws in mobility, workplace, pay, marriage, parenthood, entrepreneurship, asset ownership, and pensions. It ranges between 0 and 100; higher values indicate a more gender-equal legal environment.

Figure 11.1. Aspects of Gender Inequality**1. Labor Force Participation****2. Education****3. Adolescent Fertility Rate****4. Maternal Mortality Ratio****5. Women's Legal Rights****6. Access to Finance**

Source: World Bank World Development Indicators and Women Business and the Law Database.
 Note: Panel 1 plots the mean ratio of female labor force participation rate over the male labor force participation rate (in the corresponding population age 15 and older). Panel 2 plots gross female school enrollment ratio, as the ratio of total enrollment, regardless of age, to the female population of the age group that officially corresponds to secondary school education. Panel 3 plots adolescent fertility rate (the number of births per 1,000 women ages 15 to 19). Panel 4 plots maternal mortality ratio (per 100,000 live births). Panel 5 plots the overall index on women's legal rights from the World Bank's Women, Business and the Law index (ranging between 0 and 100, higher values indicating lower levels of legal impediments to the economic inclusion of women). Panel 6 plots account ownership at a financial institution or with a mobile-money-service provider, the percentage of female respondents who report having an account, or the percentage who have personally used a mobile money service in the past 12 months (percentage of population ages 15 or older). This chapter uses the World Bank's definition of regions, where high-income countries are grouped together and the geographical groups include only developing countries: ECA = Europe and Central Asia; EAP = East Asia and Pacific; HIC = high-income countries; LAC = Latin America and the Caribbean; MNA = Middle East and North Africa; SAS = South Asia; SSA = sub-Saharan Africa.

informal jobs (Chapter 15; Aslam, Fuje, and Rawlings 2021; Aoyagi 2021); this led to higher increases in unemployment—which have stayed persistently higher—for women than for men (World Economic Forum 2022). Furthermore, indirect channels such as childcare and homeschooling responsibilities fall disproportionately on women. The pandemic lockdowns also exposed them to increased gender-based violence (Chapter 5; Ouedraogo and Stenzel 2021).

The World Economic Forum's Global Gender Gap Report for 2021 shows that although sub-Saharan Africa has been steadily improving its performance on gender equality overall, the pandemic had detrimental consequences: there was a reversal in gender gaps in three dimensions of the World Economic Forum's gender gap measure—economic participation and opportunity; educational attainment; and political empowerment—and progress stalled in the aspect of health and survival (World Economic Forum 2021). While some progress was made in 2022 to regain the lost ground, in the Global Gender Gap Report for 2022, sub-Saharan Africa still ranked low—sixth out of eight regions globally—and even ranked the lowest on some aspects, such as educational attainment. With about 68 percent of the total gender gap bridged, the report estimates that, if the current pace holds, it will take 98 years to fully close the gap.⁴ COVID-19 and geopolitical shocks are thus stalling progress on gender equality, and a risk of reversal looms, especially as leaders focus on tackling the compounding crises.

Overall, according to the IMF's Regional Economic Outlook for sub-Saharan Africa from April 2022 (IMF 2022), the gradual economic recovery in the region is short of the pace needed to make up for the lost output. Furthermore, with the economic scarring from the pandemic more severe and more long-lasting in developing economies relative to that of the developed ones (Yeyati and Filippini 2021), sub-Saharan Africa's development gaps are likely to persist. Prolonged school closures in the region have hurt much-needed human capital accumulation and put girls at a greater risk than boys of not returning to school, despite a greater return to education than those for men; see Kattan, Montenegro, and Patrinos (2021) and Bah Diallo and others (2021). Such closures may lead to permanent scarring and weigh on medium-term growth, making it harder for sub-Saharan Africa to catch up with advanced economies. All in all, despite the region's significant progress in different dimensions over the last decades, pressing ahead with closing gender gaps holds a key to invigorating growth and counteracting the setback caused by the pandemic and compounding shocks, especially for those areas hit hardest.

The rest of this chapter provides a literature review regarding various channels of the relationship between gender equality and economic growth and then discusses their causal relationship in the context of Bertay, Dordevic, and Sever (2020).

⁴ The Global Gender Gap Index measures gender-based gaps in access to resources and opportunities in a given country, regardless of that country's development level and level of available resources. It scores on a scale of 0 to 100, and scores can be interpreted as the distance covered toward parity (that is, the percentage of the gender gap that has been closed).

POSSIBLE CHANNELS OF THE RELATIONSHIP BETWEEN GENDER EQUALITY AND ECONOMIC GROWTH

The vast literature on the positive relationship between gender equality and economic performance goes back to, at least, Boserup (1970). Other examples include Hill and King (1995); Dollar and Gatti (1999); Tzannatos (1999); Klasen (2000, 2002); Knowles, Lorgelly, and Owen (2002); Lagerlof (2003); Abu-Ghaida and Klasen (2004); Esteve-Volart (2004); Klasen and Lamanna (2009); Duflo (2012); Elborgh-Woytek and others (2013); Gonzales and others (2015); Cuberes and Teignier (2016); Hakura and others (2016); Kochhar, Jain-Chandra, and Newiak (2017); Ostry and others (2018); Hsieh and others (2019); and Ouedraogo and Stenzel (2021). Gender equality is found to yield favorable economic performance through several channels, such as productivity growth, human capital accumulation, higher female labor force participation, and improved labor market efficiency. The literature attests that some of the main channels are improving women's access to finance, education, and health services; participation in the workforce; and a level playing field in terms of pay and career prospects. Studies have also shown that gender equality improves economic performance indirectly. For example, because working women likely invest more in the education, nutrition, and health of their children, more women in paid employment facilitates human capital accumulation over time (see, for instance, Schultz 2002; World Bank 2012). Men and women also complement each other in production with different perspectives and skills (Ostry and others 2018).

Other studies show that women's participation in economic activities can improve economic outcomes by reducing income inequality, contributing to financial stability, facilitating economic diversification, improving competitiveness, and mitigating the negative effects of population aging (Steinberg and Nakane 2012; World Economic Forum 2014; Gonzales and others 2015; Kazandjian and others 2016; Kochhar, Jain-Chandra, and Newiak 2017; Sahay and Cihak 2018). In addition, new important interactions of gender equality and economic growth emerged together with other macro-critical issues. For example, women's empowerment and participation in decision making is essential for mitigating climate change (Gambacorta and others 2022; Altunbas and others 2022), and fintech is associated with more female employees in the workforce and fewer financial constraints for female-headed firms (Chapter 10; Loko and Yang 2022). Gender equality enhances growth through such channels.

Among the most visible obstacles to achieving gender equality are the prevalent legal impediments to women's economic empowerment (Chapters 12 and 13), since gender inequality in economic outcomes is not only a result of social attitudes, cultural norms, and economic policies, but also of discrimination in the law (Christopherson and others 2022; Sever 2022). A growing body of literature has shown positive associations between legal gender equality and several metrics of women's economic empowerment, including labor force participation, financial inclusion, asset ownership, and starting businesses in the formal sector (Khan 1996;

Agarwal 2003; Demirgüç-Kunt, Klapper, and Singer 2013; Gonzales and others 2015; Hyland, Djankov, and Goldberg 2020; Hyland and Islam 2021; Sever 2023). Moreover, Sever (2022) has found that abolishing gender discriminatory laws catalyzes the process of income convergence through which developing economies can catch up with the richer ones, with significant implications for sub-Saharan Africa given the still low economic development in many countries of that region.

Even though ample evidence has suggested various direct and indirect channels through which gender- and economic growth-related variables interact, the strand of literature that tries to establish a causal link between them is scarce. With a focus on a recent cross-country empirical study, the following subsection addresses the causal link between gender equality and economic growth.

CAUSAL LINK FROM GENDER EQUALITY TO ECONOMIC GROWTH

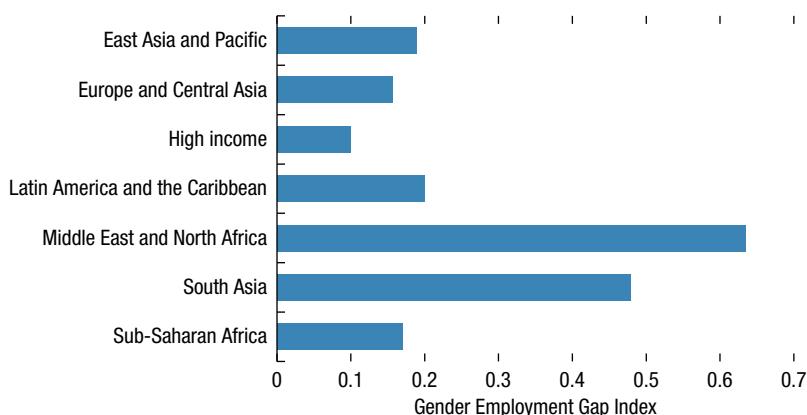
As presented, multiple theoretical frameworks associate gender equality with better economic outcomes. Various empirical analyses have supported these channels at the micro level by focusing, for example, on firm-level or single country studies.⁵ Many cross-country macro studies have also shown positive relationships between various measures of gender equality and economic growth (Cuberes and Teignier 2014 reviews the literature). Figure 11.2 shows the Gender Employment Gap Index, which is calculated in Pennings (2022) and indicates mean long-run GDP per capita gains from closing gender employment gaps at the regional level. Indicated gains are economically large at 10 percent of long-term GDP per capita, even for high-income countries. For sub-Saharan Africa, where the gender employment gaps are relatively low, there are also significant benefits to closing the gender gaps—leading to 17 percent higher GDP per capita in the long run.

There are, however, only a few studies that have tried to establish a causal effect of gender equality on economic growth, due to embedded endogeneity concerns typical to the empirical literature on development (for instance, reverse causality and omitted variable bias).⁶ The present chapter is based on the work of Bertay, Dordevic, and Sever (2020), which examined whether higher gender equality facilitates economic growth by enabling a better allocation of female labor at the industry level. By focusing on the differential effect of gender inequality on economic growth within countries between industries with varying labor compositions along gender lines, Bertay, Dordevic, and Sever (2020) address many of the endogeneity

⁵ Joshi (2017) reviews the literature, methodological challenges in this literature, and channels of relationship between gender diversity and firm performance. Single-country studies also include an increasing number of IMF country reports.

⁶ Some studies used the instrumental variable method—for instance, Klasen (2002) in a cross-country setting and Esteve-Volart (2004) at the subnational level—while others used the instrumental variable generalized method of moments technique (IV-GMM; for example, Kazandjian and others 2016) or the system-GMM estimations (for instance, Hakura and others 2016).

Figure 11.2. Long-Run GDP per Capita Gains from Closing Gender Employment Gaps



Source: Authors' calculation from Pennings (2022).

Note: The Gender Employment Gap Index used here is the full Gender Employment Gap Index—that is, the weighted average of a “better employment gap” and “other employment gap”—from Pennings (2022). The World Bank's definitions of regions are used, where high-income countries are grouped together and the geographical groups include only developing countries.

concerns in cross-country studies and also employ various robustness checks to rule out alternative explanations for the findings.

Using an industry-level cross-country analysis focusing on manufacturing, Bertay, Dordevic, and Sever (2020) tested whether increased gender equality disproportionately benefits industries with a typically higher female employee share relative to that of other industries. The authors argue that such industries should grow relatively more (and have higher productivity growth) related to both extensive and intensive margins of employment,⁷ and the effects on both margins would be more prevalent in high-female-share industries:⁸

- On the extensive margin, additional women in the labor force (due to higher gender equality) contribute to a larger pool of talent to hire from, and a larger share of newly hired women joins high-female-share industries.
- On the intensive margin, gender equality enables women to fully develop their potential in the labor market by, for example, occupying higher value-added jobs and having longer career ladders. Unlocking women's potential at work would in turn be beneficial for growth, this effect being particularly pronounced for high-female-share industries.

⁷ Improvement on the extensive margin means employment of more women in an industry (higher labor force participation by women), whereas intensive margin suggests women working longer hours (that is, less part-time or temporary work and more full-time positions generally associated with better jobs).

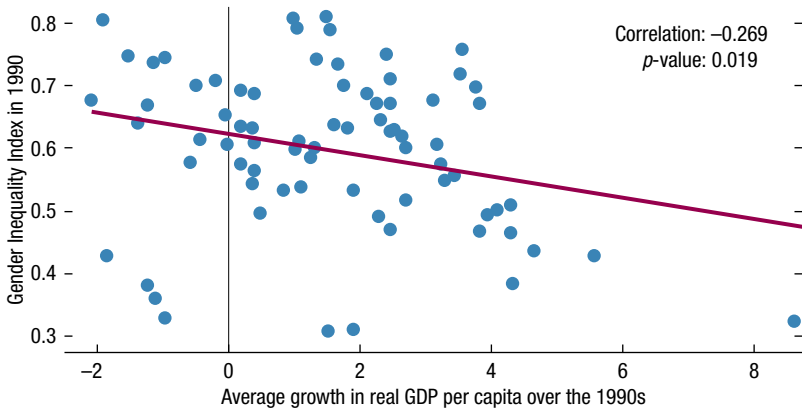
⁸ The authors define industries as high-female-share in terms of having a higher share of female workers in the industry in comparison with other industries and not necessarily having a higher share of women than men in the industry.

Bertay, Dordevic, and Sever (2020) use a difference-in-differences approach as in Rajan and Zingales (1998)—which studied the finance and growth nexus—and test whether manufacturing industries’ growth rates are differentially affected by country-level gender equality depending on the industries’ typical gender composition. The authors focus on a large sample of developing countries in the 1990s. In the empirical model, economic growth outcome is captured by industry-level value-added growth or value-added-based labor productivity growth. Country and industry fixed effects are included to control for unobserved country and industry characteristics, and possible convergence effects are captured by initial value-added share of respective industries. Finally, to explicitly address endogeneity concerns, the growth differentials of interest are captured by an interaction term between the gender inequality index and the female labor share of industries coming from Sweden in the 1980s. Namely, the identification problem addressed by this approach is that the actual industry gender compositions are generally distorted by existing gender inequalities—due to, for instance, discrimination in the labor market or gender social norms. In other words, observed industry gender compositions may not reflect their underlying compositions, which would prevail in the absence of labor market frictions. Thus, Bertay, Dordevic, and Sever (2020) use industry gender compositions of 1980s Sweden—a high-gender-equality country with few gender-related frictions—and assume that some intrinsic reasons⁹ lead some industries more than others to typically employ more female labor. This assumption is critical to the identification strategy—the other assumption being that these industry differences in gender composition carry over to other countries. The decade lag of the benchmark period in Sweden (1980s) with respect to the study sample of developing countries (1990s) aims to address their technological differences that have thus affected the production process and potentially also the nature of jobs and relative marginal productivity of the labor of women as compared with that of men.

The measure of gender inequality used is the United Nations Development Programme’s Gender Inequality Index (GII), which contains information on gender disparities in empowerment (that is, education and political power), health, and educational outcomes. The Gender Inequality Index varies between 0 and 1, and the Gender Inequality Index of a country in which women and men fare equally would be 0, so a higher value of the index indicates higher gender inequality. Figure 11.3 plots GDP per capita growth against the Gender Inequality Index in the 1990s and shows that gender inequality is indeed negatively correlated with the average growth of real GDP per capita.

As Figure 11.4 visualizes, this chapter is based on the empirical strategy by Bertay, Dordevic, and Sever (2020) to identify a causal effect of gender equality on economic growth that relies on within-country differences between industries. The baseline result suggests that the effect of gender equality on industry value-added growth is statistically significant and economically large. To give a sense of the economic

⁹ One example is the industry-specific relative marginal product of labor between men and women, which shapes women’s comparative advantage to men and thus their relative employability in that industry.

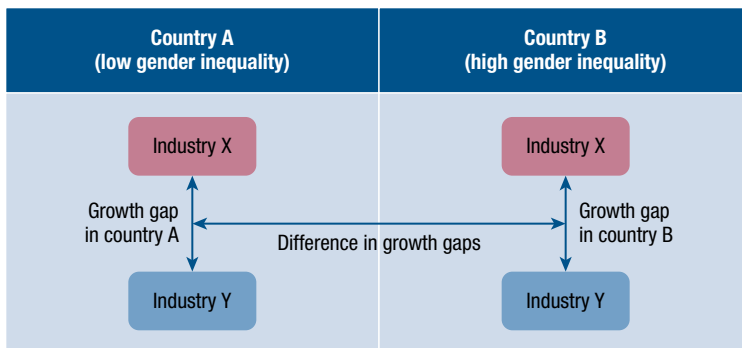
Figure 11.3. Gender Inequality and Real GDP per Capita Growth

Source: Bertay, Dordevic, and Sever (2020).

Note: Chart plots the average growth rate of real GDP per capita (from the World Bank World Development Indicators database) for emerging market and developing economies over the 1990s and Gender Inequality Index (GII) in 1990 (from United Nations Development Programme). The correlation between the two variables is about 0.3 with a p -value of 0.02.

significance of the estimated impact, the additional growth gains in an industry with relatively high female share are compared to those in a low-female-share industry, when they are located in countries with a high versus low gender inequality.

The result suggests that a high-female-share industry (at the 75th percentile of female shares as estimated from Sweden, that is, the rubber and plastics products industry) grows 1.7 percentage points faster than a low-female-share industry (at the 25th percentile of female shares, that is, nonmetallic mineral products industry), when it is located in a low-gender-inequality country (at the sample's 25th percentile of Gender Inequality Index, that is, Costa Rica) rather than in

Figure 11.4. Industry-Level Growth Differentials

Source: Adapted from the empirical strategy of Bertay, Dordevic, and Sever (2020).

Note: Industry X represents a high-female-share; industry Y represents a low-female-share industry.

high-gender-inequality country (at the 75th percentile of Gender Inequality Index, that is, Cameroon). This difference in growth gaps of 1.7 percentage points can be interpreted as the typically female-dominated rubber and plastics products industry growing 1.7 percentage points faster per year than the typically male-dominated nonmetallic mineral products industry in Costa Rica than in Cameroon.

Notably, the estimated 1.7 percentage-point growth rate is how much additional growth a high-female-share industry can achieve once it is located in a more gender-equal country, above the growth gains for a low-female-share industry. This is not to say that low-female-share industries do not grow faster in a more gender-equal environment. Indeed, the result suggests that gender equality can boost growth for all industries, but high-female-share industries can gain even more as countries move toward gender equality.

Looking at other sub-Saharan African countries below the 25th percentile of the sample in terms of gender inequality, the implied (differential) growth effects are even larger, as those countries make steps to reach to the 75th percentile of the sample (that is, the level of Costa Rica). For instance, the typically female-dominated rubber and plastics products industry could grow 2.4 percentage points faster than the typically male-dominated nonmetallic mineral products industry in Costa Rica than in Kenya and Central African Republic. The size of the growth effect becomes 3.0 percentage points when Niger is compared with Costa Rica.

Given that the average real annual value-added growth rate in the sample over the period is 2.2 percent per year, estimated growth differentials are substantial. The results thus provide valuable evidence that gender inequality has a causal effect on real economic outcomes at the industry level. In particular, they suggest that there is large room to boost industry growth in sub-Saharan African countries as they move toward a level playing field for women and men.

CONCLUSION

Gender equality is not only a human rights issue but also a critical development goal. Although remarkable progress has been made toward leveling the playing field for women over the last several decades, in various dimensions, much work remains to be done. As countries take steps to tackle the unfair treatment of women in all areas, gender equality would unlock a large but untapped economic potential, yielding favorable economic outcomes. Ensuring equal access to financial, education, health services; providing equal opportunities in the labor market; and eliminating legal impediments to women's economic empowerment are critical to attain this goal. The literature confirms that such efforts pay off in the form of more resilient, sustainable, and inclusive economic growth. In this regard, Bertay, Dordevic, and Sever (2020) made a step forward in providing causal evidence on the positive impact of gender equality on growth focusing on granular, industry-level data for emerging market and developing economies. This chapter, based on Bertay, Dordevic, and Sever (2020), discusses that gender equality disproportionately affects growth in industries that rely more on female labor.

The link between gender equality and economic performance is even more critical in the post-COVID-19 period given the scarring effects of the pandemic on both gender imbalances and economic performance. The current moment constitutes a window of opportunity for policymakers to not only address long-standing issues regarding gender inequality but also mitigate the adverse effects of the COVID-19 shock going forward. In sub-Saharan Africa—a region severely hit by the shock and with persisting, even widening, gender gaps along multiple dimensions—policies supportive of gender equality can help the countries leap forward in economic development.

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PART IV

Fair Play: Equal Rights and Women's Empowerment

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Legal Rights: Women's Economic Empowerment

Katharine Christopherson, Audrey Yiadom, Juliet Johnson, Francisca Fernando, and Lucia Gruet

In many countries, laws often reflect and perpetuate gender norms that limit women's economic participation. Removal of these impediments through legal reform has been shown to be an effective method to catalyze greater participation of women in the economy—along with the related macroeconomic benefits. Once legal barriers are removed and provisions for more equal treatment under the law are embedded, the law can also be employed as a powerful tool to incentivize women to pursue equal opportunities, change mindsets regarding the role of women, and hold institutions and individuals accountable for achieving results. Accordingly, it is imperative for countries to focus on eliminating existing legal impediments and designing appropriate incentives to increase women's participation in the economy.

INTRODUCTION

Empirical evidence shows that women's economic empowerment and the closing of gender gaps in key areas are associated with positive macroeconomic outcomes, including higher economic growth, lower inequality, increased productivity, and greater financial stability (Chapters 6–12). However, despite momentous strides toward improved opportunities for women and girls, gender gaps persist, and legal barriers remain.

It is also well established that various legal impediments in countries' domestic laws have prevented women from achieving full economic empowerment and that in many countries, laws often reflect and perpetuate gender norms that limit women's economic participation. The removal of these impediments through legal reform has been shown to be an effective way to catalyze greater participation of women in the economy—along with the related macroeconomic benefits. Once legal barriers are removed, the law can be used as a powerful tool to incentivize women's economic empowerment, change mindsets regarding women's role, and hold institutions and individuals

This chapter draws on Christopherson and others 2022.

accountable for achieving the intended results. Therefore, it is imperative for countries across all regions, including sub-Saharan Africa, to focus on eliminating existing legal impediments and designing appropriate incentives to increase women's participation in the economy.

This chapter discusses the key types of laws that impede women's economic empowerment; the impact of cultural norms, traditions, and belief systems on gender equality; and ways in which the law can be used as a tool to create behavioral changes and shifts in perceptions of women in the economy.

Sources of Legal Impediments to Women's Economic Empowerment

The categories of laws that often contain significant barriers to women's economic empowerment in sub-Saharan Africa include constitutional law, family law, property law, labor law, and tax law.

Constitutional Law and Civil Rights

Legal impediments to women's economic and political participation are often found in constitutional law, which provides civil rights and civil liberties. In some countries, the legal framework prevents women from voting and running for political office. Enshrining in countries' constitutions the right of women to equal treatment under the law, and other key constitutional rights such as citizenship and freedom of movement, should be the first step in addressing gender gaps and promoting women's economic empowerment. As the case studies in the following chapter will show (for instance, Rwanda, Cabo Verde), legally required quotas for women in leadership roles (in both public and private sectors) can significantly increase the number of women in those roles. More women in these roles helps diversify the leadership perspective and can provide a stronger support for policy agendas aimed at initiating political processes that inform constitutional reforms and strengthen civil rights movements.

Legal literacy and access to justice for women are also crucial to help address gender inequality. Access to justice for women entails the ability to exercise the same legal rights as men to sue, apply for legal aid, provide testimony, and obtain a fair trial. However, women often experience legal restrictions in exercising their rights because of gender-based discrimination. When women have adequate access to justice and their rights are appropriately enforced, it discourages gender-based discrimination.

Family Law

Family law touches on the most intimate part of people's lives, regulating marriage, divorce, adoption, child custody, and other domestic issues. For this reason, family law often reflects gender inequality in society. In several countries, the legal age to marry is lower for women than for men. Early marriage may lead to early motherhood, which hinders girls' chances of continuing their education and

thereby limits their prospects for work. Eliminating child marriage today would increase long-term annual per capita real GDP growth by about 1 percentage point in emerging and developing countries (UN Women 2020; Mitra and others 2020). This economic impact is closely connected to legal norms that provide for the husband's power over the marriage (and the wife's obedience, respectively). A study found that legal discrimination against women in the form of limits on their ability to be head of household is negatively correlated with female labor force participation (Gonzales and others 2015). Such legal discrimination reduces women's bargaining power when allocating roles within the family and their ability to pursue professional roles outside the household (Islam, Muhi, and Amin 2019; Htun, Jensenius, and Nuñez 2019). Recognizing this relationship between family law and women's economic empowerment, countries like Rwanda have enacted laws to enable women to be designated as the head of the household and have given women rights and duties equal to those of their husbands. Such measures have supported women's financial stability during and after marriage (World Bank 2021).

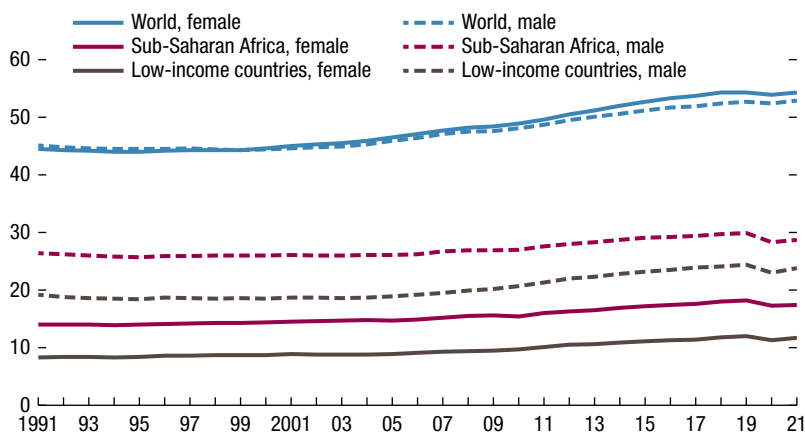
Property Law

Property law governs the ability to collect, own, manage, and transfer assets and therefore is an essential component of engaging in productive economic activities, including entrepreneurship. People need assets to sustain their livelihoods, provide for family members, build wealth, and provide collateral for credit. Laws that prevent women from owning, managing, and disposing of property and other assets can effectively hinder or significantly disincentivize them from fully participating in the economy. Such legal impediments appear in various forms. For example, some laws prohibit women from owning movable or immovable property on their own or being granted property during a marriage and after a divorce. Property can be distributed in discriminatory ways that prevent women (daughters or female surviving spouses) from inheriting property because the relevant laws follow the cultural norm of inheritance through male lineage. Because of such laws, some women do not have the same access as men do to financial services, including opening bank accounts. This in turn prevents them from building credit, borrowing and saving money, or obtaining insurance.

Labor Law

Employment is the most common way that women can participate and contribute to the economy. When women are prevented from obtaining employment, the economy loses a significant portion of potential for growth. Globally, 55 percent of adult women are in the labor market compared with 78 percent of men. Moreover, women face a 37 percent wage gap (the ratio of the wage of a woman compared with that of a man in a similar position) and a 51 percent income gap (the ratio of the total wage and nonwage income of women compared

Figure 12.1. Wage and Salaried Workers as Percentage of Total Workforce
(Modeled ILO estimate, percent)



Source: International Labour Organization ILOSTAT database. Data retrieved on January 29, 2021, from World Bank Gender Data Portal. License type: CC BY-4.0. See https://genderdata.worldbank.org/indicators/sl-emp-work-zs?geos=WLD_SSF_LIC&view=trend.

Note: ILO = International Labour Organization.

with that of men; World Economic Forum 2021) (Figure 12.1). There are often legal restrictions on women's ability to work, including laws that prohibit them from choosing a certain profession or industry according to their interests and abilities, as well as laws that require a spouse or other male relative to provide a work authorization. Studies show that such discrimination is a significant predictor for the gender wage gap (Htun, Jensenius, and Nuñez 2019). As numerous organizations have found, women continue to be paid less than men for work of equal value. Data show that fewer than half of countries have a law mandating equal pay for equal work (World Bank 2021).

The lack of protection and failure to prevent gender-based discrimination and sexual harassment in the workplace often hinder women from obtaining jobs or advancing their careers. This begins with discriminatory hiring practices and later with concerns about unfavorable treatment and dismissal of women in the workplace, especially when pregnant or after giving birth.

The lack of supportive legal and institutional frameworks providing for parental leave, childcare benefits, and eldercare benefits means there is room for improvement in addressing gender inequality. This is because women are the ones who typically bear the bulk of household responsibilities. Many countries do not yet provide for paid maternity and paternity leave; where countries do so, it is often the employer who bears the burden of cost, thereby reducing retention of female employees who become more costly than men (World Bank 2020). In contrast, generous parental leave provisions are associated with smaller wage gaps in economies with a large formal sector (Asai 2019).

Social security and pension laws affect women's retirement conditions negatively when they do not account for times of absence or part-time employment because of childcare or eldercare. The statutory retirement age may also be different between women and men. Shorter working lives of women have a negative impact on their income, contributions, and pensions. The gender wage gap also has a direct negative effect on pensions for women relative to men even when they have equal years of contributions to the social security system because women's salaries are on average lower than men's.

Tax Law

Explicit gender bias is generally easily identifiable in the relevant tax law provisions. Some examples include explicitly assigning different tax rates to male and female taxpayers; permitting tax preferences only for male or female taxpayers; or assigning joint business, asset income, or childcare deductions or allowances only to males (Stotsky 1996).

In the area of personal income taxation, the family-based method of determining income undermines the individual filing system and results in secondary earners (usually women who make less money) facing a higher marginal tax rate (Grown and Komatsu 2010; Organisation for Economic Co-operation and Development [OECD] Development Centre 2016). Therefore, providing options for individual income taxation could be a necessary legal reform to increase female labor force participation (Gonzales and others 2015). Even within income tax systems that impose a joint income tax on the household, it is possible to structure the income tax to reduce the higher effective tax rate on secondary earners, in particular through tax benefits (for instance, targeted tax reliefs) to promote the female labor supply. Other examples of implicit gender bias would be the availability of tax deductions for work expenses predominantly borne by men (for instance, uniforms or tools) but not for work expenses predominantly borne by women (for instance, the cost of daycare; Stotsky 2017).

Implicit or explicit biases may also exist in tax procedural rules, such as the inability of spouses to separately file personal income tax returns, or where the spouse's consent is required where a possibility of separate filing exists. These procedural requirements can impede women's economic empowerment.

Impact of Cultural Norms, Traditions, and Belief Systems on Gender Equality

The existing inequality between men and women under law has various root causes, including cultural norms, traditions, and belief systems that promote differential treatment of women. In addition, plural or inherited legal systems often facilitate the inclusion of legal provisions that have a gender bias. Legal reform can change attitudes and behavior and thus can be an important tool for addressing gender inequality. This section discusses the main sources of legal impediments to women's economic empowerment that usually arise from countries' broadly accepted cultural norms and belief systems.

Cultural Norms and Belief Systems

Many laws that prevent women from having the same legal status as men are derived from deep-rooted cultural norms or other belief systems. The persistence of those norms and systems can impede the advancement of women's rights. Some argue that certain beliefs are harder to change because they are considered by society to be divinely inspired and sacred. In some African countries, for example, dowry payment is prohibited by law but is still required by families and provided because of religious or traditional beliefs (OECD Development Centre 2019). Others argue that unequal treatment of women is the product of a culture of patriarchy and the consolidation of power in men, with religion serving as justification for conservative interpretations regarding gender roles (Cherif 2010). However, some countries have adopted legal reforms previously considered to be infeasible in light of prevailing norms and beliefs. For example, Côte d'Ivoire amended its marriage laws in 2019 to facilitate women's access to credit by allowing them to use land as collateral (Law No. 2019-570 of June 26, 2019, relating to marriage). The former marriage laws had restricted female spouses' rights to manage, dispose of, and inherit marital property. Similarly, Togo reformed its family code in 2014 to enable a woman to be the "head of household" in the same way as a man (Code of Persons and the Family, Article 99). The Democratic Republic of Congo passed a new family code in 2016 that did not require women to obey their husbands and allowed them to sign a contract, register a business, open a bank account, choose where to live, and get a job in the same way as a man (Law 16/2008 of July 15, 2016).

Plural Legal Systems

Many countries have plural legal systems that recognize different sources of law (for example, common law, customary law, and religious laws) (UN Women 2015). These plural legal systems accept that different legal systems can coexist, and they may overlap, complement, reinforce, and even conflict with each other at times. This can become more complex in countries where multiple legal systems exist in parallel that in practice conflict with each other.

The existence of legal pluralism affects gender equality under law in various ways. Although in some cases it helps advance women's rights by providing more avenues for legal redress, it can create a barrier in other situations. For instance, because different belief systems can contain provisions that discriminate against women, in some plural legal systems, reforming the legal framework to promote gender equality requires dialogue with ethnic and religious leaders as well as women's groups to empower each of them to challenge discriminatory laws. It also requires education and resources to navigate these complex systems (Chiongson and others 2012).

Inherited Legal Systems

Many countries that were former colonies inherited legal codes from colonial powers, which included provisions that discriminate against women. Although most European countries that were former colonial powers have phased out gender discrimination in their codes, the original codes are still in force in some

former colonies (even after these countries gained independence and later adopted several international, regional, and subregional conventions promoting gender equality). Countries that still maintain gender-biased codes inherited from the British, French, German, and Spanish legal systems, including those from the apartheid regime, could embark on reforms similar to those the former colonial powers have introduced to remove these gender impediments. For example, in Equatorial Guinea, a woman still needs her husband's permission to sign a contract because of the 1960 Spanish Civil Code. In Chad, and Niger, married women cannot open a bank account without their husbands' permission due to a 1954 ordinance from the former federation of French West Africa. Similarly, due to a Spanish decree enacted in 1957, women in 12 countries in the Spanish-speaking world are prohibited from working in mining, certain construction jobs, and the electricity sector. In addition, women of Namibia and South Africa were doubly burdened by gender bias and the racial discrimination embedded in laws of the South African apartheid state.

Changing Behavior and Creating Cultural Shifts through Legal Reform

Extensive research has shown how law and legal reform influence behavior, with research frequently relying on economic and psychological theories and empirical studies (Anderson and Whitman 1972). The goal of legal regulation is to exercise control over certain behaviors (Bilz and Nadler 2014). This can happen in various ways, including by incentivizing or disincentivizing behavior. Democratically produced legislative outcomes are an even stronger signal to people that the values expressed by the law are deemed desirable by society (McAdams 2000). As discussed in the next section, country-specific experiences in sub-Saharan Africa, where constitutional and legal changes aimed at addressing gender inequality were introduced, show the power of legal reform to modify behavior and to produce positive outcomes in supporting women's economic empowerment.

After the independence movements prevalent in sub-Saharan Africa in the 1950s and 1960s, many countries made significant progress in removing legal impediments to women's economic empowerment by adopting several international, regional, and subregional conventions that promote gender equality. The most important treaty is the 1979 Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) that has been adopted by most countries, including 46 in sub-Saharan Africa. The Beijing Declaration and the Platform for Action has been signed and ratified by all countries in the sub-Saharan Africa's region except Somalia. This treaty sets forth a plan for women's empowerment and is considered the key global policy document on gender equality. At the regional level, the Protocol to the 2003 African Charter on Human and People's Rights on the Rights of Women in Africa (the "Maputo Protocol") has been signed by 43 countries in sub-Saharan Africa, but only 21 states ratified it. This protocol was drafted by the African Union to supplement the commitments of the 1981 African Charter on Human and People's Rights (the "Banjul Charter"). Article 18 calls on states to eliminate discrimination against women

and to protect women's rights in alignment with international declarations and conventions. It also defines a multitude of women's rights, including economic, social, civil, political, cultural, and environmental rights. At the subregional level, treaties and conventions also support women's gender equality and empowerment. For instance, the Economic Community of West Africa States (ECOWAS) Treaty (1975, revised in 1993) includes a dedicated article on women and development (Article 66). Also, both the 2008 Southern African Development Community's (SADC) Gender Protocol and the 2016 East African Community Gender Equality and Development Bill, which are harmonized with international and regional agreements, aim to enhance planning, implementation, and monitoring of policies that support eliminating discrimination, empowering women, and achieving gender equality in the respective subregions. In addition, the Supplementary Act related to Equality of Rights between Women and Men for Sustainable Development in the ECOWAS Region (2015) was adopted to strengthen the harmonization of national legislation with international standards on women's rights.

Each of the countries included in the case studies in the following chapter (South Africa, Namibia, Mauritius, Rwanda, and Cabo Verde) ratified the CEDAW Convention and the Maputo Protocol. Rwanda is a signatory of the East African Community Gender Equality and Development Bill. South Africa, Namibia, and Mauritius have ratified the African Union Solemn Declaration on Gender Equality in Africa, the African Union Gender Policy, and the SADC. Cabo Verde is a member of the ECOWAS and ratified its treaty.

Following the adoption of the conventions mentioned earlier, many sub-Saharan African countries continued to make progress in eliminating legal barriers to women's economic empowerment from 2000 to 2009. Some of this success is generally attributed to the Maputo Protocol, which entered into force in 2005 and led to significant reforms in the region. This treaty may have particularly helped by accelerating the adoption of gender policies in the region because many countries amended their domestic legal frameworks to align them to the principles and standards on gender equality enshrined in the protocol (World Bank 2022). In light of this, the principle of nondiscrimination on the basis of gender is included in every constitution in the region. In addition, some countries went further by providing for women to own property in their constitutions, which helped facilitate shifts in certain social norms (for instance, the constitutions of Ethiopia, Guinea-Bissau, Kenya, Liberia, Malawi, Senegal Swaziland, and Zimbabwe; Hallward-Driemeier and Hasan 2012). However, enshrining gender equality provisions in the constitution is not enough to ensure that no gender discrimination exists under law because some African countries have plural legal systems where discrimination may still arise from customary laws or religious laws (N'Dulo 2011; Ezer 2016). An example of this, six sub-Saharan African countries (Botswana, The Gambia, Ghana, Lesotho, Mauritius, and Sierra Leone) currently exempt customary law from the application of constitutional provisions on the prohibition of discrimination (Klugman and Twigg 2015; Dinokopila and Bonno 2021). In countries in sub-Saharan Africa that have plural legal systems, it is important to ensure that laws providing for gender

equality are enforced and are given priority over conflicting customary or religious laws. This can be addressed by explicitly establishing in the constitution that it invalidates any law, including customary law, contradicting the constitution. This is the case in Kenya, where gender equality prevails despite customary law because its constitution explicitly states this (OECD Development Centre 2021).

CONCLUSION

Legal reforms to remove barriers to women's economic empowerment can be powerful instruments in promoting women's economic participation and helping countries achieve higher growth and a range of other macroeconomic benefits. In addition, legal reforms should be adopted that incentivize women's economic participation, even after legal impediments have already been removed via law reform. As noted previously, these legal reforms can entail a broad range of measures across categories of laws, including repealing laws that discriminate against women, providing parental leave for women and men, making gender pay gaps illegal, implementing regulations that require appropriate facilities and infrastructure to allow women to benefit fully from participating in the economy, subsidizing childcare, ensuring laws do not penalize secondary earners, incentivizing the training and hiring of women in traditionally male-dominated fields, and enabling women to exercise all their rights by improving access to justice.

Countries that have high levels of female labor force participation have made extensive changes to their legal frameworks over time. The case studies in the following chapter present the experiences of several African countries—Rwanda, Namibia, South Africa, Mauritius, and Cabo Verde—with legal reform to support women's economic empowerment and illustrate that the pace and nature of legal reform vary from country to country and are specific to a country's historical, cultural, and belief systems. Despite such reforms, some barriers to women's economic empowerment remain. Thus, continued attention to removing legal impediments is necessary.

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Legal Rights: Women's Economic Empowerment—Case Studies

Katharine Christopherson, Audrey Yiadom, Juliet Johnson, Francisca Fernando, and Lucia Gruet

Case studies of five countries in sub-Saharan Africa—Rwanda, Namibia, South Africa, Mauritius, and Cabo Verde—that rank high in gender equality in their respective regions demonstrate how legal reforms have been implemented in differing contexts to help achieve women's economic empowerment.

INTRODUCTION

Removing legal impediments and using the law as a tool to incentivize women's full economic participation can encourage needed behavioral changes and cultural shifts, particularly in countries with prevalent belief systems or traditions that discourage the inclusion of women in the workforce. Specifically, reforming a country's legal framework to remove discriminatory provisions and support women's economic empowerment has proven to result in positive legal, social, and economic outcomes. The case studies included in this chapter—Rwanda, Namibia, South Africa, Mauritius, and Cabo Verde—demonstrate the power of legal reform in various areas that affect women's ability to work and contribute to economic growth. Also, these countries' legal reforms have placed them in the upper rankings of reputable studies, including the World Bank's Women, Business and the Law (WBL) report; the World Economic Forum's (WEF) Global Gender Gap Report; and the Organisation for Economic Co-operation and Development's (OECD) Social Institutions and Gender Index.

This chapter builds on the analysis and methodology in Chapter 12, which describes the types of laws that often impede women's economic empowerment.

Although this chapter highlights dominant belief systems and cultural norms that have contributed to limiting women's economic empowerment, the authors do not intend to render judgment on these belief systems or norms. This chapter builds on Christopherson and others 2022.

Rwanda

Rwanda is among the top 15 countries with the lowest gender gap, according to the 2023 Global Gender Gap Report (WEF 2023).¹ It is also the second highest-ranking country in this index for the sub-Saharan Africa region. Rwanda's progress in gender equality stems in part from conscious legal reforms, which have fostered gender equality and strengthened protection against gender-based crimes (WEF 2022).

The Rwandan Constitution, which was adopted in 2003, and amended in 2015, replaced the 1991 Constitution, and enshrines in its preamble that the people of Rwanda are committed to building a state based on the principle of equality between men and women. Other key constitutional provisions include Article 17, which provides for matrimonial equality and stipulates that no woman can be married without her consent. The devastating effects of the 1994 genocide fundamentally altered Rwanda's demographic configuration, resulting in women representing 70 percent of Rwanda's total population (Women's Commission for Refugee Women and Children 1997). Therefore, it became necessary to enhance women's role in the public sphere. In addition to the Constitution, the Penal Code, Civil Code, and various other laws, including property and labor laws, establish gender rights and afford increased protection for women.

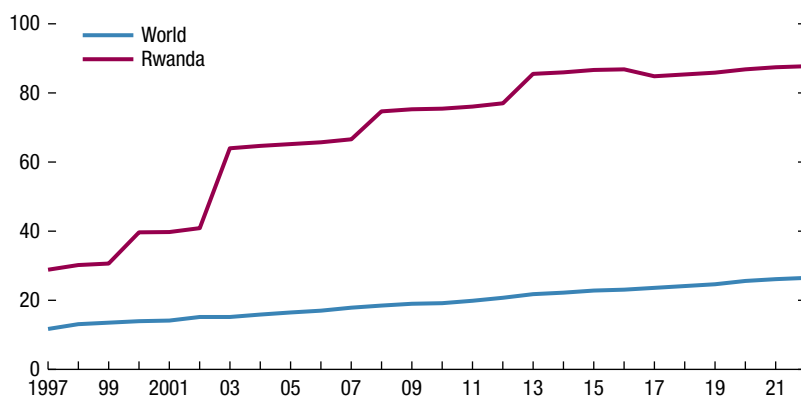
Rwanda gets high marks for women's participation in political decision-making processes. Most notably, the Constitution requires that women occupy at least 30 percent of positions in national legislative bodies (the Organic Law 03/2010/OL and Law 27/2010 outline the process for such quotas). The Constitution also designates that 24 of the 80 seats in the Chamber of Deputies (lower house of Parliament) comprise women and that at least 30 percent of elected and appointed senators must be women. As a result, Rwanda is now among the top 10 countries for women's political empowerment, with over 50 percent of women holding positions as parliamentarians and ministers (Articles 10, 75, and 80 of the Constitution). With increased female participation in politics, female parliamentarians have successfully advocated for key legal reforms aimed at strengthening gender equality in Rwanda. Before its civil war in the early 1990s and the genocide in 1994, Rwandan women never held more than 18 percent of the seats in the country's Parliament (Powley 2006) (Figure 13.1).

A strong formal legal framework also exists for women's property rights in divorce and inheritance law. For example, the Matrimonial Regimes, Liberties and Successions Law No. 22/1999 granted equal rights to male and female children to inherit from their parents, and for men and women who have married under a community property arrangement to have joint rights to marital property. Rwanda updated this law in 2016 to address gaps. For example, if

¹ The WEF's Global Gender Gap Index benchmarks the evolution of gender-based gaps among four key dimensions (Economic Participation and Opportunity, Educational Attainment, Health and Survival, and Political Empowerment) and tracks progress toward closing these gaps over time.

Figure 13.1. Proportion of Seats Held by Women in National Parliament, 1997–2020

(Percent)



Source: Inter-Parliamentary Union (www.ipu.org).

parents make land donations to their children (“umunami”)—a common traditional practice—they cannot give larger land parcels to their sons and must treat sons and daughters equally. Moreover, Law No. 27/2016 Governing Matrimonial Regimes, Donations, and Inheritance establishes, for the first time, that the surviving spouse has the same right to inherit from her husband as other heirs (Food and Agriculture Organizations of the United Nations Gender and Land Rights Database [2023]; Articles 27, 28, and 75 of the Rwandan Constitution). Law No. 43/2013 governing land in Rwanda also prohibits all forms of discrimination regarding access to land and the enjoyment of real rights (Article 4 of the Land Law 2013). Nonetheless, these laws do not protect the rights of women in customary, informal, or polygamous marriages, and male family members remain resistant to change given that inheritance is patrilineal in Rwandan customary law (Parallel Report to the United Nations Committee on the Elimination of Discrimination Against Women 2017). However, Article 39 of the Gender-Based Violence law stipulates that if a person in an informal marriage decides to formally marry someone else, the belongings of the informally married couple must be divided equally (US Agency for International Development 2016).

Rwanda has a female labor-force participation (54.8 percent) superior to the global average of about 47 percent (International Labour Organization ILOSTAT database 2022). As previously noted, the participation rate stems from the devastation of the male population during the 1994 genocide, leaving women to fill roles that men once occupied. Rwanda has enacted various labor laws to support women in the workplace, including providing for equal working conditions and equal salary for all employees irrespective of their sex (Law No. 51/2001),

maternity benefits, guaranteeing employment security for women on maternity leave (Law No. 3/2016), and nursing breaks for mothers who return to work (Labor Law No. 66/2018). Notwithstanding, there remain gaps that need to be addressed. For example, gender gaps persist in labor force participation, and occupational and industry segregation is prevalent (Labor Force Survey 2020 Gender Report). Women are also more prone to climate change shocks than men (Annex III of IMF staff report 2023/198).

Rwanda has also taken steps to address gender-based violence, which 41.5 percent of Rwandan women experience at the hands of their intimate partners during their lifetime (UN Women Global Database on Violence Against Women [2023]; 2019–2020 Rwanda Demographic and Health Survey). A national law criminalizes and penalizes several forms of gender-based violence (Law No. 59/2008), and the Prime Minister's Order (No. 001/03 of 2012) sets out the modalities through which government institutions should prevent gender-based violence. One successful government initiative has been the “Isange,” one-stop centers that provide medical, legal, forensic, psychological, and safety support to victims of gender-based violence. There are now 44 operational Isange Centers across the country, funded with assistance from the World Bank. The UN has reported that Isange centers assisted more than 49,000 victims of gender-based violence between 2015 and 2018 (United Nations 2019).

In addition to the legal framework, Rwanda has implemented several strategies and established institutional mechanisms to promote gender equality. A National Gender Policy formulated in 2010 and revised in 2021 addresses gender issues in various sectoral policies and programs, including in the areas of education for girls, women in agriculture, gender-based violence, and sexual and reproductive health (Rwanda Ministry of Gender and Family Promotion 2020). National budget considerations also factor in gender issues. For example, the Organic Law on State Finances and Property Law No. 12/2013 made gender balance in public financial management a fundamental principle. This law mandates including a gender-budget statement as one of the annexes to the national budget framework paper submitted to Parliament, and all public entities must submit activity reports to specify how plans for gender balance have been implemented.

The combined effect of these legal reforms, strategic implementation plans, and relevant institutional mechanisms have enabled Rwanda to put in place a progressive framework for gender equality, resulting in notable progress.

Namibia

The experience of Namibia illustrates how tackling impediments and undertaking legal reforms can incentivize women's economic empowerment. The Namibian Constitution—adopted in 1990 after independence from South Africa—grants women and men equal status and expressly prohibits discrimination based on sex. Moreover, Article 23 of the Constitution explicitly recognizes that “women in Namibia have traditionally suffered special discrimination and

that they need to be encouraged and enabled to play a full, equal and effective role . . . in the life of the nation.” In this manner, the Constitution promotes affirmative action to address equality for women (Constitution of the Republic of Namibia 1990, Articles 10, 23, 95).

Over the past few years, the WEF has recognized Namibia among the top 10 countries worldwide to close the gender gap (WEF 2023). According to the WEF, this progress is due largely to significant increases in women's participation in the political arena. Leading from the top is Saara Kuugongelwa-Amadhila, who was elected the first female prime minister in 2015 and remains in this position as of 2023. Although no legislated quotas exist for women in Parliament, in 2013, the leading political party (the South West Africa People's Organisation, or SWAPO), adopted a voluntary 50/50 policy for forming its parliamentary lists, for example, equal representation of men and women. Following the November 2019 elections, women held 46.8 percent of the seats in the lower house of Parliament, known as the National Assembly (Inter-Parliamentary Union [IPU] Parline, Global data on Parliaments, Namibia National Assembly data [2023]). This policy is an important first step, but as some critics have noted, the greater number of women among SWAPO parliamentarians has not translated into concrete legislative action on gender matters (Amupanda and Thomas 2019). Moreover, female representation in the upper house of Parliament (the National Council), whose members are indirectly elected by the Regional Councils, is only 14 percent (IPU Parline, Global data on Parliaments, Namibia National Council data [2023]).

Namibia has one of the narrowest gender-income gaps worldwide; the average income of a Namibian woman is about 81.6 percent of that of a Namibian man (WEF 2022). Many elements of the formal Namibian legal framework are conducive to women's economic empowerment. For example, the National Labour Act prohibits discrimination in the workplace and in employment decisions on the grounds of sex, provides for equal pay for equal work, and provides for 12 weeks of paid maternity leave and job security for new mothers (The Labour Act, Act 11 of 2007, Sections 19, 26). In 2021, women's labor force participation was 54.5 percent compared with 62.2 percent for Namibian men and the women's world average of 46.3 percent (World Bank 2022b). Also, women are relatively well represented in the salaried sector (56.5 percent compared with 66.7 percent for men) due to high levels of educational attainment, with 80 percent of girls completing secondary school and 17 percent of women completing tertiary education (World Bank 2019; World Bank 2017; WEF 2022). The Married Persons Equality Act of 1996 also gives women the right to contract, open bank accounts, register a business, and sell assets (The Married Persons Equality Act of 1996, Articles 2, 3, and 5). Data show that approximately 80 percent of women have a bank account (although this number may include joint accounts) or access to mobile money service (World Bank 2017). Despite this considerable progress, legal impediments remain to women's full financial inclusion in the economy.

Regarding assets accumulated in marriage, the Constitution provides for universal rights to acquire, own, and dispose of property (Article 16) and grants women equal rights during and at the dissolution of marriage (Article 14). The Married Persons Equality Act further establishes that the default marital property regime is “in community of property.” If the spouses divorce, the property brought to and acquired during the marriage is divided in half. Although a positive development, the Married Persons Equality Act applies to civil marriages, not marriages by customary law (that is, when a man and woman are married according to the traditions of their community). Moreover, under the Native Administration Proclamation of 1928, which still applies to the north central sector of Namibia (above the “red line”) where historically the majority of indigenous people lived² the default marital property regime for civil marriages is “out of community property”—each spouse retains his or her own property before, during, and after marriage—which is far less beneficial to women. This difference dates back to the experience of “native” men in this region who entered into civil marriages and may also have had a customary wife or wives, and who, by maintaining separate property from the civil-law wife, could preserve their separate property for their customary wife or wives. A Uniform Matrimonial Property bill, which would repeal these remaining provisions of the Native Administration Proclamation, has been pending for several years (Committee on the Elimination of Discrimination Against Women [CEDAW] 2022).

Regarding land ownership, Namibia has enacted gender-neutral legislation in the Communal Law Reform Act 5 of 2002 (CLRA), which permits women to register for communal land and to inherit land from their deceased husband (CLRA, Chapter II, Sec. 4; Chapter IV, Sec. 26). The CLRA has also mandated that every communal land board have at least four women, giving them slightly more than one-third representation on a typical board. These are important advances; however, the legacy of customary law and culture in rural communities persists. Even if the formal legal framework enables a widow to inherit communal land, widows may be subject to property grabbing or other threats by the deceased husband’s family to force them off the land (Mwetulundila 2021). Moreover, many women do not know about the CLRA’s provisions, and surveys have shown that rural women have often internalized the conservative cultural views that they should not own land. The CLRA also does not address the rights of spouses to land in divorce cases, leaving local courts to apply customary law that generally does not favor women (University of Wyoming Human Rights Clinic 2015).

Thus, despite progressive formal legislation, the existence of customary law—which primarily regulates marriage, divorce, and land rights—still affects control over assets. The Constitution of Namibia expressly recognizes the validity of common law and customary law, although only to the extent that these laws do not

² The boundary of this area was demarcated on maps by a red line, with the south and central part of Namibia below the line known as the Police Zone, where German and South African colonial administrators established police control over the territory and white settlements were concentrated.

conflict with the Constitution or other statutes (Article 66). In *Myburgh v. Commercial Bank of Namibia* (2000), the Supreme Court of Namibia held that the Constitution and Married Persons Equality Act overrode common law limitations on a woman's right to contract in civil marriages. Although this same principle should apply to conflicts between the Constitution and customary law, it has not yet been tested before the Supreme Court. Many experts advise that despite the Constitution's legislative supremacy and statutes enacted after independence, rural communities continue to follow customary law, often with discriminatory effect on women (Immigration and Refugee Board of Canada 2012). Although the Global Gender Gap Report indicated that women have “near equal rights” with regard to access to finance (land and non-land assets; WEF 2022), this claim derives from official legislation that does not fully take into account the continuing influence of customary law and culture in rural communities.

Like the other countries discussed in this chapter, Namibia has a strong national institutional framework for addressing remaining challenges. The Ministry of Gender Equality, Poverty Eradication and Social Welfare targets key issues like combating gender-based violence, and the national gender machinery (central coordinating unit for women's affairs within the Government) constitutes 11.3 percent of the national budget (CEDAW 2022).

South Africa

Since the end of the apartheid regime, South Africa has enacted a host of progressive legislation to address gender inequality. The Constitution enshrines the principle of nondiscrimination on the grounds of race, gender, sex, pregnancy, and marital status, among others (Section 9). The government has promulgated the Promotion of Equality and Prevention of Unfair Discrimination Act No. 4 of 2000 (PEPUDA) to give effect to this provision of the Constitution by establishing Equality Courts to hear cases of alleged discrimination, harassment, and hate speech. To expand access to justice, these courts have more informal procedures and are meant to be inexpensive. However, the Equality Courts have been underutilized because of inadequate information to the public and insufficiently trained court personnel (Kaersvang 2008).

Section 8 of the PEPUDA also lists specific types of discrimination that are prohibited, such as any system that prevents women from inheriting property and any policy or conduct that unfairly limits women's access to land rights, finance, and other resources. Following the landmark Constitutional Court case *Bhe v. Magistrate, Khayelitsha* (2005) that declared male primogeniture unconstitutional, the government enacted the Reform of Customary Law of Succession (Act 11 of 2009), which stipulates that widows and daughters may not be subject to inheritance discrimination. In addition, women in customary marriages are entitled to a portion of their deceased husband's estate. Other legislation promotes women's access to bank accounts, credit, and financial support, particularly for previously disadvantaged black women (Broad Based Black Economic Empowerment [Act 53 of 2000]).

Women's labor force participation is at 54.3 percent compared with 64.9 percent for men (South Africa, Quarterly Labor Force Survey 2023). These levels of women's economic engagement are made possible in part by legislation like the Employment Equity Act. Before its 2022 amendment, this Act required designated employers—firms with more than 50 employees—to create plans to achieve an appropriate representation of black people and women in their workforce and submit reports detailing progress on those plans to the government. Studies indicate that this Act restructured previously male-dominated industries that were required to recruit female workers for legal compliance (Landman and O'Clery 2020). In addition, the Employment Equity Act requires equal pay and benefits for men and women performing the same role. Nonetheless, according to the WEF, wage equality in South Africa lagged behind the rest of the world, ranking 123 out of 148 countries (WEF 2022). Other laws that support women's ability to work are the Basic Conditions of Employment Act, which provides four months of maternity leave, and the Labor Relations Act, which, as amended, provides 10 days of gender-neutral parental leave (South Africa Department of Women, Youth and Persons with Disabilities 2020). Employers are not required to pay for maternity leave, but women may apply for unemployment insurance from the Unemployment Insurance Fund. However, women's shouldering the burden of childcare remains a significant constraint on their labor force participation. Studies have shown that women spend 2.4 to 3 times more on unpaid domestic and care work compared with men (Kirkwood 2018).

One of the most serious issues facing South Africa today is widespread gender-based violence. Although it is difficult to quantify, some have estimated that gender-based violence costs South Africa approximately 1.3 percent of GDP per year, and a woman's lifetime experience of violence is associated with 35 percent lower earnings (Kirkwood 2018). The UN Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) conducted a confidential inquiry and concluded on May 17, 2021, that South Africa violated the Convention due to its failure to take sustained measures to prevent domestic violence. To address this critical problem, in 2021, the South African Parliament enacted the Domestic Violence Amendment Bill and other amendments to criminal laws that address sexual offenses. This legislation imposes reporting requirements on those who observe domestic violence and puts in place support at police stations ("victim-friendly rooms"), creates sexual offenses courts, and provides legal aid to victims of domestic violence (Inquiry concerning South Africa conducted under Article 8 of the Optional Protocol to the Convention 2021). However, legislation alone is not enough. Research in South Africa has shown that traditional notions of masculinity harmful to women—including that rape strengthens a perpetrator's masculinity—are strong. Other practices such as forced marriage through kidnapping ("ukuthwala") and "blessers" (older men supporting younger women in exchange for sex) are exploitative (Kirkwood 2018; OECD Development Centre 2019).

A need thus exists for institutional mechanisms focused on implementation and monitoring. To address this need, in addition to a national ministry dedicated to

women, the Constitution of South Africa established the Commission for Gender Equality. This independent statutory body is charged with “the protection, development and attainment of gender equality” (South Africa’s Constitution of 1996, Sections 181 and 187). The Commission for Gender Equality Act 39 of 1996 further provides for the commission’s composition, powers, and functions. This commission is a crucial voice in assessing and monitoring the South African government’s progress on gender equality in a range of areas, such as the governmental response to gender-based violence, implementation of economic empowerment programs, and gender-responsive budgeting (Commission for Gender Equality South Africa 2020, 2021a, 2021b). It is also active in analyzing and commenting on legislation relating to gender equality and making legal submissions to Constitutional Court cases involving women’s rights (Commission for Gender Equality South Africa 2017). An independent watchdog on progress toward gender equality is a valuable institutional mechanism.

Mauritius

Mauritius has a strong legal framework for enabling women’s access to economic opportunities, as evidenced by the World Bank’s WBL project (which assesses and analyzes legal barriers faced by women in various economies), affording it the highest rating (89.4 out of 100) of any African country (World Bank 2022). However, other independent assessments find Mauritius lagging (98th in the Global Gender Gap Report 2023) due to challenges in areas such as women’s labor force participation rate, earnings levels (WEF 2023), and women’s weak formal political engagement as only 20 percent of the National Assembly members are women (IPU Parline, Global Data on Parliaments, Mauritius National Assembly [2023]).

Recently, Mauritius has been increasingly taking steps to address discriminatory labor laws. These steps include the 2019 Worker’s Rights Act mandating equal remuneration for work of equal value—which includes male and female workers—that increased the duration of paid maternity leave up to 14 weeks and prohibited the dismissal of pregnant workers. In addition, the 2008 Equal Opportunities Act prohibited gender discrimination in employment and enacted civil remedies for sexual harassment in employment. Despite these efforts, females represent only about 45 percent of the total labor force (World Bank, Gender Data Portal, Labor Force Participation by Females [2023]). This statistic is also noteworthy because unemployed women are generally more educated than their male counterparts: 60 percent of women have a school certificate compared with 48 percent of men (Dabee 2021). Also, women tend to leave the labor force upon marriage. Specifically, more than half of the working-age females are outside the labor force—one of the reasons for the World Bank’s key recommendations to reform laws affecting women’s work post childbirth (Dieterich, Huang, and Thomas 2016).

The Civil Code establishes women’s rights in some areas of marriage and family law. The Civil Code stems from the Napoleonic Code and had patriarchal features, such as treating married women as “minors.” Following widespread

protests and advocacy by women's groups, the courts amended the Code to remove this inferior status in the early 1980s (Ramtohul 2015). Another victory for the women's movement concerned an immigration law that prohibited foreign husbands of Mauritian women from residing in Mauritius (but not foreign wives of Mauritian men), thereby threatening Mauritian women's family stability. The women's front brought a sex discrimination case to the United Nation Human Rights Committee about this provision, which was repealed in 1977. Revised in 2011, the amended Civil Code gave women equal status to men in the household, equal authority as legal guardians of their children, and the freedom to open bank accounts without the consent of their husbands. With respect to property rights, women and men have equal rights to own and manage property, create a will to outline the succession of their property, and attain the same inheritance rights. Laws are also in place to protect against gender-based violence, such as the amended Protection from Domestic Violence Act, 2016.

In 2022, the Mauritius Law Reform Commission, in response to the attorney general's request, reviewed discriminatory laws against women in Mauritius. The subsequent report outlined some key successes in legal reforms over the years, such as the Civil Code reforms that used to require a wife's obedience to her husband. In addition, the report makes recommendations for the way forward (Discriminatory Laws against Women in Mauritius Law Reform Commission Report 2022).

Mauritius has consciously adopted policies and action plans to improve gender equality. The government instituted a Ministry of Gender Equality and Family Welfare with a dedicated unit working on women's social and economic empowerment, including ongoing work to adopt a Gender Equality Bill. In addition, the government adopted a new National Gender Policy for 2022–30 and a renewed 2020–24 National Strategy and Action Plan on the Elimination of Gender-Based Violence in Mauritius.

Cabo Verde

According to the World Bank's WBL report 2023, Cabo Verde scores 86.3 out of 100 in terms of formal laws and regulations that affect women's economic opportunity, which is higher than the regional average for sub-Saharan Africa (71.5; World Bank 2023). Nonetheless, its lower scores on women's educational attainment and political empowerment lower its ranking in the Global Gender Gap Report (37th in the world) (WEF 2023).

Cabo Verde's overarching legal framework is set forth in its Constitution, which enshrines the principle of equality between sexes in Article 24.4 and emphasizes that the government must remove barriers to equality between citizens "especially factors of discrimination against women in the family and in the society" (Article 7).

In addition, high-level political commitment exists to achieve gender equality, including through a National Plan for Gender Equality (2021–2025). This plan

presents the government's strategic vision to achieve gender equality consistent with the latest CEDAW recommendations issued in 2019, gender-sensitive budgeting and the establishment of gender markers in the budget system, and prioritizing gender equality in the Strategic Plan for Sustainable Development 2017–2021.

The Constitution encourages women's political participation (Article 55 says: "The State shall encourage balanced participation of citizens of both sexes in political life"), but balanced participation has been more expressly prioritized with the new Parity Law. Adopted in 2019, this law established a mandatory quota of 40 percent of women on the list of candidates running for Parliament, municipalities, and other elected offices and decision-making positions in public administration (Law No 56 /Vii/2010 B.O 29 I Suplemento 9 De Março 2010; Law No 68/IX/2019 De 28 De Novembro 2019, Article 4). As a result, following the municipal elections held in October 2020, women represented 42 percent of local elected representatives in Municipal Chambers or Assemblies (National Directorate of Planning 2021). In 2021, women held 26 of the 72 Congressional seats (a total of 36 percent of the seats), an increase from 17 compared with the latest legislature (US Department of State 2021).

Because access to credit is a major constraint for entrepreneurs, especially for women, an amendment to Cabo Verde's Penal Code in 2005 made access to credit easier for women by prohibiting gender-based discrimination in financial services (Penal Code, Article 161). In addition, government programs promoting entrepreneurship also directly targeted women. Specifically, from 2015 to 2017, the government established three entrepreneurship programs in partnership with several nongovernmental organizations and the Agency for Entrepreneurial Development and Innovation. One of these programs, Support to Women's Entrepreneurship, provided training in entrepreneurship, financing for business plans, and the incubation process to support the establishment and growth of women-led start-ups (Cabo Verde National Report: Beijing+25; UN Women 2019).

Despite significant progress, there is still room for improvement in certain areas. Regarding access to justice, each citizen, female or male, has the right to access justice and the right to obtain effective protection of rights (Article 20 of the Constitution). However, women filled only two of the seven seats on the Supreme Court and remain underrepresented in the judicial sector, especially in prosecutorial positions.

Under the Constitution, female and male spouses have equal rights and duties, and amendments to the Civil Code in 1981 allowed women to freely choose where to live, get a job, sign a contract, register a business, open a bank account, and head a household (OECD 2021). However, discriminatory practices and patriarchal values regarding the role of women in family and society remain deeply rooted.

From a labor law perspective, Cabo Verde prohibits gender discrimination in employment; however, no provision exists in the law specifically mandating equal remuneration for work of equal value (Legislative Decree No. 5 of 2007). Consequently, in some sectors of the formal economy, women receive lower

salaries than men for similar work. In addition, the labor market remains highly segregated by gender, with women still overrepresented in certain sectors, such as the informal economy where salaries are generally lower and labor protections are not necessarily enforced (US Department of State 2021). As an example, a 2018 study on the rights of domestic workers showed that Cabo Verde employed 6.6 percent of people as domestic workers; although 93.9 percent of these employees were women, only 11 percent were covered by the National Institute of Social Welfare (INPS) and therefore able to benefit from a pension and health insurance plan (International Labour Organization 2021).

CONCLUSION

Significant progress has occurred in removing legal impediments to women's economic empowerment in sub-Saharan Africa. Despite these tangible advances, the level of progress in each subregion varies, and all countries should continue working to eliminate legal barriers to women's economic empowerment. For instance, countries in eastern and southern Africa tend to rank slightly better than countries in western and central Africa. The 2023 World Bank's Women, Business and the Law report measuring laws and regulations restricting women's economic opportunities showed that the average score in eastern and southern Africa was 73.8, which is slightly above the average score of western and central Africa (68.7) but below the global average of 76.5 (based on a total of 190 countries; World Bank 2022a). The case studies of Rwanda, Namibia, South Africa, Mauritius, and Cabo Verde highlighted in this chapter illustrate the positive impact of removing such legal impediments on women's economic, social, and political empowerment. As shown in the case studies, removing the existing legal barriers in sub-Saharan African countries while using legal reform to incentivize women's full economic participation is essential to promote women's economic empowerment, facilitate economic growth, and create financial stability.

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Rethinking Fiscal Law Design to Address Gender Inequality

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This chapter examines the design and adoption of tax and public financial management legislation in various countries, including those in sub-Saharan Africa, that aim to reduce gender inequality. It highlights the legal and institutional issues that countries should consider when designing gender-related fiscal policies. Reflecting on the COVID-19 pandemic and its disproportionate social and economic impact on women, the chapter explores the roles that robust budgeting and tax legal frameworks play in fiscal policies that are more sustainably rooted in gender equality. The chapter notes that although legislative reform is not the single determining factor, it is a vital element in ensuring positive outcomes and addressing gender inequality for an inclusive recovery and beyond.

INTRODUCTION

The pandemic crisis has brought the role of fiscal policies in promoting gender balance to the fore. The economic and social consequences of the pandemic affected women disproportionately (Christopherson Puh and others 2022)—either directly, because women are overrepresented in those economic sectors most adversely affected by the crisis (for instance, the services sector) or indirectly, as lockdowns forced women to leave their jobs because responsibility for home schooling, childcare, and general household chores generally fell on them.

In response, several countries, including in sub-Saharan Africa (IMF 2021a), adopted legislation that provides lifeline tax and public spending measures—in some cases limited in scope and time and in others based on a more permanent framework. Although transitional measures were crucial during the most acute phase of the crisis, the COVID-19 recovery presents an opportunity for countries to ensure that their fiscal policies are sustainably

rooted in gender equality—aimed at achieving both equality of opportunities and outcomes.¹

This chapter discusses the legal and institutional considerations countries should take into account when designing gender fiscal policies toward a gender-inclusive recovery and beyond, to create more inclusive and resilient societies. It also examines fiscal law gender-related issues in sub-Saharan African countries.

THE BROADER GENDER LEGAL FRAMEWORK

The legal framework underpinning gender considerations is multilayered. It comprises general foundational principles of equality and nondiscrimination embedded in international conventions, countries' constitutions, and (primary and secondary) legislation.

International Conventions

In addition to international conventions guaranteeing equality more generally, such as the Universal Declaration of Human Rights, several international conventions, such as the Convention on the Elimination of all Forms of Discrimination Against Women, specifically target gender discrimination. Although they do not specifically target fiscal frameworks, these legal instruments create obligations applicable to all governmental policies, including taxation and spending. The international commitment to gender equality was further cemented in the 1995 Beijing Declaration, which included a budgetary commitment for gender-impact analyses and for meeting social needs. International conventions recognize that ensuring equality may also require positive policy action (for instance, United Nations 1989). Both the Millennium Development Goals and the subsequent 2015 Sustainable Development Goals established gender equality and women's empowerment as a standalone, single goal; they also set gender-sensitive targets within other goals.

On the African continent, the 2003 Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa ("The Maputo Protocol") outlines a range of women's rights including economic, social, civil, and political as well as cultural and environmental rights. Within Africa's subregions, multiple legal instruments address women's empowerment and gender equality, such as the Southern African Development Community's Gender Protocol (2008), the East African Community Gender Equality and Development Bill (2016), the Economic Community of West Africa States (ECOWAS) Treaty (1975, revised in 1993), and the Supplementary Act relating to Equality of Rights

¹ The concept of "gender equality of outcomes" is a function of the policy objective pursued and its gender impact in a particular area. For instance, if the policy objective is to have equal representation on boards, then gender equality of outcomes is achieved through such representation. If the policy objective is to incentivize tax savings through the tax system, and the resulting tax benefit is found to accrue predominantly to men, then there is no gender equality of outcomes. This does not mean that, at an individual level, each person (whether male or female) should receive the same tax benefit, but rather that, as a group, one gender should not be systemically disadvantaged.

between Women and Men for Sustainable Development in the Economic Community of West Africa States Region (2015; Organisation for Economic Co-operation and Development [OECD] 2021b).

Constitutional Provisions on Gender and Nondiscrimination

Most countries' constitutions prohibit discrimination, and many also specifically prohibit discrimination based on gender (for instance, Central African Republic, Germany, India, Italy, Tunisia) or contain provisions enshrining broad principles of gender equality (Ghana, Liberia, Malawi). Although no specific references are usually made to fiscal measures, constitutional obligations are generally binding across all state actions and government policies, including tax and public financial management frameworks. Some constitutions incorporate gender balance as a budgetary principle (Austria, Kenya); require the government to implement policies promoting gender equality and to enact laws necessary to achieve it (Ecuador, Morocco, Uganda); create public entities responsible for promoting or coordinating gender equality policies (for instance, the National Human Rights and Equality Commission in Kenya). Ghana's constitution requires the state to ensure full integration of women in economic development.

Legislation

Countries' laws and regulations often prohibit gender-based discrimination generally or in specific policy areas, such as in the labor market (Christopherson Puh and others 2022; IMF 2021a). In addition, these laws and regulations promote, enforce, and monitor measures to address gender inequality—for instance, regarding violence against women, employment and economic benefits, marriage, and family (see UN Sustainable Development Goals Global Indicator Platform 5.1.1²). Other legal measures that support fiscal policies to promote gender considerations include laws that assist in the collection and maintenance of gender-disaggregated data; these so-called equality laws encompass multiple areas of social, political, and economic life (for instance, public contracts, quotas for women in civil service management positions). Other laws promote access to information. For example, in Morocco, Law 31-13 makes information systematically accessible to women and guarantees access, in a timely manner and by all possible means, to data relating to political participation at the level of the parliament, local authorities, and political parties.

The legislation also delineates roles and responsibilities of public entities and promotes their coordination on gender-related issues. While the executive branch of government is generally responsible for administering and promoting gender policies, parliamentary oversight (including through specialized committees in the budget review and approval process) can offer independent review of such policies.

² United Nations Sustainable Development Goals Indicators Metadata Repository, SDG Indicators (un.org).

The judiciary, in turn, may be called on to review the application of laws and policies to individuals and corporations and give appropriate recourse where gender inequalities are identified in defiance of legal norms. Supreme audit institutions may include gender issues in their audits of government policies, either as part of regular assessments or in targeted reviews.³ The legal framework defines the responsibilities of different actors, for example by establishing who leads the design of gender-related policies (for instance, Ministry of Finance, Ministry of Women's Affairs, or another specialized agency) and the role of spending ministries. Fiscal—tax and public financial management—legal frameworks and institutions thus support fiscal policies that aim to reduce gender inequalities. The remainder of this chapter focuses on the role of these fiscal legal frameworks, both in the context of the COVID-19 pandemic and toward a sustainable recovery.

LEGAL UNDERPINNINGS FOR GENDER-BASED BUDGETING AND PUBLIC FINANCIAL MANAGEMENT TOOLS

Gender budgeting has gained prominence in the response to the pandemic (IMF, United Nations Development Programme, and UN Women 2021). As outlined in more detail in Chapters 11 and 12, gender budgeting is an approach to government accounting that uses fiscal policy (tax and spending measures) and public financial management tools to promote girls' and women's development and address the negative macroeconomic outcomes caused by gender gaps and inequalities. Its role in advancing gender considerations plays out in different phases. First, gender budgeting can help governments understand where inequalities lie as well as how, and to what extent, women are being affected in different areas (for instance, in the labor market, health, and education). Second, it can inform the design and implementation of fiscal measures aimed at mitigating the adverse effects of the pandemic on women and girls and help them to get back in the workforce, which in turn would give countries an economic lift during the recovery phase. Third, gender budgeting provides tools for regularly monitoring and evaluating the impact of gender-related policy measures.

Gender budgeting legal frameworks can help address gender inequalities by providing countries with the foundations for gender-related fiscal measures (Alonso-Albarran and others 2021). Gender budgeting legal frameworks consist of the overarching public financial management legislation; public financial management tools and obligations related to gender budgeting implementation; and tax and sectoral legal instruments regarding expenditure policies. Legal reform in this area has led to provisions in constitutional frameworks, public financial management laws, and executive decrees. Although the breadth and scope of gender

³ See, for instance, the Austrian Court of Audit's review of the gender impact of the country's employment tax policies (Austrian Court of Audit 2017).

budgeting practices vary widely across countries, legal frameworks support the implementation of an extensive toolkit that includes ex ante gender-impact assessments, gender-budget statements, budget circulars, budget planning and programming tools, budget execution rules, and reporting and audit requirements to ensure greater accountability. Law can also support efforts to strengthen gender-disaggregated statistics, which can allow for sound identification and assessment of gender-related policies and help remove potential gender biases from policies, regulations, and budgets.

In some countries, legal frameworks enacted *before* the pandemic provided for gender budgeting tools that hardwired a gender perspective into the design of fiscal packages *during* the crisis. Such legal frameworks—in some cases embedded in the country's constitution—have allowed for or required the incorporation of a systematic gender perspective throughout the budgetary process. Here are several examples:

- In 2018, Canada enacted the Gender Budgeting Act, requiring that the principle of gender equality be promoted in the federal budget and that gender and diversity be considered in taxation and resource allocation decisions, including with respect to direct spending. To enhance transparency and accountability, the Act also requires that information on the impact of government decisions on gender be made available to the public. In response to COVID-19, all ministerial proposals contained a Gender-Based Analysis Plus assessment, required by the legislation. Overall, 77 percent of the measures taken during the crisis were gender balanced, and 18 percent were directed specifically to women. Such measures included entrepreneurship programs for women, support to shelters for women and sexual assault centers, and increases in child benefit allowances.
- In Iceland, the 2016 Public Finance Act provides a robust framework for gender budgeting. It mandates that the Minister of Finance, in consultation with the ministry responsible for gender equality, leads the formulation of a gender budgeting program to be incorporated in the drafting of the budget bill. Further, the Act mandates that the budget bill clearly outline its impact on gender-equality targets. During the pandemic, line ministries were instructed to estimate the number of jobs and gender ratio created by their proposed investment measures, with the Ministry of Finance providing its own estimation and ensuring consistency on proposals. An overall Gender Impact Assessment, with regard to social and labor market measures, was introduced for the first time as part of the budget bill (Birna Baldursdottir 2020).
- In Mexico, the 2012 Federal Budget Framework Law requires that all institutional budget proposals detail the measures that promote gender equality and the eradication of violence against women or any other form of gender discrimination. Further, the Federal Budget Framework Law demands that performance evaluation systems incorporate specific indicators that allow an evaluation of the gender impact of budget programs. As mandated by law, during the COVID-19 crisis the 2021 annual budget law included an annex detailing the resources allocated to budget programs targeted to promote

gender equality, including the responsible units for their execution.⁴ Another key component of Mexico's framework is an additional parliamentary oversight procedure. A working group headed by the Speaker of the Lower House of Parliament assessed the governmental measures taken during the COVID-19 pandemic on a weekly basis to ensure that they were designed and implemented in a gender-responsive way.

- In Rwanda, the Organic Law on State Finances and Property declares “gender balance” as a fundamental principle in the use of public funds. It also requires that during the budget preparation process, the budget framework paper, approved by cabinet and submitted to parliament, includes a gender-budget statement. All ministries, departments, and agencies must submit annual implementation reports. In line with this framework, the 2020/21 Gender Budget Statement identified gender priorities, including for agriculture, health, infrastructure, trade, and industry.⁵ Rwanda's constitution also established the Gender Monitoring Office in 2003 to monitor the compliance of gender-related commitments across public, private, and nongovernmental institutions.
- Uganda's constitution requires the government to take affirmative action in favor of marginalized groups, including women, and further provides that the state must provide facilities and opportunities to enhance the welfare of women, which promotes the advancement and empowerment of women, and ensures their equal opportunity in political, economic, and social activities. The constitution also provides for the promulgation of the Equal Opportunities Act and the establishment of the Equal Opportunities Commission, which is mandated to eliminate discrimination and inequalities on the basis of gender. More specifically, the Public Finance and Accountability Act provides that the annual budget must specify measures taken to ensure equal opportunities for both women and men. An analysis must be done by all government agencies—and must be accompanied by a certificate issued by the Minister of Finance in consultation with the Equal Opportunities Commission—on the extent to which the budget considers gender and equity. This is a requirement at the formulation and approval stages of the budget cycle—guided by a budget circular to government agencies and institutions that includes gender-responsive budgeting guidelines. A national program to support women's economic empowerment through entrepreneurship was a source of resilience during the pandemic (Willman, Atamanov, and Myers 2022).
- At the local level, several countries (Argentina, Austria, Germany, Italy, Liberia, Norway and Spain) have a legal mandate for gender budgeting, which reinforced the adoption of gender-responsive fiscal measures during the pandemic.

Other supporting mechanisms in the legal framework are crucial to the effectiveness of gender-responsive budgeting. Laws that promote strong institutional capacity and mandate the collection of sex-disaggregated data are important in

⁴ See Mexico, Ministry of Finance 2021.

⁵ See gender mainstreaming in Rwanda's fiscal year 2020/21 budget (Republic of Rwanda 2020).

carrying out legal and policy measures. In some countries, the statistics law was amended to require that all data-collecting institutions analyze and publish statistics by gender, for example Israel. In other countries, laws on equality set the framework for collecting gender-disaggregated data, enabling these countries to formulate gender-responsive policies and track progress made toward gender balance, for example, Thailand.

The legal framework in some countries supported the scrutiny of gender-related policy measures and promoted accountability during the pandemic, mostly through two mechanisms:

- *Parliamentary scrutiny of COVID-19 support packages.* Oversight by parliament or special committees helped to determine whether women's and girls' specific needs were being addressed and gender equality promoted. In most jurisdictions, parliaments exercised this scrutiny throughout the budget process, such as in the revisions of medium-term expenditure frameworks, in the approval of annual and supplementary budgets, and during the review of budget execution reports. For example, under both the Canadian Gender Budgeting Act and the Korean National Finance Act, budgetary plans and reports submitted during the pandemic by the executive to the legislature laid out the impact of the budget on gender equality.⁶ In Austria, the Parliamentary Budget Office analyzed gender budgeting and gender equality and gave disaggregated gender data to parliament, allowing its members to adequately debate gender issues.⁷ Canada and Iceland established parliamentary committees to deal with COVID-19 responses and worked with ministries to review the impact of proposed fiscal measures on gender equality.
- *Gender-related audits by supreme audit institutions.* Gender audits can help determine the degree of government compliance with national legislation related to gender equality, examine the gender impact of government programs, and make recommendations on the design and implementation of policy measures. If broadly formulated, the legal mandate of Supreme Audit Institutions would grant appropriate powers and tools to plan, define, and undertake a gender audit. In Austria, guidelines for the Austrian Court of Audit recommend that performance audits include gender considerations such as an impact analysis or the relevance of gender objectives. In Rwanda, all public entities are required to submit an audit report detailing how the programs for gender equality have been implemented. Mauritius formed a parliamentary gender caucus, through the amendment to the Standing Orders and Rules of the National Assembly, to conduct gender audits and ensure the promotion and attainment of gender equality by carrying out gender assessment of government policies.⁸ The Office of the Auditor

⁶ Examples include in Canada Annex 4, titled "Gender, Diversity, and Quality of Life Statement" of the 2021 Budget proposal. Such assessments include the COVID 19 "influence map."

⁷ See official page of the gender assessments published by the Austrian Parliament in 2021 (Austrian Parliament 2021).

⁸ See Mauritius's National Gender Policy 2022–2023 (Republic of Mauritius, Ministry of Gender Equality and Family Welfare 2022).

General of Uganda announced a performance audit of nationally agreed-upon targets, to assess the increase of intimate partner violence against women during the COVID-19 crisis (IMF 2021a; UN Women 2021)

Beyond the crisis, the legal framework can support the implementation of gender budgeting tools and their enforceability. Gender measures taken during the COVID-19 pandemic by countries with ex ante and comprehensive gender budgeting legal frameworks will bear effects beyond the crisis. These frameworks can help governments introduce and institutionalize gender budgeting elements in their fiscal policymaking. Without binding legislation—and absent strong political commitment and social movement mobilization⁹—governments may easily disregard gender budgeting efforts in times of crisis (Polzer, Nolte, and Seiwald 2021).

Female representation in decision-making public institutions is important to ensure that decisions are made with a balanced gender perspective during the budget process. In sub-Saharan African countries, some constitutions require a specific quota of women representatives in legislative bodies and government offices; these quotas are crucial in implementing policies that promote gender equality. Kenya's constitution provides that the State should take legislative action and other measure to ensure that no more than two thirds of the members of elective and appointed bodies are of the same gender.¹⁰ In Rwanda, in addition to a general commitment to ensuring gender equality as a fundamental principle, women must occupy at least 30 percent of the positions in decision-making bodies.¹¹ Low levels of female representation or participation in the planning process may sometimes create a challenge for mainstreaming gender into national and subnational budgets; these low levels of participation often mean that women's views and concerns are not reflected in the budgets, and so the budgets, in turn, are not responsive to women's issues (Kusambiza 2013).

TAX-LAW DESIGN CONSIDERATIONS FOR ADDRESSING GENDER EQUALITY

The following legal-design issues emerge in the design of gender-sensitive tax measures, including in the context of the COVID-19 crisis.¹²

- *The choice of legal instruments.* Depending on the choice of measure, countries have introduced tax measures in response to COVID-19 through primary or

⁹ For instance, in 2021, Argentina, while not having comprehensive gender budgeting legislation, increased its budget dedicated to closing inequality gaps to 12.5 percent of its total budget envelope.

¹⁰ Article 27 of the Constitution.

¹¹ Article 10(4) of the Constitution.

¹² Gender-responsive measures seek to remove gender-differentiated effects and systemic barriers by directly and explicitly targeting a particular gender. Gender-sensitive measures are not explicitly targeted at a particular gender; instead, they are designed to have a positive impact on women's economic security, for instance, because they target sectors in which women are overrepresented.

secondary law instruments (laws or regulations), either as standalone measures or as part of omnibus crisis bills. Some measures have also been introduced through administrative actions (such as administrative guidelines), aiming to achieve greater flexibility and swifter dissemination of information to taxpayers (IMF 2021b).

- *Sectoral measures or specifically targeted measures.* Several countries introduced sectoral measures to further protect households and/or small businesses. Examples include tax benefits or incentives for sectors in which women are overrepresented, such as health care, education and other service sectors (IMF 2021a). Other countries introduced measures designed to provide immediate tax relief and/or assist women to remain in the workforce—for instance, giving or increasing childcare allowances (Canada) or maintaining the possibility to offset childcare costs through tax credits even if parents had to reduce their working hours (UK).
- *Temporary or more permanent measures.* Tax measures aimed at restoring fiscal space for crisis recovery have in some countries been temporary, while others have taken more permanent measures (IMF 2021a). For instance, contributions on personal or corporate income or taxes on wealth have been introduced as one-off measures (Argentina) or as a new and more permanent feature of the tax system (Bolivia). Given that these types of measures can be targeted to those with high incomes or wealth—that is, those with the greatest ability to pay—hence increasing the redistributive impact of tax systems, they can be expected to have positive impacts on gender equality to the extent that women are overrepresented in the lower income brackets.

Regardless of the type of measure, the delivery of timely, easily accessible, and tailored taxpayer guidance is important. Providing legal certainty is even more important in times of crisis, when taxpayers—as well as tax administrations—must rapidly adjust to changing economic and social circumstances. The UK, for instance, has issued various guidelines on COVID-19–related tax measures, including working-from-home tax relief claims (linked to the increase of household expenses) and childcare allowance claims.

The economic recovery from COVID-19 offers an opportunity for countries to ensure greater gender balance in tax-law design. Although most countries' tax laws are written in a gender-neutral¹³ way, without explicitly differentiating based on gender,¹⁴ they may still exacerbate gender inequalities because of underlying

¹³ According to the United Nations Children's Fund (2017) the term "*gender neutral*" is anything—a concept, an entity, a style of language—that is unassociated with either the male or the female gender. The nature of systemic and embedded or internalized bias is such that, unfortunately often, what is perceived to be gender neutral is in fact gender blind."

¹⁴ According to the World Bank (2015), 16 out of 173 countries had tax systems that directly favored men. A range of discriminatory tax laws have persisted in many countries (such as an explicit or implicit tax deduction or a credit to the male head of household) along with legal barriers restricting women's ability to work.

socioeconomic differences between men and women or because of different household structures. Faced with such implicit gender biases, countries that introduce or maintain gender-neutral tax laws may not be doing enough to remove gender-differentiated effects, and positive policy action may be required—whether in the form of gender-responsive or gender-sensitive measures. A better understanding of how current tax rules affect genders differently is a necessary first step because tax-law reform is one possible response, among other policies, to address implicit gender biases. Therefore, when (re)designing tax laws, governments should consider the following strategies to address gender inequality:

- *Identify existing gender biases in the tax system.* A “gender balance in tax” legal diagnostic of existing laws and procedures would be a useful first step in designing more gender-effective reforms.¹⁵ The objective would be to identify explicit and implicit gender biases¹⁶ in the substantive tax laws and tax procedures of a country, against the background of its legal, social, and cultural environment, supported by gender-disaggregated data to the extent available. The diagnostic can then inform the choice of appropriate policy measures—including whether they should involve tax, spending, other structural measures, or a combination of them—and inform their legal design in support of the country’s domestic and international commitments to gender equality. For example, a gender analysis of the personal income tax system could consider the following main issues: level of tax relief for minimum basic living expenses; the impact of flat-rate and presumptive forms of taxation; joint taxation of adult couples; and the “tying” of social benefits to income (UN Committee of Experts on International Cooperation in Tax Matters 2018). Other areas of focus could include corporate taxation (for instance, tax incentives in sectors that are male dominated), property taxation (for instance, real estate or assets predominantly owned by women subject to higher tax rates; Niesten and Hyland 2022), and inheritance and indirect taxes. According to the UN Inter-Agency Task Force on Financing for Development (2018), guidelines and methodologies for medium-term revenue strategies, tax policy assessment frameworks, and tax administration diagnostic tools should eliminate gender biases.
- *Ensure effectiveness of gender-related tax policies through tax law design.* Although the specific policy mix will vary according to countries’ circumstances and policy objectives, anywhere tax measures are involved, law reform will likely be required. This would generally be the case, for instance, if the objectives were to increase tax progressivity; to move from family-unit to individual-unit taxation, for instance to eliminate tax-induced work disincentives for secondary (predominantly female) earners, including social security disincentives to spousal work; and to design a broad-based, low-rate VAT, with appropriately targeted exemptions or reduced VAT rates.¹⁷ More generally, tax reforms can

¹⁵ See Stewart 2017.

¹⁶ See Stotsky 1996.

¹⁷ On specific tax policy considerations to address gender gaps, see Delgado Coelho and others 2022; and IMF, United Nations Development Programme, and UN Women 2021.

help to secure additional revenues for public financing of services and infrastructure to help address gender inequalities.¹⁸

- *Address tax biases arising from other policies.* Implicit and explicit biases arising from the tax system may also exacerbate gender biases arising from other policies or circumstances. For instance, the tax system may broaden the so-called pension gap if pension savings are tax favored, that is, if that tax benefit mostly accrues to men earning higher salaries. The pension gap is further widened if the country's statutory retirement age for women is lower than for men.¹⁹ Addressing such biases may therefore require reforms beyond the tax system to labor legislation and/or social protection frameworks, to encourage women's labor force participation and pay equality, including in retirement. A diagnostic of existing biases could involve specific considerations and therefore examine the country's labor force structure and household structure. The impact of social security contributions on gender outcomes as a priority for further work has already been highlighted (OECD 2022). For example, South Africa noted the potential for social security contributions (for instance, earned income tax credits) as a possible means to improve female labor force participation. More generally, a highly progressive personal income tax, such as in South Africa, can be seen as correcting for gender biases in the labor market (Delgado Coelho and others 2022; OECD 2022).
- *Educate taxpayers.* This communication can be particularly important for empowering women. Gender-responsive and gender-sensitive tax reforms should be communicated clearly, with information carefully adapted to the target audience. This includes information on tax procedural obligations that are often noted as a disincentive for women to engage in business activities. For example, in Rwanda it has been noted that the understanding of tax procedures and tax liabilities is limited for some taxpayers, including businesswomen (Rwanda Private Sector Federation 2019). The complexity of tax procedures—such as the use of multiple tax forms, which are not always available in all official languages—may increase tax compliance burdens, in particular on women entrepreneurs. This is principally due to lower levels of education, resulting in some women taxpayers having to rely on help from a third person to complete tax obligations. Countries should also consider drafting laws and regulations in gender-neutral language, to better communicate the scope and application of tax provisions.²⁰ Box 14.1 presents key considerations on the use of gender-neutral language in tax laws.

¹⁸ Tax financing of specific initiatives is different from standard revenue earmarking through legislation, which usually causes excessive budgetary inflexibility and inefficiencies.

¹⁹ The term *pension gap* refers to women typically living on lower income in retirement than men do, at least partly driven by the gender pay gap; see OECD 2021a.

²⁰ UN Women (n.d.) offers examples for the English language.

Box 14.1. The Use of Gender-Neutral Language in Tax Laws

Gender-inclusive language, or gender-neutral language, is language that avoids bias toward a particular sex or social gender and therefore is less likely to convey gender stereotypes.

Gender discriminatory language is commonly found in tax legislations, especially in the wording of various provisions applicable to individuals. Examples include: (a) references to “husband” and “wife” instead of “spouse,” “married man” and “married woman” instead of “married person,” and/or “the taxpayer and his wife” instead of “the taxpayer and their spouse” and (b) the use of male pronouns such as “he” and “his” instead of “he/she,” “him/her,” or “they/their.”

When drafting tax legislation, drafters should consider:

- Avoiding gender-specific nouns when making generic references and, when possible, using a gender-neutral alternative.
- Avoiding the use of masculine reference words, by (1) omitting the masculine reference word; (2) using plural forms for both nouns and reference words; (3) using “they/their” to refer to singular nouns (that is, using the singular “they”); and (4) using the passive voice.
- Using both feminine and masculine forms, either as separate words or via slashes.

The choice of drafting strategy should be informed by the nature of the text and the structure as a whole; however, gender-inclusive writing should not impede the text’s readability.

Sources: UN Women (n.d.); and authors’ compilation.

Note: The examples provided above are for drafting in the English language. See UN Women (n.d.) for examples in French and other languages.

A high-level review of select sub-Saharan African tax laws raises the following general observations:²¹

- *Many tax laws do not formally discriminate by gender.* In several countries, including Ghana (Grown and Valodia 2010), Liberia, Rwanda, Uganda, Namibia, and Lesotho, the tax law does not explicitly discriminate against women: individuals are not identified by gender or by marital status, and they may file taxes separately. Gender-neutral language is generally used (for instance, the term *spouse*, rather than *husband and wife*, is used), and the personal tax system may be described as marriage neutral—that is, if the income of an individual remains unchanged before and after marriage, the tax paid before marriage will remain the same after the person gets married. In some cases, single-unit taxation applies; however, sometimes the language used is not gender neutral (for instance, in Senegal and Madagascar).
- *Several countries have taken active steps toward eliminating discriminatory tax-law provisions.* In Kenya, a recent amendment to the Income Tax Act

²¹ For the purposes of this chapter, income tax and tax procedure legislation from selected sub-Saharan African countries are presented through the country examples. This review focused on: (1) the use of gender-neutral language, (2) household-unit or single-unit taxation, (3) benefit from personal or child allowances, and (4) spouses’ joint or separate filing and assessment.

abolished a discriminatory provision on personal income tax relief for insurance premiums received. Until 1995, South Africa levied higher taxes on “married women” than “married persons,” which primarily applied to married men—with the former taxed more and the latter less than an unmarried person; a woman qualified for the married-person treatment only as an exception, for example, if she was able to prove that she was the main income earner in her household (Valodia, Smith, and Budlender 2001). A single tax structure available to all taxpayers, regardless of their gender or marital status, was subsequently introduced.

- *However, explicit gender biases remain in tax legislation.* In the Democratic Republic of the Congo, the income of a married woman living with her husband is deemed to be the income of the husband and the tax is assessed on him. In Kenya, an explicit provision in the Income Tax Act titled “wife’s income” provides for assessment in the name of the husband in case of joint filing. It also establishes joint and several liability of the wife for the payment of her share of the tax even though the assessment is not made in her name. A similar provision titled “income on married women (and minor children)” is also found in the Malawi Taxation Act. In the Republic of Congo and in Gabon, income is taxed on the household as a unit, which, according to the tax code, comprises the head of the family, “his” spouse, and dependents, with separate assessments made only in exceptional cases. This is also the case in Comoros and Chad but with a broader range of exceptions.²² In some cases, the term “head of the family” is expressly used to describe the husband (for instance, in Benin and Guinea). Other discriminations exist in the form of joint and several liability of the wife for the payment of income tax assessed in the name of the husband. In the Central African Republic, a woman living with her husband is jointly and severally liable for the payment of personal income tax established in his name.
- *Despite formal gender equality under the tax law, structural issues may lead to implicit discriminations.* In Guinea, although a single-unit taxation option is available, incentives encourage a household-unit taxation option, in the form of allowances for the wife (or wives) in favor of the husband. In addition, minor or infirm children are in principle considered dependent on the husband. Personal income tax systems imposing higher tax burdens on single-earner households may also lead to implicit discrimination where those households are predominantly headed by women (see Valodia, Smith, and Budlender 2001, where South Africa is mentioned as an example).
- *In some cases, tax laws contain positive discrimination measures.* Examples include Equatorial Guinea, where single women with more than three dependents under the age of 18 are exempt from the individual tax; Mozambique, where municipal personal tax is levied on resident individuals between ages 18

²² See Delgado Coelho and others 2022.

and 60, but women in domestic service are exempt from it;²³ and Mali, where women who have had at least four pregnancies are exempted from the regional and local development tax.

- *A series of other tax-relevant positive measures are identified in tax and related legislation.* Uganda recently amended the Tax Appeals Tribunal Act to provide that at least 40 percent of the members of the tribunal shall be women. In Kenya, no more than two-thirds of the members of the National Land Commission—which is responsible for the assessment of tax on land and premiums on immovable property—may be of the same gender; this rule reflects a constitutional requirement for balanced gender representation in national committees.

CONCLUSIONS: THE ROLE OF FISCAL LAW AND INSTITUTIONS IN ADDRESSING GENDER INEQUALITIES

Robust public financial management legal frameworks can support the design, implementation, and monitoring of gender-balanced fiscal policies. Law reform can help introduce and institutionalize gender budgeting practices, possibly through gradual implementation. Although broad differences in gender budgeting practices occur across countries, public financial management legal frameworks could: (1) introduce gender balance as one of the principles of the budgetary system, (2) require the preparation of gender-budget statements and/or ex ante gender impact assessments, (3) require the inclusion of instructions on gender classification and tracking in budget circulars (or similar instruments) for budget preparation, (4) require the use of gender performance-based indicators for budget execution, (5) require that in-year and end-year budget execution reports incorporate gender-related information, (6) require ex post gender-impact assessments, (7) strengthen the legal mandate of Supreme Audit Institutions to conduct gender-related audits, and (8) strengthen enforcement mechanisms such as sanctions—a phase of gender budgeting often overlooked.²⁴

Legislation itself cannot ensure successful reform; however, it is an important element for determining effective outcomes in implementation and enforceability. If instead countries rely on high-level political commitments rather than binding laws, the risk is that—absent a robust institutional framework and other prerequisites—the gender perspective will be disregarded, particularly in times of crisis.

Legal considerations also help policymakers to address implicit gender biases in the tax system. While explicit gender biases should generally be addressed by removing discriminatory tax laws, addressing implicit gender biases arising from

²³ See Delgado Coelho and others 2022.

²⁴ This list is based on the common gender budgeting practices and legal provisions inventoried by Alonso-Albarran and others (2021).

or exacerbated by the tax system may require a more calibrated policy response. Understanding the gender impact of current tax rules is an important first step in this respect. A gender balance in tax legal diagnostic can usefully complement gender budgeting tools—not only in the response to a crisis but also on a permanent basis—to achieve equality of opportunities and outcomes. More generally, improving tax capacity to raise additional revenues can help to reduce gender inequalities; these additional revenues can generate additional financing for public spending on the services and infrastructure needed to address those inequalities.

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PART V

The Power of Policies: From Fiscal to Fintech

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Tax and Expenditure Policy for Gender Equality

Brooks Evans, Alexander Klemm, Carolina Osorio-Buitron, and Mauricio Soto

This chapter focuses on how tax and expenditure policies can improve the lives of women and girls in sub-Saharan Africa. Not only are tax policies important for generating revenue, they also interact with structural barriers women face—some of which are unique to Africa. Eliminating tax incentives in sectors where women are underrepresented and removing explicit biases in the tax system (and laws that interact with it) enhance economic efficiency and gender equality. Higher carbon prices could help mitigate the effect of climate externalities on African women and girls. On the expenditure side, the closing of gender gaps requires using domestic and external financing; public expenditure needs to be calibrated in terms of magnitude and composition, as it can have a large impact in the short and long term. These efforts include major spending items, such as health, education, social protection, capital expenditure, and the public sector wage bill. As fiscal space tends to be more limited in the region, many sub-Saharan African countries must be judicious in how to allocate spending that is adequate, efficient, and sustainable to maximize impact.

INTRODUCTION

How governments raise taxes and spend revenues has profound impacts on gender equality. This chapter considers these interactions from the perspective of sub-Saharan Africa, ascertaining similarities and differences with other regions. Social norms, regulatory barriers, and capacity and enforcement constraints in sub-Saharan Africa sometimes differ from those of other developing economies. Such differences can affect the relationship between fiscal policy and gender equality—hence, the policy implication expects to address gender disparities.

Research support was provided by Kardelen Cicek.

From the tax policy side, the key general messages on the interaction between tax and gender, as summarized in Coelho and others (2022), are as follows:

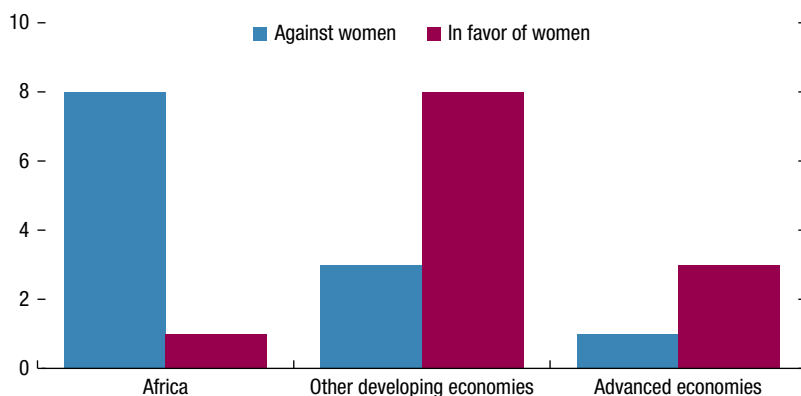
- Gender discrimination in income tax codes and the legal environments for property and asset ownership is declining.
- The taxation of labor income has the largest impact on gender inequality. Progressive and individualized systems can help reduce gender employment gaps. The unequal distribution of capital income between men and women is exacerbated wherever the taxation of capital income is more lenient than for labor income (dual systems).
- Excises penalize (or subsidize) women when tax rates deviate from the social cost of consuming harmful goods, and consumption patterns differ by gender.
- Reducing selected consumption tax rates is likely inefficient and ineffective, while a broad-based value-added (or retail sales) tax (VAT) with minimal exemptions can mobilize sufficient revenues to finance gender-inclusion programs.

General insights from studies on expenditure policy are that it can economically empower women and girls via multiple channels:

- Social protection and labor programs can reduce poverty, enhance human capital formation, improve employability, and empower women and girls.
- Education spending not only improves learning for girls and women but also tends to lead to higher-paying jobs and more informed decisions regarding nutrition and health care.
- Health expenditure improves life quality and productivity of mothers and their children.
- Wage bill policy can play an equalizing role in both labor income and employment in the public sector.
- Capital expenditure can reduce unpaid work via improved water and sanitation, while digital access can boost employment.

The rest of this chapter is structured as follows. The following two sections cover tax policy, first focusing on explicit discrimination in the law across regions—both with respect to the tax system and ownership rights. This comparison highlights unique characteristics of the (sub-Saharan) African region—in terms of social norms and the economic and regulatory framework—which are essential to understanding implicit gender biases in tax systems. Second, implicit tax biases in Africa are considered, with a description of how the different taxes interact with underlying economic characteristics or circumstances to either reinforce or correct gender inequality. The following sections then cover how expenditure policies can further improve female economic empowerment via multiple channels. The final section concludes and formulates key policy implications for African countries.

Figure 15.1. Number of Countries with Explicit Discrimination Provisions in Labor Tax Codes



Source: IBFD.

TAX POLICY

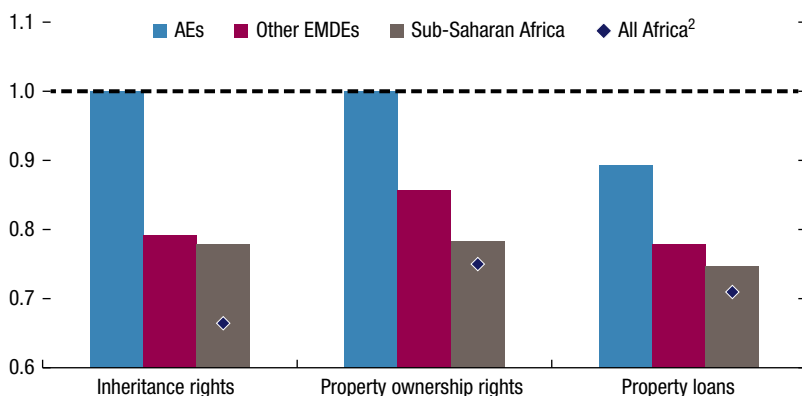
Explicit Gender Discrimination in the Tax System

Although explicit gender biases in tax systems have receded in most parts of the world, they remain more common in some African countries. The cases of explicit gender differentiation have been found mostly in labor income taxation, where there is differential treatment of husbands and wives, or where the “household head” is legally defined as a man. In Africa, this type of explicit tax bias remains, whereas globally tax provisions that discriminate against women often have been replaced by tax provisions that support them (Coelho and others 2022). Eight countries in Europe and Asia abolished tax laws explicitly biased against women, and the number of countries that adopted income tax laws in favor of women increased from 4 to 12. In Africa, South Africa moved toward making the system notionally neutral.¹ Other countries with tax systems that explicitly discriminate against women have refrained from reform (Figure 15.1).

Currently, gendered laws cover tax provisions, rates, and thresholds as well as set forth who is responsible for filing and paying taxes. In Tunisia, only the head of the household—legally defined as a man—is entitled to a special allowance and a child tax deduction. In multiple African countries, the responsibility for filing taxes explicitly lies with husbands. In Kenya and in the Democratic Republic of the Congo, the reason is that a married woman’s income belongs to her husband. In the Republic of Congo, Gabon, Comoros, and Chad, labor income is taxed at the household level, and the law explicitly refers to the husband.

¹ Until 1995, South Africa levied higher taxes on “married women.”

Figure 15.2. Gender Gaps in Ownership Rights and Property Loans in 2019¹
(Regional average; 1 = gender equality)



Sources: World Bank; and IMF staff calculations.

Note: AEs = advanced economies; EMDEs = emerging markets and developing economies.

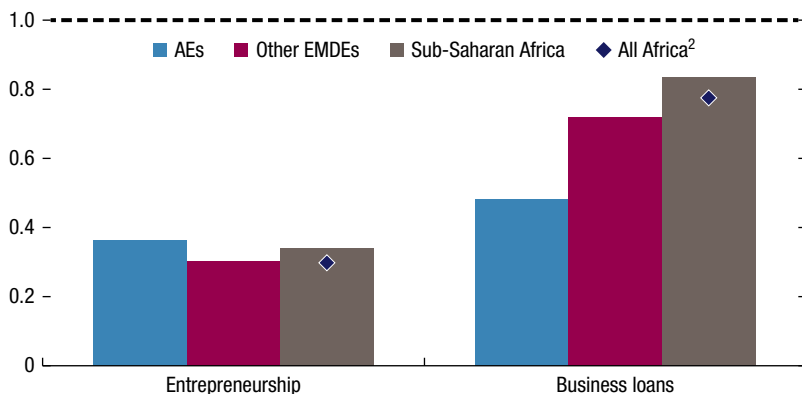
¹The gap in loans represent the ratio of the percent of women with access to finance relative to the percent of men. Property loans include borrowing to buy a dwelling or land.

²Includes six countries from North Africa for loans and eight for rights.

Africa also has a high concentration of countries with legal frameworks that restrict women's property rights. As of 2020, 66 economies worldwide legally limited women's property rights—including asset ownership and inheritance rights—with more than 40 percent of those countries located in Africa (see also Chapters 12 and 13). Figure 15.2 displays (normalized) regional average indicators of gender gaps in asset ownership rights—a *de jure* measure—as well as access to property loans—a *de facto* measure. For these indicators, a value of 1 is regarded as a benchmark of gender equality. Property rights laws are more disadvantageous for women in African countries than in other developing countries—notably if north African countries are included in the calculation—and they are associated with the wider gender gap in property loans. Legal restrictions to the property rights of women limit their access to external finance, as banks require collateral from those who seek financial assistance. Although in several African countries the marriage of a daughter entails a transfer of wealth from the man's family, these transfers do not (necessarily) benefit the bride directly, nor do they help to narrow gender gaps in wealth, given the myriad legal and cultural barriers to female asset ownership. For instance, in Botswana, customary rules entail lower inheritance entitlement shares for girls, and wives require their husbands' authorization to own property or open bank accounts, as married women are viewed as children (Rametse and Huq 2015).

Female entrepreneurship in Africa is undermined by ownership constraints that inhibit access to external finance (see Chapter 2). For the average African country, less than 40 percent of firms have at least one woman among the principal owners. While this phenomenon appears pervasive across the world

Figure 15.3. Gender Gaps in Entrepreneurship and Business Loans in 2019¹
(Regional average; 1 = gender equality)



Sources: World Bank; and IMF staff calculations.

Note: AEs = advanced economies; EMDEs = emerging markets and developing economies.

¹The gap in entrepreneurship represents the share of firms with a woman among the principal owners. The gap in loans represents the ratio of the percent of women accessing loans relative to the percent of men. Business loans capture borrowing to start, operate, or expand a business or farm.

²Includes one country from North Africa for entrepreneurship and six countries for loans.

(Figure 15.3), the underlying forces in Africa are unique. Moreover, some scholars have argued that although female entrepreneurship is opportunity driven in developed economies, women become entrepreneurs to survive dire economic circumstances in developing economies (Adebayo 2015). As documented previously, women in some African countries are excluded from credit markets because they cannot own property or instruments for mortgaging. Other loans appear more equally distributed across genders, both in absolute terms and compared to other regions (Figure 15.3), likely reflecting new financing alternatives among female entrepreneurs, such as crowdfunding and peer-to-peer lending (Ojong, Simba, and Dana 2021). Notwithstanding these positive developments, the gender bias in collateralized lending continues to undermine female entrepreneurship, as the market for alternative financial vehicles remains small.

Moreover, myriad social norms and institutional factors deter female entrepreneurship in the region. The process of starting a business in many African countries is lengthy, cumbersome, and costly, and these constraints can place additional burdens on female entrepreneurs (Amine and Staub 2009). For example, household and childcare responsibilities mean that female entrepreneurs have less time available than their male counterparts do to deal with bureaucratic procedures (as in Tanzania). Many African entrepreneurs also operate in a context of pervasive corruption, and females may be less able than males to deal with corrupt officials, because females cannot meet disproportionately higher bribery demands

(as in Uganda and Cameroon) without putting themselves at risk (as in Nigeria). Poor public transportation systems primarily affect the mobility of female entrepreneurs, as cars are most often owned by men (as in South Africa). Moreover, cultural forces contribute to the entrepreneurship gender gap in the region, as the legitimacy of female-owned businesses is often questioned or undermined—for instance, because female entrepreneurship may be perceived as a threat to husbands' authority.²

Legal, institutional, and sociocultural hurdles impede the formalization and growth of female-owned businesses. In most African countries, female entrepreneurs tend to be concentrated in micro, small, and medium enterprises, mostly in the informal sector—where entry requirements are low (Vossenbergh 2013). Female-owned businesses tend to be concentrated in the retail and services sectors, as women sometimes tend to shy away from high-tech concerns (Scarborough 2012). The high degree of informality and sociocultural headwinds do not set up women for success. Empirical evidence for some countries suggests that, under those conditions, businesses operated by women are more likely than those operated by men to fail.³ This result stands in stark contrast with studies from countries in other regions, showing that businesses are more profitable when management includes women. A potential interpretation for this difference is that the gender-based restrictions and barriers characteristic of sub-Saharan Africa are preventing women from fulfilling their entrepreneurial potential.

Implicit Bias in Tax Systems

Implicit biases are more difficult to identify and address than explicit biases. Explicit bias refers to the case where the tax code discriminates individuals on the basis of gender, including by making reference to family roles, such as husbands and wives. In theory, this type of bias is not difficult to identify or address, but the resolve to address it may face political opposition in countries with gendered social norms. When tax systems are notionally neutral, they may still entail an implicit bias—either because they provide different incentives to men and women or because their incidence across genders is considerably unequal, given their economic characteristics.⁴

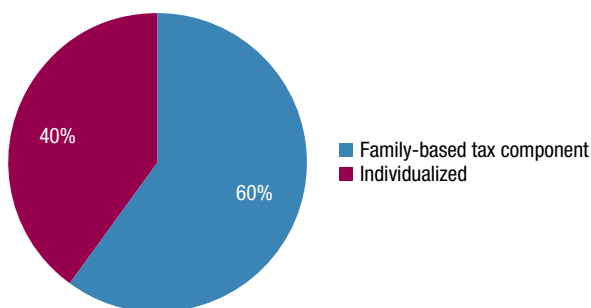
Labor income tax systems in many sub-Saharan African countries entail implicit biases against women. As is well known, family-based tax regimes tend to discourage (formal) employment by secondary earners, who are often women. In *individualized tax systems* all agents are subject to the same tax rate

² For a detailed discussion and review of the empirical literature, see Etim and Iwu 2019; Ojong, Simba, and Dana 2021.

³ Muhamad (2016) and Langevang and others (2015) provide empirical evidence for Ethiopia and Ghana, respectively.

⁴ Stotsky (1997, 1) defines *explicit bias* as “specific provisions of the law or regulations that identify and treat men and women differently.” The line between explicit and implicit differentiation is not always clear, though. For example, charging a tax for a product or service consumed by only one gender comes quite close to explicit bias and could be included in broader definitions.

Figure 15.4. Implicit Bias in Labor Income Tax Systems across Sub-Saharan Africa in 2020
(Percent)



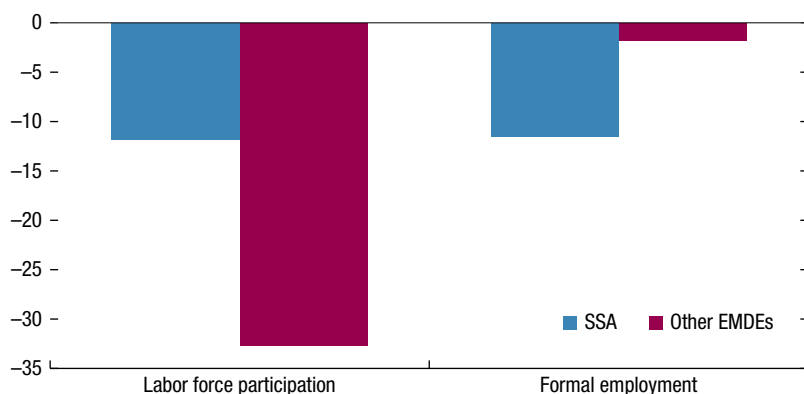
Source: IBFD.

schedule, so tax-induced work disincentives are the same for men and women. In contrast, where income taxes are filed at the household level or the tax code includes provisions (allowances, tax credits, deductions) to the family, secondary earners tend to be discouraged from working. Where the tax rate schedule is progressive, family-based components in the tax code entail a cost for secondary earners who are considering entering the labor force or working more hours, because the primary earner “uses up” the lower income brackets, and any tax benefits to the family may be lost. About half of sub-Saharan African countries require spouses to file taxes jointly, provide tax provisions at the household level, or both (Figure 15.4).

Individualization of personal income taxes is not enough to level the playing field for African women in the labor market. Theoretically, individualizing personal income taxes would increase female labor force supply. While there is scope to introduce such reforms in half of the countries in the region, their effects may be limited. Sub-Saharan African countries collect a small share of revenues through the personal income tax. At 2.5 percent of GDP, the average country in the region collects significantly less than do OECD countries (8 percent of GDP) and the average developing economy (3 percent of GDP). The low personal income tax take in sub-Saharan Africa is partly explained by the low number of contributors. In some cases, high exemption thresholds explain the low coverage of the tax, with eight sub-Saharan African countries having very high exemption thresholds that range from 1.5 to more than 4 times the average wage.⁵ But there are also cases in which the tax system is too “punitive” at the low end of the distribution—where women tend to be overrepresented⁶—pushing potential

⁵ Among these economies, only Zambia has an individual regime.

⁶ The female population is overrepresented at the bottom of the income distribution because they tend to exit the labor force more frequently, work fewer hours, and work in lower-paying jobs than do men.

Figure 15.5. Regional Average Gender Gaps in Labor Markets, 2019*(Female-to-male rate; 0 = gender equality)*

Sources: World Bank; and IMF staff calculations.

Note: SSA = sub-Saharan Africa; EMDEs = emerging market and developing economies.

taxpayers outside of the formal work force. A regional comparison of gender employment gaps shows that the labor force participation gap in the average developing country is 3 times wider than that in sub-Saharan Africa, where the female labor force participation rate is just 10 percentage points lower than for males. However, in terms of the probability of formal employment, women in Africa are at a disadvantage relative to women in other regions, where men and women are, on average, just as likely to be employed in the formal sector (Figure 15.5).

Making personal income taxes more progressive can reduce gender gaps in the quantity and quality of employment. Progressive tax systems can narrow the net income pay gap—through redistribution and by encouraging lower-income individuals to seek formal employment. Because of a weak (or lacking) social safety net and pervasive corruption, taxpayer morale in most African countries is low, which puts a premium on tax policies that affect the bottom of the distribution. Several tax systems in sub-Saharan Africa seem to lack sufficient progressivity and have very low exemption thresholds. About 20 countries require individuals who earn less than 30 percent of the average wage to pay tax, and 4 of these countries do not even have an exemption threshold (Cameroon, Guinea-Bissau, Republic of Congo, and Eritrea). In addition, social security rates are very high in some cases. Because these contributions are paid by all employees regardless of their income level, high rates can severely undermine the progressivity of the tax. High employer rates may also discourage formal employment from the demand side. In the Republic of Congo (total) rates are as high as 44 percent.

The tax system can exacerbate gender inequality in entrepreneurship and property rights, potentially creating trade-offs for tax policy design. As documented earlier, legal frameworks that restrict women's ownership rights, as well as the myriad socioeconomic and cultural factors that hinder female entrepreneurship,

imply that capital income is primarily concentrated among African men. Against this backdrop, the dual-income tax system implemented by most economies—whereby capital income is, on average, taxed at lower rates than labor income—exacerbates explicit gender discrimination. Moreover, the lower and differential taxation of capital income has made overall income tax regimes less progressive (IMF 2017, 2021), disproportionately hurting women, who are concentrated at the low end of the income distribution. The alternatives are a comprehensive income tax system whereby both labor and capital incomes are subject to the same (progressive) tax rate schedule, or where this is too complex, given higher administration costs, simply raising the capital income tax rate.

In the African context, some features of the tax regime both strengthen implicit gender biases and make the tax system more distortionary. About 60 percent of the countries in the region allow interest-payment deductions at the personal level. These tax provisions result in inefficiently high levels of debt and exacerbate gender disparities, because they primarily benefit men, as men have more access to funding than do women. The gender gap is even more pronounced for collateralized lending, given the legal barriers women face to own mortgageable properties. Hence, countries with mortgage deductions (for example, Ghana, Kenya, South Sudan, and Nigeria) are particularly biased against women. Furthermore, women still face multiple *de jure* and *de facto* restrictions to own land and engage in certain economic activities. African women's rights to own land remain elusive, even as female-headed households grow (Agarwal 1994), because such rights tend to be defined through women's relationship with men. For example, women are unable to inherit land where the law considers it to be the property of the husband's lineage. Studies indicate that rural women work informally in the agricultural sector in larger shares than do men, even though women rarely own the land on which they work (Ogunlela and Mukhtar 2009).⁷ Against this backdrop, countries that provide tax incentives and provisions to the agriculture sector implicitly discriminate against women. Although some agriculture-specific policies can be justified on economic grounds—such as the zero-rating of agricultural inputs (for instance, Lesotho, Kenya, and Uganda) or expensing rules to encourage investment (for instance, Côte d'Ivoire, Malawi, and Nigeria)—tax holidays are highly distortive. Business owners in the agriculture sector benefit from reduced rates or exemptions in income tax in at least 17 sub-Saharan African countries.

A single-rate VAT system without exemptions is less prone to create gender disparities, and it can help mobilize revenues to finance programs that foster gender equality. A system with VAT-rate differentiation across goods creates implicit biases if consumption patterns differ by gender across goods subject to different rates. In contrast, the incidence of a single-rate regime would differ across genders only to the extent men and women have different consumption shares in income. VAT-rate reductions and exemptions have been shown to make the system more

⁷ For instance, in Zimbabwe only 3 percent of women hold land in the small farming sector and about 10 percent in the large-scale commercial sector. In Benin, women hold 11 percent; and in Tanzania, 25 percent (Cotula 2007).

inefficient, as they increase administration and enforcement costs (see Ebrill, Keen, and Perry 2001). The empirical evidence of pass-through of VAT reductions on to consumer prices is mixed, given that the effect depends very much on market structure characteristics (Rossouw and Ross 2020; Jurga, Yates, and Bagel 2020). Moreover, the econometric literature examining the gender incidence of sales taxes is scarce for several reasons, including because severe data challenges prevent measuring consumption at the individual level and, thus, by gender. The exception is feminine hygiene products, the taxation of which has attracted attention in the debate of improving their access by poor women. No African country has a single-rate VAT system; hence, from a fairness perspective, feminine hygiene products should be taxed at the reduced rate applied to other necessities. All African countries for which data are available do so, except for Egypt. Either way, a broad-base VAT—with minimal exemptions and a statutory rate that applies to most goods—is preferable because it is highly effective in mobilizing revenues, which can then be used to finance programs that protect women and lower-income individuals much more effectively, including by strengthening the social safety net.

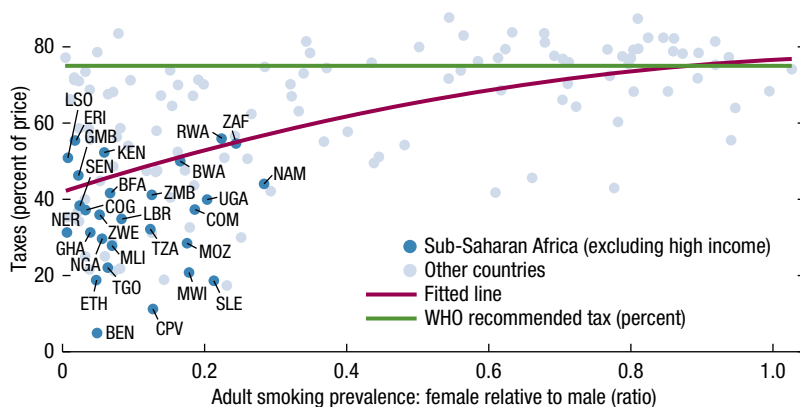
In contrast, excises can play a meaningful role in amplifying or mitigating gender inequality. Excises are corrective taxes; they are often introduced to discourage the consumption of harmful goods or services that may also entail a social cost—a so-called externality. As explained by Coelho and others (2022), excises—if they do not fully internalize the social cost—can implicitly create gender biases. What is more, if the excise is too low compared with the externality, and consumption patterns differ between men and women, then the gender consuming the good more heavily is implicitly subsidized. The opposite happens when the excise is higher than the externality. Tobacco is an example of an excise often set below its estimated externality, which financially benefits men because, on average, they smoke more than women do (Crawford and Le Nestour 2019). The implicit subsidy on male tobacco consumption is particularly large in sub-Saharan Africa, where the excise is one of the lowest compared with the externality, and smoking rates for men are lower than those for men (Figure 15.6).

Extreme weather events are particularly costly for women and girls in sub-Saharan Africa (see also Chapters 9 and 20). The high concentration of vulnerable groups and the large population share that depends on agriculture make weather events in Africa an important source of negative income shocks.⁸

Due to cultural norms, women are often expected to forgo educational and formal employment opportunities to engage in unpaid household work, such as caring for the sick or injured after extreme weather events and health epidemics.⁹

⁸ Agriculture, the main source of livelihoods and income for most people in Africa, is weather dependent. The average share of the employment in agriculture in Africa was estimated at 51 percent between 2011 and 2016. The agricultural sector is the primary employer in Africa, particularly in rural areas, which host most of the population (African Development Bank Group 2018).

⁹ Climate change appears to spur disease environments, particularly in the tropics and where the world's poorest populations currently reside—Africa and Asia. See Brody, Demetriades, and Esplen 2008; World Health Organization 2014.

Figure 15.6. Tobacco Consumption by Gender and Excises*(Female-to-male rate; 0 = gender equality)*

Sources: World Bank; and World Health Organization.

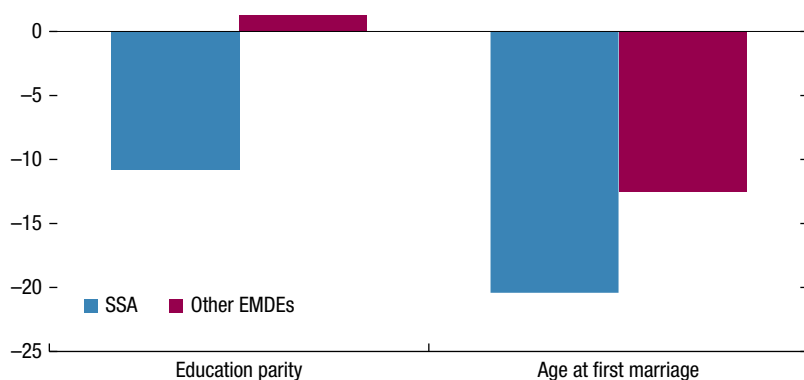
Note: Figures uses International Organization for Standardization (ISO) country codes. WHO = World Health Organization.

In this context, women bear higher costs than do men, as households respond to climate-driven income shocks by reallocating the time and talents of women to smooth consumption—in ways that are detrimental to them (Dercon and Krishnan 2000; Hoogeveen, van der Klaauw, and van Lomwel 2011; Corno and Voena 2016). Recent studies present evidence that poorer (liquidity-constrained) households in sub-Saharan Africa even marry their daughters at earlier ages when faced with income shocks caused by an epidemic (Archibong and Annan 2020a), floods, and droughts (Corno and Voena 2016), and that following such shocks, educational attainment of school-aged girls falls relative to that of boys (Archibong and Annan 2020b). In many African countries, marriage involves a transfer of wealth from the groom's family to the bride's—the so-called bride price (Rajaraman 1983; Wendo 2004). Countries with bride-price payments include Ghana, Kenya, Nigeria, Rwanda, Senegal, South Africa, Tanzania, and Uganda. There is evidence that, in some African countries, a lower age at first marriage for girls is associated with lower years of schooling and higher fertility rates (Corno, Hildebrandt, and Voena 2017; Corno and Voena 2016; Archibong and Annan 2020a; Mbaye 2020). The female-to-male ratios of school enrollment and of age at first marriage (normalized to zero as the equality benchmark) are, on average, lower in sub-Saharan Africa than in other developing countries (Figure 15.7). Meanwhile, adolescent fertility rates are much higher—99 per 1,000 habitants in sub-Saharan Africa compared with 28 in the rest of the developing world.

Therefore, a more ambitious agenda for carbon taxes, notably among large emitters, could help empower women in (sub-Saharan) Africa. The region contributes the least to global fossil fuel emissions—the average country in sub-Saharan Africa produces 0.8 ton of CO₂ per person per year, compared with a

Figure 15.7. Regional Average Gender Gaps in Education and Marriage Age, 2019

(Female-to-male rate; 0 = gender equality)



Sources: World Bank; and IMF staff calculations.

Note: SSA = sub-Saharan Africa; EMDEs = emerging markets and developing economies.

global average of 4.8 tons.¹⁰ However, the continent experiences some of the most extreme weather events in terms of drought, flooding, heat waves, and viable land use. Climate-change mitigation measures, including carbon taxes, are being adopted globally. Although more than 30 carbon tax programs have been adopted across the world, as of 2020 only the European Union and 5 Nordic countries have managed to raise carbon prices (mostly through higher taxes) to their recommended level (World Bank 2021). Though it is widely recognized that a more ambitious climate agenda is necessary, the discussion has failed to internalize the costs of climate-related shocks on women and girls in sub-Saharan Africa in terms of forgone education and formal employment opportunities. Supporting women and girls is macro-critical because they represent a significant share of the continent with the fastest-growing population. Policy options require swifter actions by the largest emitters—especially the United States, China, and India—which are key to global success. Moreover, consideration should be given to raising the recommended level of carbon prices. Carbon taxes primarily curb pollution and emissions but should also aim to reduce mortality rates from environment-related occupational risks (OECD 2020) as well as the economic costs that extreme weather events impose on the African population—and, in particular, women.

¹⁰ The exceptions are South Africa, which is highly developed and coal dependent, as well as the low-population and oil-rich countries Libya and Equatorial Guinea. These countries produce 5 to 10 times more per capita CO₂ than the regional average. In contrast, the country with the largest population, Nigeria, has below-regional average emissions (0.7 CO₂ ton per capita), and most other countries in the region have negligible emissions, regardless of their economy and population size.

Expenditure Policies to Economically Empower Females

Expenditure policy can economically empower women and girls via multiple channels. Public spending can have direct and indirect effects on women and girls, which can in turn improve their social welfare and economic output. Spending on health and education can improve human capital; social protection and labor policies can mitigate economic shocks; government employment and compensation policies can close gender gaps; and capital spending on water, sanitation, and digitalization can reduce unpaid work and improve knowledge of job opportunities. Although not all spending areas can be examined in detail, the following section presents the findings in an actionable way. The authors note main issues that affect women and girls, such as gender gaps, and various spending measures that have a direct and indirect impact are presented.

Social Protection and Labor

Social Assistance/Social Safety Net and Labor Market Programs

Given the prevalence of poverty and that a large share of women in the region work in the informal sector, social assistance and labor market programs can play an important role in enhancing social outcomes, including those for women. The region has the highest poverty rate, with 40 percent of the population below the extreme poverty line in 2018. Further, women typically work under precarious conditions. An estimated 89 percent of women in sub-Saharan Africa are employed in the informal sector (UN Women 2015). Less than 10 percent of the region's population is covered by social insurance.

Social safety nets and labor market programs can foster gender equality through several channels.¹¹ The primary objective of social safety net programs is to reduce current and future poverty. These programs can affect a broad range of household- and individual-level outcomes differently, depending on individuals' characteristics. Men and women respond differently to social safety nets and labor market programs and can benefit from these programs in different ways, depending on the general context as well as on the program's design and implementation. Social safety nets and labor market programs can foster gender equality by reducing female poverty, increasing human capital accumulation, improving female labor market participation and employment opportunities, and contributing to female empowerment.

- *Poverty reduction.* Women and girls are more likely to live in poor households and therefore may disproportionately benefit from the poverty-reduction effects of social safety nets. Even when social safety nets do not include gender-specific elements, programs may benefit women relatively more than do other forms of social protection because women are typically not tied to

¹¹ Note that the terms *social assistance* and *social safety nets* are used interchangeably.

formal employment, and their participation in formal labor markets tends to be lower than that of men.

- *Human capital accumulation.* Social transfers can foster investments in children's health and education—both of which are key determinants of human capital accumulation. Conditional cash transfer programs increase school enrollment and attendance rates and reduce dropout rates, but these effects are not consistently higher for girls or boys across countries. Household characteristics and national contexts—which lead to differences in the perceived opportunity costs of and returns to schooling across genders—can bring about different effects of social safety nets on girls and boys and help to increase education relatively more for the gender that is more disadvantaged in a particular context (UNICEF 2021). Cash transfer programs in many countries condition benefits on the practices that mothers adopt during pregnancy, at delivery, and postpartum. The evidence suggests that these conditionalities are associated with increased prenatal visits, skilled attendance at birth, and delivery at a health facility, and reduced incidence of low birth weight and maternal mortality. Conditional cash transfer programs also tend to have a positive impact on children's growth, but this effect tends to be similar across genders.
- *Employment opportunities.* Labor market programs can help close gender gaps by increasing women's employability and access to paid jobs. Women are generally overrepresented in the under- and unemployed, and therefore active labor market policies may disproportionately benefit them; however, the empirical evidence is not conclusive on differences across genders in terms of the effects of active labor market policies. Public works programs can create temporary employment opportunities for women who may otherwise be out of the labor market, increasing female labor force participation especially among young women (UNICEF 2021). Many public work programs include female quotas and women-friendly working conditions, but the transition to more stable employment can be difficult (World Bank 2014).
- *Female Empowerment.* Although the concept of empowerment is broad and hard to measure, the available evidence suggests that cash transfers can increase female empowerment by improving women's decision-making power, including for marriage and family planning. Many social safety nets target benefits to women, thus increasing their control over resources and improving their say within the household. Social safety nets also reduce domestic partner violence against women.

Social safety nets can affect gender relationships within the family and the community, regardless of whether program designs include gender-specific elements. Social safety net interventions provide resources to the household—not only cash but also employment opportunities and information—and can affect bargaining power within the family as well as household and individual consumption, production, and investment decisions. Even when they do not target women explicitly, social safety nets may offer important resources and opportunities to women. If not properly designed, such programs may reinforce existing gender

inequalities. Many social safety net programs deliver benefits to women because they are perceived to be more likely to spend resources on children's food, education, and health. Indeed, paying cash transfers directly to women has been shown in some contexts to lead to greater spending on children's needs, strengthening the impact of social safety nets on future poverty reduction (UNICEF 2021). However, this will not be the case everywhere and depends on gender norms and the roles assigned to women by society. In some cases, cash transfers that target women, especially those transfers conditional on children's school attendance or health checkups, can add to the burdens of poor mothers and reinforce traditional gender roles. Social safety nets can be more effective at achieving gender-relevant impacts if they are thoughtfully designed with this aim.

Pensions

Public pension schemes protect women from old-age poverty. Given that women's life expectancy is higher than that of men, pension income into old age is particularly important. With high informality and low pension coverage in most sub-Saharan African countries, noncontributory social pensions can be considered as fiscal space permits, as well as ensuring elderly women can access social safety net programs. There can be positive knock-on effects of pension spending, as women receiving a pension are more likely than men to invest in their grandchildren and support their daughters' job-seeking (Kabeer 2012).

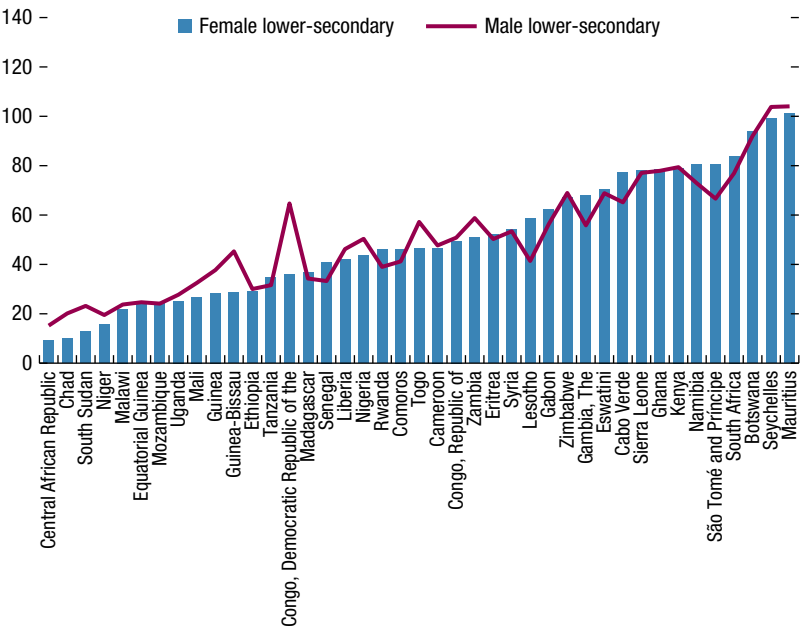
Education

Gender gaps in education remain relatively high in sub-Saharan Africa. The gross enrollment ratio of girls and boys has narrowed over time but persists, particularly at the tertiary level. Completion rates in many countries in the region still have gender gaps (Figure 15.8). Various measures, such as lowering school distance and cost, can improve access and learning outcomes while enhancing health conditions at schools (Evans and Yuan 2022). Public spending on education offers various indirect benefits, such as higher earnings and better health outcomes. Government spending on education can substantially improve the social welfare of females now and intergenerationally.

Health

Although, in general, health outcomes in sub-Saharan Africa have improved, women and girls tend to fare worse there than in other parts of the world. The maternal mortality rate has declined since 2000, but it remains, on average, above that of other regions and is nearly seven times higher than that of advanced economies (Figure 15.9). Adolescent fertility, which places girls and their potential offspring at greater risk, has been declining, though it is nearly six times higher than that of advanced economies and consistently above the low-income country average. Further, health care expenditure, per capita, is the lowest of all comparators; therefore, this spending area likely needs both increases and better efficiency.

Figure 15.8. Secondary Education Completion Rates
(Percent)

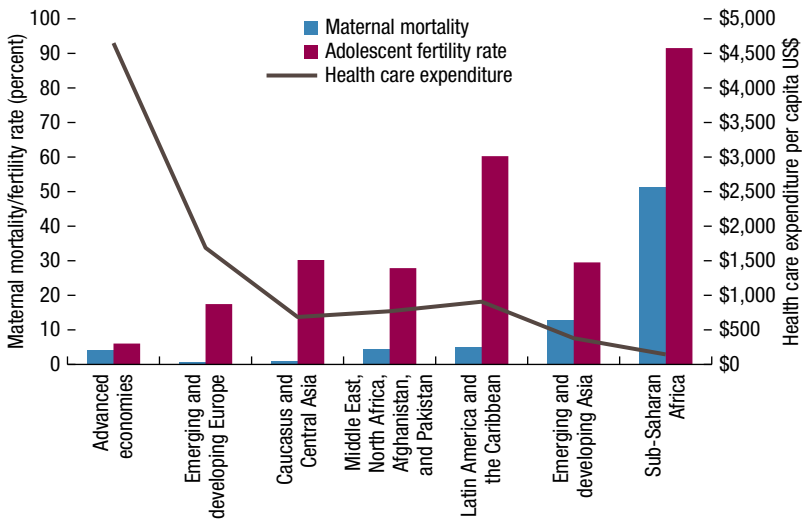


Source: Britt and Egerer 2023.

Multiple spending measures can support female health—and human capital formation more generally. Prenatal care has been shown to improve the health of mothers as well as the cognitive ability of their offspring. Further, conditional cash transfers contingent on mothers and daughters visiting a health clinic deliver health benefits. Spending on the health of women and girls can support their economic situation considerably—for instance, via improved productivity—and can save lives and boost quality of life. Moreover, health care must begin before birth, as maternal health affects health at birth, and in utero deprivations can compromise postnatal health care.

Capital Expenditure

In developing countries, women spend considerable time on unpaid work for tasks such as collecting water and finding fuel, and a lack of digital access can decrease knowledge of employment opportunities. About two billion people lack access to adequate sanitation; one billion, to drinking water. The situation in sub-Saharan Africa is severe: the region has 2.5 times the world average of hospitals without sanitation (Djantchiero and White 2022a, 2022b). In addition to creating poor health outcomes, these infrastructure gaps disproportionality burden women and girls, who are most likely to engage in unpaid work such as water

Figure 15.9. Selected Health Indicators

Source: Britt and Egerer 2023.

and fuel collection. Women also tend to have less access to the internet than men do, which can negatively affect learning and job searching.

Public investment spending can play a strong role in female economic empowerment in sub-Saharan Africa by reducing unpaid work, improving health, and increasing employment opportunities. Such spending includes that on water access and portability, sanitation, energy, and digitization. These investments directly affect women and girls in many ways, giving them a higher chance of engaging in paid work, attaining better educational outcomes, realizing more future earnings, and knowing about available jobs.

Public Sector Employment

Women tend to be paid less and represent a smaller share of paid employment relative to men. Research covering more than 30 countries worldwide showed that, on average, women in the public sector earned less than men did, although employment shares were more equal (Evans, Le, and Soto, forthcoming). Because the public sector is the largest formal sector employer in many sub-Saharan Africa countries, public sector employment can play an important role in closing gender gaps. In the private sector, women are paid about 70 percent of what men make, but the pay gap in the public sector is much lower, controlling for skills and other factors. A more explicit focus on gender-inclusive wage bill spending may narrow gaps. Countries with gender-disaggregated data, such as Costa Rica, the Dominican Republic, and Honduras, have smaller gender gaps for pay and seniority, and other governments require gender-disaggregated public sector employment data in their

budgets. Improved public sector employment policies could have positive knock-on effects for the economy. Over time, if women in the public sector have more positions of power and higher pay, then the pay gap in the formal private sector may also improve.

CONCLUSION

This chapter has reviewed the multiple interactions between tax and spending policy and gender in the context of sub-Saharan Africa. Of course, the optimal policy choices must be tailored to each country's circumstances, but some general conclusions can be drawn that might serve as a first inspiration for potential reforms that would improve gender equality.

The key tax policy implications for sub-Saharan Africa are as follows:

- Explicit biases in the tax system are still more common in sub-Saharan Africa than elsewhere and should be removed. In some cases, the biases are in other laws (for example, on ownership or inheritance) that interact with the tax system and should be addressed directly there.
- There is scope to individualizing the personal income tax in about half of the countries in the region; however, as long as labor market informality remains high and tax bases narrow, such a reform is unlikely to have meaningful effects, since the tax is paid by very few people. Personal income tax should therefore be combined with other reforms, such as those aimed at bringing more people into the tax, notably by encouraging formal employment. Apart from tax administration measures, policy options include making the tax and social security system more progressive; if absent, introducing a simple small businesses regime may also encourage (female) entrepreneurs to formalize.
- Low capital income taxes that disproportionately benefit men could be raised. Given severe enforcement constraints, this change can be most easily achieved by abolishing preferential tax treatments. Policy options include abolishing mortgage and other interest deductibility under personal income taxes as well as limiting interest deductibility under the corporate income tax.
- The economic justification for differential rates in the VAT/sales tax regime is weak. A single-rate regime with minimal exemptions is a powerful revenue mobilization tool; hence, preserving the integrity of the tax is important. Its higher revenue yield can provide the necessary resources to fund social programs that foster gender equality, such as by strengthening the social safety net, which may also encourage formal employment.
- Excises can help correct and mitigate gender biases in many ways. If set at a level that internalizes harmful externalities, excises are gender neutral. They can also soften the disproportionate adverse effects of climate-induced income shocks on African women and girls (higher carbon taxes, especially by large emitters).

On the spending side, key considerations are as follows:

- Social protection and labor programs are critical to overcoming the high levels of informal employment and poverty. These spending programs can (in)directly target females, thereby reducing poverty, boosting human capital, improving employment, and furthering empowerment (for example, by enhancing women's decision-making role in the household).
- Education spending can narrow or close enrollment and completion gaps, which, in turn, can improve future earnings and health.
- Health expenditure saves lives, including those of mothers and children, while enhancing their well-being and productivity.
- Capital spending on water, sanitation, energy, and closing the digital divide improves female welfare, reduces unpaid work, and increases employment access.
- Gender-sensitive wage bill policies can narrow wage, employment, and seniority gaps—while also possibly having a positive impact on the private sector.

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Gender Budgeting Practices in Sub-Saharan Africa

**Lauren Keating, Laura Gores, Carolina Rentería,
Vincent Tang, and Nino Tchelishvili**

In the context of constrained fiscal space and exacerbated gender inequalities resulting from the COVID-19 pandemic, gender budgeting remains important for governments worldwide. Gender budgeting not only signals a government's commitment to advancing gender equality, but it prompts more efficient resource use through the improved design of responsive fiscal policies. Countries in sub-Saharan Africa were some of the first adopters of gender budgeting, with Rwanda and Uganda continuing to demonstrate regional leadership. This chapter uses country self-assessment data from an IMF Gender Budgeting Survey to analyze the current state of gender budgeting practices in 34 sub-Saharan African countries. It examines strong practices across the regions and highlights areas for advancing practices.

INTRODUCTION

As the other chapters in this book illustrate, the COVID-19 pandemic intensified gender inequalities in sub-Saharan Africa, much as it did in the rest of the world, and it continues to hamper the region's growth potential (IMF 2021). The pandemic threatens progress toward addressing these inequalities. Although many governments provided immediate fiscal policy responses to support health sector spending needs and stabilize economies, the crisis hit the world's vulnerable population, especially women, particularly hard. In addition to women shouldering additional unpaid care work, sub-Saharan African women are more likely to work in the informal sector and outside of the reach of COVID-19 response measures. Meanwhile, many girls have left the school system permanently, and violence against women has increased (OECD 2020).

Well-structured fiscal policies supported by sound public financial management tools can potentially improve the conditions for women and men and advance government efforts toward greater gender equality—a practice known as gender budgeting. As defined in Alonso-Albarran and others (2021, 5) “gender budgeting incorporates a gender lens into the budget process to ensure that governments are aware of the impact of their choices on gender outcomes. Gender budgeting is not just about funding explicit gender equality initiatives. It also entails analyzing fiscal policies and budgetary decisions to understand their impact—intended and unintended—on gender equality and using this information to design and implement

more effective policies.” Originating in Australia in the 1980s, more than 100 countries now practice some elements of gender budgeting. Certain sub-Saharan African countries were early architects in developing gender budgeting (South Africa), while others have emerged over the years as regional leaders (Rwanda, Uganda). Some countries are recent adopters (Togo), and some have embarked on their first efforts (Sierra Leone).

This chapter provides an overview of gender budgeting practices in 34 sub-Saharan African countries using data obtained from an IMF survey completed by national officials.¹ This chapter focuses on the existence of gender budgeting practices in the region and not on their quality or effectiveness. It presents the status of gender budgeting in sub-Saharan African countries, as self-reported by government officials, and compares sub-Saharan African gender budgeting practices with countries in other regions. An examination of the challenges and key success factors sub-Saharan African countries have faced when implementing gender budgeting follows this analysis. The chapter concludes with a short summary and discusses next steps for the region.

STATUS OF GENDER BUDGETING IN SUB-SAHARAN AFRICA

Since the 1990s, gender budgeting initiatives have taken hold across sub-Saharan Africa. In 1995, South Africa launched its first Women’s Budget Initiative, becoming one of the earliest adopters of gender budgeting internationally and inspiring the introduction of gender budgeting in countries across the region (Kolovich 2018). For some countries in the region, however, gender budgeting momentum has waxed and waned, including South Africa’s. Many factors account for this waning, including changes in political support and limited capacity (Polzer, Nolte, and Seiwald 2021).

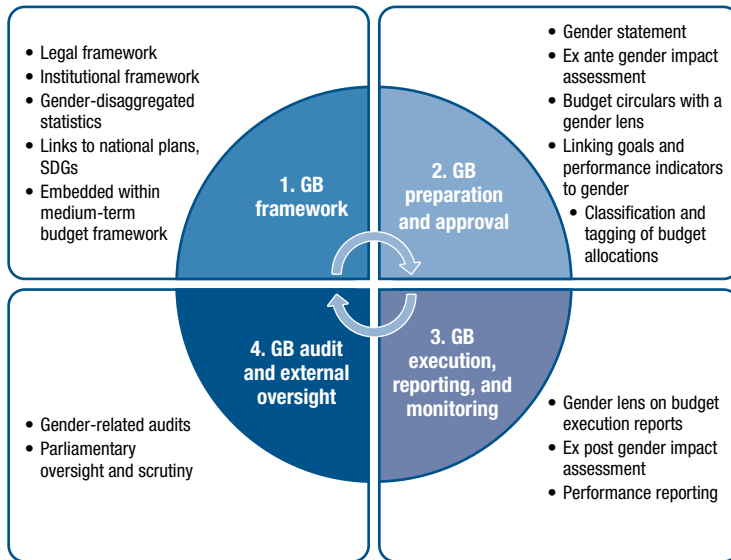
Today, some sub-Saharan African countries have more advanced gender budgeting practices, while others are just starting out. For example, Rwanda and Uganda have successfully embedded gender budgeting tools into existing budgeting procedures, and, as a result, gender budgeting has reputedly influenced changes in fiscal policies (Stotsky, Kolovich, and Kebhaj 2016). At the same time, many countries in the region, such as Togo, are making a strong start. Each of the 34 sub-Saharan African countries that responded to the IMF’s survey have introduced at least one gender budgeting practice or tool. The IMF created the survey in 2018 and collected answers over 2019–21, with some additions and updates in 2023. Following the IMF’s gender budgeting framework described in Box 16.1 and the concepts presented in the IMF’s gender budgeting in the G20 working paper by Alonso-Albarran and others (2021), the 12 public financial management instruments covered by the IMF survey and their associated 23 questions used in this analysis examine and compare gender budgeting tools and practices across

¹ Government officials from 34 sub-Saharan African countries participated in the online self-assessment multiple-choice survey. Survey results are based on country declarations and are static in that they capture the existence of a practice or tool at a specific point in time (<https://cvent.me/m950my>).

Box 16.1. IMF Gender Budgeting Framework and Survey

The IMF applies a holistic approach to gender budgeting in which each phase of the budget cycle incorporates a gender perspective. Gender budgeting tools and practices support the integration of gender into the four phases, from budget planning and preparation to auditing and evaluation, as illustrated in Figure 16.1.1. This approach has evolved over time, beginning with a conceptual framework outlined in the IMF's assessment of gender-budgeting practices in G7 countries (IMF 2017) and further adapted in the gender budgeting in G20 Countries working paper (Alonso-Albarran and others 2021).

Figure 16.1.1. A Holistic Approach to Gender Budgeting throughout the Budget Cycle



Source: Alonso-Albarran and others 2021.

Note: GB = gender budgeting; SDGs = Sustainable Development Goals.

The IMF developed a gender budgeting survey based on this framework. The survey captures information on gender budgeting practices from countries globally and is the basis for this analysis. The survey was issued in 2018 with answers collected over 2019–23 and includes information on fiscal policies related to gender equality, the existence of gender budgeting practices and tools, and challenges as well as success factors.

Government officials from 34 sub-Saharan African countries participated in the online self-assessment multiple-choice survey. Survey results are based on country declarations and are static in that they capture the existence of a practice or tool at a specific point in time. However, it is worth noting that gender budgeting practices can vary over time, and what countries report one year may not necessarily be practiced to the same extent in the next. Additionally, although IMF staff have conducted a general validation process, actual practices may differ. Responses at the central government level provide the basis for this chapter.

TABLE 16.1.

Count of Gender Budgeting Practices in Sub-Saharan Africa

1–8 Tools and Practices	9–15 Tools and Practices	16+ Tools and Practices
Botswana	Benin	Rwanda
Burundi	Burkina Faso	Tanzania
Central African Republic	Cameroon	Togo
Chad	Congo (Republic)	Zimbabwe
Comoros	Cote d'Ivoire	
Congo (Democratic Republic)	Ethiopia	
Gabon	The Gambia	
Lesotho	Kenya	
Malawi	Liberia	
Namibia	Madagascar	
São Tomé and Príncipe	Mauritius	
Seychelles	Mozambique	
Swaziland	Niger	
	Senegal	
	South Africa	
	Uganda	
	Zambia	

Source: IMF Gender Budgeting Survey.

Note: Country tools and practices are based on country declarations in the IMF gender budgeting survey and this figure reflects the existence of tools at a point in time and not their quality or effectiveness. The IMF created the survey in 2018, and answers were collected over 2019–21, with some updates in 2022.

sub-Saharan African countries. Annex 16.1 outlines the 12 instruments and corresponding questions about tools and practices (23 total).

The number of gender budgeting practices and tools applied across the region is considerable. Table 16.1 highlights the number of gender budgeting practices and tools that each sub-Saharan African country has declared to have adopted. Half of the countries surveyed reported they have 9 to 15 tools or practices, while four countries have 16 or more tools or practices. The latter are Rwanda, Tanzania, Zimbabwe, and Togo. Countries with the lowest tool count or practices are those with policies to promote gender equality but no additional gender budgeting practices, including Gabon and Comoros.

Sub-Saharan African countries have adopted a higher number of gender budgeting tools and practices in the first two phases or pillars of the budget cycle than in the last two. Table 16.2 outlines the 12 instruments under each of the four pillars of the IMF's gender budgeting framework and identifies which country has at least one tool or practice in place for each. Across the four pillars, sub-Saharan African countries are more likely to have gender budgeting practices in legal and institutional frameworks and budget preparation. Although sub-Saharan African countries have fewer practices in the downstream pillars, most countries have started to integrate gender budgeting considerations into parliamentary oversight.

The Legal and Institutional Framework for Gender Budgeting

Almost half of sub-Saharan African countries surveyed have laws or regulations that enshrine a gender perspective into the budget process. While many countries have included gender budgeting requirements in their regulations, only seven include requirements in an organic or a public finance law. For example, Uganda

has included this requirement in the 2015 Public Financial Management Act. Enshrining gender budgeting into law is considered a good practice because it improves the likelihood that gender budgeting will continue over time and across political cycles (see also Chapter 14).

Nearly 80 percent of countries surveyed have some form of institutional framework in place for gender budgeting. Gender equality is an important issue for all ministries and departments. While the central budgetary authority leads and coordinates gender budgeting efforts, line departments and the ministry or agency for gender equality have play essential roles in developing sector strategies for gender equality and implementing programs. An institutional framework that outlines roles and responsibilities for gender budgeting is critical. For example, South Africa introduced in 2019–20 a Gender Responsive Planning, Budgeting, Monitoring, Evaluation and Auditing framework detailing how the government will incorporate gender mainstreaming into public finances. Similarly, Ethiopia's 2012 National Gender Responsive Budgeting Guidelines provide information on the national legal, policy, and institutional framework for gender equality and describe key steps for mainstreaming gender into the national budget cycle.

In most sub-Saharan African countries, the primary institution responsible for gender budgeting is the ministry of finance or the ministry or agency for gender equality. For example, the responsibility for gender budgeting reform in The Gambia is shared between the Ministry of Finance and Economic Affairs and the Ministry of Gender, Children and Social Welfare, while in Ethiopia, the Ministry of Finance is the central entity in charge of steering gender budgeting reform; however, a Gender Affairs Directorate supports the gender responsiveness review of budget submissions during budget hearings. Some countries have taken a different approach. Senegal distributes the responsibility for gender budgeting more widely to include units within departments or agencies as well as an independent agency. For gender budgeting to be most effective, there must be strong leadership from the ministry of finance with support from a ministry or agency responsible for gender equality (Downes and Nicol 2020).

The majority of sub-Saharan African countries collect gender-disaggregated data. These data are necessary for understanding a country's gender inequalities and provide the foundation for many gender budgeting tools, for example, ex ante and ex post gender impact assessments. Applying gender-disaggregated data in the budget process varies widely. Some countries reported using the data occasionally or frequently to develop gender-related performance indicators and targets, to inform decision making on policies and new legislation, and to inform analyses on gender issues. Only half of the governments collecting these data publish the statistics.

A Gender Perspective in Budget Preparation

Each surveyed sub-Saharan African country has gender equality policies in place to respond to country-specific challenges. National or sectoral gender equality policies can help ensure adequate consideration of equality goals during budget planning and prioritization. In sub-Saharan Africa, a national development plan or individual sectorial or ministerial plans often integrate these policies, providing a foundation for gender budgeting practices. For example, Zimbabwe officially

adopted gender budgeting in 2007 with the objective of achieving gender equality and inclusive growth in line with national commitments (Zimbabwe Ministry of Finance and Economic Development 2023). Translating policies into specific goals with measurable indicators is a good practice for measuring a country's progress toward closing gender equality gaps.

Nine sub-Saharan African countries have government-wide gender equality performance targets, while seven have sector-specific targets. Setting performance targets for gender equality goals and policies improves monitoring and tracking of a country's progress in key areas. For example, Rwanda has performance measures and targets in its National Strategy for Transformation I (NST I 2017–24). Scaling up Early Childhood Development Center services at the village level is one of these targets.

Over two-thirds of countries surveyed link gender equality goals to programs. Half of these countries also cost goals on an annual or medium-term basis. Linking gender equality goals with program objectives allows governments to track implementation of programs expected to advance gender equality and identify areas where more attention is needed. Simultaneously, costing programs linked to gender equality goals can help ensure proposals are prioritized within allocated expenditure ceilings. For example, Rwanda links costed subprograms, whenever applicable, to the achievement of the National Gender Policy and NST I (2017–24).

Only one-quarter of sub-Saharan African countries produce an annual gender budget statement; nevertheless, the countries that produce these statements are likely to publish them. A gender budget statement is a gender-specific policy and accountability document that a government crafts to show how the annual budget aims to improve gender equality. The statement takes stock of where a country stands with reaching gender equality goals, describes new budget proposals and previous actions expected to advance progress toward equality goals, and can include an analysis of aggregate spending on gender equality or beneficiaries. Publishing a gender budget statement is a good practice for improving accountability and receiving feedback on gender budgeting practices from civil society. Zimbabwe's Ministry of Finance and Economic Development leads gender budgeting and has recorded progress to date in published National Gender Budget Statements since 2022. This statement outlines gender-specific expenditures that the government will track along with budget implementation. Similarly, Togo has made considerable efforts toward implementing gender budgeting in recent years, including introducing gender budget statements (Box 16.2).

Just under half of surveyed sub-Saharan African countries issue guidelines for gender budgeting procedures in budget call circulars. Issuing detailed instructions in the budget circular not only signals that the government takes gender budgeting seriously but it also ensures that gender equality perspectives adequately inform budget submissions. These guidelines allow for a coordinated and comprehensive approach that improves efficiency during budget preparation and negotiation. Uganda has refined its budget circular instructions over time and adjusts them after each budget cycle. For example, after learning that sectors were providing blanket plans for addressing gender inequalities in their Budget Framework Papers, the Ministry of Finance made its budget circular instructions

Box 16.2. Case Study of a Recent Gender Budgeting Adopter: Togo

Togo has undertaken ambitious budget reforms in recent years. The government of Togo undertook flagship reform by transitioning from line-item to program budgeting in 2020, setting the stage to introduce gender budgeting. In 2023, the Togolese prepared their third Gender Budget Statement (GBS).

The process of introducing gender budgeting in Togo started in 2018. The government achieved results through a collaborative approach. A small group of gender focal points in line ministries reports to the Prime Minister and the Ministers of Finance, Women, and Planning. The authorities intend to formalize this structure by creating a gender budgeting unit within the Ministry of Finance, spearheading gender budgeting across government. Regular communication with the public has been key to gathering stakeholder support for the process and improving accountability for gender reforms.

A 2020 decree on the publication of budgetary documents paved the way for a GBS. Later that year, the authorities created a roadmap for gender budgeting, defining methodological, legal, and organizational aspects and gender budgeting tools. Key milestones in the roadmap included: (1) integrating the process in the budget calendar, (2) providing gender budgeting instructions and templates in the budget circular, (3) formulating budget allocations based on a gender impact assessment, and (4) publishing a GBS. Togo published its first GBS with the 2022 budget law, covering an initial group of six pilot ministries that had analyzed their budget programs using the OECD Development Assistance Committee (DAC) gender marking methodology. The GBS shows the results of this analysis and highlights how the budget contributes to the national gender equality policy.

At the outset of the 2022 budget process, the Prime Minister's budget call circular gave instructions to the ministries, with templates to assist them in identifying gender sensitive budget allocations. Budget negotiations and the discussions in Parliament involved the gender focal points in the ministries. Training for government officials and members of Parliament and a media campaign to inform the public preceded the presentation of the 2022 GBS. To monitor the budget execution of gender-related spending, disaggregated gender-related spending information will supplement quarterly and annual performance reports.

Togo is now rolling out gender budgeting to a wider group of line ministries. Challenges that lie ahead include the introduction of quality management processes, to ensure that the expanding number of line ministries consistently apply the gender marking method, and of in-depth ex ante gender impact assessments. These measures will help increase the budget impact in operationalizing the government's gender equality agenda.

Source: Togolese Republic, Ministry of Economy and Finance, Gender Responsive Budget Document (DBSG) 2022; DBSG 2023.

more specific the following year, requiring sectors to outline detailed actions in an annexed template (Stotsky, Kolovich, and Kebhaj 2016).

Eight sub-Saharan African countries conduct ex ante gender impact assessments on some or all new government policy initiatives. In some countries, individual spending ministries or departments (Mauritius and Tanzania) conduct these assessments, while in others, the ministry or agency responsible for gender equality conducts the assessments (Republic of Congo, Cote d'Ivoire, and Kenya). It is a good practice for the government official in the line ministry responsible for the program or policy to conduct the assessment early in the program or policy design. Of these countries, only four have a common gender impact assessment methodology—which

is a good practice because it enables consistency and comparability across ministries and allows for streamlined quality checks. In some countries, the ministry or agency for gender equality has the sole responsibility for quality control of these assessments (Republic of Congo and Cote d'Ivoire), while other countries share responsibility with the ministry or agency for gender equality, ministry of finance, and gender units in line ministries (Tanzania and Zimbabwe). For these assessments to inform budget decisions, it is a good practice for the ministry of finance to conduct quality controls and integrate the assessment into budget negotiations or briefings.

Many sub-Saharan African countries report they classify their budgets according to programs, although their definition, usage, and effectiveness of programmatic classification can vary widely. In general, a programmatic budget structure could make it easier to integrate gender equality issues into budget decisions and allow for more effective tracking of expenditures that likely advance gender equality. In nearly all countries reporting they classify their budgets according to programs, the countries incorporate gender into either stand-alone gender programs (16 percent), other individual programs (37 percent), or both (47 percent).

Gender in Budget Execution and Monitoring

A third of sub-Saharan African countries report on the execution of gender-related expenditures. Monitoring and evaluating the implementation of gender-related programs or policies sheds light on whether funds achieve gender equality objectives while improving accountability and transparency. For an example of Rwanda's effective monitoring approach, see Box 16.3. Eight of the surveyed countries report on gender-related expenditures, while six compare approved with actual gender-related expenditures.

Only five sub-Saharan African countries perform ex post gender impact assessments of budget expenditures. Ex post gender impact assessments are a key tool for evaluating the effects of revenue and expenditure initiatives on gender equality. These assessment findings inform and improve the design of existing and new policies and programs. For example, the South Africa's Department of Trade and Industry's ex post analysis of expenditures to small, medium, and micro-enterprises observed that spending on these enterprises was a small percentage of the Department's budget despite women's overrepresentation as owners. This assessment resulted in the Department increasing allocations to support and promote women business owners (Stotsky, Kolovich, and Kebhaj 2016). Ex post gender impact assessments can also be used in reporting—for example, Burkina Faso uses these assessments in an annual report on the implementation of the Sustainable Development Goals.

Audit and Parliamentary Oversight of Gender Budgeting

Parliamentary and legislative oversight of gender equality initiatives is a robust practice across sub-Saharan African countries. Over two thirds of sub-Saharan African countries' legislatures and parliaments are involved in gender equality issues—likely a result of nearly 60 percent of countries having established a designated committee for gender issues. For example, Namibia's Standing Committee on Gender Equality, Social Development and Family Affairs addresses gender issues

Box 16.3. Sub-Saharan African Success Story: Rwanda's Gender Monitoring Office

Rwanda has some of the most prominent gender budgeting practices in the region. In addition to a solid legal framework, clear focus on gender-disaggregated data collection, and extensive use of gender budget statements as a tool for integrating gender in the budget preparation, Rwanda has built a strong system of monitoring and evaluating results.

Rwanda established a Gender Monitoring Office (GMO) in 2007 with a mandate to ensure compliance and effective implementation of various national, regional, and international commitments, including on gender responsive budgeting. The GMO conducts field visits to monitor quality and conformity with gender budget statements and audits results that line ministries achieve. The GMO then produces an Annual Report that identifies achievements and existing challenges and supports planning for the next fiscal year (Gender Monitoring Office 2020).

For example, in its Annual Report 2019–2020, the GMO's assessment of FY2018/19 gender budget statements found positive results, including increased retention and performance of boys in school, improved health care for women, and increased access to clean water. The latter not only saved women quality time and households money, but women were also actively involved in water management committees. The report also highlighted areas for improvement, including that a wide gap in the gender distribution of employment, especially in leadership positions, remains in all districts audited. The GMO called for increased efforts from all levels to address this gap.

To further facilitate monitoring, Rwanda developed a gender management information system, which is a database for national gender targets and indicators that is expected to provide sex-disaggregated data wherever it is needed for gender responsive decision making, planning, and budgeting.

Source: Gender Monitoring Office, Government of Rwanda 2021. <https://www.gmo.gov.rw>.

across party lines and promotes gender equality to improve the status of women. Establishing a gender committee ensures that the legislature remains involved and can reinforce accountability on gender budgeting reforms. Across sub-Saharan Africa, the role of the legislature varies, with 32 percent discussing the budget impact on gender equality or conducting hearings on gender equality issues. Far fewer discuss the tax policy impact on gender equality (9 percent). For example, Ethiopia's Standing Committee for Women, Youth and Children of the Parliament makes recommendations to reinforce the effectiveness of the budget in the attainment of gender equality goals before legislative approval. Additionally, Uganda's Parliament has played a critical role in promoting and enforcing gender and equity budgeting. At the budget approval level, Parliament receives a Certificate of Gender and Equity Compliance when various budget products are satisfied. In this process, champions from the Uganda Women Parliamentary Association call on colleagues to critically vet Budget Framework Papers for gender and equity responsiveness.

National audit institutions are less involved in reviewing or assessing gender equality policies. Only five sub-Saharan African countries stated that the national auditor produced a report on the performance of gender programs or activities. Independent evaluations of gender programs and activities are essential for transparency and accountability, identifying implementation challenges, and incorporating

TABLE 16.2.

Sub-Saharan Africa Countries' Gender Budget Practices												
	1. Legal and Institutional Framework			2. Budget Preparation					3. Budget Execution and Monitoring		4. Audit and Parliamentary Oversight	
	Legal Framework on Gender Budgeting	Institutional Framework	Fiscal Data Disaggregated by Gender	Gender Statements	Ex ante Gender Impact Assessments	Budget Circular Containing Specific Guidelines on Gender Objectives	Linking Gender Goals to Programs and Performance Indicators	Budget Classification Incorporates a Gender Perspective	Budget Execution Reports Incorporate Gender Perspective	Ex post Gender Impact Assessments	Gender Related Audits	Parliamentary Oversight
Benin												
Botswana												
Burkina Faso												
Burundi												
Cameroon												
Central African Republic												
Chad												
Comoros												
Congo (Democratic Republic)												
Congo (Republic)												
Cote d'Ivoire												
Ethiopia												
Gabon												
Kenya												
Lesotho												
Liberia												
Madagascar												
Malawi												

Mauritius												
Mozambique												
Namibia												
Niger												
Rwanda												
São Tomé and Príncipe												
Senegal												
Seychelles												
South Africa												
Swaziland												
Tanzania												
The Gambia												
Togo												
Uganda												
Zambia												
Zimbabwe												

No Gender Budgeting Tool or Practice	At least one Gender Budgeting Tool or Practice
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Source: IMF Gender Budgeting Survey.

Note: This table outlines the 12 instruments under each of the four pillars of the IMF's gender budgeting framework and identifies which country has at least one tool or practice in place for each; see Annex 16.1 for a list of associated questions behind each instrument. If a country answered yes to any one of the associated questions, then the square was colored green. Note that for the indicator on "linking gender goals to programs and performance indicators," the green marking indicates countries that reported "yes" to questions 14 and/or 19 only (If gender equality goals are set, are they linked to programs? Are there performance targets related to gender equality?). This does not necessarily entail program-based budgeting. As a reminder, country tools and practices are based on country declarations in the IMF gender budgeting survey, and this figure reflects the self-reported existence of tools at a point in time, not their quality or effectiveness.

lessons learned to strengthen future policy designs. For example, Uganda's Supreme Audit Institution, with support from the International Organization of Supreme Audit Institutions' Development Initiative, conducts an audit of government efforts toward the elimination of intimate partner violence against women.

COMPARISON OF SUB-SAHARAN AFRICA'S GENDER BUDGETING PRACTICES WITH OTHER COUNTRIES

The level of gender budgeting practices of low-income and developing countries (LIDC) and emerging market economies (EME) within sub-Saharan Africa is similar to those outside the region. Beyond sub-Saharan Africa, the survey collected data on gender budgeting practices from 71 other countries worldwide, including 10 LIDCs and 49 EMEs. Figure 16.1 compares the average results for sub-Saharan African countries with other LIDCs and EMEs. The average Gender Budgeting Index² for sub-Saharan African countries is slightly above other LIDCs and EMEs, but both fall under the "basic" gender budgeting practice category. Consistent with the finding from Alonso-Albarran and others (2021), no country has reached an overall "advanced" practice level.

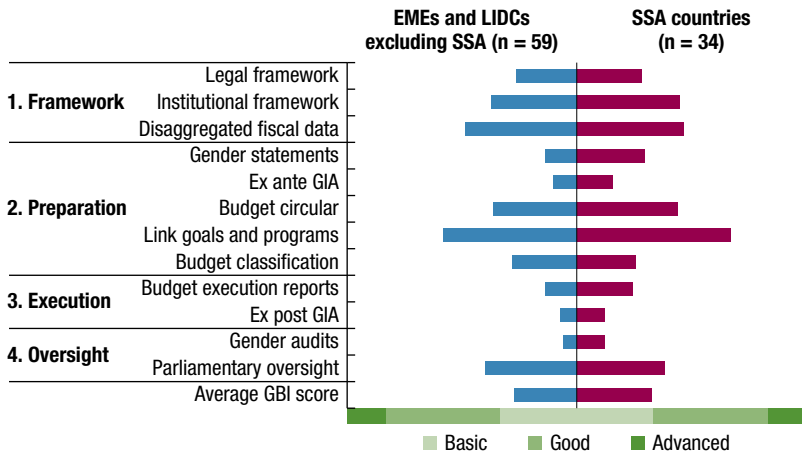
Observers note that LIDC and EMEs within and outside sub-Saharan African countries demonstrate good practices regarding institutional frameworks, disaggregated fiscal data collection, budget circular, linking goals and programs, and parliamentary oversight. Sub-Saharan African countries slightly outperform other LIDC and EMEs in select practices in budget preparation (gender statements, ex ante gender impact assessments, budget circular) and in budget execution reports and gender audits. Consistent with the findings of the G20 paper, the greatest weaknesses are in ex post and ex ante gender impact assessments and gender audits.

Sub-Saharan Africa is comparable with other regions in implementing gender budgeting. Figure 16.3 shows that most regions slightly outcompete the others in one or two areas but, broadly, all regions follow a similar pattern. For example, sub-Saharan African countries slightly outperform other regions regarding gender audits, budget execution reports, and linking goals and programs, although the first two are still at the basic practice level.

While advanced economies fare better in certain gender budgeting practices, they face similar challenges as LIDC and EME in many areas. Figure 16.2 shows that advanced economies outperform LIDC and EMEs in disaggregated fiscal data, gender statements, and ex ante gender impact assessments. However, these countries face similar challenges as LIDCs and EMEs regarding gender audits, and they score similarly or lower than LIDCs and EMEs in certain practices, specifically budget execution reports.

² The Gender Budgeting Index was introduced in the IMF's Gender Budgeting Practices in G20 Countries Working Paper by Alonso-Albarran and others (2021) and is an index calculated by assigning a score (from zero to three) to 23 key questions within the survey that relate to the 12 public financial management instruments used in a budget cycle. These are the same questions that were used to count practices in Table 16.2. For a list of the questions see Annex 16.1. The unweighted average of the 23 questions is equal to the overall country Gender Budgeting Index.

Figure 16.1. Public Financial Management Tools Used to Support Gender Budgeting Using the Average Scores of the Gender Budgeting Index



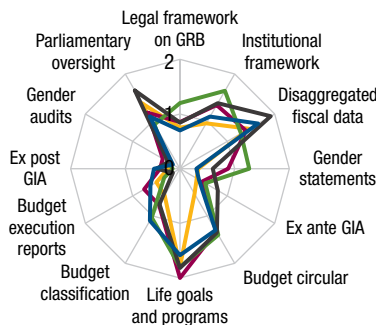
Sources: IMF Gender Budgeting Survey; and IMF staff.

Note: EMEs = emerging market economies; GBI = Gender Budgeting Index; GIA = gender impact assessments; LDCs = low-income developing countries; SSA = sub-Saharan Africa. The bars are the average scores of the GBI of each country group. For a description of this index, see footnote 2.

Figure 16.2. Gender Budgeting Index Scores by Region and Income Group

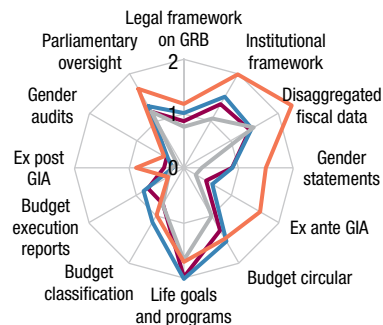
1. Gender Budgeting Index Scores by Region

AFR (n = 34) APD (n = 15)
EUR (n = 20) MCD (n = 16)
WHD (n = 20)



2. Gender Budgeting Index Scores by Income Group

AFR (n = 34) LDC (n = 10)
EME (n = 49) AE (n = 12)



Sources: IMF Gender Budgeting Survey; and IMF staff.

Note: Since the publication of the IMF's Gender Budgeting Practices in G20 Countries by Alonso-Albarran and others (2021), additional countries have been added to the IMF Gender Budgeting Database; specifically, eight countries have been added to the Western Hemisphere Department region and four to the African Department region. AE = advanced economy; AFR = Africa; APD = Asia and Pacific; EME = emerging market economies; EUR = Europe; GIA = gender impact assessments; GRB = gender-responsive budgeting; LDC = low-income developing countries; MCD = Middle East and Central Asia; WHD = Western Hemisphere.

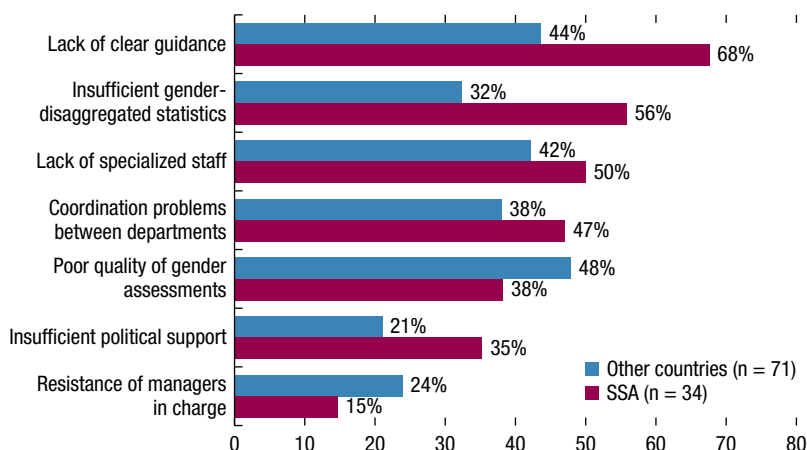
Development partner support is likely a key factor in the increase in gender budgeting practices across sub-Saharan Africa. The survey results for sub-Saharan African countries align with the literature highlighting that international organizations and financial and technical support by development partners to LIDCs culminated in greater gender budgeting adoption in these countries. Indeed, IMF technical assistance and supporting organizations' capacity initiatives to sub-Saharan African countries in this public financial management reform have increased over recent years.

IMPLEMENTATION OF GENDER BUDGETING IN SUB-SAHARAN AFRICA—CHALLENGES AND SUCCESS FACTORS

Sub-Saharan African countries cite lack of clear guidance as the most pressing challenge when implementing gender budgeting. As outlined in Figure 16.3, 71 percent of sub-Saharan African countries' survey responses identified a lack of clear guidance on incorporating gender considerations into the annual budget as one of four major challenges when implementing gender budgeting. This factor is considerably higher than responses from all other countries. A cause of this challenge may be related to a lack of guidance provided in budget circulars. Indeed, less than half of surveyed sub-Saharan African countries include details on applying gender budgeting in a budget circular. Where these guidelines exist, they sometimes come from individual spending ministries to subordinate departments and, therefore, could be inconsistently applied or not enforced, compounding this challenge.

Other primary challenges identified in the region include insufficient gender-disaggregated statistics and coordination problems among departments. Despite nearly 70 percent of governments in the region collecting some disaggregated data,

Figure 16.3. Major Challenges Encountered when Successfully Implementing Gender Budgeting
(As a percentage of the sample)

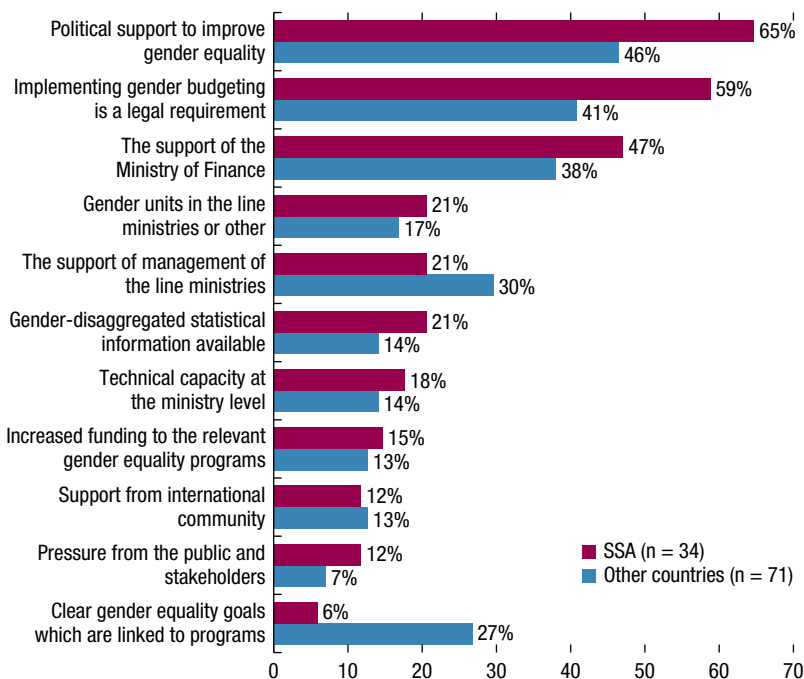


Source: IMF Gender Budgeting Survey.

sub-Saharan African countries have noted that insufficient gender-disaggregated statistical information inhibits them from adequately costing new gender initiatives. Additionally, challenges for roughly half of sub-Saharan African countries include a lack of specialized staff and coordination with departments with competencies in gender policies during budget preparation. Some countries have dedicated coordination units to overcome these challenges. For example, Uganda's Ministry of Gender established a national coordination mechanism to ensure systematic integration of gender equality and women's empowerment issues in development plans, policies, programs, and the budget. In Togo, a coordination mechanism between the budget directorate, the ministry of women, and gender focal points in line ministries ensures a consistent application of gender budgeting across public sector entities.

The top three success factors for implementing gender budgeting in sub-Saharan African countries align with the rest of survey respondents' answers—political support, binding legislative requirements, and support from the ministry of finance (Figure 16.4). A general understanding is that for gender budgeting to be successful, change needs to come alongside a high-level commitment. This finding is consistent with the literature highlighting that strong political

Figure 16.4. Most Important Factors for the Successful Implementation of Gender Budgeting
(As a percentage of the sample)



Source: IMF Gender Budgeting Survey.

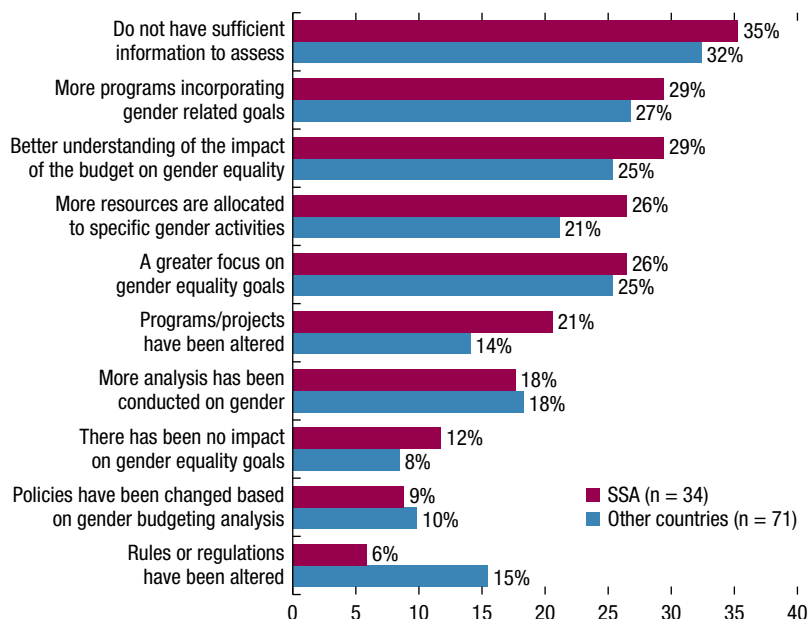
leadership, solid legal foundations, and clear direction from the ministry of finance add the necessary weight for gender budgeting reforms to take hold (for instance, OECD 2020; IMF 2017; Kolovich 2018).

THE IMPACT OF GENDER BUDGETING ON GENDER EQUALITY

Across sub-Saharan African countries, gender budgeting has primarily increased the number of programs that have gender-related goals, increased resources allocated to gender activities, and provided a better understanding of the budget impact on gender equality (Figure 16.5). The direct impact of implementing gender budgeting on gender equality is not easily measured. An increased awareness of gender equality goals may have brought to light key equality gaps and could have resulted in a redistribution of resources to address challenges. Additionally, gender budgeting has increased the understanding of the intended and unintended impact of the budget on either improving or undermining gender equality in sub-Saharan African countries, which could have led to modifying programs and projects during their development. Of note, rules or regulations seem to have been altered in other countries more frequently in response to gender budgeting than in sub-Saharan African countries.

The introduction of gender-related fiscal policies has accompanied progress in gender budgeting implementation. Figure 16.6 shows that nearly all sub-Saharan African countries surveyed have introduced paid parental leave for mothers, a key

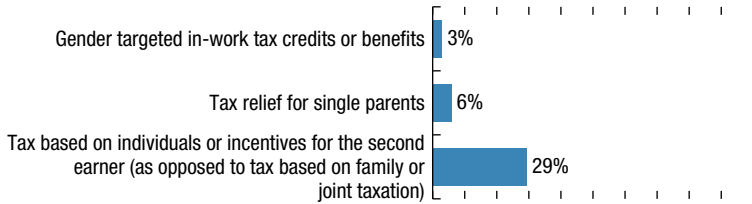
Figure 16.5. Impact of Gender Budgeting on Gender Equality
(As a percentage of the sample)



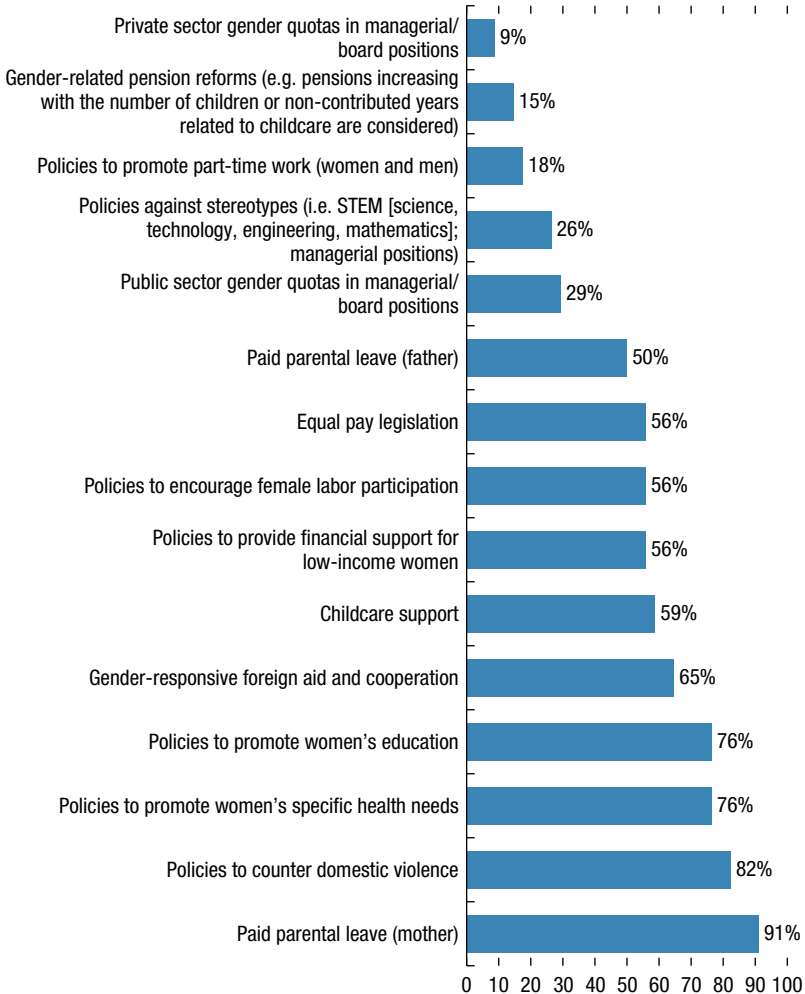
Source: IMF Gender Budgeting Survey.

Figure 16.6. Gender-Related Fiscal Policies Introduced by Sub-Saharan African Countries
(As a percent of the sample)

1. Tax Policies



2. Expenditure and Regulatory Policies



Source: IMF Gender Budgeting Survey.

Note: This illustration is based on a list of common fiscal policies that intend to promote gender equality, and it is not intended to be an exhaustive list.

determinant for women's continued labor force participation. Additionally, policies to counter domestic violence, promote women's specific health needs, and increase access to education for girls are among the top fiscal policies in place across the region. While over 50 percent of countries in sub-Saharan Africa have introduced at least 10 specific gender-related expenditure or regulatory policies, tax policies have been less widely adopted. The tax policy lag is consistent with findings that gender budgeting has typically focused on expenditures and overlooked taxes (Kolovich 2018).

CONCLUSION

The budget is a core policy instrument of the government; integrating gender equality objectives will ultimately result in better budgetary decisions. Although gender budgeting reforms require significant effort, consistent incremental steps to implementation over the medium term with feedback loops for learning and improvement can help ensure this reform's success. That is, countries should not be afraid to get started.

Sub-Saharan African countries have made substantial progress in adopting and implementing gender budgeting practices over the past decades. Good practices involve gender budgeting frameworks, linking gender equality goals with programs, and parliamentary oversight. Given the status of gender budgeting in sub-Saharan Africa, developing and publishing gender budget statements and improving guidance in budget call circulars is a suggested way forward. Only one quarter of sub-Saharan African countries surveyed produce an annual gender budget statement. As a strategic document, a gender budget statement tracks progress made toward gender equality objectives and identifies the programs or policies in place to close equality gaps. These statements, therefore, not only increase accountability, they also improve annual budget prioritization and planning. Sub-Saharan African countries are readily able to develop these statements since the information to formulate a statement has already been collected in many cases. That is, each country surveyed has gender equality policies in place and over two thirds of surveyed countries stated that they link gender equality goals to programs.

Sub-Saharan African countries should ensure their budget call circular is specific to overcome the lack of clear guidance on incorporating gender considerations into the annual budget. A budget call circular should explain to line departments and agencies what is required of them during the budget cycle, provide details on meeting the requirements, and include any gender budgeting templates to submit with their budget submissions. To complement these instructions, training on requirements for government officials regarding gender budgeting should be incorporated into broader budget training. Less than half of sub-Saharan African countries surveyed issued guidelines for gender budgeting procedures in a budget call circular. Issuing guidelines can begin with the ministry of finance having a clear plan for gender budgeting, which, if improved, could result in a more efficient budget process.

ANNEX 16.1

ANNEX TABLE 16.1.1

Instruments and Corresponding IMF Gender Budgeting Survey Questions			
12 “Instruments” or “Pillars”		23 Questions	
1	Legal framework on GRB	Q9	Is there a legal provision for gender budgeting? (select all that apply)
2	Institutional framework	Q7	Does the government have a gender budgeting framework?
		Q10	Is there a specific institution with primary responsibility for gender budgeting reform? (select up to two responses)
		Q13	Is there a specific organization that coordinates discussions on gender issues during budget preparation?
3	Gender statements	Q15	Does the government produce and publish a gender budgeting statement?
4	Ex ante gender impact assessment to assess the impact of proposed policies on gender equality	Q17a	Does the government conduct ex ante gender impact assessments on new government policy initiatives?
		Q17c	Is there a common methodology for ex ante gender impact assessments?
		Q17d	How are gender impact assessments used in the budget preparation? (Example of response option: It is used to identify the budget lines with gender impact (quantitative or qualitative))
5	Budget circular containing specific guidelines on gender objectives	Q16a	Does the budget circular or guidelines include details on the application of gender budgeting? (select all that apply)
6	Linking gender goals to programs and performance indicators	Q2	Does the government have policies to promote gender equality? (select all that apply—response example: Yes, gender equality policies are part of the national development / strategic plan)
		Q14	If gender equality goals are set are they linked to programs?
		Q19	Are there performance targets related to gender equality?
7	Budget classification incorporates a gender perspective	Q21b	If yes, how is gender incorporated into the program classification? (21a: Is the budget classified according to programs?)
		Q22	Are budgetary allocations tagged with any type of budget classifiers to identify their linkage to gender objectives?
8	Fiscal data disaggregated by gender	Q25a	Does the government collect data disaggregated by gender?
		Q25b	Is this data published?
		Q25c	Indicate how often the data is used in decision making (multiple responses “inform budget decision making,” “inform decision making on policies,” “inform decision making on new legislation,” and then options from Very Frequently, Frequently, Occasionally, Rarely, Never)
9	Budget execution reports incorporate gender perspective	Q26	Do budget execution reports issued by the government in-year include information on gender related expenditures?
		Q27	Does the government's end of year annual financial statement include information on the following?
10	Ex post gender impact assessments to assess the impact of policies on gender goals	Q28	Does the government conduct ex post gender impact assessments of budget expenditures?
		Q29	Indicate how often the gender impact assessments (ex ante and ex post) have influenced decision making in the following areas (list of areas such as informed annual budget decision making, and options from Very Frequently, Frequently, Occasionally, Rarely, Never)
		*adjusted to 28	
11	Gender-related audits	Q32	Has the National/Supreme audit institution in the past three years produced gender-related audit reports?
12	Parliamentary oversight	Q31	Has the legislature/Parliament in the last three years undertaken any of the following activities in relation to gender equality?

Source: Alonso-Albarran and others 2021.

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Gender Inequality and Care Work: Valuing and Investing in Care

Mehjabeen Alarakhia, Zahra Sheikh Ahmed, and
Tanima Tanima

This chapter explores the extent of and trends in care work in sub-Saharan Africa and makes a case for gender-responsive care policies. Care work, which encompasses both unpaid and paid care, is often undervalued and disproportionately falls on women and girls. Available time-use data show that caring responsibilities vary substantially with women's personal and labor market characteristics, and employed women—if their paid and unpaid work activities are added together—have much longer workdays. The unequal distribution of care work hinders women's empowerment, perpetuates gender stereotypes, and contributes to economic and social inequalities. Care work should be a critical component of the development agenda; there is a substantial need for better data, analysis, and research to inform gender-responsive care policies and measure progress. The authors outline the current state of care services and infrastructure in sub-Saharan Africa and present a framework for transformative care policies in the region. The chapter concludes by underscoring the significance of prioritizing care policies and infrastructure to achieve gender equality, inclusive development, and resilience in the face of crises.

INTRODUCTION

Care work—essential for sustainable economic development, social well-being, and the future of decent employment—involves the activities and social relations required to satisfy the productive and reproductive needs of all human beings. This includes direct care for children, the elderly, people living with disabilities, and people facing illnesses, as well as indirect care or domestic work, such as providing food and other necessities for a household. Care work encompasses two spheres of work—unpaid care work, including both direct and indirect, that is provided by individuals for the household or community, and paid care work that is usually direct care performed for pay or profit.

Care is a universal right and an essential public good. Despite its importance to society and the economy, care work is often not recognized as skilled work or

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a public issue. Households, governments, the private sector, and the community should share the responsibility for providing care work. Yet, most of the care work worldwide is undertaken by unpaid carers at a household level, and women and girls shoulder a disproportionate responsibility for it. Although unpaid care work can be rewarding, it is often drudgery, and its unequal distribution by gender forms a major barrier to women's empowerment in economic, political, and social spheres, and it hampers the overall well-being of women and girls. Compared with their prevalence in the labor market, women are also overrepresented in paid care jobs, especially domestic work jobs. Complex socioeconomic norms precipitate the disproportionate distribution of unpaid care work by gender and the feminization of care work. Care work is viewed as an extension of women's "natural" role as caregivers and thus perceived as a women's issue.

Despite data limitations, a few trends in sub-Saharan Africa are clear. First, women and girls make up most of the unpaid care workforce. According to estimates from the International Labour Organization (ILO), Addati and others (2018), and Charmes (2019), women and girls spend as much as 3 to 3.4 times more on unpaid care work than do men and boys. Second, responsibilities of women to provide unpaid care work hinders their economic, social, learning, and leadership opportunities. Women in sub-Saharan Africa cite unpaid care work duties as the top reason for not participating in the labor force (Addati and others 2018). In addition, women in households with dependents (children or elderly people) work fewer hours in paid employment than do women in households with no dependents. Third, on average, women's workdays—when both paid and unpaid work hours are added together—are much longer than men's. Time poverty among employed women is significant and can lead to or exacerbate income poverty when women and households are unable to outsource care services (Zacharias, Antonopoulos, and Masterson 2012). Fourth, when women engage in paid work, their unpaid care responsibilities do not decrease proportionately. This affects their overall well-being, the quality of care enjoyed by caregivers and care recipients, and women's overall productivity both in paid and unpaid work. Finally, paid care work is a very important source of employment for women, and most women in paid care jobs are domestic workers with informal working arrangements, low pay, and poor working conditions.

Why do we need better care policies and investment in care infrastructure in sub-Saharan Africa?

For better social, economic, and political growth of women and girls. Considerable research has identified unequal gender distribution of care work as a key barrier to women's social, economic, and political empowerment. Socially prescribing women and girls as caregivers undermines their basic human rights and limits their opportunities and capabilities. Better care policies and infrastructure are key for alleviating gendered inequalities and increasing women's quality of life.

For poverty reduction, which includes better health and education outcomes for all. Care work is systematically linked with economic inequality, poverty, and

precarity. Care work is largely performed by women and girls within the home in an unpaid manner; thus, it affects marginalized households the most. Marginalized households lack essential infrastructure, such as clean water and fuel, so women spend more time on indirect care work. Moreover, low-income households are unable to afford quality childcare services or domestic care services. This constrains women from doing paid work or from working for more hours in paid jobs. Thus, evidence shows that better care systems, especially public childcare provisions, are fundamental to overcoming poverty and reducing inequalities (Halim, Perova, and Reynolds 2023). Further, better quality care services enhance quality of life as well as health care and education outcomes for care recipients, especially children (Staab 2015).

For supporting inclusive economic growth and development. Investment in care systems linked to labor market policies can create jobs, increase women's productivity in both paid and unpaid work, facilitate women's participation in the labor market, and remove inequalities in the labor force and at the household level. Women's entry into the labor market and increased productivity lead to large economic dividends; this is linked to the fact that paid care work is an important source of employment for women, and it is poorly remunerated, undervalued, and frequently carried out in precarious conditions. Quality employment also generates more economic activity and higher tax revenues for the government. Overall, better care systems support inclusive economic growth and development.

For creating a resilient society and economy. The COVID-19 pandemic made even more evident the deficiency in how care systems are set up. Sub-Saharan Africa is vulnerable to crises, including but not limited to internal and external conflicts, environmental degradation, and climate change. Crises tend to increase care needs, make care work more arduous, and compound the injustices related to its unequal distribution. Such effects are felt most by those pursuing subsistence livelihoods. Robust care infrastructure and care systems help families, communities, and societies weather these crises.

In light of these factors, development practitioners and policymakers are increasingly recognising the need for better care policies and care infrastructure. This chapter discusses the state of care work and care policies in sub-Saharan Africa and highlights the need for more data and analysis to further the care agenda. The rest of this chapter is organized as follows: The second section uses available data in sub-Saharan Africa to measure trends and determinants of care work, both paid and unpaid, and discuss the importance of data and time-use surveys (TUSs); this section constitutes the core of the chapter. The third section, "Transformative Care Policies," discusses what transformative care policies look like and presents an overview of the state of the care economy in the region. The fourth section, "From Better Data to Better Measures of Progress," makes the case for better data, statistics, and research on care for informing gender-responsive care policies and measuring progress. The final section contains a summary and considers avenues forward.

MEASURING CARE WORK IN SUB-SAHARAN AFRICA: PAID AND UNPAID

All over the world—and sub-Saharan Africa is no exception—the family or household provides most unpaid care work. Surveys on time use and the labor force are complementary sources of data for understanding the extent, trends, and determinants of unpaid care work. Although the paid care sector in sub-Saharan Africa is small, it is essential to the functioning of society. This section mainly draws on estimates from the ILO (Addati and others 2018), compiled using the latest TUSs and labor force surveys in all sub-Saharan African countries where data were available. First, the authors measure the magnitude and determinants of unpaid care work, composed of own-use production work of services¹ and volunteer work in households producing services according to the 19th International Conference of Labour Statisticians Resolution I concerning work, employment, and labor underutilization. Second, the authors assess labor market characteristics of unpaid carers. Third, they examine care jobs and care workers. Fourth, they discuss data gaps and TUSs.

Extent of and Trends in Unpaid Care Work

The average woman in sub-Saharan Africa² spends 249 minutes per day on unpaid care work; figure 17.1 presents average time spent by women on unpaid care work by country and category (Charmes 2019). In comparison, the average man spends 87 minutes per day on unpaid care work, which is about 40 percent of the time spent by women on the same activities. Figure 17.2 presents ILO estimates of time spent on unpaid care work by men, by country and category. Both men and women spend most of their time on domestic services for own final use within households (for example, indirect care work, housework), indicating such services' importance for all adults. Caregiving services to household members (that is, direct care work) make up the next largest category of unpaid care work and community services. Help to other households (for example, volunteer work, a mixture of direct and indirect care work) makes up the smallest category of unpaid care work.

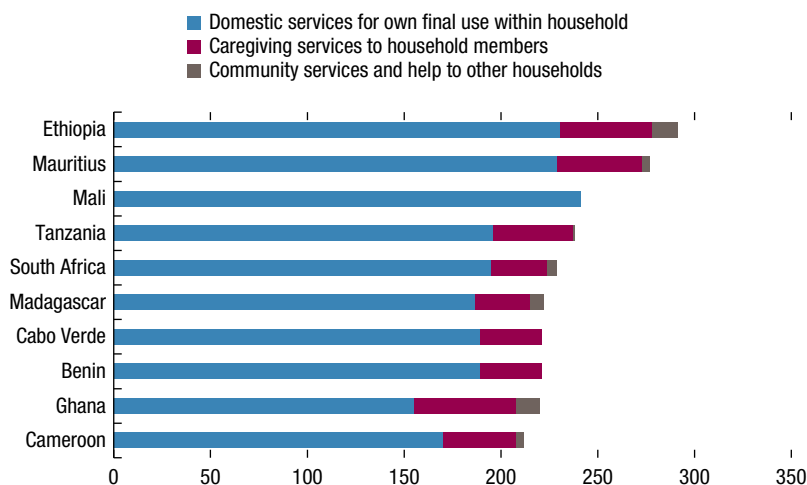
The average time spent on unpaid care work varies considerably across countries. Men in Mali spend the least time (21 minutes) and men in Cabo Verde spend the most time (246 minutes). It is worth noting that Cabo Verde is characterized by a difference in data collection methodology, so the authors

¹ The ILO refers to own-use production work as “activities performed to produce goods or provide services intended for final use by the producer, their household and/or family” (ILO 2023).

² In this subsection the authors calculate the regional averages for time use by using weights according to country population; however, readers should be careful with these estimates because the authors use data from only 10 countries in the region: Ethiopia, Mauritius, Mali, Tanzania, South Africa, Madagascar, Cabo Verde, Benin, Ghana, and Cameroon.

Figure 17.1. Women's Time Spent in Unpaid Care Work, by Country and Categories

(Minutes per day)

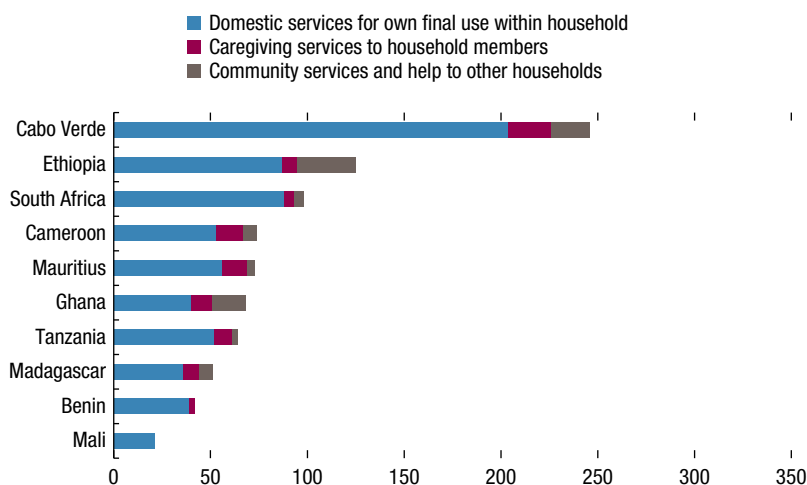


Source: Charmes 2019.

Note: Refer to Annex 17.1 for survey year and characteristics.

Figure 17.2. Men's Time Spent in Unpaid Care Work, by Country and Categories

(Minutes per day)



Source: Charmes 2019.

Note: Refer to Annex 17.1 for survey year and characteristics.

urge caution about comparing Cabo Verde with other countries.³ Women's unpaid care work ranges from 212 minutes in Cameroon to 291 minutes in Ethiopia. Mali has the largest gender divide, with women performing 92 percent of total unpaid care work. Even in Cabo Verde, which has the smallest gender divide in time allocation, women do 66.6 percent of the total unpaid care work.

The large gender gap in time spent on unpaid care work has two main and direct effects. One, women participate much less in paid work. In sub-Saharan Africa, women work only 0.66 hour for every 1 hour spent by men in paid work. Second, the total working hours spent by women per day is more than that spent by men. If the total time spent working, both paid and unpaid, is added together, women spend 432 minutes per day working, while men spend only 361 minutes, on average, which means that women have less time available for other activities such as education, participating in civic activities, or leisure. This outcome—referred to as *time poverty*⁴—can lead to or exacerbate income poverty for women and households. It can also lead to poor-quality care for recipients and affects the health and well-being of the unpaid carers. Research confirms that women's labor force participation shrinks the gender gap in unpaid care work, but the decrease in time spent by women in unpaid care work does not lessen in proportion. Rost, Bates, and Dellepiane (2015) conducted a study in Ethiopia, Philippines, Uganda, and Zimbabwe; the researchers found that women spend 10 to 44 fewer minutes on primary unpaid care work per 60 minutes of paid work; however, women's participation in paid work has no effect on secondary care work or supervision of dependents.

Gender gaps in time use are not uniform; they vary based on numerous socioeconomic factors and individual characteristics, which highlights that intersectionality matters. Overall unpaid care work done by women and girls is higher in rural areas, for less educated women, for married women, and for women living in households with children. Men in rural residence and men living with children spend more time on unpaid care work, similar to women. However, the addition of a child to the household increases women's care duties more than men's care duties. Interestingly, more educated men spend more time on unpaid care work, indicating that education may influence social norms. In Ethiopia, Ghana, and Tanzania, young women spend as much or more time on unpaid care work as do adult women. Children are also significant providers of unpaid care work in Africa, and girls perform more work than do boys.

³ The survey in Cabo Verde was conducted by asking respondents stylized questions about their activities, not including paid work. Thus, one cannot reconstitute the complete schedule of an individual's day.

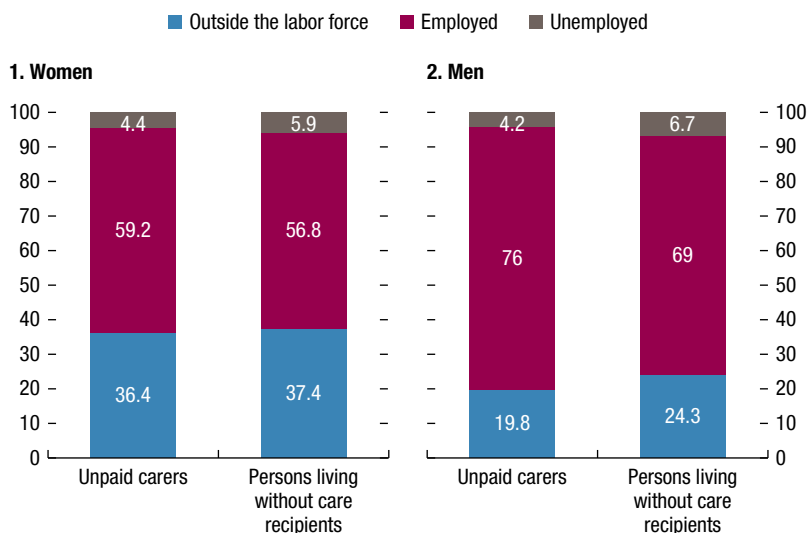
⁴ Although several understandings and definitions of the term time poverty exist, the authors here conceptually use it to mean “the potential failure of individuals to fulfill their labor commitments while simultaneously meeting some minimum needs of personal maintenance” (Zacharias 2023).

Unpaid Care Workers and the Labor Force

The authors now compare people living with care recipients at home (children under 15 years or elderly people), a proxy for people with high unpaid care work responsibilities, with people living without care recipients. This assessment continues to use ILO estimates (Addati and others 2018), based on data from national labor force surveys and household survey microdata, which covers 24 countries and approximately 70 percent of sub-Saharan Africa's population. As shown in Figure 17.3, approximately 36.4 percent of women with direct care responsibilities are outside the labor force, compared with 19.8 percent of men living with care recipients. Conversely, 24.3 percent of men living without dependents are outside the labor force, which is much more than men with care responsibilities. In this way, men living with care recipients experience a *labor force premium*, which provides evidence for the “male as breadwinner” model.

Labor force and household surveys from 21 African countries covering 61 percent of the region's population show that the most important reasons cited by women for being out of the labor force was unpaid care work (34.4 percent), followed by personal reasons—that is, being in education or being sick or disabled (33.8 percent). Sub-Saharan Africa's share of people ages 15 to 24 is also substantial (20 percent), and a large proportion of these young people are not in employment, education, or training (NEET). Based on data

Figure 17.3. Unpaid Carers and Persons Not Living with Dependents, by Gender and Labor Force Status
(Percentage)



Source: Addati and others 2018.

Note: Refer to Annex 17.1 for survey year and characteristics.

from nine⁵ countries in east and southern Africa, Perry (2022) found that women are more likely to be NEET than men. Importantly, care work increases the probability of NEET status for women, especially for women 20 to 24 years old. Taken together, these findings suggest that women's labor force participation is hurt by caring responsibilities.

Income and residence also affect labor force participation decisions. Women living without care recipients are less likely than women living with care recipients to participate in the labor force. This is most likely explained by the fact that in resource-poor environments, women must search for paid work when there are dependents to look after. Further, households in sub-Saharan Africa often consist of extended family members—an arrangement that allows for care responsibilities to be shared. Unpaid carers are also less economically disadvantaged in rural areas, where 34.7 percent of women with care recipients at home are outside the labor force, compared with 41.6 percent of their urban counterparts.

In sub-Saharan Africa, whether a woman resides in a household with dependents alters the intensity and quality of her employment. In households with dependents, especially with young children who are less than six years old, women work fewer hours in paid employment; men see no such trend. Using data from 23 countries, which constitute 72 percent of the total employed population in sub-Saharan Africa, the authors find that the gender gap in working hours per week was 5.2 hours for employed men and women living in households with no children. This gap rises to 6.8 hours for employed men and women living in households with one child and increases further to 7.2 hours for employed men and women in households with two children. It is a well-documented fact that women's wage employment is low in sub-Saharan Africa, but it dips even lower for employed women carers (13.8 percent). Most employed women living with dependents work in the informal sector (92.2 percent), mostly in own-account and contributing family work, which is slightly higher than employed women who live with no dependents (88.6 percent). Informal jobs come with significant downsides, but one reason women prefer them is for the flexibility they allow for juggling unpaid care work and paid work (Ceita 1999 cited in González and Grinspun 2001; Marcucci 2001; Verceles and Beltran 2004; Ramirez and Roses 2005).

Care Jobs and Care Workers

A serious deficit exists in care service provisions in the region. Sub-Saharan Africa has the lowest proportion of paid care jobs among all regions in the world, although there is considerable variation across countries. The care workforce generally includes (1) workers in education, health, social work, and domestic work; (2) care workers in other sectors (such as nurses in factories); and (3) non-care workers in care sectors (such as cooks and cleaners in a hospital or school). In sub-Saharan Africa, about 10.7 percent of employed women, compared with only 5.4 percent of employed men, are employed in care jobs. Duffy (2021)

⁵ Botswana, Ethiopia, Kenya, Malawi, Mozambique, Namibia, Rwanda, South Africa, and Uganda.

estimates that the care employment constitutes as much as 18 percent of total employment in South Africa and as little as 3.5 percent of total employment in Mozambique.

Such feminization of care work holds true across the world, and care work is often an entry point into the labor force for women in low-income countries. In sub-Saharan Africa, only about half of the care workers in care sectors are women, although this statistic hides significant vertical and horizontal segregation: the majority of the high-status care jobs are held by men. Agriculture is the largest sector in most sub-Saharan countries and women are overrepresented in agricultural jobs. For instance, about three out of every four people in Mozambique work in agriculture and 58 percent of agriculture workers are women (Duffy 2021).

On average, care work is underpaid, unprotected, and undervalued. These characteristics are related to its perception as an extension of women's care roles, which lends it low status and low social recognition. However, not all care workers earn low wages. Hierarchies and differences exist among care workers in terms of remuneration, working conditions, and status. Some care professions are legally and socially regulated, which makes them high status and better paying. For example, medical doctors earn 55 percent more than nurses and 89 percent more than midwives in South Africa.

Domestic work is an important sector of care work in the region. About a third of sub-Saharan Africa's care workforce, 2.3 million people, are domestic workers. This is a strongly feminized sector: 80 percent of all domestic workers are women. Informal working arrangements, low remuneration, and long working hours are common characteristics of domestic work in the region. Domestic work happens behind closed doors and comes with much higher risk of sexual harassment and physical violence. These workers are isolated and unable to organize, aspects that affect their bargaining position and contribute to their lower pay and long working hours. The pay penalty is particularly prevalent in this sector, for example, as found in South Africa (Budlender 2011). Given the relational nature of care work, it is also difficult to threaten to withdraw services or go on strike.

Data Gaps and TUSs

Unfortunately, much is left to be understood, especially about unpaid care work. For example, men's and women's contribution to unpaid care work has fallen slightly across the globe from 1997 to 2012; however, overall gendered distribution of unpaid care work remains largely the same. This global estimate is based on 23 countries with time series data, which include only 3 African countries (Benin, South Africa, and Tanzania). Thus, the authors cannot generalize this trend to sub-Saharan Africa. Similarly, income and time poverty move together and have an inverse-U relationship across the globe, but it's unclear whether this holds true in sub-Saharan Africa. Also, links between time poverty and inequality, agricultural productivity, income poverty, and labor force participation are not yet understood.

Lack of nationally representative and comparable time-use data remains a big part of the challenge. The authors use data from 10 countries in sub-Saharan

Africa, where comparable data from TUSs were available between 1998 and 2019. Since then, a few more countries in the region have conducted TUSs, some of which are still forthcoming. Yet most countries in the region have not conducted such surveys. To a great extent, this is because these surveys are costly and complex to conduct and require literate respondents. Time-use data are most reliably collected based on time diaries—that is, through an inventory of all activities conducted in the previous 24 hours. However, this effort requires literacy and numeracy. Asking stylized questions on a one-week reference period is still common practice in some TUSs. This approach avoids the problem of literacy and numeracy but often results in imprecise estimates. Further, there are problems with harmonizing definitions across countries and surveys and with adequately capturing activities that are simultaneously undertaken—for example, watching television while caring for a child.

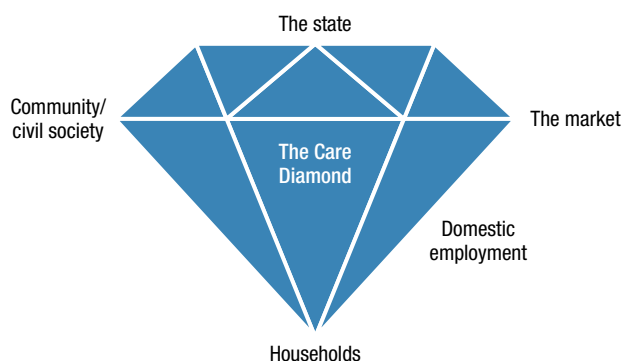
TRANSFORMATIVE CARE POLICIES

This section discusses the ILO's 5R framework, a policy approach to care work, applied to sub-Saharan Africa. The authors also introduce the African Care Economy Index (ACEI), which measures social recognition and state support for care work in African countries and reveals significant deficits in care policies across the region.

The 5R Framework

The care diamond expresses the social division of care in a society among four key institutional actors: families and household, markets, the state, and the community and other civil society organizations (Figure 17.4; Razavi 2007). Although most unpaid care work is concentrated at the household level, paid care work

Figure 17.4. The Care Diamond



Source: UN Women Training Center. <https://portal.trainingcentre.unwomen.org/product/introduction-to-care-work-and-care-economy/>.

Note: Reproduced with permission from UN Women.

forms the other three points of the diamond. Governments are primary duty bearers for the provision of care services and care infrastructure. Care is both a right to which people should have access and a function that some people perform. From a rights-based perspective, care policies ensure that everyone has the right to receive and provide quality care.

The ILO's 5R framework is a human rights-based and gender-responsive approach to public policy for care work. The framework recommends that policy should *recognize*, *reduce*, and *redistribute* unpaid care work; *reward* paid care work; and guarantee *representation* for care workers through social dialogue and collective bargaining. It aims to build a virtuous cycle to mitigate inequalities in care work, improve the socioeconomic lives of unpaid carers and paid care workers, and address barriers that keep women from entering the labor force. By extension, it works to improve quality of care for care recipients.

As a first step, more widespread *recognition* of the importance of care work at both the international and national level is required. Care work is necessary for all; requires time, energy, and skills; and comes with an opportunity cost in terms of time not spent on other activities. There was a landmark resolution in 2013 by the 19th International Conference of Labour Statisticians for inclusion of unpaid work and household production in System of National Accounts. However, national-level legislation and policy efforts across all countries in sub-Saharan Africa are far from including unpaid care work in national statistics. Foremost, there is a need for robust data at a national level, especially time-use data. Further research is required to measure all forms of care; understand links with GDP, poverty, and labor force participation and mitigating factors; and track progress in building care policies, care infrastructure, and care services.

Reducing unpaid care work requires public investment in both (1) social care services infrastructure and (2) physical rural infrastructure that is of high quality and affordable and accessible to all. Social care infrastructure—physical, human, and financial—supports universal social care services for children and people living with disabilities as well as universal health care. It transforms a substantial amount of unpaid domestic care work to paid social care work. In the absence of state involvement, such care services are expensive and quality services are available only to the high-income minority. This situation enhances gender, class, and intergenerational inequalities and leads to increased reliance on informal labor and migrant domestic workers. Building better physical rural infrastructure lessens unpaid care work by reducing the time required to deliver indirect care work. This includes but is not limited to building roads and providing access to clean water, sanitation facilities, electricity, digital technology, and the internet. Protection of natural resources such as land and water is equally important in ensuring adequate and inclusive care infrastructure for all. The formal private sector also has a role to play in protecting workers and providing decent working conditions so these employees are not overworked, undervalued, or underpaid.

Redistributing care work calls on two types of redistribution: between women and men and between all actors in the care diamond—households, the state, the

private sector, and the community. Given the gender-based division of care labor, all relevant actors must aim to eliminate discriminatory social norms and gender stereotypes that assign care work to women by promoting positive masculinities to encourage men's increased participation in care work. Such transformation of mindsets cannot happen in isolation and requires changes across society. Efforts must be culturally sensitive and emphasize strengthening the family unit and community bonds in sub-Saharan Africa. Furthermore, the state must actively take up responsibility for providing care as a universal right, in terms of both access to and quality of care services. Households usually bear the ultimate responsibility for filling care deficits and assume care responsibilities when public provisions are lacking. Important areas of intervention include labor market regulation and provisions of public care services. The state also has an active role to play in encouraging and incentivizing the private sector to offer gender-inclusive family-friendly employment policies to maintain work–life balance and maternity and paternity leave; they must support carers at the workplace. The last point of the care diamond, the community, can also support care work in various ways, from formalized entities to unofficial community-based networks.

Paid care work is systematically undervalued and underpaid. Policy efforts should seek to adequately *reward* paid care work, recognizing that it requires skills and resources. This includes regulating formal employment terms and conditions so that they reflect the principles of equal pay for equal work and decent working conditions. It is crucial to recognize that most of the labor force in sub-Saharan Africa, especially care workers, is concentrated in the informal sector. Thus, the state must focus its efforts on formalizing care systems and ensuring that social protection schemes include informal workers and migrant workers. Laws and policy measures must also ensure safe working conditions for paid care workers. Promoting *representation* of workers through collective bargaining and social dialogue is an equally important focus area. Decades of privatization and austerity have eroded collective bargaining power (Montague-Nelson 2022), but collective action remains an important forum in which paid care workers, especially women, can engage with employers and governments to discuss and negotiate working conditions and participate in decision-making processes.

Although not officially a part of the 5R framework, there is an emerging need to recognize the significance of *resilience* as an important characteristic of care systems. The COVID-19 pandemic shed light on the importance of care in the face of crisis. Sub-Saharan Africa is prone to various crises, including conflicts, climate change, and public health emergencies. Essential care services are needed more than ever in crisis situations, which often exacerbate the gendered distribution of care work. Thus, policy efforts need to promote resilient care systems in the face of spiraling care needs and demands on women and girls, and such efforts must address how crises affect food and energy shortages as well as forced migration.

The State of Care Services in Sub-Saharan Africa and the ACEI

There are large deficits in the coverage of care policies across sub-Saharan Africa, and all countries are far from realizing universal care provisions. Legislation for maternity leave, ranging from 8 to 17 weeks, exists in 23 African states.⁶ A few countries offer paid paternity leaves, but they are considerably shorter and range from 1 day to 10 days. There is a distinct lack of socialized childcare, elder care, and care for people living with disability; only a few countries regulate private provision of care in these areas. Socialized health care is also generally lacking and inadequate in size and accessibility. Notably, Kenya has recently set precedence by being the first country in the region to start work on a comprehensive care policy, though many countries do have legislation that addresses specific aspects of care.

ACEI is a tool developed by the African Women's Development and Communication Network (FEMNET) to measure social recognition and state support for care work in Africa. The index is composed of 10 basic metrics related to legislation, policy, and government expenditure. They are maternity and paternity leave, socialized childcare, socialized care for the elderly, socialized care for people living with disability, socialized health care, socialized food production, COVID care measures, domestic worker protection, care grants and subsidies, and family care leave. The index assigns a score out of 30, and a score of 18 and above is considered a passing grade. FEMNET analyzed and assigned a measure of the index in 54 African countries (Valiani 2022). All 54 countries scored less than 8, much below the passing grade. Burkina Faso (7.2), Ethiopia (6.3), Zimbabwe (6.0), South Africa (5.7), Kenya (5.7), and Ghana (5.5) were the top-scoring countries, with index values higher than 5. The Gambia (0.9), Nigeria (0.9), and Eritrea (0.3) were the lowest-scoring countries, with index values less than 5.

FROM BETTER DATA TO BETTER MEASURES OF PROGRESS

Care work should be understood contextually because it is deeply influenced by social norms, cultural practices, and historical factors specific to a given society or region. In this section, the authors emphasize the importance of a pan-African feminist framing of care and consider the role of TUSs and qualitative data. This section also discusses the exclusion of care work from GDP and income estimates and the need for reliable and sex-disaggregated data to inform gender-sensitive policies and programs in the region.

⁶ Algeria, Angola, Benin, Botswana, Burkina Faso, Burundi, Cabo Verde, Cameroon, Central African Republic, Chad, Comoros, Republic of Congo, Democratic Republic of the Congo, Côte d'Ivoire, Djibouti, Egypt, Equatorial Guinea, Eritrea, Eswatini, Ethiopia, Gabon, The Gambia, and Ghana.

A Context-Specific Understanding of Care Work

Economists, feminists, and feminist economists have been discussing care since the 1960s, as highlighted in Ossome and Naidu (2021). Care work is central to human societies and essential for the functioning of families, societies, and economies. Care is defined by interdependence and interconnectedness, given the care needs of a person over a lifetime, from childhood to old age, and given the prevalence of sickness and disabilities. This interconnectedness is similar to the South African ethos of “Ubuntu,” or “togetherness.” Ubuntu highlights that “an authentic individual human being is part of a larger and more significant relational, communal, societal, environmental, and spiritual world” (Mugumbate and Chereni 2020).

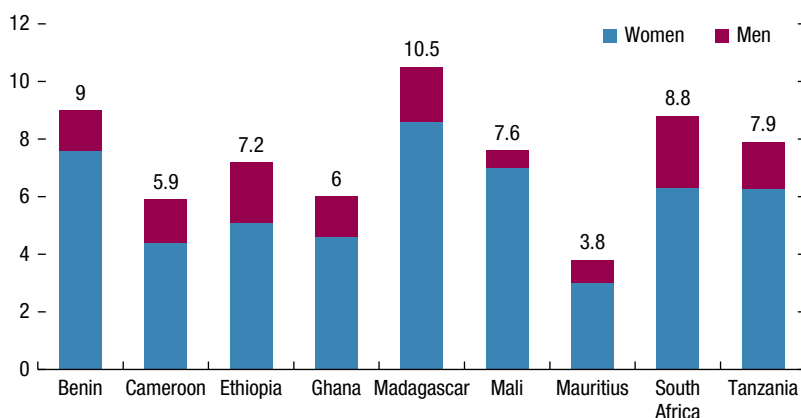
Precisely due to this nature of care, it is heavily influenced by social norms that define *what* is good care, *who* is the caregiver, and *who* is the care recipient. Thus, a discussion and understanding of care must acknowledge how culture, religion, and history have contributed to the formation of these norms in the African context. Conceptual understandings of care, especially at a policy level, do not include a pan-African feminist framing, which is unsurprising given that care has not been a policy priority in the region. A growing body of thought and research on care work and the care economy in Africa is highlighting the need to define care for the region that considers more than just childcare, domestic work, and health care. The extended values of Ubuntu that embrace care for informal networks within the community, including extended families and friends and humanitarian work, should be considered. Furthermore, care work could extend to traditions of well-being, such as Indigenous healing practices, environmental protection, and spiritual care.

TUSs constitute a key source of quantitative data on unpaid care work. Although difficulties in implementing TUSs have been briefly discussed, their effectiveness in the African context must be further interrogated. For instance, what qualifies as a “household”? The answer is important because this unit is used to link the number of dependents and amount of care work needed. These contextual considerations must be balanced with concerns for international comparability. Much can be gained by conducting TUSs that are at least regionally harmonized—something that has not been achieved among the few such surveys that have been conducted so far. Furthermore, while TUSs give a quantitative estimate of time invested in unpaid care work, they are insufficient to understand the consequences and patterns that emerge from the observed time-use trends. Documenting real stories of people’s, especially women’s and girls’, lived experience is required to deeply understand the extent of care labor and its value to the household and community. Detailed community studies can also provide a better understanding of the local environments in which unpaid care work takes place (Folbre 2018).

Incorporating Data in National Statistics and Measures of Economic Progress

Unpaid care work generally compensates for a lack of public expenditure on care infrastructure and services. Despite its substantial magnitude and contribution to the economy, unpaid care work is excluded from the measurement of GDP, the

Figure 17.5. Value of Unpaid Care Work as a Percentage of GDP, US Dollar PPP 2011, by Gender and Country
(Percent of GDP)



Source: Addati and others 2018.

Note: Refer to Annex 17.1 for survey year and characteristics. PPP = purchasing power parity.

main indicator of national income. Although GDP is often used as an indicator of development, it underestimates overall economic activity and undervalues social well-being. Figure 17.5 presents ILO estimates (Addati and others 2018) for the value of unpaid care work as a percentage of GDP for nine sub-Saharan African countries. These estimates are calculated by assigning a monetary value to time spent on unpaid care work by costing time based on minimum wages in the country, known as the opportunity cost approach. Women's unpaid care work ranges from 8.6 percent in Madagascar to 3 percent in Mauritius, and men's contribution ranges from 2.5 percent in South Africa to 0.6 percent in Mali. Although there are concerns that such estimates may be imprecise, they provide at least a lower bound to the economic value of unpaid care work. These concerns also overlook the reality that despite its considerable contribution, unpaid care work is currently valued at zero.

Another important consideration that has received less attention is valuation at the household level. Unpaid care work is not accounted for in household income or consumption estimates (Folbre 2015). Market-centric discussions about women's labor supply and labor market participation fail to recognize that unpaid care work is an essential complement to market income. Thus, market-based income measures ignore the opportunity cost of reduced time for unpaid care. Investment in time-saving devices and basic infrastructure could significantly improve women's productivity, increase their participation in paid work, and enhance household living standards. Such payoffs to public investments are understated when nonmarket work is not valued.

Lack of reliable and sex-disaggregated data on unpaid care work is a major barrier to gender-sensitive policy and programming. Efforts across the world and

especially in sub-Saharan Africa still have a long way to go. Although there are some general trends in unpaid care work, there is considerable heterogeneity in the state of the care economy across nations. Thus, a more localized and contextual understanding of unpaid care work and care needs is needed. Such efforts are also important for tracking overall progress toward the care agenda at the national and regional levels and for sharing learnings across the region.

CONCLUSION

Women and girls play a significant role in providing care services across sub-Saharan Africa, in both an unpaid and a paid capacity. Care responsibilities have a negative impact on the ability of women to participate in the economy; form a barrier to their social, learning, and leadership opportunities; and affect their well-being and social status.

Most care services are provided by the family or household in an unpaid capacity, while paid care workers deliver a smaller, but important, proportion of care services. Women in sub-Saharan Africa spend more time on unpaid care work than do men, which leads to time poverty for employed women as well as gender gaps in labor force participation and time spent on paid work. Even as women enter the labor force, their unpaid care work responsibilities do not reduce proportionately. Thus, employed women work a “double shift”—one at work and one at home. Women from varied socioeconomic backgrounds have varying responsibilities for care work, and gender gaps in time use are not uniform. Hence, policymakers must employ an intersectional approach when devising care policies and care infrastructure. Unfortunately, any understanding of unpaid care work in sub-Saharan Africa is severely limited by data unavailability. TUSs are the most important source of data on unpaid care work, and few countries in the region have conducted a TUS in the past three decades.

Care jobs, especially paid domestic work, are a big source of employment for women. The paid care workforce is very small in sub-Saharan Africa and suggests deficiencies in care service provisions. Women’s care jobs, on average, are characterized by low pay, informal contracts, and difficult and unsafe working conditions. Most of these jobs are domestic work. The role of social norms in observed gender gaps cannot be overstated. Gender stereotypes of unpaid care work and the social association of women’s “natural” inclination toward caring explain to a great extent the high level of feminization prevalent in paid care employment.

Policymakers first need to recognize the care agenda as an important policy priority. Universal provision of quality and accessible care services is still far from reality on the continent. Public investment in social care and physical rural infrastructure is required to reduce time spent on unpaid care work. Redistributing care work between men and women and between social actors—households, markets, the state, and the community—is also an important part of the care agenda. Policies must focus on adequately rewarding paid care work and promoting representation of paid care workers through collective bargaining and social dialogue.

Care work must be understood in a context-specific manner—one that considers social norms and the extended values of Ubuntu. Because unpaid care work is excluded from the measurement of GDP, as such, GDP underestimates overall economic activity and undervalues social well-being. Efforts to collect reliable and sex-disaggregated data on care work, as well as including such data in national statistics, are necessary for tracking progress at the national and regional levels. TUSs are a key source of quantitative data on unpaid care work, but their implementation in sub-Saharan Africa must be carefully examined to balance contextual considerations with concerns for international comparability. A key priority for the care agenda is to strengthen the production and dissemination of data and statistics on care work and care-related policies.

ANNEX 17.1

ANNEX TABLE 17.1.1.

Time-Use Survey Characteristics, by Country						
Country	Year	Period	Type of Survey	Sample Size	Survey Instrument	Mode of Data Collection
Benin	2015	1 month	Module of household survey	13,026 individuals	One diary	Interview
Cameroon	2014	1 month	Module of household survey	4,988 households	One diary	Interview
Cabo Verde	2012	3 months	Module of household survey	3,390 households	One diary	Interview
Ethiopia	2013	1 month	Stand-alone	52,262 individuals	One diary	Interview
Ghana	2009	2 months	Stand-alone	9,297 individuals	One diary	Interview
Madagascar	2001	2 months	Stand-alone, sub sample of household survey	7,749 individuals	One diary	Interview
Mali	2008	2 months	Stand-alone	2,249 individuals	One diary	Interview
Mauritius	2003	2 months	Module of household survey	6,480 households	One diary	Interview
South Africa	2010	3 months	Stand-alone	30,897 individuals	One diary	Interview
Tanzania	2014	4 quarters	Module of household survey	10,553 individuals	One diary	Interview

Source: Charmes 2019.

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Fintech, Female Employment, and Gender Inequality

Boileau Loko and Yuanchen Yang

Fintech—new technology in finance that delivers financial services digitally—promises to promote financial inclusion and close the gender gap. Using a novel fintech data set, this study shows that fintech adoption significantly improves female employment and reduces gender inequality. The effect is more pronounced in firms without traditional financial access. Fintech not only increases the number and ratio of female employees in the workforce but also mitigates the financial constraints of female-headed firms. However, poor institutions and governance weaken such benefits.

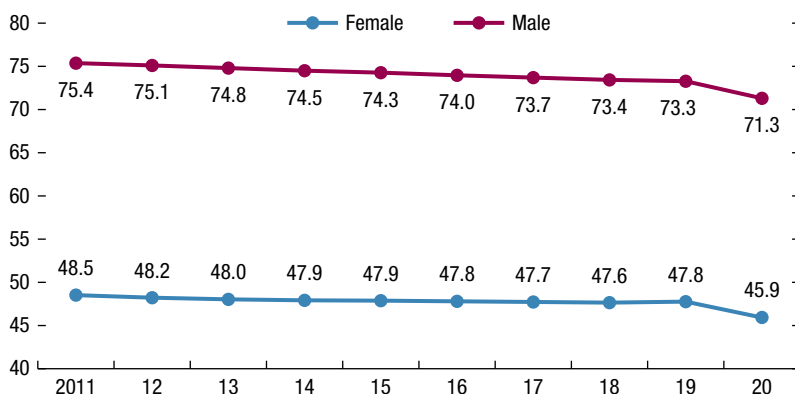
INTRODUCTION

Gender equality lays the foundation for a peaceful, prosperous, and sustainable world (United Nations Development Programme 2020). Despite progress made over the past decades, as highlighted in Chapters 1 and 2, many challenges remain—women experience higher poverty levels, unemployment, and other economic hardships (IMF 2021). In the global financial system, women continue to be underrepresented at all levels, from depositors and borrowers to managers and regulators (Sahay and Cihak 2018).

Much hope exists that new technology in finance (fintech), which has spread quickly worldwide, will unlock vast potential for stimulating economic growth and increasing social welfare. Fintech could also help reduce gender inequality and promote female employment, particularly in the current context, where the COVID-19 pandemic has exacerbated the existing gender gap in employment (Figure 18.1). According to the World Bank, the current global employment rate is less than 46 percent for women but 71 percent for men. Boosting female employment could generate substantial growth benefits (IMF 2021; International Labour Organization 2022) and lay the foundation for other forms of gender equality, such as income and social status, while boosting macroeconomic outcomes.

Loko and Yang's study (2022) was among the first to shed light on the link between fintech and gender inequality measured by female employment. Previous research had focused on specific case studies; however, Loko and Yang used a cross-country data set covering 114 economies and a range of gender inequality indicators at the micro level. They showed that women—traditionally marginalized by the formal financial system—can be included and experience improved welfare in the new fintech ecosystem.

Figure 18.1. Employment Rate by Gender
(Percent)



Sources: International Labour Organization; and author calculations.

Note: Female (male) employment represents the labor force participation rate of female (male) population ages 15+ based on modeled International Labour Organization estimate.

Using a novel fintech indicator and firm-level employment data, this study¹ shows that fintech adoption significantly improves female employment and reduces gender inequality worldwide and in the sub-Saharan African region. Specifically, a 1 percent increase in the scale of fintech usage is associated with a 1.4 percent increase in the number of female workers and a 0.4 percent increase in the ratio of female to total employees in the sample firms. The economic significance is pronounced, given that female employees represent only 32 percent of employees in the sample. Fintech not only increases the number and ratio of female employees in the workforce but also mitigates the financial constraints of female-headed firms. The effects are more pronounced in firms without traditional financial access. However, the digital divide and poor institutions weaken such benefits.

Admittedly, a significant gender divide exists in accessing fintech services that can be ascribed largely to differences in attitudes toward privacy and technology (Chen and others 2021) and technological and institutional factors. This study identifies technological, legal, and regulatory barriers that have constrained fintech usage and proposes pathways to build a more gender-inclusive financial ecosystem.

BACKGROUND

Using digital platforms, fintech could easily cross physical barriers and expand financial services to geographically marginalized communities. With these digital platforms making big data available, fintech firms can process borrower information more efficiently and overcome information asymmetry. Unlike their traditional

¹ This chapter draws upon Loko and Yang 2022.

counterparts with heavier compliance and capital requirements, fintech firms have lighter regulations, enabling them to operate nimbly in certain market segments, lend with less collateral, and better support the economy. When viewed through a gender lens, the benefits from fintech could therefore be significant.

First, fintech can leverage digital financial tools to increase access to and use of financial services, benefiting populations disproportionately excluded from the traditional financial system (Sahay and others 2020). According to the World Bank Group's 2017 Global Findex report, more than 25 percent of women still do not use or have access to the financial system, and more than 70 percent of female-owned small and medium enterprises have inadequate or no access to financial services (World Bank 2017; Demirgüç-Kunt and others 2018). Developing fintech-enabled services will likely lead to greater convenience, privacy, and security for the traditionally unbanked or underbanked female population.

Second, fintech can better evaluate the creditworthiness of individuals whom the traditional financial system may have previously marginalized due to no or minimal credit history. Using alternative data—for example, information generated by and about consumers on digital platforms—fintech helps loan providers make lending decisions without relying on credit reports or scores. Many loan applicants, including female applicants with no credit reports or credit scores, would benefit from these innovative measures to assess credit risk and model creditworthiness.

Third, fintech can facilitate access to financing, especially for female-headed households and businesses. Worldwide, an estimated \$300 billion financing gap exists for formal, female-owned small businesses (International Finance Corporation [IFC] 2017). Without such access, women face difficulties collecting and saving income, growing their own businesses (Sahay and others 2015), and pulling their families out of poverty. Many fintech-based platforms operating on “big data, small credit” propositions can contribute to women's economic empowerment and entrepreneurship by targeting small and medium enterprises, lowering interest rates, and relaxing collateral requirements.

Most studies analyzing the nexus between fintech and inequality have focused on income inequality (Suri and Jack 2016; Asongu and Nwachukwu 2018; Demir and others 2020; Zhang and others 2020; Chinoda, Mashamba, and Vivian 2021). Several studies have pointed to either finance or technology as a positive force for improving female employment. Based on a sample of 48 African countries, Ngoa and Song (2021) concluded that information and communications technology (ICT) significantly stimulates female labor market participation, and financial development enhances the effect. Focusing also on Africa, Asongu and Odhiambo (2019) showed that promoting ICT beyond certain thresholds is necessary for ICT to mitigate inequality and increase female participation in the economy. Studies focused on Europe and Asia also found a positive impact of finance and technology on female employment (Nassani, Aldakhil, and Moinuddin 2019; Chen and others 2021). However, few, if any, studies have examined the intersection of female employment with fintech. This gap derives partly from the lack of data and partly because of the difficulty in establishing causality.

Identifying the causal effects of fintech development on female employment is challenging because of well-known endogeneity concerns, namely the potential correlation between explanatory variables and the error term that arises from omitted variable and simultaneity (Yang 2021). Building on the seminal work of Rajan and Zingales (1998), this study makes progress on establishing causality by including an array of controls and interacted fixed effects (country-industry and year) that allow accounting for a range of omitted variables.

As fintech expands access to financial services and credit, its adoption should disproportionately help firms with financial constraints, high-tech firms facing greater information asymmetry and thus higher borrowing costs, and firms without existing financial access. The study provides detailed evidence on economic mechanisms through which fintech development affects female employment by including several interaction terms between fintech and firm characteristics such as financial constraint, high-tech intensiveness, and loan access. In this way, the model specification captures the rich dynamics between fintech and firm variables, allowing for more reliable statistical inferences.

LINK BETWEEN GENDER INEQUALITY AND FINTECH

Empirical Strategy

To estimate the relation between fintech and gender inequality, Loko and Yang (2022) constructed the following baseline model, as shown in equations (18.1) and (18.2):

$$Gender_{i,j,t} = \beta_0 + \beta_1 Fintech_{i,t-1} + \beta_2 X_{i,t-1} + \beta_3 I_{i,j,t-1} + \eta_{i,k} + \mu_t + \varepsilon_{i,j,t} \quad (18.1)$$

$$Gender_{i,j,t} = \beta'_0 + \beta'_1 Lending_{i,t-1} + \beta'_2 Capital Raising_{i,t-1} + \beta'_3 X_{i,t-1} + \beta'_4 I_{i,j,t-1} + \eta_{i,k} + \mu_t + \varepsilon_{i,j,t} \quad (18.2)$$

where $Gender_{i,j,t}$ refers to the level of gender inequality of country i , firm j , in year t , measured by the number of female employees and the ratio of female employees over female and male employees in the sampling firms. $Fintech_{i,t}$ captures the fintech development of country i in year t , measured by the volume of alternative finance, which can be further classified into digital $Lending$ and digital $Capital Raising$. $X_{i,t-1}$ is a vector of country-level controls, including the natural logarithm of per capita GDP, GDP growth rate, and trade openness. $I_{i,t-1}$ is a vector of firm-level controls, comprising firm size, firm age, export dependence, foreign ownership, and sector specialization. All explanatory variables are lagged by one year to mitigate endogeneity concerns.

Variable $\eta_{i,k}$ accounts for country–industry fixed effect that absorbs variations in the financial environment between countries and industries, such as systematic differences in economic development, government policies, and industry-specific reforms. The μ_t denotes year fixed effect that picks up any variation in the

outcome that happens over time and is not attributable to other explanatory variables. Standard errors are clustered at the country and industry levels to account for heteroskedasticity.

The coefficients of interest are β_1 , β'_1 , and β'_2 , which are associated with fintech variables. If they are positive and significant, it suggests that a higher level of fintech development is associated with higher female representation in the workforce, hence a lower degree of gender inequality. If they are negative and significant, a negative correlation between fintech development and gender equality can be inferred.

As discussed earlier, identifying the causal effects is a challenge because of the potential correlation between right-side variables and the error term arising most notably from omitted variables and reverse causality. On the one hand, omitted variables could bias the estimation that results from traditional cross-country regressions. Unobservable country or industry characteristics related to both fintech and female employment are left in the error term, making statistical inferences hard to draw. On the other hand, a higher female employment rate could increase the use of fintech.

In their pioneering work, Rajan and Zingales (1998) proposed a fixed-effect identification strategy with interaction terms. They showed that better-developed financial markets lead to higher economic growth in industries heavily dependent on external finance. Loko and Yang (2022) built on this work and established the following model that extends their empirical framework to the fintech setting. By estimating various forms of the model, this study examines the effects of fintech on gender inequality, as shown in equation (18.3):

$$\begin{aligned} Gender_{i,j,t} = & \alpha + \beta(Fintech_{i,t-1} \times Firm_j) + \gamma Fintech_{i,t-1} + \rho Firm_j + \delta I_{i,j,t-1} \\ & + \eta_{i,k} + \mu_t + \varepsilon_{i,j,t} \end{aligned} \quad (18.3)$$

where $Firm_j$ is firm-level financial constraint, loan access, digital infrastructure, and other variables that capture economic mechanisms and help with identification. Note that only additional explanatory variables that vary both with country and firm need to be included. All explanatory variables are lagged by one year to mitigate simultaneity concerns.

One key virtue of the model is that it allows using interacted fixed effects (country–industry and year) to control for a range of omitted variables. Thus, the model treats country and firm characteristics in ways that previous cross-country empirical studies could not correct and will be less subject to criticism about the model's specifications. When interpreting the results, the focus is on the signs and economic significance of β . A positive (negative) and significant coefficient indicates that fintech exerts a disproportionately positive (negative) effect on firms with financial constraints, high-tech intensiveness, loan access, and internet access. In addition, including various interaction terms clearly illustrates the specific mechanisms through which fintech affects female employment. These mechanisms are firmly grounded in economic theories, thus effectively addressing concerns of reverse causality.

Data

The data on fintech adoption derive from the Cambridge Alternative Finance Benchmark, which contains the volume of finance through digital platforms from the world's 191 jurisdictions spanning 2011–20.² The benchmark is based on an online survey hosted by the Cambridge Centre for Alternative Finance Judge Business School in partnership with the University of Agder (for the EU report), the University of Chicago Booth School of Business (for the Americas study), the University of Sydney Business School, the University of Tsinghua Graduate School at Shenzhen, Shanghai Jiaotong University Law School (for the Asia-Pacific regional study), and Nesta (for the UK report).

Data on female employment come from the World Bank Enterprise Survey (WBES). Since the 1990s, this renowned firm-level survey has covered a representative sample of firms in the world's major economies. The WBES is a standard establishment-level survey representative of the nonagricultural, non-extractive private sector, covering registered establishments with five or more employees. The database covers various business environment topics, including access to finance, corruption, infrastructure, crime, competition, and performance measures. However, given the incidence of agriculture in female employment, the WBES database that only includes manufacturing and service industries has certain limitations.

In addition to the main variables of interest, the country-level control variables derive from the IMF's World Economic Outlook (WEO) database. To estimate mediating effects, the study merges the existing data set with the Worldwide Governance Indicators proposed by Kaufmann and Kraay in 1999. The indicators report on six broad dimensions of governance—voice and accountability, political stability, government effectiveness, regulatory quality, rule of law, and control of corruption—for more than 200 countries and territories over the period 1996–2020. The Women, Business and Law Index derives from the World Bank's annual report under the same title that analyzes laws and regulations affecting women's economic opportunity in 190 economies.

The sample period is 2011–20, the overlapping years of all previously referenced databases.

² Loko and Yang (2022) chose the current fintech indicator over other fintech-related indexes, including Global FINDEX compiled by WB and FAS compiled by IMF for the following reasons: (1) They are published in waves and the only available years are 2011, 2014, and 2017, making it difficult to perform reliable panel-based analysis; (2) the fintech landscape has been changing rapidly since 2017, making it preferable to use the latest data available to reflect the most recent developments; (3) Global FINDEX and FAS do not make the distinction between lending and equity financing, which is economically important given the vastly different natures of, and incentives offered by, debt and equity financing. The final index on alternative finance includes financial channels and instruments that emerge outside the traditional financial system. Mobile money and internet banking that are often operated by traditional banks are thus not included.

Results

Table 18.1 provides the baseline regression results. In column (1), equation (18.1) is estimated using the number of female employees as the dependent variable and the level of overall fintech finance as the independent variable. Dropping missing values leads to a sample of 22,631 firms. The fintech coefficient is positive and significant at the 1 percent level. The result appears consistent with the hypothesis that fintech development is associated with a significant increase in female employment. Specifically, a 1 percent increase in the volume of transactions through fintech platforms is associated with a 1.4 percent increase in the number of female full-time employees in sample firms.

A likely explanation is that with easier financial access enabled by fintech, firms have more financial resources to expand their businesses, make investments, and boost production. Since capital and labor are complements in the production

TABLE 18.1.

Fintech Development and Female Employment				
Variables	(1) Female Employees	(2) Female Ratio	(3) Female Employees	(4) Female Ratio
Fintech	1.363*** (0.456)	0.375*** (0.112)		
Lending			−0.143* (0.083)	−0.047*** (0.011)
Capital Raising			0.776*** (0.112)	0.152*** (0.019)
GDP	2.931** (1.318)	0.642* (0.330)	−1.344*** (0.114)	−0.334*** (0.020)
GDP Growth	−0.627* (0.329)	−0.094 (0.084)	0.098** (0.046)	0.060*** (0.006)
Openness	6.764** (2.774)	1.543** (0.702)	−0.023 (0.826)	0.595*** (0.110)
Sales	0.354*** (0.005)	−0.003*** (0.001)	0.366*** (0.006)	−0.006*** (0.001)
Age	0.081*** (0.010)	−0.003 (0.002)	0.086*** (0.012)	−0.002 (0.003)
Export Share	0.109*** (0.041)	−0.014* (0.007)	0.027 (0.050)	−0.021** (0.009)
Foreign Ownership	0.306*** (0.033)	0.025*** (0.006)	0.311*** (0.043)	0.027*** (0.008)
Sector Specialization	0.068 (0.448)	−0.056 (0.076)	0.020 (0.434)	−0.056 (0.077)
Constant	−29.434*** (8.811)	−6.675*** (2.161)	−3.488*** (0.945)	0.228 (0.155)
Country–industry Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	22631	26447	14631	17021
R ²	0.390	0.263	0.393	0.270

Sources: IMF World Economic Outlook database; Loko and Yang 2022; World Bank Enterprise Survey; and author calculations based on data from Cambridge Alternative Finance Benchmark.

Note: *, **, and *** denote significance levels of 0.1, 0.05, and 0.01, respectively.

process (Allen 1968), increased investments create more demand for laborers, including female laborers (Benmelech, Bergman, and Seru 2011).

In column (2), the dependent variable Female Employees was replaced with the ratio of female employees over total employees and we found a significantly positive correlation between fintech development and female ratio. The results indicate that fintech adoption not only leads to more jobs for women but also raises the ratio of female to male workers.

Next, the fintech indicator was disaggregated into fintech lending and fintech capital raising instruments. Interestingly, opposite signs were observed on the estimated coefficients. Despite the overall positive influence of fintech on female employment, the correlation between fintech lending and female employment is negative and significant. In contrast, the coefficient on capital raising is positive and significant at the 1 percent level, consistent with the hypothesis that equity-like instruments are more effective tools to mitigate financial distress than debt-like instruments. The results highlight the importance of distinguishing between different fintech tools in estimating the economic impact.

GENDER INEQUALITY, FINTECH, AND FIRM CHARACTERISTICS

How does fintech disproportionately increase the number of female employees? First, fintech reduces firms' financial constraints. With more financial resources made available through fintech, employers might be able to hire female workers who require on-job trainings, maternity leave, flexible hours, and other benefits (Liu and others 2017; Lim and Zabek 2021; Kalev and Dobbin 2022). If this hypothesis is true, a more pronounced effect should occur in firms with financial constraints. This study tests this hypothesis by interacting the fintech indicator with a firm's financial constraint; Table 18.2 presents the results. Consistent with the hypothesis, the coefficient on the interaction term is positive and significant (column 1), suggesting a stronger effect for financially constrained firms. Thus, fintech promotes female employment by providing firms with more financial resources to hire more employees, especially female employees.

Fintech adoption could disproportionately benefit female-led firms, which are more likely to hire female workers (West and Sundaramurthy 2019). Numerous studies have documented a gender divide in financial access (Organisation for Economic Co-operation and Development [OECD] 2016). Women are less likely than men to obtain the financing needed to start a business because of a lack of collateral guarantees, a lack of credit history, or the bank's practice of gender discrimination. With a lower collateral requirement and alternative ways to establish credit worthiness, fintech is expected to provide more convenient access to finance for female borrowers. The results, reported in Model 2 of Table 18.2, confirmed the conjecture. The effect of fintech is positive and significant in female-led firms, suggesting that fintech can contribute to a more equal distribution of financial resources between genders.

TABLE 18.2.

Gender Inequality, Fintech, and Firm Characteristics				
Variables	(1)	(2)	(3)	(4)
	Female Employees			
Fintech	0.444* (0.238)	0.406* (0.236)	0.555** (0.282)	0.727** (0.291)
Fintech × Financial Obstacle	0.016*** (0.006)			
Fintech × Female-Led		0.014** (0.006)		
Fintech × Small Business			0.017*** (0.005)	
Fintech × Service Sector				0.044** (0.018)
Financial Obstacle	−0.272*** (0.088)			
Female-Led		−0.173* (0.093)		
Small Business			0.175** (0.075)	
Service Sector				−0.424* (0.255)
Age	0.186*** (0.011)	0.186*** (0.011)	0.174*** (0.011)	0.189*** (0.011)
Export Share	0.004*** (0.000)	0.004*** (0.000)	0.003*** (0.000)	0.003*** (0.000)
Foreign Ownership	0.008*** (0.000)	0.008*** (0.000)	0.008*** (0.000)	0.008*** (0.000)
Constant	−5.524 (3.601)	−4.981 (3.580)	21.222* (12.399)	16.140 (12.873)
Country–industry Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	25,120	25,120	25,120	25,142
R ²	0.150	0.150	0.173	0.096

Sources: IMF World Economic Outlook database; Loko and Yang 2022; World Bank Enterprise Survey; and author calculations based on data from Cambridge Alternative Finance Benchmark.

Note: *, **, and *** denote significance levels of 0.1, 0.05, and 0.01, respectively.

A stronger effect of fintech on small businesses should also be expected. Small firms are often unable to pledge collateral due to the lack of collateral assets (Nguyen and Qian 2012). In the meantime, small firms tend to hire more women than larger firms do, as the latter prefer more educated workers (Paik 2008). This study tests this hypothesis by interacting the fintech indicator with a dummy that indicates small business. The positive and significant sign on the interaction term in Model (3) of Table 18.2 confirms the stronger impact on small businesses.

In addition, more women cluster in the service industry because of their comparative advantages, for example in interactive tasks (Georgieva and others 2020). In contrast to manufacturing sector firms, service sector firms have more intangible assets such as goodwill and trademarks. The limited collateral value of intangible assets restricts the use of traditional financial instruments such as bank loans (Hsu, Tian, and Xu 2014). The traditional financial system often discriminates and marginalizes service sector firms, which are thus more likely to face financial constraints.

Consistent with this hypothesis, the coefficient on the interaction term in Model (4) of Table 18.2 is positive and significant, suggesting that fintech adoption brings additional benefits to firms in service sectors.

In summary, fintech promotes female employment mainly through a favorable allocation of financial resources to firms that are more female-labor intensive, and more likely to have financial constraints, such as female-led and small firms and firms in service sectors.

GENDER INEQUALITY, FINTECH, AND COUNTRY CHARACTERISTICS

The impact of fintech development on female employment could vary with the heterogeneity in countries' institutional quality. To test this hypothesis, sample firms were divided based on the institutional quality of the country that hosts their headquarters. The study examines four governance dimensions—female business law, government effectiveness, regulatory quality, and rule of law. Table 18.3 displays the results and shows that women living in countries with better legal protection are likely to benefit more from fintech.

TABLE 18.3.

Gender Inequality, Fintech, and Country Characteristics		
Variables	Stronger Law Protection	Weaker Law Protection
	(1)	(2)
	Female Employees	
Fintech	1.441*** (0.452)	0.027 (0.025)
GDP	3.238** (1.312)	0.391*** (0.091)
GDP Growth	−0.714** (0.327)	0.064*** (0.005)
Openness	7.533*** (2.766)	1.873** (0.753)
Sales	0.393*** (0.006)	0.269*** (0.008)
Age	0.083*** (0.011)	0.069*** (0.018)
Export Share	0.095** (0.047)	0.090 (0.077)
Foreign Ownership	0.285*** (0.037)	0.291*** (0.074)
Sector Specialization	0.001 (0.458)	0.607** (0.251)
Constant	−31.661*** (8.754)	−7.905*** (1.017)
Country–industry Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
Observations	15,849	6,782
R ²	0.406	0.370

Sources: IMF World Economic Outlook database; Loko and Yang 2022; World Bank Enterprise Survey; and author calculations based on data from Cambridge Alternative Finance Benchmark.

Note: *, **, and *** denote significance levels of 0.1, 0.05, and 0.01, respectively.

In untabulated results, the details of which can be found in the original paper, countries with greater government effectiveness, regulatory quality, and the rule of law experience greater welfare improvement via fintech. To some extent, institutional quality could increase the risk aversion of the country's investors, discouraging the development of fintech innovations and thus limiting its impact on gender inequality.

The sample countries are also examined in regional groups, based on the geographical classification in WBES. The results, summarized in Table 18.4, point to a greater effect of fintech on female employee ratio in sub-Saharan Africa than in other regions, as suggested by the larger magnitude of the coefficient in the sub-Saharan Africa subsample. Fintech positively affects female employment in Asia and the Pacific, Europe, and Central Asia whereas it negatively affects female employment in Middle Eastern and northern African countries.

Interestingly, a positive association exists between fintech and female employment in sub-Saharan Africa. With less developed financial markets, sub-Saharan African countries are home to fewer entrenched players than advanced economies. As a result, they may offer more opportunities for innovation, as disrupting the equilibrium faces less resistance.

TABLE 18.4.

Gender Inequality and Fintech in Different Regions					
	Sub-Saharan Africa	Asia and Pacific	Europe and Central Asia	Middle East and North Africa	Latin America and Caribbean
Variables	(1)	(2)	(3)	(4)	(5)
	Female Ratio				
Fintech	0.142*** (0.041)	0.041*** (0.006)	0.104*** (0.016)	-0.032*** (0.004)	-0.012 (0.012)
GDP	-9.370*** (2.664)	-0.068*** (0.015)	0.043 (0.096)	0.032 (0.023)	0.006 (0.026)
GDP Growth	0.966*** (0.276)	0.004 (0.017)	-0.037 (0.063)	0.000 (0.002)	0.008 (0.011)
Openness	-0.607*** (0.171)	0.066* (0.036)	0.661* (0.373)	0.063* (0.033)	-0.091 (0.197)
Sales	-0.001 (0.001)	0.001 (0.002)	-0.007*** (0.001)	-0.003 (0.002)	-0.005** (0.002)
Age	-0.008** (0.004)	-0.012** (0.006)	0.000 (0.003)	0.010** (0.005)	0.002 (0.007)
Export Share	-0.026* (0.014)	0.006 (0.020)	-0.034*** (0.011)	0.061** (0.024)	0.030 (0.028)
Foreign Ownership	-0.014 (0.010)	0.055* (0.033)	0.049*** (0.009)	0.058* (0.031)	0.028* (0.016)
Sector	-0.059 (0.068)	0.135** (0.064)	-0.293*** (0.024)	-0.114 (0.101)	0.261*** (0.062)
Constant	-1.451*** (0.416)	-0.170 (0.365)	-0.379 (0.327)	0.547*** (0.143)	0.328 (0.312)
Country-industry Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	6,614	3,765	10,766	2,629	2,673
R ²	0.136	0.283	0.296	0.084	0.087

Sources: IMF World Economic Outlook database; Loko and Yang 2022; World Bank Enterprise Survey; and author calculations based on data from Cambridge Alternative Finance Benchmark.

Note: *, **, and *** denote significance levels of 0.1, 0.05, and 0.01, respectively.

Taken together, the findings not only confirm the previous estimates about the association between fintech and female employment but also show that prevailing governance ineffectiveness, poor regulatory quality, and weak rule of law associated with less developed countries constitute major obstacles to fintech adoption in these economies.

CONCLUSION

Fintech development leads to significant welfare improvements for women. It not only increases the number of female employees in the workforce but also raises the ratio of female relative to male employees. This study also sheds light on the economic mechanisms—fintech provides easier financial access to firms with financial constraints, especially female-led firms, small firms, and firms in service sectors that traditionally hire more female workers.

Weak institutions reduce the positive effect of fintech. Fintech can significantly increase female employment in countries with good governance, law, and regulations, while it has weaker benefits in countries whose institutional quality is below the median. At the regional level, the effect of fintech is positive in sub-Saharan African, Asian and Pacific, and European countries.

These results provide important policy implications. First, closing gender gaps in digitalization is critical to fully reap fintech benefits on gender equality. In most countries, unequal access to mobile phones and other electronic devices creates financial inclusion gaps. For example, according to the OECD (2018), worldwide, 327 million fewer women than men have a smartphone and can access the mobile internet. The findings indicate that the inequality-reducing effects of fintech are significantly weaker in firms without access to the internet compared to firms with such access. Thus, the digital divide must be addressed, for example, by investing in technological innovation and increasing the supply of digital infrastructure, to fully leverage fintech benefits. Second, policymakers need to promote good governance, laws, and regulations to ensure that fintech effectively reduces gender inequality.

Valuable avenues of research exist that are worth exploring: Does fintech help reduce firms' earning inequality in addition to the gender employment gap? What are the fintech-related distributional effects and welfare implications on female-led households and female entrepreneurs who start businesses? If banks and fintech lenders compete on credit provision, how will that affect consumers and investors? Do the new forms of financing introduced by fintech demand new forms of regulation? Answering these questions allows a comprehensive approach to building more gender-inclusive economies.

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Digitalization and Gender Equality in Political Leadership in Sub-Saharan Africa

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This chapter examines the impact of digitalization on people's perceptions of women as political leaders in 34 sub-Saharan African countries. We find that being a social media or internet user is linked to a higher likelihood of supporting gender equality in political leadership; however, the intensive margin of usage does not appear to be significant. Furthermore, women's perceptions of gender equality in political leadership are more sensitive than men's to internet and social media use. The chapter recommends policies for improving information and communications technology (ICT) infrastructure and investing in technological education.

INTRODUCTION

Increasing gender diversity in political leadership is critical not only for fairness and inclusion but also for potential economic benefits.¹ First, diverse political leadership can bring a variety of perspectives and experiences to policy- and decision-making, resulting in more innovative and effective solutions to problems. This is especially important when dealing with economic issues because different leaders may have dissimilar ideas about how to promote economic growth and development. For example, Perkins, Phillips, and Pearce (2013) found that in more ethnically diverse nations, a female national leader is associated with higher GDP growth. Second, diverse political leadership can improve a political system's credibility and legitimacy, which can be important for fostering economic confidence and attracting investment. A diverse group of leaders perceived to represent the population's interests can help to build trust and support for the government and its policies. Third, diverse political leadership can contribute to the creation of a more inclusive and equitable society, which in turn can aid economic development. More people, regardless of their background, participating in the political process and having access to leadership roles can lead to greater prosperity and opportunity for all members of society.

¹ For further reading on the subject and empirical evidence, see Hessami and da Fonseca (2020), Anderson (2022), and the references therein.

This is especially important for marginalized groups, particularly women, who may face economic barriers and would benefit from having representatives who understand and advocate for their needs.²

Nonetheless, despite the importance of gender diversity in political leadership, gender disparities in political participation are widespread around the world. Perkins, Phillips, and Pearce (2013), for example, show that from 1950 to 2013, less than 5 percent of national leaders in 188 countries were female. According to the World Economic Forum's Global Gender Gap Report, the largest gender gap today is in political participation; this report monitors progress toward gender equality in four areas: economic participation and opportunity, educational attainment, health and survival, and political empowerment. In the 2022 edition, the political empowerment gap was found to be the largest of the four, with a score of just 22 on a scale of 0 to 100 (where 100 represents total parity). The gender gap in political participation is especially pronounced in Africa. According to the World Economic Forum data, the average score for sub-Saharan Africa on the political empowerment index is just 21.3, the lowest of any region in the world, although heterogeneity exists between countries within the region.³ This means that women in sub-Saharan Africa are even less likely than the global average to hold positions of political influence. This finding highlights the need for increased efforts to promote gender equality in the political sphere and to ensure that women, particularly in Africa, have equal opportunities to participate in decision-making and leadership. Even though women have historically been successful leaders, stereotypes and traditional barriers persist.⁴

Changes in perceptions of women as leaders constitute one likely mechanism for increasing female political participation. Gender stereotypes and biases can influence how women are perceived and treated in positions of leadership, preventing them from achieving important roles in the political sphere. For instance, society may judge women in political leadership positions more harshly than

² According to Cowper-Coles (2020), more women in politics seem to be associated with lower levels of corruption. Hence, having more women in political leadership in Africa could be an asset as the region seeks to lower corruption.

³ According to the World Bank's World Development Indicators, a significant disparity exists among African countries in terms of the proportion of parliamentary seats held by women as of 2020. Despite efforts in Rwanda (63.8 percent of parliamentary seats), Senegal (43 percent of parliamentary seats), and South Africa (42 percent of parliamentary seats), countries such as Nigeria (less than 4 percent of parliamentary seats) and Benin (less than 8 percent of parliamentary seats) continue to lag significantly. More recently, the August 2022 elections in Kenya highlighted persistent threats to women, with around a dozen of them being seriously and physically assaulted as they considered running for political office (Akwiri and Miriri 2022).

⁴ Women have led as chiefs, monarchs, and politicians for centuries. Nefertiti (-1370 to -1330) and Cleopatra (-51 to -30) reigned as pharaohs in ancient Egypt, and the Candaces of Meroe (-170 to 314) reigned as queens in Ethiopia. More recent examples include queen mother Yaa Asantewaa (1840 to 1921) of Ghana, who led the war against the British colonial authority; Aba, who organized a successful revolution against the British tax regime in Ghana; Queen Nzinga of Angola (1624 to 1663), who led an army of women against the Portuguese colonialists; and Zewditu, who ruled as Empress of Ethiopia from 1916 to 1930 (Loiseau and Nowacka 2015; Deloitte 2019).

men, and voters may perceive female candidates as less competent or likable. Changing these perceptions and challenging gender stereotypes may help to increase the number of women in positions of leadership in government and parliament. Efforts could include raising awareness of existing biases as well as initiatives to promote more positive images of women as leaders—for example, by exposing society to real-life examples of successful female leaders and highlighting their accomplishments or providing training and support to women seeking leadership roles.

Digitalization, through increased internet and social media usage, may be able to help change perceptions of women as political leaders. The internet and social media platforms offer myriad ways for people to access information locally and worldwide, connect with others, and participate in public discourse. These tools can help women showcase their ideas, experiences, and leadership skills as well as challenge traditional gender stereotypes and biases. Social media platforms, for example, can be used to amplify the voices of female politicians and leaders—and to promote their work and accomplishments. This promotion can help increase their visibility and credibility, and it can challenge the notion that women are incapable of holding positions of leadership.

Furthermore, the internet and social media can provide a platform for women to network and share resources and support.⁵ This is especially important for women who may face barriers to entering or advancing in leadership roles, and online channels can help to foster a sense of community and solidarity among women in positions of leadership. Online channels can also expose users to the positive work experiences of women in leadership roles—not only locally but also globally—helping to show that when women are in positions of political leadership, they can deliver exceptional results.

To shed light on the interplay between internet and social media usage and perceptions of women as political leaders in Africa, the authors conducted an empirical analysis using survey microdata from the Round 7 vintage of the Afrobarometer covering 34 sub-Saharan African countries. The authors took advantage of specific questions that asked people about their views on female political leadership and their use of the internet and social media. The authors then created indicator variables for gender equality in political leadership and for internet and social media usage, as well as categorical variables that measure the

⁵ Individuals have been able to spread information about social movements around the world via social media and the internet since the early 2000s. Some of these social movements include #Bring-BackOurGirls in 2013, which spread the heartbreaking news of girls kidnapped from school by Nigerian rebels, and #HeForShe in 2014, which invited men and others to stand in solidarity for women, garnering the attention of more than a billion social media users. Other critical movements include the 2012 #DelhiGangRape, the 2015 #sendeanlat (#tellyourstory in English), and the 2017 #MeToo (first used in 2006). In March 2020, a young female African journalist launched one of the most successful African activism movements with #vraiefemmeafricaine (“genuine African woman” in English), which ironically refers to the perception that African women should continue to be bound by traditional laws to reflect their “Africanness.” The hashtag was used by millions of users, both male and female, from other continents.

intensity of their use. Regressions were estimated with the indicator that measures gender equality in political leadership as the outcome variable and the internet and social media usage as the main independent variables. These regressions estimate how much use of the internet and social media, and their intensity, are related to the likelihood of people agreeing that women should have the same chance as men to be elected political leaders. The regressions control for a wide range of explanatory factors, including gender, age, education, religion, geographical region, and exposure to other media.

The results show that being a social media or internet user is associated with a higher likelihood of agreeing that women should have equal chances as men to be elected political leaders. According to the authors' estimates, more people using the internet is associated with a 3.2 percentage point increase in the proportion of people who support that women should have equal opportunities in political leadership as do men; for social media usage, the figure is 3.5 percentage points. The findings also support the notion that women are more likely than men to support gender equality in political leadership. Having a female identity is linked to an increase in the likelihood of supporting women having equal opportunities as men in political leadership positions by slightly more than 12.5 percentage points. The results also support that women's perceptions of gender equality in political leadership are more sensitive than men's to internet and social media use. Finally, the authors find no compelling evidence to support the idea that the more people who use the internet and social media, the more likely they are to support women having equal opportunities as leaders as do men—that is, the intensive margin of usage does not appear to be significant.

These results are closely related to those of Shamaileh (2016), who found evidence that internet access improves women's, but not men's, perceptions of female political leadership ability in the Arab world. The authors contribute to the discussion by broadening the analysis to a wide range of sub-Saharan African countries, investigating the impact not only of internet but also of social media usage, and by considering the effects of the intensity of internet and social media use. This research also speaks to that of Yang and others (2024), who investigated the role and the impact of digitalization on gender employment and job loss in Latin American countries. Their findings suggested that higher levels of digitalization are associated with higher female employment and lower job loss for both men and women. The current chapter contributes to the discussion by demonstrating another nontrivial benefit of digitalization that has the potential to help close gender gaps in political participation, which is currently the largest gender gap.

This chapter also adds to a growing body of literature on the effects of the internet and social media on women's empowerment (Crittenden, Crittenden, and Ajjan 2019). The internet and social media offer a larger platform for women to find and follow role models; increase and create entrepreneurial activities, including marketing and advertisement (Bailur and Masiero 2017; Delacroix, Parguel, and Benoit-Moreau 2019; Camacho and Barrios 2022); and network (Suseno and Abbott 2021). More precisely, women could also be able to connect with others who share the same concerns and stories, making it easier for them to stand together and claim their rights. It has also been found that through social media, women have more opportunities to

boost their self-esteem and receive informal training for self-development and career advancement (Delacroix, Parguel, and Benoit-Moreau 2019). The authors' contribution to this literature is to shed light on the potential impact of social media and the internet on gender diversity in political leadership.

The rest of the chapter is structured as follows. The second section summarizes the data and gives an overview of the relationship between internet and social media usage and perceptions of female political leadership in sub-Saharan Africa. The third section technically describes the regression analyses; the fourth presents the main findings. The fifth section validates the main conclusions via robustness tests, and the sixth offers concluding comments.

DATA AND FACTS

The authors use survey microdata from the Afrobarometer's Round 7, completed in 2019.⁶ The Afrobarometer is a nonprofit organization that collects and analyzes high-quality survey data on African attitudes, experiences, and aspirations. Its data measure the social, political, and economic atmosphere in more than 30 countries in Africa. The data collection process involves face-to-face interviews with a randomly selected, nationally representative sample of 1,200 to 2,400 adult citizens in each country. Annex Table 19.1.1 lists the countries included in the Round 7 sample and the number of individuals surveyed in each country.

The main outcome variable is based on a question about whether people agree with men only as leaders or women as leaders as well (Question Q16). The question is phrased exactly as follows:

Which of the following statements is closest to your view? Choose Statement 1 or Statement 2.

Statement 1: Men make better political leaders than women, and should be elected rather than women.

Statement 2: Women should have the same chance of being elected to political office as men.

Respondents were asked to state whether they (1) Agree very strongly with Statement 1, (2) Agree with Statement 1, (3) Agree with Statement 2, (4) Agree very strongly with Statement 2, or (5) Agree with neither. The authors of this chapter developed an indicator for agreement with women as leaders that has a value of 1 if respondents agree or agree very strongly with Statement 2 (option 3 or 4) and a value of 0 otherwise. In these regressions, this indicator is used as the dependent variable.

The first treatment variable of interest is based on a question about how often people use the internet (Question Q91B). Respondents must select one of the following alternatives: (0) Never, (1) Less than once a month, (2) A few times a month, (3) A few times a week, or (4) Every day. The authors then created an indicator for internet usage that has a value of 1 if individuals reported some internet access (options 1–4) and a value of 0 if they never use it. The second treatment

⁶ Visit <http://www.afrobarometer.org> for detailed information about the data.

TABLE 19.1.

Fractions Agreeing with Gender Equality in Political Leadership			
	All	Women	Men
Internet Usage			
Never	0.687	0.737	0.628
Some Usage	0.754	0.830	0.697
Less Than Once a Month	0.713	0.770	0.666
A Few Times a Month	0.738	0.810	0.688
A Few Times a Week	0.765	0.848	0.702
Every Day	0.759	0.833	0.700
Social Media News			
Never	0.691	0.741	0.634
Some Usage	0.759	0.839	0.698
Less Than Once a Month	0.707	0.768	0.659
A Few Times a Month	0.750	0.838	0.686
A Few Times a Week	0.761	0.844	0.698
Every Day	0.767	0.846	0.706

Sources: Afrobarometer; and authors' calculations.

Note: All figures were calculated using the individual sample weights provided by the survey.

variable under consideration is based on a question about how often people get news from social media such as Facebook or Twitter (Question Q12E). The alternatives are the same as in the previous question. Similar to the previous indicator, the authors created one for social media news access and investigated the results further for both treatment variables by taking the intensity of usage into account.⁷

The authors begin by getting a bird's-eye view of how use of the internet and social media influences people's perceptions of women having an equal chance as men to be elected political leaders, using the variables described previously. Table 19.1 documents the fractions of individuals in the sample that agree with gender equality in political leadership; these figures are calculated for the entire sample and separately for men and women. The authors also do it for both the extensive and intensive margins of internet and social media usage. Five facts stand out:

1. Most people in this sample believe that women should have the same opportunity as men to lead. This is true not only for the entire sample but also for men and women separately. All figures are greater than 50 percent, regardless of frequency of internet and social media use.
2. Being a social media or internet user increases the likelihood of agreeing that women should have the same chance as men to lead. Around 75 percent of

⁷ According to the data, not all internet users also use social media. For instance, among women, 34.6 percent are internet users, but only 29.7 percent use social media. For men, the figures are 45 percent (internet) and 39 percent (social media). Therefore, assuming that both groups are the same may be misleading, which is why the authors have considered both. Remarkably, the data also show that across all age and education groups, women are less likely than men to use the internet and social media. This confirms the International Telecommunication Union's (ITU) findings that the digital gender gap persists in low-income countries (<https://www.itu.int/itu-d/reports/statistics/2021/11/15/the-gender-digital-divide/>). Finally, the authors observe that use of the internet and social media increases with education level and decreases with age.

users support gender equality in political leadership; this drops to 69 percent among those who do not use it.

3. Women are more likely than men to agree on female leadership. For both the internet and social media, and regardless of intensity of use, the proportion of women who are positive about female–male equality in political leadership is always greater than the proportion of men.
4. Women's perceptions of female–male equality in political leadership are more sensitive than men's to the use of the internet and social media. Using the internet increases the proportion of women by approximately 9 percentage points while increasing the proportion of men by only 7 percentage points. In terms of social media, these figures are 10 and 6 percentage points for women and men, respectively.
5. The more frequently people use the internet and social media, the more likely they are to agree that women should have the same chance as men in political leadership. This is true across the board as well as for men and women separately. When compared with using them less than once a month, using the internet and social media every day adds about 5 to 6 percentage points to the fraction.

The authors recognize that numerous factors could play an important part in shaping perceptions of female–male equality in political leadership (for instance, age, education, religion, geographical region, and exposure to other media). The following section describes regression exercises that systematically control the estimates for many of these key features, providing more accurate figures to validate the documented facts.

REGRESSIONS

The authors used the entire Round 7 sample, which includes respondents from the 34 African countries covered by the survey, to estimate the marginal effects of internet and social media usage on people's perceptions of women having the same chance as men to be elected to positions of political leadership in sub-Saharan Africa. Linear probability models were estimated using ordinary least squares with standard errors clustered at the regional level.⁸ The regressions have been controlled for a rich set of variables that includes age (six groups), education (four levels), religion (three kinds), an employment indicator, urban/rural area, and dummies for geographical regions.⁹ To control for wealth levels, the authors included a lived poverty index provided by the Afrobarometer that is an average index of five poverty items. To control for the influence of other

⁸ The Afrobarometer data include information on the region within each country where people live. The authors therefore take advantage of this more granular level of aggregation when controlling the regressions and estimating the standard errors.

⁹ The age groups are 18–25, 26–35, 36–45, 46–55, 56–65, and age 65 and over. The education levels consist of no formal education, primary, secondary, and post-secondary. Religions include Christian, Muslim, and others.

kinds of media, the authors added three more indicators of news access through radio, television, and newspaper.¹⁰ Finally, two new categorical indexes were added: one that measures satisfaction with democracy and another that assesses an individual's perception of how the government handles promoting equal rights and opportunities for women.

To formalize the regressions, the authors indexed individuals by i and let y_i be the indicator for agreement with women having the same chance as men to be elected political leaders, I_i the indicator for internet or social media usage, F_i a female dummy variable, and $\mathbf{X}_i$ the vector of controls described previously. Then, the first regression specification estimated can be stated as

$$y_i = \alpha + \beta I_i + \gamma F_i + \boldsymbol{\theta}' \mathbf{X}_i + \varepsilon_i \quad (19.1)$$

where ε_i is the error term. Positive estimates for β imply that being a social media or internet user increases the likelihood of agreeing that women should have the same chances as men of being elected as political leaders (fact 2). Positive estimates for γ imply that women are more likely than men to agree on gender equality in political leadership (fact 3). The authors ran a second specification that interacts the internet or social media indicator with the female dummy, which can be formalized as

$$y_i = \alpha + \beta I_i + \gamma F_i + \delta(I_i \times F_i) + \boldsymbol{\theta}' \mathbf{X}_i + \varepsilon_i \quad (19.2)$$

Positive estimates for δ indicate that women's perceptions about gender equality in political leadership are more sensitive than men's to the use of the internet and social media (fact 4). Finally, the authors estimated a third specification that explores the intensity of internet or social media usage, letting $I_{i,k}$ be the usage indicator associated to the intensity k . Then, the third regression is given by

$$y_i = \alpha + \sum_k \beta_k I_{i,k} + \gamma F_i + \boldsymbol{\theta}' \mathbf{X}_i + \varepsilon_i \quad (19.3)$$

Higher values for β_k , with k 's associated with higher usage intensities, indicate that the more frequently people use the internet and social media, the more likely they are to agree that women should have equal chances to be elected as political leaders (fact 5).

MAIN RESULTS

Table 19.2 summarizes all the regression results. The column labels correspond to the regression specification numbering from the previous section. To conserve space, only the estimates for the independent variables of interest are presented—internet or social media usage indicators, the female dummy, the interaction between the usage indicator and the female dummy, and the

¹⁰ These three indicators were developed in the same manner as the social media news indicator described in the previous section.

TABLE 19.2.

Ordinary Least Squares Regression Results						
	Internet			Social Media		
	(1)	(2)	(3)	(1)	(2)	(3)
Usage Indicator, I_i	0.032*** (0.007)	0.026*** (0.010)		0.035*** (0.007)	0.024*** (0.009)	
Female Dummy, F_i	0.127*** (0.007)	0.121*** (0.008)	0.127*** (0.007)	0.126*** (0.007)	0.118*** (0.008)	0.126*** (0.007)
Interaction, $I_i \times F_i$		0.013 (0.011)			0.023** (0.010)	
Intensity of Use, $I_{i,k}$						
Never			<i>base</i>			<i>base</i>
Less Than Once a Month			0.020 (0.016)			0.009 (0.017)
A Few Times a Month			0.027** (0.012)			0.034*** (0.013)
A Few Times a Week			0.039*** (0.009)			0.034*** (0.009)
Every Day			0.032*** (0.009)			0.041*** (0.009)
Observations	40,042	40,042	40,042	40,044	40,044	40,044
R^2	0.100	0.100	0.100	0.100	0.100	0.100

Sources: Afrobarometer; and authors' calculations.

Note: Estimates are weighted using the individual sample weights provided by the survey. Standard errors in parentheses are robust and clustered at the regional level.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

intensity of usage categories. Estimates have been weighted using the individual sample weights provided by the survey, and standard errors are robust and clustered at the regional level.¹¹

The results of the internet and social media exercises all point in the same direction and confirm three facts documented earlier. The estimates for the usage indicators are positive and significant, validating the fact that being a social media or internet user increases the likelihood of agreeing that women should have the same chances as men of being elected as political leaders (fact 2). According to regression specification (19.1), using the internet increases the share of people who agree that women should have equal chances of being elected to political leadership positions by 3.2 percentage points, while using social media increases the share by 3.5 percentage points. The female dummy coefficient estimates are positive and significant across all regression specifications, corroborating the fact that women are more likely than men to agree on gender equality in political leadership (fact 3). Overall, having a female identity is linked to an increase of slightly more than 12.5 percentage points in the proportion of people who agree that women should have the same chances as men of being elected to political leadership positions. The estimates for the interaction terms are also positive

¹¹ The authors used variance inflation factor tests to assess multicollinearity in the regression models. All the independent variables of interest had variance inflation factors less than 3, indicating that multicollinearity for those variables is not severe enough to warrant corrective measures.

TABLE 19.3.

Wald Test Results for Intensity of Use (Ordinary Least Squares Regressions)				
Test	Internet		Social Media	
	F-statistic	Significance Level	F-statistic	Significance Level
Less Than Once a Month <i>Equals to</i>	0.14	0.7092	2.13	0.1448
A Few Times a Month <i>Equals to</i>	0.91	0.3420	0.00	0.9481
A Few Times a Week <i>equals to</i>	0.99	0.3195	0.69	0.4081
Every Day <i>All Equalities Above</i>	0.66	0.5757	1.29	0.2773
Simultaneously				

Sources: Afrobarometer; and authors' calculations.

Note: Wald tests results on the equality of all subsequent usage intensity categories.

despite being more uncertain in the case of the internet. This finding supports the fact that women's perceptions about gender equality in political leadership are more sensitive than men's to internet and social media use (fact 4).

All the estimates for usage intensity categories are positive and increasing, in general, with intensity level; however, to determine whether there is an increasing relationship between the intensity of use and the likelihood to support women having equal chances as men of being elected as political leaders, the authors first had to determine whether the usage intensity categories are statistically distinct from one another, so Wald tests of simple and composite linear hypotheses about the intensity of use categories were performed (Table 19.3). Every two subsequent categories were tested to see if they are equal. For instance, the first row displays the results of the test to see if the coefficients of "Less than once a month" and "A few times a month" are equal. The authors also tested for all equality conditions simultaneously (last row). All tests have significance levels greater than 10 percent, implying that the equality hypotheses at reasonable significance levels cannot be rejected. These results do not support the notion that the more people use the internet and social media, the more likely they are to support women having equal chances as men to be elected as political leaders (fact 5). In other words, these findings support that the extensive margin of usage, rather than the intensive, is important.

ROBUSTNESS ANALYSIS

The authors further performed a series of robustness tests to assess if the estimation method or specific countries influenced the main results; Gomes and Meyimdjui (2023) provided the detailed results of these tests. The authors began by re-estimating the models with logistic regression rather than ordinary least squares. The results confirmed the findings of the last section: (1) the estimates for the usage indicators are positive and significant; (2) the female dummy

estimates are positive and significant; (3) the estimates for the interaction terms between usage indicators and female dummy are positive and significant; and (4) the estimates for usage intensity categories are positive and increasing, in general, with intensity level, and all Wald tests for the equality of subsequent categories have significance levels greater than 10 percent. Though the interpretation of the point estimates from logistic regressions differs from that of ordinary least squares, the logistic regressions' average marginal effects have the same sign as their point estimates.

The next set of robustness tests involves re-estimating the models while excluding each country individually. The goal is to demonstrate that the main results are not influenced by any particular country. In practice, the authors estimated 204 additional regressions, as there are 2 treatments of interest, 3 regression specifications, and 34 countries in the sample. Overall, regardless of which country the authors excluded from the sample, the results confirmed the findings of the previous section. These outcomes provide confidence in the reliability of these findings.

CONCLUSION

Increasing gender diversity in political leadership is critical for promoting equity, inclusion, and economic benefits. Despite the importance of gender diversity in political leadership, gender disparities in political participation exist globally, particularly in sub-Saharan Africa. Changing perceptions about female leadership abilities and challenging gender stereotypes may aid in increasing the number of women in leadership positions in government, parliament, and other spheres of society. This chapter shows that digitalization, as measured by internet access and social media use, is associated with increased perceptions of sub-Saharan African women having equal chances as men to be elected as political leaders. Because of this significant positive spillover from digitalization, as well as its other well-known benefits, this paper recommends policy measures to improve information and communications technology infrastructure, with a focus on the lowest cost of energy. It does not go without encouraging measures to improve people's education, which gives them solid prerequisites to learn and benefit from modern technologies such as the internet and social media.

ANNEX 19.1. DATA DETAILS

ANNEX TABLE 19.1.1.

Sample Description		
Country Name	Number of Individuals	Percent Distribution
Benin	1,200	2.62
Botswana	1,198	2.61
Burkina Faso	1,200	2.62
Cabo Verde	1,200	2.62
Cameroon	1,202	2.62
Côte d'Ivoire	1,200	2.62
Eswatini	1,200	2.62
Gabon	1,199	2.62
The Gambia	1,200	2.62
Ghana	2,400	5.24
Guinea	1,194	2.61
Kenya	1,599	3.49
Lesotho	1,200	2.62
Liberia	1,200	2.62
Madagascar	1,200	2.62
Malawi	1,200	2.62
Mali	1,200	2.62
Mauritius	1,200	2.62
Morocco	1,200	2.62
Mozambique	2,392	5.22
Namibia	1,200	2.62
Niger	1,200	2.62
Nigeria	1,600	3.49
São Tomé and Príncipe	1,200	2.62
Senegal	1,200	2.62
Sierra Leone	1,200	2.62
South Africa	1,840	4.02
Sudan	1,200	2.62
Tanzania	2,400	5.24
Togo	1,200	2.62
Tunisia	1,199	2.62
Uganda	1,200	2.62
Zambia	1,200	2.62
Zimbabwe	1,200	2.62
Total	45,823	100

Sources: Afrobarometer; and authors' calculations.

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PART VI

Looking Ahead: New Opportunities

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Gender and the Green Economy: What Conditions for a Gender-Transformative Transition to the Green Economy in Africa?¹

Elena Ruiz Abril

This chapter intends to inform current discussions about what constitutes, and most importantly, how to realize gender-transformative green transition paths in sub-Saharan countries. The transition to a green economy will create many new jobs around the world, including in sub-Saharan Africa, but there is a risk that women and girls will not share its benefits. This chapter concludes that women can benefit from the green transition in the region if countries adopt strong policies and programs to make it happen. Strategies should aim at facilitating a transition in which women and girls leapfrog to high-productivity green jobs. This chapter offers a menu of policy interventions to make that happen, with a focus on getting women ready for the transition, leveling the playing field to remove structural obstacles that women face to access green jobs, and accelerating their participation in the green economy while leveraging green finance and green economic instruments. Action on both short- and long-term strategies needs to start now to prevent the green transition from leaving women and girls behind.

INTRODUCTION

The transition to a green economy will create many new jobs around the world, including in sub-Saharan Africa. But will women share in these new jobs, and will the economic transformation help them move into higher-paid, more stable jobs that require more education and skills? The short answer is “yes”—provided that countries adopt strong policies and programs to make it happen.

The transition to the green economy will be a major trend shaping socio-economic outcomes worldwide over the next 50 years. This transition will create new economic opportunities, spawn new jobs, and spur the adaptation

¹ This chapter is based on the findings of the 2021 joint report “Green Jobs for Women in Africa” by UN Women and the African Development Bank Group. The chapter incorporates findings from other research by UN Women in sub-Saharan Africa on different aspects of women’s participation in agriculture, as well as recent research and policy work supporting countries in integrating gender considerations in green transition policies.

of existing jobs. One projection is that 24 million new jobs will be created worldwide just to accommodate the changes that are necessary to contain global warming to 2° C (International Labour Organization 2018a).

Like in any transition, there will be winners and losers. Some economic sectors will expand, others will shrink, and there will be associated employment redistribution. The gender impact of this transition could take different forms depending on whether women are over- or underrepresented in sectors that will grow or reduce their weight in the economy.

A focus of feminist analysis of climate change over the years has been on identifying how climate change affects men and women differently and the consequences of climate change with regard to gender inequalities. Much less attention has been paid to how the transition to the green economy, mitigation, and low-carbon paths could create opportunities for women's empowerment. This chapter strives to shed some light on this topic for sub-Saharan Africa.

This chapter examines what needs to happen so that women and girls do not miss out on the opportunities to be created by the transition to the green economy in Africa. It will look at what policies and programs need to be put in place now so that women and girls can access the green jobs that will be generated in the region in the coming decades and for the accessed jobs to be high quality, high productivity, and decent. The answers to those questions intend to inform current discussions about what constitutes and, more importantly, how to realize gender-transformative green transition paths in sub-Saharan countries.

OPPORTUNITIES FOR WOMEN IN THE GREEN ECONOMY IN SUB-SAHARAN AFRICA

The International Labour Organization (2018b) projected that the greening of the world economy (to hold global warming to 2° C) would create 24 million new jobs and that job growth would take place across nearly all sectors of the economy. Globally, the shift to a green economy to combat global warming would create 4 percent more jobs in agriculture than if business continued as usual, 20 percent more jobs in forestry, and 10 percent more jobs in transport. A range of sectors will create green jobs in sub-Saharan Africa, with energy, construction, and agriculture accounting for the largest numbers.

Across sub-Saharan Africa, employment data show that women are well positioned in some of the sectors where green jobs will be created but are underrepresented in others that will create the higher-quality jobs. The agricultural sector in sub-Saharan Africa employed 53 percent of working women in 2023. Most women's work in agriculture is via self-employment.

Sub-Saharan Africa has the world's highest rate of women involved in entrepreneurial activity at 26 percent; women entrepreneurs contribute as much as 13 percent of the continent's GDP (Dushime 2022). Forestry and tourism, where the growth of green jobs is also expected in the region, are also important areas of female employment. However, women are underrepresented in key sectors of the green economy that will offer the best jobs. Those sectors most likely to create

higher-end green jobs include energy (especially wind and solar), transportation, construction, and some niche areas of services (for instance, green advisory). Women are overrepresented in agriculture, waste management, and certain areas of renewable energy (biomass), which are likely to create mostly lower-end jobs. The energy mix will depend to a certain degree on country policies. A positive development, however, is that even in sectors where women are not well represented, they are finding niches, often as small women-led businesses in indirect jobs in green construction, renovation, or energy efficiency (see examples in Box 20.1).

Box 20.1. Women Greenpreneurs in Sub-Saharan Africa

Across sub-Saharan Africa, women entrepreneurs and women-led businesses are entering the green economy. These are some examples of leading women “greenpreneurs” in the region.

Solar Sister is a nonprofit organization that recruits, trains, and supports women entrepreneurs and supplies them with durable, affordable solar-powered products and clean stoves, thus leading to the development of clean energy businesses and markets in last-mile rural African communities. Through this model, implemented in Tanzania, Uganda, and Nigeria, the Solar Sister women entrepreneurs earn income by selling the clean energy products directly to people without access to power in their communities, which helps eradicate energy poverty. The business model of Solar Sister is to eradicate extreme energy poverty while empowering African women. Solar Sister invests in local women entrepreneurs to start, grow, and sustain successful clean energy businesses by providing them with a comprehensive training package and ongoing mentoring in the business, technology, and leadership skills necessary to kickstart a sustainable clean energy business. As of 2023, more than 8,500 Solar Sister Entrepreneurs have reached 3.9 million people in Africa with clean energy.

Solar Mamas is a women-led business that has installed solar panels in rural coastal communities in several countries in sub-Saharan Africa, including Tanzania, where less than 4 percent of the population has access to the electricity grid. As of 2016, the group of 13 women with low literacy trained in a six-month course to become solar engineers at India's Barefoot College (including building, installing, and maintaining solar panels and batteries) and had installed more than 800 solar units. Recipients pay a monthly fee set to be less than the average monthly cost of kerosene, the most common source of light. The panel installer receives half of the fee, and the other half goes to community projects. With support from UN Women, the Solar Mamas model has also been developed in Liberia where, as of 2017, the Solar Mamas have electrified over 425 homes and structures in their hometowns. According to the India-based nongovernmental organization, every woman trained by the program electrifies an average of 50 homes in her village.

In Senegal, **E-Cover** is a woman-led business that recycles used tires into rubber aggregates, tiles, shoe soles, and flooring for businesses, households, sports complexes, schools, and town halls. In 2020, the company collected more than 834 tons of tires and employed 25 people. They have recently secured funding to expand through WIC Capital, an investment fund for women-led micro, small, and medium enterprises in West Africa.

In Kenya, **Pine Kazi**, a women-led fashion design business, uses pineapple waste produced after harvest to craft ecofriendly shoes. Their model encourages pineapple farming communities to move from a linear model of plant-harvest-dispose to a circular model of plant-harvest-reuse. They connect waste managers with product designers to use pineapple-leaf waste as clothing material and create green jobs for unemployed youth. Pine Kazi was a winner in the African Development Bank's Fashionomics Africa Initiative (<https://fashionomicsafrica.org>), which supports producers of sustainable fashion.

Focus on Agriculture

In the short term, given the high share of women in the sector's workforce in sub-Saharan Africa, most green job opportunities for women in the region will be in agriculture. The agriculture, forestry, and fisheries sector, as well as the entire system of food production, is both the highest greenhouse gas-emitting sector in sub-Saharan Africa and the most vulnerable to climate change.

The agriculture sector will need to shift fundamentally to more sustainable practices to reduce greenhouse gas emissions and to enhance resilience to climate change. Whether this is done by promoting a more efficient use of resources in conventional agriculture or via a shift toward conservation and organic farming, such changes will require financial investments, research, and capacity building, and will demand new skills and create new jobs. In the short term, the shift to conservation agriculture may even cost jobs because it requires less labor. In the medium term, on the other hand, it may produce better-paying jobs for more skilled workers. Evidence shows that green practices in agriculture help increase workers' incomes by lowering input needs, increasing yields, and fetching higher prices or a combination of these factors. Returns on economic activities in green agriculture can be high where jobs are linked to green certification and labeling, which can help ensure sustained access to profitable global markets.

The combination of a transition toward green agriculture and the process of digitalization currently underway in the region, which has accelerated during and in the post-COVID-19 period, can also bring new opportunities for high-end green jobs in the sector. Recent research in East Africa shows that developing innovations and digital technologies that support agroecological practices, such as blockchain technologies for quality assurance and supply management and digitized certification of standards in organic farming, can create new jobs. However, the same study shows that most of the job opportunities in high-value products and services are taken by educated youth and youth who reside near towns (UN Women 2021a).

Climate-resilient agriculture, conservation agriculture, and organic farming may offer quick-win opportunities for female smallholder farmers to move into green jobs because the transition to green jobs can be realized in a short time with relatively small investments for training and other skills-enhancement strategies. Access to higher-end jobs linked to digitalization and innovation trends in agriculture will require longer-term strategies that emphasize investment in education among other things.

However, even to realize short-term opportunities for women in green jobs in agriculture, several structural obstacles to their participation in rural value chains need to be addressed. Despite women's prominent role in agriculture, they tend to be employed mostly in informal, vulnerable jobs at the early, low-productivity stages of agricultural value chains: 89.7 percent of employed women in Africa are in informal employment when agriculture is excluded, and 92.3 percent of employed women in Africa are in informal employment when agriculture is included (International Labour Organization 2018b). Research shows that the gender gap in agricultural productivity does not arise

because women are less efficient farmers but because they experience inequitable access to agricultural inputs, including family labor, high-yield crops, pesticides, fertilizer, and land (UN Women 2019). Unequal access to male family labor is one of the most important factors in Ethiopia, Malawi, and Tanzania. In Tanzania, lack of male family labor explains virtually the entire gap in agricultural productivity, whereas in Ethiopia and Malawi, it accounts for about 45 percent of the agricultural productivity gap (UN Women 2019). One of the primary explanations for these gaps is women's household care responsibilities, along with lower access to cash income related to heavy demands on their time performing unpaid farm labor for their husbands. Women may also be unable to scale up to the level required for high-value crops if they are constrained by plot size. UN Women's research in East Africa shows that women, as well as youth with disabilities and youth who reside in rural and remote areas, are excluded from opportunities in agriculture because of problems with accessing land, lack of access to finance, lack of capacity development efforts near their area, lack of access to digital technologies, and low digital literacy (UN Women 2021a).

Social norms are an important constraint to women's access to high-productivity jobs in agriculture. In Malawi, Uganda, and Tanzania, women's high burdens of unpaid care and domestic work leave them less able than men to invest their time in agricultural work. UN Women's research (2023) shows that, in northern Senegal, as many as 45 percent of rural women are responsible for caring for a family member with a disability or chronic illness; female farmers spend up to 12 hours per day on direct and indirect care work, which substantially limits the time that they can spend farming. Women are also less able to work on their own self-managed plots of land because of social norms that create the expectation that they will work on plots that are controlled by or jointly with their husbands before working on their own, particularly in polygamous households. These norms reduce the amount of time that women have available for their own plots and their likelihood of investing in higher-value, higher-maintenance crops (UN Women 2019).

Overall, user- and women-centric, multifaceted interventions that address these constraints and facilitate women-led business development in agribusiness and entrepreneurship can enable them to realize the gains associated with the green transition in the agriculture sector in the region.

Focus on Entrepreneurship and Green Finance

In the short and medium term, most opportunities for women's participation in the green economy in the region will arise via self-employment and entrepreneurship. Women make up 58 percent of Africa's self-employed population and are more likely to become entrepreneurs than men. Sub-Saharan Africa has the world's highest rate of women involved in entrepreneurial activity at 26 percent (Mastercard 2022). In the short term, most opportunities for green jobs will arise through entrepreneurship and women-led businesses in agriculture, forestry, waste management, circular economy, or renewable energy. Although women are

already claiming green jobs in these areas as entrepreneurs and owners of small and medium enterprises (see Box 20.1 for examples), lack of adapted finance remains an important constraint to take their economic activities to scale.

The gender gap in access to finance in the continent is estimated at \$42 billion (African Development Bank Group 2023). This gap has widened in recent years. Despite high rates of women entrepreneurs, women find it particularly difficult to secure financing for their businesses. Today, in the region, only 37 percent of women have a bank account, compared with 48 percent of men. This gap is strongly linked with social and cultural norms as well as access to land, which is especially important as collateral for loans to small businesses owned by women (Chapter 15).

Despite the tremendous growth of green finance in the past decade, women entrepreneurs and women-led businesses continue to be excluded from this source of financing. In 2021, green finance exceeded \$720 billion, including green bonds, green loans, venture capital and private equity funding for green tech, green initial public offerings, and green acquisitions, which use funds to buy companies that bring environmental benefits (Barry 2022). Lack of sex-disaggregated data makes it difficult to fully assess the extent of the gender gap in access to green finance or the actual share of green finance that trickles down to women-led businesses, but the use of standard financial products and conventional mechanisms to deliver green finance points to the same barriers that women-led businesses face to access commercial finance. Most of the of green finance is channeled via commercial banks and investment vehicles that are not well adapted to small businesses, where a large proportion of women-led entrepreneurial activity lies. A recent report by the United Nations Secretary-General's Special Advocate for Financial Inclusive Finance for Development (2023) claims that green finance has focused primarily on opportunities for traditional players (for instance, banks and institutional investors), with less attention on tools and pathways for microsegments, including households and micro, small, and medium enterprises.

Only a fraction of green finance is gender-responsive, and just a small percentage reaches organizations for women in the Global South. Taking public finance as an example, and using the Organisation for Economic Co-operation and Development (OECD) Development Assistance Committee (2016) methodology to assess gender-responsiveness of official development assistance,²

² The Development Assistance Committee gender equality policy marker is based on a three-point scoring system to qualitatively track the financial flows that target gender equality. Not targeted (score: 0) means that the project or program has been screened against the marker but has not been found to target gender equality. Significant (score: 1) means that gender equality is an important and deliberate objective but not the principal reason for undertaking the project or program. Principal (score: 2) means that gender equality is the main objective of the project or program and is fundamental in its design and expected results; that is, the project or program would not have been undertaken without this gender equality objective. This allows the OECD to identify gaps between the Development Assistance Committee's donor policy commitments and financial commitments. For more information, see the Development Assistance Committee gender equality policy marker (OECD 2019).

about 31 percent of total climate aid was gender-responsive; just 3 percent had gender equality as a principal objective, whereas 28 percent integrated gender equality as a secondary objective. Across regions, the gender focus of climate aid in Africa seems to be higher than in other regions with six African countries among the top 10 recipients of gender-responsive climate aid (Ethiopia, Kenya, Mali, Mozambique, Tanzania, and Uganda). Overall, however, the proportion of gender-responsive aid that went to southern civil society organizations amounted to just 2 percent (OECD DAC Network on Gender Equality 2016). The role that women's organizations can play in climate change adaptation and mitigation is restricted because of limited green finance flows to women's organizations. Women's access to green finance has been a priority in the advocacy of the global women's movement over recent years. The Women's Environment and Development Organization (2019) identifies women's access to green finance as an essential element of gender-just climate action to achieve systemic change in addressing climate issues.

Gender-responsive finance concentrates on adaptation rather than mitigation and focuses on the primary and secondary sectors. According to the OECD, adaptation and projects with cobenefits appear to offer the most potential for integrating gender considerations, currently reporting 27 percent and 43 percent of finances as gender-responsive, respectively. On the other hand, only 7 percent of tracked mitigation finance was gender-responsive. This is consistent with the emphasis of feminist analysis and climate action on the impact of climate change on women and gender equality and less so on how transition to the green economy and mitigation paths could generate opportunities for women. By sectors, agriculture, forestry, and other land use and industry had the highest gender-responsiveness rates: 35 percent and 28 percent, respectively. Almost none of the aid to energy and transport, critical sectors in the generation of green jobs in the future, was reported as targeting gender equality as a principal objective (Women's Environment and Development Organization 2019).

WOMEN'S OBSTACLES TO PARTICIPATE IN THE GREEN ECONOMY IN SUB-SAHARAN AFRICA

Women face several barriers that may limit their full access to green jobs in the coming years. Many of the obstacles mentioned regarding agriculture or entrepreneurship also apply to green jobs in other sectors. These include barriers to women's and women-led businesses' access to land, finance, and technology. In addition, skills deficits are barriers for women to access green jobs in the short term. Addressing this, via access to key assets and investments in skills, will be important in the short term and will realize some quick wins. However, addressing skills deficits will not be enough to ensure that men and women enjoy equal access to green jobs. Skills-oriented interventions might be sufficient to facilitate access to green jobs for men and boys. However, for women and girls, structural barriers from social norms will need to be addressed in parallel.

Discriminatory social norms underlie many of the obstacles that women face in accessing green jobs—and jobs generally—including barriers to land access, finance, and skills-building opportunities. They also constrain women's participation in certain sectors by dictating which ones are considered male-oriented areas of employment and which are perceived as offering acceptable jobs for women. More critically, the unequal distribution of unpaid care work between men and women is a significant barrier to labor market access for women. In sub-Saharan Africa, women living alone have a higher labor force participation rate (92 percent) than women who are married (77 percent) and women with children (80 percent). Women are still responsible for the vast majority of sub-Saharan Africa's unpaid domestic and care work and spend more than twice the time on household chores and care responsibilities as men. Indicatively, in Tanzania and South Africa, women's share of unpaid care work is 79 and 70 percent, respectively (Charmes 2019). In Senegal, women spend an average of four hours per day on housework and childcare compared to less than 30 minutes for men (Agence Nationale de la Statistique et de la Démographie, Agence Française de Développement, and UN Women 2022). In Benin, women spend, on average, three hours and 41 minutes per day on unpaid care work compared to 42 minutes for men (Charmes 2019). In Mali, women spend an average of 21.6 hours per week on unpaid work compared to 5.7 hours for men (Observatoire National du Dividende démographique Mali and UN Women 2022).

In rural areas particularly, water and wood collection are time-consuming tasks done largely by women and girls, which limit their opportunities to earn money. Women's disproportionate care work burden limits their ability to participate in the labor market overall, and it has been shown to affect women's ability to enter green areas of employment. For example, a study on skills required for green sectors across 32 countries concluded that without policy interventions, the energy transition is expected to promote more employment opportunities for men than for women (International Labour Organization 2019). According to the International Renewable Energy Agency (2019), lack of adequate work-life balance, care solutions, and flexible working measures is a critical impediment to retaining women in the wind energy industry.

GENDER IN GREEN TRANSITION POLICIES IN SUB-SAHARAN AFRICA

Integrating gender equality into mitigation and adaptation policies helps accelerate sustainable, inclusive, and equitable country-driven climate action. Nationally Determined Contributions, which are at the heart of the Paris Agreement, are policy documents that state efforts by each country to reduce national emissions and to adapt to the impacts of climate change. Integrating gender considerations in Nationally Determined Contributions is an iterative process to meet gender equality goals through climate action, including by allocating budget for gender activities. According to the International Union for Conservation of Nature (2021), countries have altogether increased their attention to gender, but still a

quarter of Nationally Determined Contributions is gender blind at the global level. About 94 percent of sub-Saharan Africa Nationally Determined Contributions mentioned gender or women—they consider the impact of climate events on women's livelihoods, acknowledge women's contributions to climate adaptation, and commit to women's participation in decision-making roles.

Similarly, the National Adaptation Plans, from the United Nations Framework Convention on Climate Change (2021), identify medium- and long-term adaptation needs and develop and implement strategies to address those needs. This is an iterative process implemented in a country-driven, gender-sensitive, participatory approach. As of April 2022, of 44 countries that had submitted National Adaptation Plans, 17 are from sub-Saharan Africa (UNFCCC 2023). All National Adaptation Plans integrate gender considerations to varying degrees. This helps ensure that adaptation planning processes address persistent gender inequalities by recognizing gender differences in adaptation needs and capacities, promoting gender-equitable participation and influence in adaptation decision-making, and ensuring gender-equitable access to financial resources and other benefits because of climate adaptation investments. The greatest increase has happened since 2020, which coincides with the establishment of the enhanced Gender Action Plan. Most National Adaptation Plans recognize women as a vulnerable group, whereas fewer identify women as a key stakeholder group in the adaptation planning processes.

Finally, Long-Term, Low-Emissions Development Strategies are policy instruments, in line with the Paris Agreement, that identify long-term, low-emissions paths of economic development and propose changes to the economic development model of a country. Because of their long-term outlook, these policies are ideal for identifying opportunities for women to participate in the green economy (in the new low-emissions scenarios identified by the strategy) while there is still time to address some of the structural barriers that women face to access those jobs. In Africa, Ethiopia and Burkina Faso have implemented Long-Term, Low-Emissions Development Strategies that have integrated gender analysis (see Box 20.2 for more on the strategy in Burkina Faso).

Beyond the Paris Agreement-related instruments, countries are developing policies to inform their transition to a green economy, both sectoral and nonsectoral, which offer opportunities to integrate gender considerations. At the sectoral level, sustainable energy and energy reform, together with agriculture, forestry, waste, or sustainable transport policies, are important instruments. Other policies with a nonsectoral focus include national green jobs strategies, such as the ones in Senegal or Ghana, and green finance strategies. Most policies lack analysis of gender co-benefits associated with those economic reforms, and gender considerations are lacking from the policy formulation process. Data constraints are an important limitation in this process because limited sex-disaggregated employment data at the subsector and position level in many countries in sub-Saharan Africa make it difficult to quantify specific gender-differentiated employment impacts associated with the transition. A narrow approach to gender analysis, focused on skills and access to finance without understanding the role that social

Box 20.2. Gender in the Burkina Faso Long-Term, Low-Emissions Development Strategy

The Government of Burkina Faso led the process of forming a Long-Term, Low-Emissions Development Strategy. With technical support from the Global Green Growth Institute and UN Women for the gender analysis, two workshops were held in 2022 to identify gender co-benefits, negative impacts, and mitigating actions associated with the strategy and to assess employment effects arising from the low-emissions scenarios. The workshops included participants from all sectoral and nonsectoral ministries concerned with the strategy as well as representatives from civil society and women's groups. Despite a lack of data to identify quantitative distributional employment impacts among men and women, workshop participants discussed in depth the challenges and opportunities for women's participation in green jobs associated with the low-emissions scenarios—particularly in agriculture, forestry, waste management and energy. As a result, the workshop participants adopted a recommendation to establish a national Program on Green Jobs for Women and Youth, and gender targets were included in the Long-Term, Low-Emissions Development Strategy performance indicators.

Source: UN Women and Global Green Growth Institute. Forthcoming. "Guidance Note to Integrate Gender in LT-LEDs: The Case of Burkina Faso."

norms play in influencing women's access to assets and the labor market, also limits the reach and effectiveness of recommendations in some of these policies.

Although integrating gender in green transition policies is important, the impact of such policies will also depend on the extent to which they are aligned with and integrated into governments' medium- and long-term priorities and expenditure frameworks. Gender-responsive budgeting, which is applied to different degrees in many countries in sub-Saharan Africa, can help establish the link between gender policy priorities in relation to the green economy and public expenditure (Chapters 11 and 12); gender-responsive budgeting can also help prioritize public resource allocations for operationalizing and implementing many of the gender commitments in sectoral and non-sectoral green economy national policies. However, effectiveness will largely depend on the countries' capacity to fully implement gender-responsive budgeting. UN Women (2021b) researched government compliance with the 10 percent allocation of agriculture budget as part of the Maputo declaration³ in seven countries in eastern and southern Africa; the study showed that, although all of the countries except Burundi had produced guidelines on gender-responsive budgeting, the governments' nonadoption of those guidelines was a barrier that prevented women from fully benefiting from the 10 percent budget allocation.

³ The Maputo Declaration is a declaration on Agriculture and Food Security in Africa that was signed by African Heads of State and Government in Maputo, Mozambique in July 2003. The declaration called for a pan-African flagship program to enhance agriculture production and bring about food security on the continent. The declaration also called for a re-investment of at least 10 percent of national budgets to improve food security, reduce poverty, and spur rural development.

POLICY RECOMMENDATIONS

Recommendations for policies to leverage women's green job opportunities fall into three categories. First, policies need to get women and girls ready for the green economy through education, skills development, capacity building, and institutional support of women-led businesses. However, a major conclusion is that policies on skills development, which are the focus of most policy interventions to promote green jobs, will not be enough for women and girls to benefit from green economy opportunities and to access a fair share of the jobs to be created. Thus, policies need to level the playing field by addressing structural barriers that women face in accessing green jobs. Finally, policies should accelerate action to promote a gender-responsive green economy so that change can happen within a reasonable time frame.

Recommendations to Get Women Ready for the Green Economy

Reducing horizontal segregation in the labor market to allow women access to all green jobs on equal terms with men will require a combination of education policies, strong female role models, and support mechanisms for women in male-dominated sectors. Initiatives should focus on:

- *Addressing gender segregation in education and promoting women's participation in science, technology, engineering, and math.* If women are to reap the benefits of the green transition in sub-Saharan Africa, strong leadership and investment are needed to improve their access to education in science, technology, engineering, and math and to careers at all levels of the education system, from primary to tertiary, including vocational education and training.
- *Reskilling and upskilling.* Retooling the labor force for the future of work is, according to the African Development Bank, a priority for Africa in the strategy for recovery from economic shocks associated with COVID-19. The report recommends that governments scale up efforts to retrain and reskill the labor force as quickly and as broadly as possible to facilitate workers' transition from low-productivity, obsolete sectors and jobs to new and emerging ones. This recommendation is particularly important in the context of women's access to green jobs in the region. Reskilling and upskilling strategies will be needed to ensure that women can access new green job opportunities in sectors where they are already well positioned, but the greening of jobs will require new skills—for example, in agriculture and services. Training offers should reflect the current and future needs of the labor market, economy, and environment.
- *Developing women's networks in male-dominated sectors.* Establishing professional women's networks and business associations in key sectors of the green economy with higher barriers to entry by women (for instance, construction, energy, and transportation) are a mechanism to support existing female professionals and to encourage others to join. Such networks can

nurture female role models and enhance women's ability to participate in those sectors. These networks can also be instrumental in advocating for corporate policies adapted to women's needs, facilitate investments in women-owned businesses and start-ups in these sectors, and be a source for mentoring and coaching of women.

- *Supporting the transition toward the formal economy*, which can facilitate women's movement into better-paying green jobs with better working conditions. This will be particularly critical in sectors such as agriculture, forestry, or waste and, to a lesser extent, tourism.

Recommendations to Level the Playing Field for a Gender-Responsive Green Economy

Addressing social norms and improving the social environment to enable women's participation in the green economy is crucial. This change requires both overcoming biases and expectations around the distribution of unpaid care work and overturning laws that limit women's access to the resources and means to undertake economic activities. To this end, specific recommendations include:

- *Removing legal barriers and addressing gender discrimination in legislation.* Key areas for reform include (1) amending legislation to ensure women and men have equal rights to ownership of immovable property (land) and equalizing the inheritance rights of sons and daughters; (2) introducing legislation (and enforcement mechanisms) to address sexual harassment in the workplace, particularly in male-dominated areas of the green economy (energy, construction, transport, and others); (3) amending legal provisions that do not allow women to open bank accounts or to move freely in the same way as men, which is required for doing business and for holding jobs on an equal basis with men; and (4) removing legal restrictions on women's access to specific jobs and tasks. Additional policy and legal provisions to incentivize the development of corporate policies for female employees can also play a critical role in certain sectors.
- *Balancing men's and women's responsibilities for care.* Addressing inequalities in the distribution of unpaid care can have an important impact on enabling women to take paid jobs, including in the green economy. Governments should invest in expanding care services, investing in infrastructure, and implementing policies to recognize, reduce, and redistribute unpaid care. Recognizing women's unpaid care work is particularly important in sub-Saharan Africa, where a large part of care work is informal. Changing social norms through education and communication and involving men and boys are important means to effect change in this area in the medium term.
- *Investing in role modeling and focusing on youth to change stereotypes about acceptable jobs for men and women.* Many of the jobs in the green economy will be in new occupations, not yet socially assigned to men or women. Thus, opportunities exist for women to claim new jobs as engineers, archi-

sects, energy-efficiency advisers, drivers of green buses, or green innovators across many different fields. The many women-led small and medium enterprises installing solar panels across the region illustrate how to pierce the “glass walls” and enter male-dominated sectors such as construction. Perceptions evolve slowly, but governments, civil society, and community leaders can take the lead through education, social communication, and advocacy by leveraging positive role models of women working in male-dominated sectors and by addressing young men and women from an early age. In the past, involving men and boys, including traditional leaders, in efforts to change social norms has proven effective in other domains such as gender-based violence prevention.

Recommendations to Accelerate Action for a Gender-Responsive Green Economy

Affirmative action to integrate gender targets and objectives into existing and new economic policy instruments can accelerate women’s access to green jobs. Through regulation, governments can expand opportunities for women in areas of the green economy that have the potential to generate many green jobs but have high entry barriers for women; these regulations can thus increase the speed with which change takes place. New economic instruments can also open new ways to assign economic value to women’s unpaid work on behalf of the environment—thus contributing to a truly transformative green transition from a gender point of view. Specific recommendations include:

- *Leveraging green economy instruments for women’s employment.* As countries explore and identify the appropriate combination of taxes, subsidies, incentives, and other economic instruments to encourage a transition to the green economy, these measures can be designed in a way that supports women’s employment objectives. Specific provisions can be incorporated into the design of these instruments to carve out employment targets for women-led businesses, or for women directly, particularly in sectors of the economy with high employment generation potential and in which women’s share is low (energy and transport).
- *Prioritizing women’s access to green finance and redressing gender biases in the green finance architecture.* The growth of green finance offers a unique opportunity to accelerate climate action. But women’s limited access to green finance restricts not only their contribution to climate change mitigation and adaptation but also their access to jobs in the green economy. For example, although carbon credits are designed to compensate for mitigation activity, across Africa, grassroots women, who are at the forefront of climate change mitigation through their (unpaid) work preserving ecosystems with important carbon sequestration functions, do not access carbon credit markets (see Box 20.3 for examples). There is a need to address the gender gap in green finance and to rethink the green finance architecture to make it

Box 20.3. Women's Contribution to Climate Change Mitigation and Climate Finance: The Missing Link

Across sub-Saharan Africa, women-led initiatives in forestry and other ecosystems play a key role in carbon sequestration. This box provides a few examples. Despite promising experiences such as the Mikoko Pamoja project described here, only a small fraction of women are accessing the economic rewards derived from carbon credit markets.

In Casamance, Senegal, the association Poumoulindiana, a women's economic interest group, seeks to reforest more than 50 hectares of mangrove trees per year. Mangrove forests are powerful tools for reducing greenhouse gas emissions; they store 50 times more carbon in their soils than tropical forests and 10 times more carbon than temperate forests.

In Mali, UN Women's Climat et Energie Mali (Mali Climate and Energy) project supports women's participation in the biomass value chain through training in the production of improved stoves and sustainable fuels (500 women trained; 150,000 improved stoves and 119 tons of sustainable charcoal briquettes manufactured). Other initiatives include reforestation; management of tree nurseries; improved and efficient carbonization techniques; and establishment of women-led forest management mechanisms. The project has led to the generation of carbon credits linked to the activities to reduce biomass consumption (estimated to be 23,689 tons of reduced emissions of CO₂ equivalent per year or 165,822 tons of CO₂ equivalent over seven years).

The Mikoko Pamoja project in Kenya is an example of a mechanism to reward communities financially for sustainable forestry activities. As the world's first project to fund mangrove conservation through the sale of carbon credits, it uses certification to compensate the community for reducing greenhouse gas emissions through the conservation and expansion of mangrove forests. Since 2017, the project has protected 117 hectares of natural mangrove forest and has planted 10 hectares of mangroves. The plan is to plant 2,000 trees annually over 20 years. The carbon benefits of the Mikoko Pamoja project are conservatively estimated at 2,500 tons of CO₂ per year. In return, in a two-year period, the community received a total of \$24,000 in income that has been reinvested in local projects determined through community consultation.

more inclusive and gender-responsive. Steps in the right direction include improving women's participation in decision-making spaces; developing specific facilities to support grassroots women's access to carbon credit markets and other green finance mechanisms; improving women's information, knowledge, and capacity to access complex processes around green finance; and collecting sex-disaggregated data to track green finance flows along gender lines. In the meantime, leveraging current momentum around impact investments and gender lens investment to stipulate gender and women employment targets in green bonds and other private finance instruments represents more immediate opportunities.

CONCLUSION

The transition to the green economy in sub-Saharan Africa can be harnessed by countries to promote gender-transformative development paths. Strategies should aim at facilitating a transition in which women and girls leapfrog to high-productivity green

jobs and skip the slow climb from one low-productive job to one that is slightly better. The extent to which these promising paths will be realized will depend on the combination of policies and the level of ambition. But whatever the policy combination, action needs to start now—on both short- and long-term strategies—while there is still time to prevent the green transition from being severely biased against women and girls.

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"This 20-chapter book provides an in-depth understanding of the nexus between gender equality and economic development in sub-Saharan Africa. As a female leader in fiscal policy and tax administration, I find it refreshing reading key parts of this book. It is a must read for practitioners, researchers, and policymakers who wish to navigate the fine lines between fiscal, tax-policy design, budgeting, expenditures, and gender."

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